NESTED-SPACE COSMOLOGY

The Geometry of a Universe

Regions, boundaries and what an interior observer can learn

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Abstract

NSC proposes a geometric account of the world seen from inside a region. Its organizing hypothesis is that approximately spherical, sourced regions can form a nested hierarchy: a child inherits physical laws and state data, while its boundary and unresolved surroundings contribute to what an interior observer measures. The atlas develops that hypothesis through explicit actions, matched spacetimes, inherited fields and observable relations.

The sphere provides the simplest isotropic closed interface. The isoperimetric inequality makes its energy advantage precise for positive isotropic surface tension at fixed volume. The physical programme then asks which gravitational sources preserve, deform or replace that geometry. Finite and infinite operator constructions establish normalized inheritance and controlled regional reduction. Their spacetime realization is developed through sourced interfaces and causal maps, with the contributions of earlier authors identified alongside the equations that enter the construction.

The central worked family uses Einstein gravity, a scalar coupled to a four-form, and charged membranes. One action relates regional vacuum energies, supports a constrained three-region initial atlas and determines an evolving protected parent. The inherited scalar supplies both a vacuum floor and an oscillating excitation. Its perturbations grow; conservation ties local concentration to an ambient reservoir; finite nonlinear examples form contracting structures. The same evolved sources determine clocks, lensing, image delays and freely falling motion. Scale-independent relations connect these local measurements to the parent vacuum, and a motion-calibrated optical forecast passes its withheld synthetic test.

Further atlas branches connect stable material interfaces to gravitational scattering, unresolved fluid motion to resolved momentum flux, and a geometry-derived quantum interaction to the cosmohedron energy kernel. Effective-gravity calculations determine when an eliminated or averaged source has vacuum form and when it does not. The quiet-cap distance law is tested against 596 Pantheon+ light curves and disagrees with their brightness-independent contrast by -0.1046 ± 0.0144 magnitudes.

The resulting book connects geometry, inherited state, stress, structure and interior observation. Its completed results establish specified mathematical and physical constructions. Its cosmological completion requires one preparation and population history to fix the remaining freedoms and survive joint observational tests. The absolute vacuum scale, early thermal history, matter asymmetry and a geometry-based derivation of quantum probabilities remain explicit parts of that programme.

I. Spherical geometry and inherited law

Geometry is the organizing object. This part states the spherical hypothesis, proves the declared operator forms of inheritance, and determines what survives when unresolved regions are eliminated. The physical measure and state enter alongside the operator: they fix the energy and pressure that a local response can represent.

1. The central geometric proposal and the route through the atlas

The central proposal of this book is that the physical world accessible inside a region can be the local expression of a larger nested geometry. A region contains fields and observers; its surroundings and boundary determine admissible states, exchange and gravitational response. What an interior observer describes as matter, expansion or a delayed signal must therefore be calculated from that connected system.

Douglas's organizing intuition is spherical inheritance through resolution. A smooth envelope can be selected by a simple balance even when the fields it encloses have fine structure. Resolving that structure changes the local description; it need not change the law. Repeating the relation suggests an atlas of parents, children and descendants rather than a collection of unrelated objects. The mathematical question is how to carry geometry, physical state and conservation through each transition.

The intended dark-sector account has two distinguishable channels. Fields inherited from a parent can reside and cluster inside the observed region. Unresolved surrounding geometry can instead contribute through a boundary or effective gravitational response. Both can be electromagnetically unseen in a stated model. Their stress, causal access and observational signatures decide how they enter the interior description. The phrase "outside or inherited" names this programme; the equations identify the actual channel.

The book follows one continuous route:

Spherical geometry → inherited law and state → physical interfaces → vacuum and clustering → clocks, light and gravitational measurements.

Parts I and II establish the geometry, inheritance and causal attachments. Part III gives the central scalar/four-form source history. Part IV follows its energy into concentrations and their surroundings. Part V derives the interior observer's measurements. Part VI develops compact-object, fluid and quantum connections. Part VII determines what effective gravitational sources these operations actually produce. Part VIII brings the results together and states the remaining physical closures. The detailed characteristic-field development and reproducibility material are retained in the appendices.

An atlas can contain alternative branches within one book. Each branch retains its action, state and domain, and a connection between branches is made by a matching or reduction equation. The main scalar history and the later static material-interface example are therefore presented as distinct worked constructions. Their shared geometric questions are visible without assuming that their different wall sources already form one cosmic history.

This synthesis builds with other researchers' mathematics. The table names the contributors at the points where their constructions enter; the subsequent chapters give the primary sources and the calculated connection.

Mathematical contribution	Contributors and role in this atlas
Surface-energy selection	Isoperimetric and Wulff theory; Neumayer supplies the isotropic/anisotropic distinction
Sourced interfaces and energy balance	Israel, Brown--York and Hayward supply matching, boundary stress and mass/work laws
Cosmological attachment	Einstein--Straus, with Mars, Mena and Vera, supplies exact mass compensation in an expanding parent
Surrounding gravitational response	Shiromizu, Maeda and Sasaki; Maartens and Koyama; Garriga and Tanaka supply explicit bulk-to-local examples
Regular material interiors	Mazur and Mottola; Visser and Wiltshire; Jampolski and Rezzolla supply gravastar, stability and nested-interface precedents
Vacuum transitions and scalar response	Brown--Teitelboim, Bousso--Polchinski, Hirano and Kaloper--Sorbo supply charged flux and scalar couplings
Regular preparation and continuation	Coleman--De Luccia and Dong--Harlow supply the geometric preparation framework
Scalar concentration and growth	Seidel--Suen, Moroz--Penrose--Tod, Hwang--Noh and the cited scalar-field literature supply cooling, profiles and perturbation methods
Light and gravitational waves	Frittelli--Newman, Perlick, Seitz--Schneider--Ehlers, Pani and collaborators, and Martel--Poisson supply observer and perturbation machinery
Flow and quantum geometry	Hamilton--Lisle, Visser, Bredberg and collaborators, the cited OpenAI flow result, and Arkani-Hamed--Figueiredo--Vazao supply the separate flow and energy-kernel constructions

The atlas's contribution is the explicit work connecting these ingredients: compatible sourced regions, inherited state transport, conserved supply, finite evolution and linked observer predictions. A result is

stated where its derivation closes. A remaining connection is stated as a specific mathematical problem, so that later work can extend the same argument.

The first motif contains two spatially separated compact regions in a common parent, unequal children, and a further generation. Figure 1 distinguishes that proposal from a particular calculated attachment.

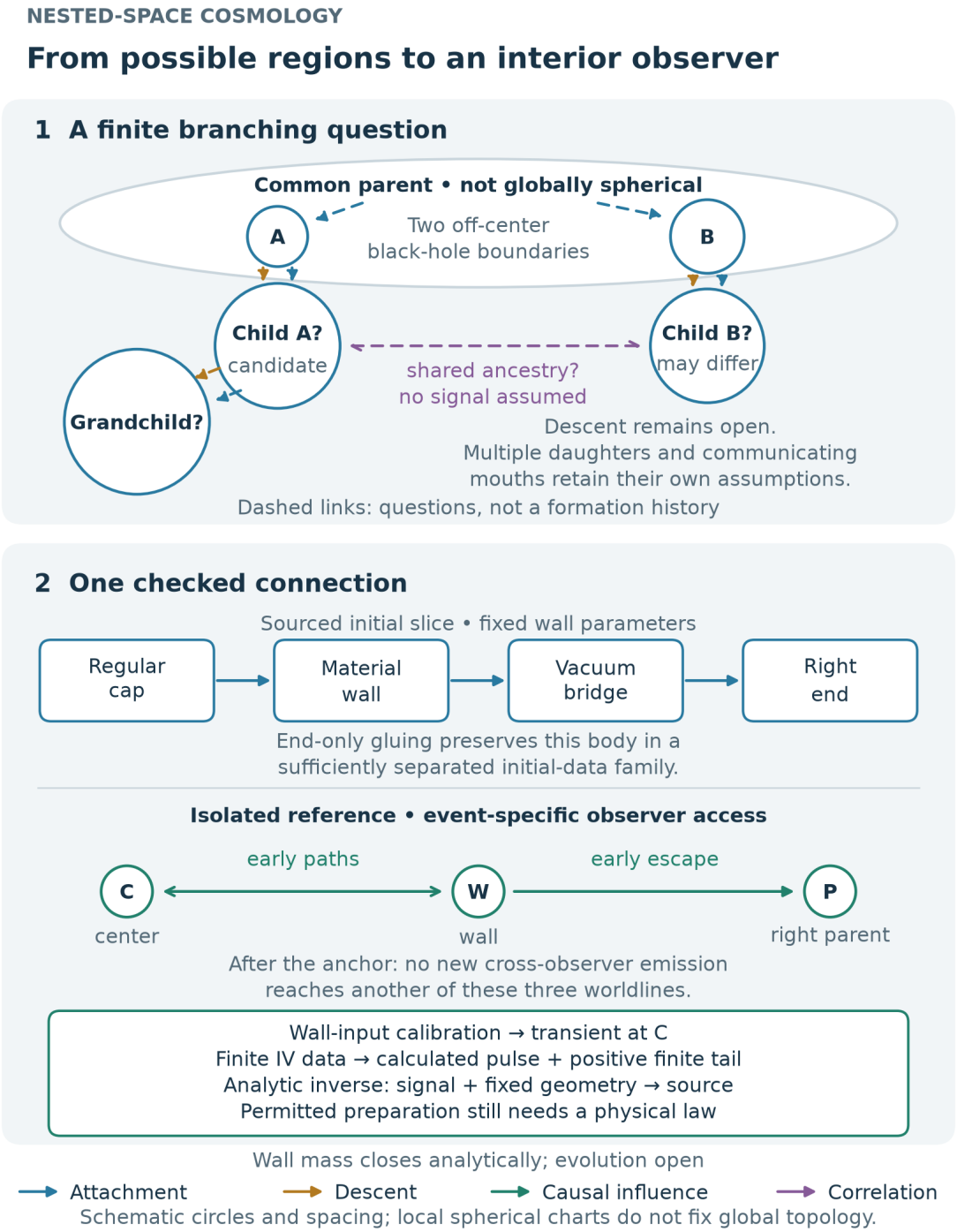


Figure 1. Possible descent (upper panel) and the calculated cap--wall--bridge connection with its observer/field reference (lower panel). Colors distinguish attachment, descent, influence and correlation. Circles show the regional hypothesis; their spacing does not specify a global embedding or scale. The figure record identifies the evidence for each connection.

2. Why begin with the sphere? Energy, relaxation and inheritance

The sphere is the starting geometry because isotropic surface energy has an exact minimizer. In Euclidean three-space, area A and enclosed volume V obey the established isoperimetric inequality

$$A^3 \geq 36\pi V^2.$$

Equality selects a ball, up to its position and size. For constant positive surface tension γ, the energy $E_{surface}=\gamma\ A$ is minimized by a sphere at fixed V. The isotropic case and its anisotropic Wulff generalization are stated in Neumayer, Section 1.1. This is the precise energetic basis for the spherical organizing idea.

A simple relaxation model also displays the direction of change. Let h be the sum of the outward principal curvatures, $\bar{h}$ its surface average, and choose normal velocity $v_n=-M\gamma(h-\bar{h})$, with positive mobility M. The first variations of volume and area give

$$\frac{dV}{dt} = 0$$
$$\frac{dE_{\text{surface}}}{dt} = -\mathcal{M}\gamma^2 \int_{\partial V} (h-\overline{h})^2\, dA \leq 0.$$

Indeed, the integral of $h-\bar{h}$ vanishes, and subtracting $\bar{h}$ in the area variation leaves the displayed square. This specifies an admissible dissipative law and its energy budget. Volume-preserving mean-curvature flow has its own developed theory, beginning with Huisken's 1987 treatment. This illustrative surface law is not substituted for the relativistic membrane equations used later in the atlas.

Simplicity here has a mathematical meaning: no angular direction is preferred by the surface energy. Finer resolution of a smooth sphere does not add a new sustaining force. Direction-dependent tension, rotation, gravity and driving contribute additional terms to the energy and dynamics. They determine whether a sphere persists, deforms or is replaced by another configuration. Entropy production alone fixes neither a universal geometry nor a maximum rate of dispersal.

NSC's geometric hypothesis. Persistent approximately spherical regions form the envelopes of a physical hierarchy. Across admissible parent--child transitions, a shared law is inherited while scale, state and resolved structure change. Repeated spherical inheritance is the proposed architecture to develop and test throughout this book.

The energy theorem supplies one reason for that architecture. Its application to every universe or an unlimited spacetime hierarchy would add a universality claim requiring a physical transition and composition law. The finite and infinite operator results below establish their stated forms of inheritance; the sourced geometries establish finite physical connections. The atlas owner records how those constructions develop the hypothesis.

A spherical boundary is a two-dimensional surface with area $4\pi R^2$. A spatial interior may be a three-ball; a closed spatial universe may have three-sphere topology. The local spherical spacetime form is

$$ds^2 = g_{ab}(x)\, dx^a dx^b + R(x)^2 d\Omega^2.$$

Here g_{ab} is the Lorentzian metric on the two-dimensional orbit space and R is areal radius. The atlas uses signature $(-+++)$. Proper radial distance, areal radius and an embedding diagram are different data. Several compact regions in one parent generally require a nonspherical parent even when each local interface has a spherical approximation. The next constructions specify which geometry is used at each level.

3. The same argument across operator, spacetime and observer descriptions

The inherited relation, the physical geometry and the interior response are three stages of one programme. The first supplies a precise law to transport. The second supplies the regions, sources and states on which that law acts. The third calculates how an observer distinguishes the resulting histories.

Stage	Mathematical owner	Role in the book
Inherited law and spectral relations	Preserved foundation	Operator identities, boundary maps and source constructions
Regional transfer and geometric feedback	Finite spherical companion	A specified action/state example of transfer and memory
Sourced spacetime and observation	Atlas evidence and this manuscript	Compatible attachments, physical inheritance, evolution and observer relations

The book connects these stages where an actual field equation, state, clock and matching condition identify their mathematical objects. Elimination of unresolved modes is used both in the inheritance calculator and in physical boundary problems; its physical interpretation follows from the operator and state supplied in each case.

Earlier constructions retain their source-bound editions and attribution. The present synthesis begins with geometry and works outward to what an interior observer can measure. The source of expansion, the supply of localized structure and the meaning of a quantum mode are all pursued through that same sequence.

4. Four arrows that must not be confused

Each region in the atlas carries a geometry, source or state, clock convention, dimensionless parameters and validity domain. Each connection states the borrowed mathematics and the additional condition still needed.

Connection	Mathematical question	What it does not establish by itself
Descent	Where do a child's initial data originate?	A traversable path
Geometric attachment	Which boundaries share admissible metric and source data?	Formation or stability
Causal influence	Can an intervention reach an observer within the stated domain?	A nonzero field response without state and coupling
Correlation	Which information is shared through state or ancestry?	Communication between siblings

Two regions may inherit correlated data without ever exchanging a later signal. An attached interior can lie beyond the causal reach of a specified parent observer. Conversely, communicating mouths require their own causal and material assumptions; they cannot be inserted into a disconnected offspring model as an unlabeled extra arrow.

The multiple-daughter geometry of Barrabès and Frolov (hep-th/9511136v1), the charged semiclassical multi-mouth route of Emparan et al. (2012.07821v2), and quantum-state coupling distributions discussed by Giddings and Turiaci (2004.02900v2) remain separate library entries. A state's coupling distribution is not itself a signal exchanged between its branches. Their source cards and unresolved connections are recorded in the mathematical library.

5. A precise finite inheritance construction

Derivation and calculation: exact finite record, frozen snapshot, and proof and domain.

The foundation proposes an inherited normalized law, $T_\Theta^* D_\Theta = D_\Theta$. One controlled meaning is available without postulating an infinite cosmos. Fix a Hermitian d-dimensional matrix H, a nonzero link B, a positive scale ratio Ω, and a finite depth N. For regional labels $j=0,\ldots,N-1$, define a block Hermitian matrix by the following entries where their indices exist:

$$(J_{m,N})_{jj} = \Omega^{m+j} H$$
$$(J_{m,N})_{j,j+1} = \Omega^{m+j} B.$$

The lower off-diagonal blocks are the adjoints. Let U_m consistently relabel the equal-sized regional frames. Then

$$J_{m+1,N} = \Omega U_m J_{m,N} U_m^\dagger$$

$$\Omega^{-1} U_m^\dagger J_{m+1,N} U_m = J_{m,N}.$$

Each block acquires the same extra factor, so the identity follows directly. For fixed depth,

$$\mathrm{spec}(\Omega^{-m} J_{m,N}) = \mathrm{spec}(J_{0,N}).$$

This is a finite realization of normalized inheritance under chosen scaling data. It does not derive Ω, establish a physical transition through a horizon, or show that the universe selects this law. Changing N changes the boundary problem and can change the dressed spectrum. The finite identity is not a proof of cosmic natural selection.

The shared law also does not fix identical states. A fermionic covariance $0 \le C \le I$ evolves as

$$C(t) = e^{-iJt} C(0) e^{iJt}.$$

Regional occupations and cross correlations can differ under the same J. Thus “same laws,” “same state,” and “same observed constants” require separate arguments. A physical transition would have to determine the relevant state and dimensionless data, rather than obtaining them from the normalization identity alone.

Douglas's proposed geometric interpretation is that an approximately spherical region can be the resolved envelope of finer nested structure. Repeated transitions preserve the inherited law while changes of depth, state and surroundings compound in the measured response. The next section's nested reduction provides a finite calculator for that compounding. The following theorem extends this to an infinite operator and a derived mode-count scaling relation. Its spacetime geometry, physical scale ratios and observer resolution remain calculation targets in the atlas programme.

5.1. An infinite inherited law with a controlled local response

Take a rooted tree with $b \ge 2$ children per vertex, an inherited positive Hermitian $d \times d$ matrix H, link B, and $\Omega > 1$. At depth n, the diagonal block is $\Omega^n H$ and the block to each child is $\Omega^n B$; reverse links are adjoints. Set

$$h = \lambda_{\min}(H)$$
$$c = \sqrt{b/\Omega}\,\|B\|$$
$$h > 2c.$$

On square-summable regional fields specify the finite-energy form domain

$$\mathcal{Q} = \left\{ u : \sum_{v} \Omega^{|v|} \|u_v\|^2 < \infty \right\}.$$

Theorem. The corresponding positive closed form defines a self-adjoint finite-energy realization D with compact inverse. Every isolated child subtree is Ω D after relabeling, so normalized inheritance holds at unlimited depth. Its root response, for $\lambda \geq 0$, satisfies

$$R(\lambda) = \left[H + \lambda I - \frac{b}{\Omega} B R(\lambda/\Omega) B^{\dagger} \right]^{-1}.$$

For $\gamma = (h + \sqrt{h^2 - 4c^2})/2$ and $r = c^2/\gamma^2 < 1$, this response is unique in the positive class $R(\lambda) \leq (\gamma+\lambda)^{-1} I$. Retaining depths $0,\ldots,N$ gives

$$0 \leq R_N(\lambda) \leq R(\lambda)$$
$$\|R(\lambda) - R_N(\lambda)\| \leq r^{N+1}/\gamma.$$

If B is invertible, every added generation increases the response strictly. The eigenvalue-counting exponent is

$$\lim_{E \to \infty} \frac{\log N_D(E)}{\log E} = \frac{\log b}{\log \Omega}.$$

Proof. Let S multiply depth-n coordinates by $\Omega^{n/2}$, and let C have diagonal H and child blocks $\Omega^{-1/2} B$. Cauchy--Schwarz bounds the child-sum map by c, hence $(h-2c)I \leq C \leq (\|H\| + 2c) I$. The form $q[u] = \langle Su, CSu \rangle$ is closed on Q, with a norm equivalent to the complete graph norm of S. Its operator has

$$D^{-1} = S^{-1} C^{-1} S^{-1}$$
$$\mathrm{Dom}(D) = \{ u \in \mathcal{Q} : CSu \in \mathcal{Q} \}.$$

The diagonal entries of S^{-1} tend to zero, with finite multiplicity at each depth, so the inverse is compact. This selects the finite-energy Friedrichs realization; essential self-adjointness under other possible conditions at infinity is not assumed.

Relabeling a child multiplies every block and its closed form by Ω. Its root inverse is therefore $\Omega^{-1} R(\lambda/\Omega)$. Eliminating the b children yields the recursion, including all descendant responses. Finite-support fields are a form core. The variational expression for $\langle f, (D+\lambda I)^{-1} f \rangle$ is the supremum of $2\mathrm{Re}\langle f, u \rangle - q[u] - \lambda \|u\|^2$ over that domain. Successive finite-depth restrictions increase to this supremum, proving Loewner and norm convergence of the root matrices.

Since $h - c^2/\gamma = \gamma$, every Schur denominator is bounded below by $(\gamma+\lambda)I$. The inverse identity bounds the outgoing difference of two responses by $c^2/(\gamma+\lambda)^2$ times their incoming difference. Its uniform factor $r < 1$ proves uniqueness in the stated class. Iterating $N+1$ times from a zero terminal response gives the depth error. Inverse order and positive congruence give monotonicity, strictly when B is invertible.

Finally, the form bounds compare D with positive constant multiples of S^2. The eigenvalue Ω^n of S^2 has multiplicity db^n. Min--max bounds $N_D(E)$ between geometric sums at constant multiples of E, proving the exponent. This completes the proof. The full domain argument and calculator give its detailed implementation.

For a second-order spectral interpretation, $d_s=2\log b/\log\Omega$; a first-order interpretation instead uses the counting exponent itself. These are spectral dimensions of this operator. A spacetime or embedding dimension requires its own geometric construction.

For $H=3$, $B=1$, $b=3$, $\Omega=4$, the exact static response is

$$R(0)=\frac{2}{3+\sqrt{6}}$$
$$D_{\mathrm{eff}}(0)=\frac{3+\sqrt{6}}{2}=2.724744871391589.$$

Thus the inherited coefficient3 changes by -0.275255128608411. Depths0 through6 give $R_6(0)=0.367006802721088$, with analytic depth bound 3.94041×10^{-8}; the closed-form difference is 3.54235×10^{-8}. Rounding is separate from this bound. The second-order spectral dimension is $\log3/\log2=1.584962500721156$. Four small controls compare full finite trees, noncommuting links, branch/scale factors, the exact scalar root and spectral bounds. They support the calculator; the infinite result follows from the proof. A metric-dependent action and physical source identification remain needed to connect this response to cosmological Lambda.

5.2. Physical measure, pressure and a measurable nesting ratio

An inherited canonical operator does not alone specify the physical measure. For an explicit three-dimensional lift, give each depth-n edge length Lq^n and section Aq^{2n}, with $0<q<1$. Half of each incident edge's volume supplies the vertex kinetic weight; the root has a fixed parent port. Direct expansion gives

$$m_0=\frac{AL}{2}(q^{-3}+b),\quad H=\frac{A(b+q^{-1})}{Lm_0},\quad B=-\frac{A}{Lm_0q^{3/2}},\quad \Omega=q^{-2}.$$

The positive margin is $A(\sqrt{b}-q^{-1/2})^2/(Lm_0)$, so Section 5.1 applies for $bq\neq1$. Its physical static root compliance is

$$\chi_0=\frac{R(0)}{m_0}=\frac{L}{A\max(b,q^{-1})}.$$

Under a common spatial dilation, $L\mapsto aL$ and $A\mapsto a^2A$. Thus $M(a)=a^3M_0$, $K(a)=aK_0$, $D(a)=a^{-2}D_0$, and $\chi(a)=\chi_0/a$. The length-only action instead has $\chi(a)=a\chi_0$; the same operator scaling can conceal different physical measures. The owned radial spatial metric $h=r^2(Q^2dx^2+d\Omega^2)$ supplies the volume scaling directly: $\int r^3Q\varphi_i\varphi_j dx$ scales as a^3 and $\int(r/Q)\varphi'_i\varphi'_j dx$ as a at fixed Q.

Pressure proposition. Declare the kinetic-plus-gradient action

$$S_\phi=\frac{1}{2}\int dt\left[\frac{a^3}{N}\dot\phi^T M_0\dot\phi-Na\phi^T K_0\phi\right].$$

Let $T_0=\pi^T M_0^{-1}\pi/2$, $G_0=\varphi^T K_0\varphi/2$ and $V=a^3V_c$. Its energy at fixed canonical data is $E=a^{-3}T_0+aG_0$. Lapse and scale variation, or direct Legendre transformation followed by $p=-\partial E/\partial V$, gives

$$\rho=\frac{E_{\mathrm{kin}}+E_{\mathrm{grad}}}{V}$$
$$p=\frac{E_{\mathrm{kin}}-E_{\mathrm{grad}}/3}{V}.$$

Both energies are nonnegative. For every nonzero finite-energy state,

$$\rho + p > 0$$
$$\rho + 3p = 4E_{\mathrm{kin}}/V \geq 0.$$

The form domain is preserved by multiplication by these positive scale factors, so the proof also holds for the infinite hierarchy. Source-free Hamiltonian evolution gives $\dot{\rho}+3H(\rho+p)=0$ in homogeneous proper time. This sector cannot be a nonzero vacuum source. If it alone sources a flat Einstein cosmology without a supplied volume term, it predicts $q_{dec}=2E_{kin}/(E_{kin}+E_{grad})\in[0,2]$. The full atlas's gravitational correlations and nonminimal terms require their own metric variation; this proposition does not replace them.

Dynamic root observable. With zero initial data and a known compact root force, measure the root field in proper time. Its Laplace response is $\chi(s)=R(s^2)/m_0$. Differentiating the exact Schur equation at zero gives

$$Z(s) = \chi(s)^{-1} = \chi_0^{-1} + \mu_* s^2 + O(s^4)$$
$$\mu_* = \frac{m_0}{1 - r/\Omega}.$$

Independently, summing the entire static profile's kinetic weight gives $\mu_*=m_0/(1-q/b)$ for $bq>1$ and $m_0/(1-bq^3)$ for $bq<1$. Consequently the response scale $\tau_*^2=\chi_0\mu_*$ obeys

$$\frac{\tau_*^2}{L^2} = \frac{q^{-3} + b}{2(b - q)} \quad (bq > 1)$$

$$\frac{\tau_*^2}{L^2} = \frac{q^{-2} + bq}{2(1 - bq^3)} \quad (bq < 1).$$

For $b=3$, $q=1/2$ and $q=2/5$ give the same static compliance $L/(3A)$ but different dimensionless dynamic values: 2.2 and 3.581730769230769. The expression is strictly decreasing in q on the admitted branches, so an independently known b,L and calibrated force permit a unique nesting-ratio inference. An unknown source or length would restore degeneracy. The scale τ_* is a response coefficient, not a first-arrival time.

The derivation and independent checks include the profile sum, differentiated Schur recursion, an explicit $O(s^4)$ Laplace remainder bound and root-work accounting. These are predictions of the stated geometric action, with no observational calibration. A compatible cap--wall/parent metric and its gravitational source remain necessary for a cosmological Lambda prediction.

5.3. A response shape that recovers the nesting ratio

For the scalar volume action above, let mu_k denote the kth root spectral moment and set g=b*abs(B)^2. Direct multiplication on two inherited levels gives

$$\mu_1 = H, \quad \mu_2 = H^2 + g, \quad \mu_3 = H^3 + gH(2 + \Omega).$$

All moments exist because every finite power of D applied to the root has finite support. The positive root spectral measure yields

$$R(\lambda) = \lambda^{-1} - \mu_1\lambda^{-2} + \mu_2\lambda^{-3} - \mu_3\lambda^{-4} + \varepsilon(\lambda)$$
$$0 \le \varepsilon(\lambda) \le \mu_4\lambda^{-5}$$

$$\mu_4 = (H^2 + g)^2 + gH^2(1+\Omega)^2 + \Omega^2 g^2.$$

For the wave response, lambda=s^2. Write an unknown constant gain and uniform clock factor into its measured coefficients beta2, beta4, beta6, beta8 multiplying s^-2, s^-4, s^-6, s^-8. Then

$$\Theta = \frac{\beta_2\beta_6}{\beta_4^2}$$
$$\Psi = \frac{\beta_2^2\beta_8}{\beta_4^3}$$
$$\Omega = \frac{\Psi - 1}{\Theta - 1} - 2$$
$$q = \Omega^{-1/2}.$$

Gain, root kinetic normalization and uniform clock calibration cancel. At b=3, q=1/2 gives (Theta,Psi)=(49/25,169/25); q=2/5 gives (617/242,13343/968). The recovered ratios are exactly 1/2 and 2/5, despite the equal static compliance. At fixed b, Theta alone is strictly monotone in q; the pair determines q even without a known b.

There is also a held-out consequence. The fifth coefficient beta10 must obey the parameter-free relation

$$\Upsilon = \frac{\beta_2^3\beta_{10}}{\beta_4^4} = \Theta^2 + (\Theta-1)(1+\Omega)^2 + \Omega^2(\Theta-1)^2.$$

A positive tree with altered child--grandchild links can retain the first four response coefficients and violate this fifth one. The relation thus tests continuation of the inherited law at the next level, using a quantity not used in the inversion.

This is a conditional inverse of the specified transfer function. An arbitrary unknown force spectrum does not cancel. Finite noisy data need their own coefficient errors. The complete proof and controls keep these requirements explicit. This connects the inherited operator to an observable relationship, with its physical atlas identification still to be established. The compact calculation and source hashes retain these exact model examples alongside the separately scoped atlas composition below.

6. What an unresolved surrounding region can leave locally

Split a specified operator into retained and complementary blocks *A,D* with link *B*. For an admissible complex spectral parameter *z*, elimination gives the usual self-energy

$$\Sigma(z) = B(zI - D)^{-1}B^{\dagger}.$$

The retained resolvent is $[zI\text{-}A\text{-}\Sigma(z)]^{-1}$. Nested elimination is associative when the required inverses exist. For the finite inherited chain its normalized recurrence is

$$\Gamma_j(x) = xI - H - \Omega^{-1}B\Gamma_{j+1}(x/\Omega)^{-1}B^{\dagger}.$$

The factor $1/\Omega$ is essential. This is borrowed finite linear algebra applied to the declared family, not an additional force law. To interpret Σ as a cosmological exterior response requires a physical operator, field state and boundary map. Sections A.24–A.25 give a separate finite Hilbert-space realization of this reduction, without identifying its complementary modes with a cosmological exterior.

6.1. The local effective Lambda target

The intended cosmological target is a local effective Λ arising from inherited Schur memory: surrounding structure that a region does not resolve contributes through $\Sigma(z)=B(zI-D)^{-1}B^{\dagger}$ to its local response. A vacuum-form local gravitational part would be written $T^{eff}_{\mu\nu}=-(\Lambda_{eff}/\kappa_4)g_{\mu\nu}$, with $\kappa_4=8\pi G$. Identifying the physical exterior operator, its state and boundary map, and obtaining that local part from the causal response remain the missing connections. The finite Schur identity fixes the reduction; it does not yet provide a measured cosmological value. The Schur-memory owner keeps this target and its retained spectral and geometric dictionary.

II. Physical regions, interfaces and causal access

A nested drawing becomes a spacetime through its sources and junctions. The constructions in this part assemble regular centres, material interfaces, shared parents and named observers. They show both successful attachments and precise exclusions. Causal access is established before a boundary or remote region is used as a source for an observer.

7. Start with two holes, without manufacturing two children

The explicit parent benchmark uses time-symmetric Brill–Lindquist vacuum initial data,

$$h_{ij} = \psi^4 \delta_{ij}$$
$$K_{ij} = 0$$
$$\psi = 1 + \frac{m_A}{2r_A} + \frac{m_B}{2r_B}.$$

The recorded choice is $m_A=1$, $m_B=1.5$, separation $d=12$. It has outer ADM mass 2.5 and two inner-end masses 1.0625 and 1.5625. These are three asymptotically Euclidean ends of an initial slice. Extra ends do not by themselves become cosmological children. The companion hole supplies nonspherical structure, so an exact isolated Schwarzschild collar cannot simply be assumed at either puncture. The parent record and owning analysis retain the construction and checks.

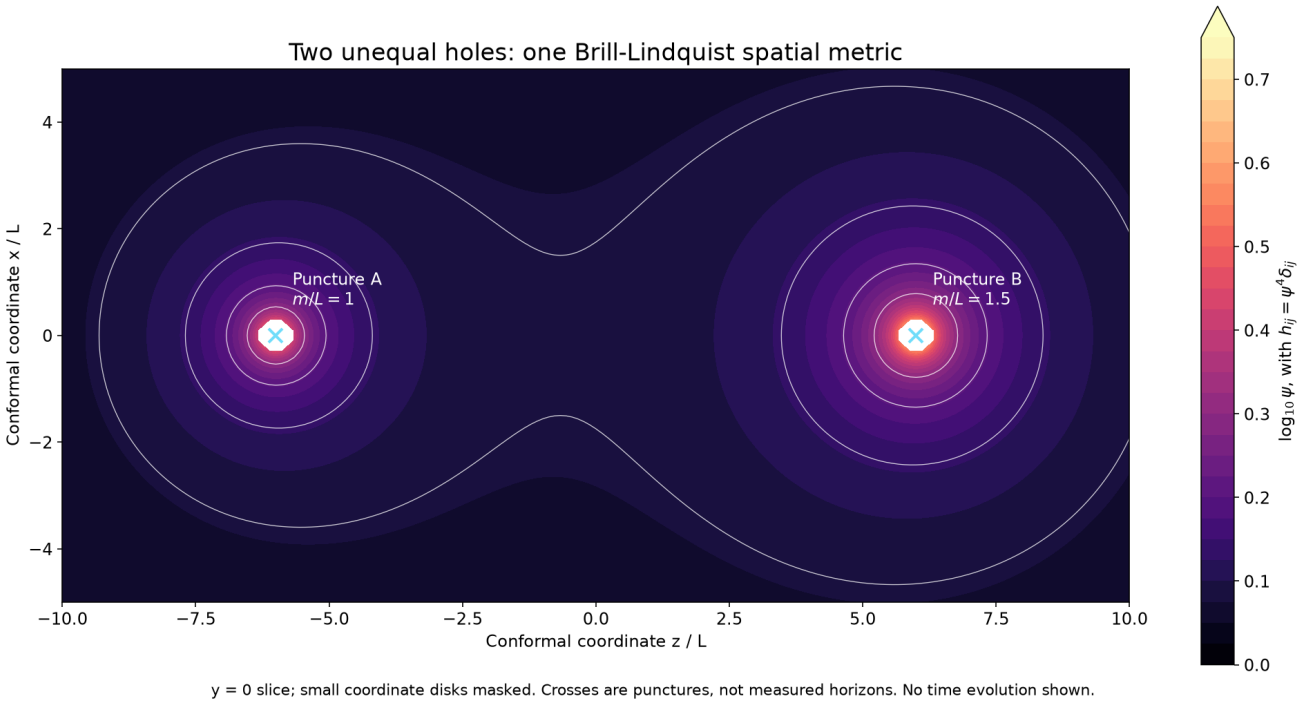


Figure 2. The explicit two-hole initial-data benchmark. This is a nonspherical common parent with unequal ends, not a solved evolution that creates children. Local spherical models must earn an attachment to it or belong to a separate parent family.

The alternative parent route imports an existence result instead of re-solving its gluing equations. Chruściel, Corvino and Isenberg (0909.1101v1) provide sufficiently separated scalar-flat bodies with retained interior regions and a common Schwarzschild outer end. Their more general vacuum-end theorem (1004.1355v1) allows appropriate non-time-symmetric ends. These establish separate families, not a modification that leaves the displayed Brill–Lindquist data unchanged. No explicit sufficient separation is certified here.

8. A failed attachment can sharpen the map

Derivation and calculation: radiation junction, moving-boundary conservation/control, and source assumptions.

The first attachment screen pairs a closed radiation FLRW region with a vacuum exterior and a pure constant-tension wall. For a comoving boundary, $a=A\sin\eta$, $t=A(1-\cos\eta)$, and $R=a\sin\chi$. The junction condition would require, over an interval of changing R,

$$k^2R^4 - 2\cos\chi\, kR^3 + 2MR - A^2\sin^4\chi = 0.$$

For nonzero $A\sin\chi$, this cannot be a polynomial identity on that interval. The recorded $\chi=\pi/3$, $M=0.4A$ test exposes a source incompatibility, not a failure of all interiors.

Could a moving wall repair the same model? Intrinsic conservation answers before a new simulation is needed. Fixed pure tension has zero intrinsic stress divergence, so the Codazzi relation requires zero net normal energy flux. In the fluid rest frame,

$$T(U,n) = (\rho + p)\gamma^2 v.$$

A vacuum exterior contributes no flux and radiation has $\rho+p>0$; therefore $v=0$. Motion alone cannot rescue these assumptions. This is a compatibility application of an established conservation law.

Allowing a different surface source, exterior state or flux would define a different candidate requiring its own test.

9. A distinct fixed-source wall that does connect

For the next branch, take de Sitter inside and Schwarzschild outside,

$$f_- = 1 - H^2R^2$$
$$f_+ = 1 - \frac{2M}{R}.$$

In $G=c=1$ units let $\kappa=4\pi\sigma>0$, with surface pressure $P=-\sigma$. The signed angular extrinsic curvatures obey

$$\beta_- - \beta_+ = \kappa R$$
$$\beta_\pm = \varepsilon_\pm \sqrt{\dot{R}^2 + f_\pm}\,.$$

Here the dot is wall proper time and $\varepsilon_\pm$ fixes the normal orientation. Squaring and then forgetting those signs would lose physical branches. The useful explicit expressions are

$$\beta_- = \frac{M}{\kappa R^2} + \frac{\kappa^2 - H^2}{2\kappa} R$$

$$\beta_+ = \frac{M}{\kappa R^2} - \frac{\kappa^2 + H^2}{2\kappa} R$$
$$\dot{R}^2 = \beta_-^2 - f_-\,.$$

We hold $H\ell=1$, $\kappa\ell=2$, $M/\ell=0.4$ fixed. On the tested expanding segment $1.2\le R/\ell\le1.8$, $\beta_->0$ and $\beta_+<0$. The wall proper duration is 0.3718706103ℓ. Direct unsquared and pressure checks, including a separate Christoffel-based calculation, agree within the recorded numerical residuals. The wall's areal-radius rate is not a local three-speed. This published wall family is classified more fully in Section 11; the source has not been retuned to make the segment work.

10. A sourced initial slice and a conditional common parent

Derivation and calculation: slice constraints, parent assembly, slice derivation, and end-only gluing argument.

At $R_w=1.5\ell$, the interior is a flat expanding de Sitter cap: a regular spatial three-ball with $h=dl^2+R^2d\Omega^2$, $R'=1$ and $K^i_{\ j}=-H\delta^i_{\ j}$. We use $K=-L_T\,h/2$, where T is the future unit slice normal. Define $V=T(R)$. The cap has $V_-=HR$. Its slice determines the wall boost,

$$\gamma = \frac{\beta_- - HR\dot{R}}{1 - H^2R^2}$$
$$\gamma v = \frac{\dot{R} - HR\beta_-}{1 - H^2R^2}.$$

The exterior inherits

$$R'_{+} = \gamma\beta_{+} - \gamma v\dot{R}$$
$$V_{+} = \gamma\dot{R} - \gamma v\beta_{+}.$$

At this event $\gamma=1.00926203$, $v=0.13516599$, $R'_{+}=-2.02778609$, and $V_{+}=1.90925372$. The exterior slice is therefore not time symmetric at the material wall.

A fixed smooth slicing function tapers $V(R)$ to zero at $R_0=\ell$, before the Schwarzschild throat $2M=0.8\ell$. In the transition,

$$R' = -\sqrt{f_{+} + V^2}$$
$$K^{\theta}{}_{\theta} = -\frac{V}{R}$$
$$K^{l}{}_{l} = -\frac{dV}{dR}.$$

These data describe a slice of the same vacuum Schwarzschild spacetime. They do not interpolate the matter source. Its mass identity is

$$M = \frac{R}{2}(1 + V^2 - R'^2).$$

Where $V=K=0$, the slice continues smoothly through a bridge using $R=2M\cosh^2 u$, with the cap on the negative-u side and the selected parent end on the positive-u side.

The distributional shell constraints are equally necessary. The surface energy and radial momentum on this slice are $E_{\Sigma}=\sigma\gamma$, $J_{\Sigma}=\sigma\gamma\, v$, giving

$$[R'] = -4\pi R E_{\Sigma}$$
$$[V] = 4\pi R J_{\Sigma}.$$

Their recorded residuals are below 2×10^{-15} in normalized units; independent finite-difference bulk residuals are below 10^{-10}. These are implementation checks, not a global stability estimate.

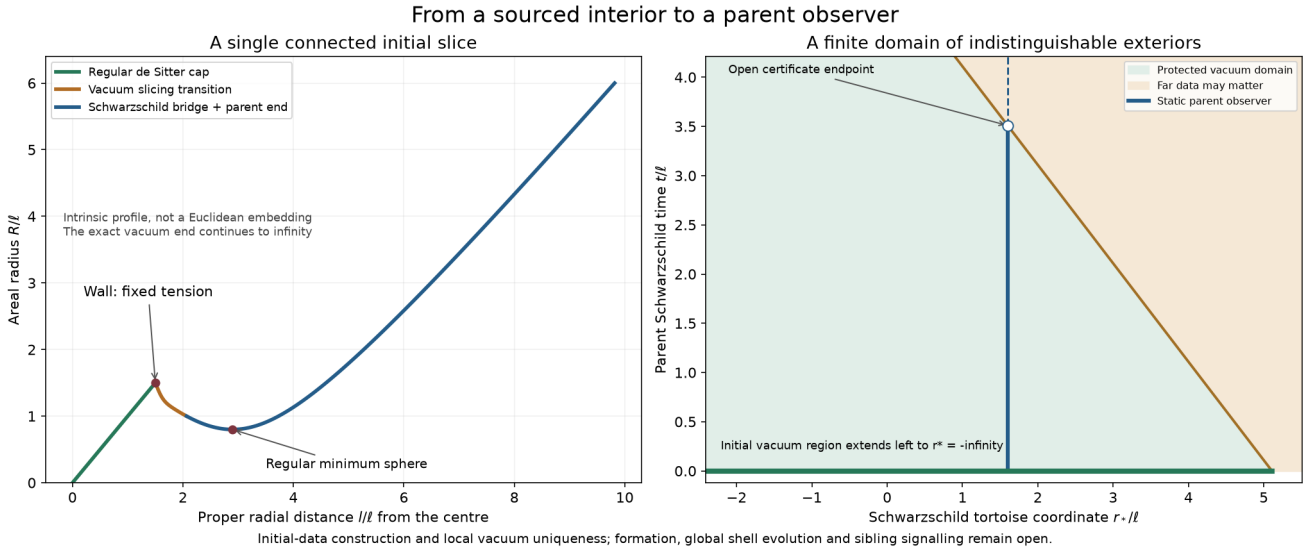


Figure 3. The verified initial connection. Spatial distances are not ages. Because |R'|>1 on part of the chosen slice, the radius profile should not be interpreted as a Euclidean surface of revolution. The protected right-exterior observer interval is a separate causal-domain statement.

The selected right end of this body *A*, of mass *0.4ℓ*, can be used with the selected right end of an ordinary complete Schwarzschild bridge *B*, of mass *0.6ℓ*, in the imported vacuum-end gluing theorem. It acts on smooth vacuum ends and preserves an open overlap. Reattaching unchanged interiors on identical full *(h,K)* data leaves the original shell delta source intact and creates no seam source. We do not apply a smooth global-data corollary directly to a thin shell.

The historical v4 conditional initial-data family has one designated common parent end **and the untouched other end of** *B*: two asymptotic ends in total. That second end has not become a cosmological child. Sufficiently large separation is required but no numerical threshold or explicit multi-body development has been calculated. The achievement is a compatible sourced initial connection, not a regular formation history, a general Einstein–shell evolution theorem or a stable cosmological future.

10.1. Two unequal sourced children in the same theory

The second body can now carry its own cap and wall. Keep $H\ell=1$, $k\ell=2$, $\xi=1/6$, quiet field $\varphi=0$ and $R_w/\ell=1.5$ for both bodies. Use masses $M_A/\ell=.4$, $M_B/\ell=.6$, with vacuum slicing tapers ending at $R_{0,A}/\ell=1$ and $R_{0,B}/\ell=1.3$. Both stops lie outside their respective throats. The mass changes the object state; the taper changes the slice, with the action held fixed.

For B the signed factors are $\beta_-=151/120$, $\beta_+=-209/120$ and $dR/d\tau=\sqrt{40801/120}$. Its wall stress projections, bulk and shell constraints, geometric mass and regular bridge pass the same checks as A. Its local normal-frame speed is $v=.161244881293$, while its areal rate exceeds unity. Each sourced body has one regular centre and one exact AE end. The unequal-input record reuses A's completed moments and evaluates B once; 42 focused controls pass.

Chrusciel, Corvino and Isenberg's end theorem therefore accepts two sourced inputs at sufficiently large separation. Their selected ends become one parent end, without B's former extra end. An exact quiet local solution neighbourhood can also be patched from the already known source solutions and an auxiliary smooth vacuum development, retaining wider identical annuli and trimming staggered cuts inside their common vacuum neighbourhood. The local argument uses smooth vacuum uniqueness only off the walls. It supplies no numerical separation threshold, uniform evolution interval, evaluated connector or general uniqueness theorem for matter shells. Their observer clocks and the cosmological reduction still need that geometry.

10.2. A retained grandchild and an explicit second boundary

A different input A2 now puts two transitions on one explicit spherical initial slice. It keeps the same quiet conformal-scalar and membrane law. The regional metric functions are $f=1-2M/R-H^2R^2$:

Region, ordered outward	M/ℓ	$H\ell$	Connection
Inner regular cap	0	1	Centre to inner wall $R/\ell=.8$
Kottler host	.01	1	Left static branch, black-hole throat, right branch and expanding patch to outer wall $R/\ell=1.5$
Outer Schwarzschild region	.4	0	Left branch, throat $R/\ell=.8$, right AE end

Both positive-tension walls obey the same signed equations,

$$\beta_- = \frac{f_- - f_+ + (kR)^2}{2kR}$$
$$\beta_+ = \beta_- - kR$$
$$\dot{R}^2 = \beta_-^2 - f_- .$$

The expanding inner and outer factors are respectively $(517/640,-507/640)$ and $(727/600,-1073/600)$. The derivative of the same law gives the temporal Israel jump, while the mean worldtube curvature satisfies the normal-force condition. The equal-vacuum inner membrane remains a material source. All supplied vacuum values and masses are part of this state, not outputs selected by a cosmological observation.

In each bulk the assigned slicing function gives

$$R' = \varepsilon\sqrt{f+V^2}, \quad R'' = M/R^2 - H^2R + VV_R, \quad K^l{}_l = -V_R, \quad K^\theta{}_\theta = -V/R.$$

Smooth tapers connect the walls to static bridges and the host's expanding flat radial profile. Signed y with $R=r_b+y^2$ resolves each minimum. The host's two horizons are $r_b/\ell=.020008009615388216$ and $r_c/\ell=.989845863754293$. The expanding profile crosses the latter with $R'=1$. Both Hamiltonian/momentum constraints and both wall distributions are retained. The slice ends analytically at its right AE infinity; $R_{cut}/\ell=4$ is the finite calculation domain.

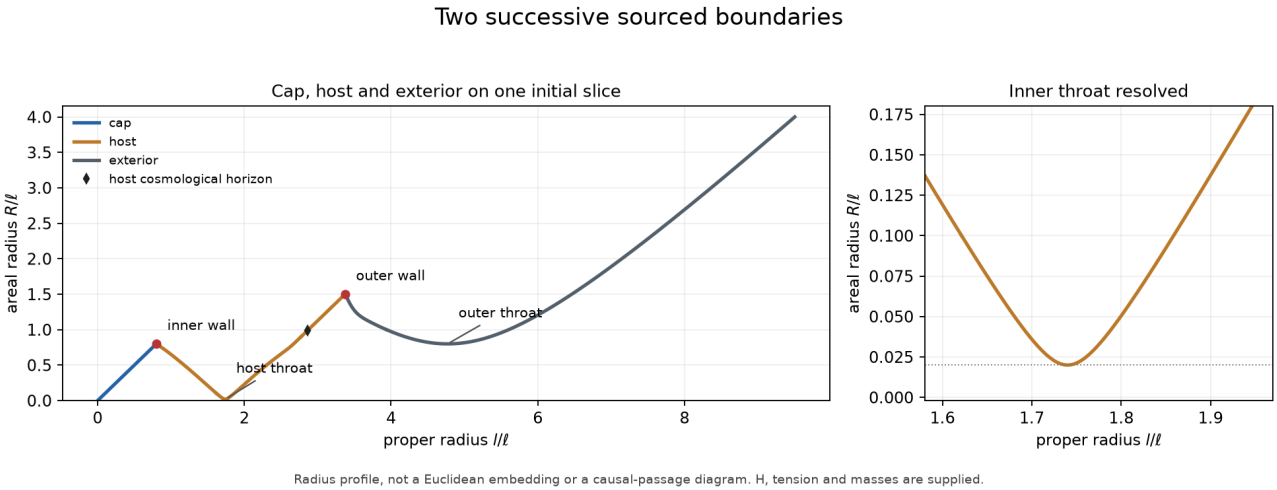


Figure 4. One initial radial chain with two sourced boundaries. The inset resolves the host's positive-radius minimum. Proper radius versus areal radius is not a Euclidean embedding or a causal-passage diagram. The same membrane law holds at both walls; regional state and slicing data are supplied.

Explicit regular bulk charts are available. At a simple horizon C, let $\kappa_C=(1-3H^2C^2)/(2C)$ and $R=C+\kappa_C uw$. Then

$$ds^2 = F du^2 + 2\,du\,dw + R^2 d\Omega^2$$
$$F = \frac{w^2}{R}\left(\frac{1}{C} + H^2(R-C)\right).$$

This is the algebraic form of Lake's coordinates, equations (5)--(10). Independent warped-curvature algebra gives $R_{\mu\nu}=3H^2 g_{\mu\nu}$; the radial determinant is -1. Finite local charts do not identify different cosmological regions or remove the $R=0$ singularity. The single Killing time has $T_l=V/f$ and future normal $N^T=R'/f$, negative on the left static branch. Section 10.5 now matches both finite worldtube segments to this same initial slice using regular overlapping coordinates; a static time alone does not cover the host.

The two-level calculation and slice derivation retain all assumptions and one Gauss 32/64 comparison. Twenty independent controls pass. Its exact vacuum end can replace A in the end construction, giving a conditional initial-data family with two unequal sourced children and a retained grandchild. This establishes finite initial-data composition, not unlimited generations, formation, stability or inherited detector states.

10.3. A central proper-clock and frequency observation

Specify a radial inward geometric-optics flash from an ideal frequency standard comoving with the inner material wall, with negligible backreaction. The receiver follows the cap centre. Set flat-cap time $t=0$ at emission; it is the centre's proper time. For $R_w=.8\ell$ and $H\ell=1$, the null path and wall boost give

$$t_{\mathrm{rec}} = -H^{-1}\log(1-HR_w) = \ell\log 5$$

$$1+z = \frac{\gamma(1+v)}{1-HR_w} = \frac{\beta_- + \dot{R}}{(1+HR_w)(1-HR_w)} = 3.7463933761061194.$$

Here $\gamma(1+v)$ is the material wall's Doppler factor. Its initial speed $v=-.280886551267$ is inward relative to the cap frame while its areal radius increases. The host affects these factors through the junction. The flash stays inside the wall: its inward radial speed is null, the wall is timelike, and the expanding wall root persists to every finite cap time.

For the supplied ruler $\ell=10$ *km*, central receipt takes *53.6850701* microseconds after the specified emission-time origin. The recorded protocol agrees with independent 80-digit rational evaluation to 1.45×10^{-15} in the frequency ratio and 1.30×10^{-16} in t_{rec}/ℓ. These measure roundoff, not input, model or experimental uncertainty. The causal derivation uses no new field evolution. The source frequency standard and vacuum scale are explicit inputs. The result supplies a conditional measurable relation, not an observation of this emitter or an outer-wall signal through the host bridge.

10.4. Radar and a wall clock recover a physical junction

The atlas has a geometry inverse using an ideal negligible transponder and a calibrated proper clock on its quiet membrane. A central clock sends a radial pulse at t_minus and receives its immediate return at t_plus. The transponder is a test device; the scalar's transparent boundary law is unchanged. Work inside the static cap, where f=1-H^2*R^2>0 and the optical radius is r_star=artanh(HR)/H. In central proper time T, the reflection event obeys t_minus=T-r_star and t_plus=T+r_star.

Let D be half the round-trip interval, Z=dt_plus/dtau the calibrated wall-clock redshift, K=dt_plus/dt_minus the echo stretch, and s=d(log Z)/dt_plus its local receiver-time drift. For beta=f*T_tau>0 and v=R_tau, normalization gives beta^2-v^2=f. Thus

$$Z=\frac{\beta+v}{f}=\frac{1}{\beta-v}$$
$$K=\frac{\beta+v}{\beta-v}$$
$$f=K/Z^2.$$

The Doppler motion cancels. More generally, send and return times are null coordinates for a radial patch with unique light paths. In their radar coordinates T=(t_plus+t_minus)/2 and D=(t_plus-t_minus)/2,

$$ds_2^2=C(T,D)(-dT^2+dD^2)$$
$$1=C\frac{dt_-}{d\tau}\frac{dt_+}{d\tau}$$
$$C=K/Z^2.$$

This measures the radial metric on the transponder worldline even in a time-dependent geometry, without imposing a gravitational field equation. Its angular-radius function and off-worldline metric remain undetermined. The separate quiet-cap inverse may fail while this readout remains valid; an explicit evolving-metric control checks that case. For the declared regular vacuum-cap family, put x=sqrt(1-f) and u=artanh(x):

$$H=\frac{u}{D}$$
$$R=\frac{Dx}{u}$$
$$\beta=\frac{Zf+Z^{-1}}{2}$$
$$v=\frac{Zf-Z^{-1}}{2}.$$

The zero-curvature limits are H=0 and R=D. This reconstructs the supplied cap curvature; it does not generate it. The signed constant-tension junction has beta=a/R^2+cR, where a=M_host/k and c=(k^2+H_host^2-H^2)/(2k). Differentiate the null clock and the same wall dynamics to obtain

$$\beta_R = Z(s - H^2 R)$$
$$c = \frac{\beta_R + 2\beta/R}{3}$$
$$a = \frac{R^2(\beta - R\beta_R)}{3}.$$

For an independently specified host curvature, retain every positive root of k^2-2ck+H_host^2-H^2=0 and set M_host=ka. An unknown host curvature leaves a continuous ambiguity; a larger known host curvature can leave two positive tensions with identical cap-side clock histories. The ordinary Schwarzschild host and the equal-curvature inner host each give a unique positive root for the states used here.

The original one-level reference at R/ell=.5 and the nested inner wall at R/ell=.8 supply the following conditional readouts:

Readout	One-level reference	Nested inner wall
D/ell	.549306144334055	1.09861228866811
Z	2.62549201161791	3.74639337610612
K	5.16990622730208	5.05276679827145
s*ell	-.433158428652097	1.06171003724629
Z after .04 ell of receiver time	2.58466102249367	3.90852634473171

The first local readout and its derivative recover H*ell=1 and k*ell=2 in both states, with M_host/ell=.4 and .01 respectively. These are different established states under one membrane law; no common-parent clock history is assumed. After inference, integrate dt_plus/dR using only H,a,c to predict the later row without additional calibration. An independent proper-time evolution agrees within 3.56e-15 in Z, and an 80-digit quadrature checks the nested result. These are numerical comparison indicators, not physical error bars.

A changed material law beta_new=beta+.5*(R-R0)^2/ell^2 has exactly the same first local readout. Its later Z differs by .0002413953 and .0001163056, while the corresponding short-interval surface density stays positive. The finite prediction therefore tests continuation of the constant membrane law. It cannot separate the two tension branches when their full cap histories agree.

At ell=10km the later receiver interval is **1.33425638079 microseconds**. For an explicitly assumed independent 100ppm bound on D,Z,K, monotonic analytic input bounds give the nested Lambda_cap*ell^2 in [2.99837630274001,3.00162434110127]. This is a synthetic instrument bound, with float64 endpoint evaluation, not an astronomical measurement, a total model uncertainty or a new effective cosmological constant. An uncalibrated beacon or unknown reflection delay restores degeneracy. The full proof and domain and compact calculation bind the source bytes and independent controls. The next section supplies a finite two-wall host history. A collective parent and the physical origin of the cap scale remain separate work.

10.5. Both walls in one host history

Continue the two walls of Section 10.2 through $0 \le \tau_i/\ell \le .04$, retaining all three regional metrics, the common tension and the original slicing functions. Each $\tau_i = 0$ is its own intersection with that initial slice. Equal elapsed wall proper times do not define simultaneous events. The scalar stays zero; this continuation supplies the geometry on which a subsequent field can propagate.

Fix the host bridge at $T=0$ and its tortoise coordinate by $r_*(.5\ell)=0$. The original oriented slice then determines $T_i = V/f$ and both wall origins. With $p=T+r_*$ and $q=T-r_*$,

$$p_I = \frac{1}{R' - V}$$
$$q_I = -\frac{1}{R' + V}.$$

The second expression stays regular through the right cosmological horizon. At the host black-hole root r_b, the Lake chart of Section 10.2 uses $q=-log(|u|/\ell)/\kappa_b$ and $R=r_b+\kappa_b uw$, where $\kappa_b=f'(r_b)/2$. Signed bridge coordinate $R=r_b+y^2$ gives leading terms $u=a\ y$ and $w=y/(\kappa_b a)$ smoothly through the neck. Their signs identify the two branches; independently resetting a wall's host clock would lose the shared embedding.

Let $\beta_h=\beta_+$ at the inner wall and $\beta_h=\beta_-$ at the outer wall. The signed junction motion and proper normalization give

$$p_\tau = \frac{1}{\beta_h - \dot{R}}$$
$$q_\tau = \frac{1}{\beta_h + \dot{R}}.$$

The incoming host characteristic has constant p at the inner wall and constant q at the outer wall. Its intersection with the original slice defines a source label l_{foot}. The arrival maps obey

$$\frac{d\tau_{\text{inner}}}{dl_{\text{foot}}} = \frac{\beta_h - \dot{R}}{R' - V}$$
$$\frac{d\tau_{\text{outer}}}{dl_{\text{foot}}} = -\frac{\beta_h + \dot{R}}{R' + V}.$$

The numerator is evaluated at the receiving wall event and the denominator at the initial foot. These maps convert the original spatial source labels into arrival proper times. They do not assign emitter-clock redshifts or ages to whole regions.

At the two distinct endpoint events $\tau_i=.04\ell$, the calculation gives:

Quantity in ℓ units	Inner wall	Outer wall
Wall areal radius	.822921192107	1.567950150020
Initial incoming-foot areal radius	.780076707080	1.466056964357
Original source-label interval	[.8, .829362369820]	[3.338689615754, 3.372632651397]

At $\tau_i/\ell=(0,.02,.04)$ the arrival derivatives are $(1.334617,1.362318,1.391007)$ for the inner wall and $(-1.143723,-1.178627,-1.214567)$ for the outer wall. These are three evaluated cuts, not extrema of the derivative. The largest recorded 32/64 quadrature change is 1.31×10^{-12}, in the arrival derivative. Independent second-order wall evolution agrees in T/ℓ within 3.34×10^{-16}; separate null integration agrees within 3.50×10^{-15}. The wall metric-normalization residual is below 1.14×10^{-13}. These are numerical comparison indicators, without a physical input-uncertainty estimate.

The future orientation has $du\le0,dw\ge0$ for $F\ge0$. Throughout these segments the inner wall has $u,w<0$ and the outer wall has $u,w>0$; the wall-growth bounds preserve these signs over the full proper-time intervals. Inner-to-outer propagation would require increasing u; outer-to-inner propagation would require decreasing w. Neither causal edge exists through this constructed host patch. This is a domain-specific result, without an assertion about additional global extensions.

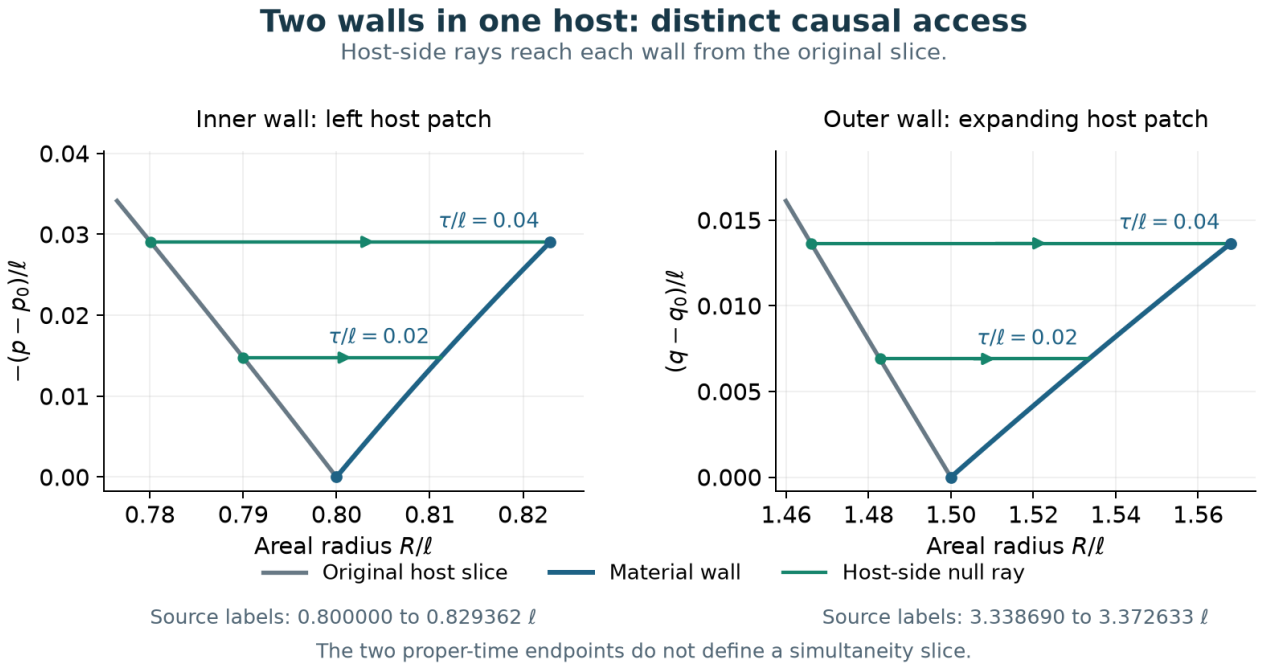


Figure 5. Two host null-coordinate panels of the common embedding. Horizontal constant-p and constant-q rays connect the original slice (grey) to the inner and outer worldtubes (blue), respectively. Displayed coordinate offsets position each panel; both host origins remain fixed by one initial slice. The marked endpoints use separate wall proper clocks and do not define a common time slice. The figure record binds these views to the finite calculation.

The shared-history derivation and compact calculation retain the common chart convention, signed normalization and finite comparison indicators. The history locates the next field-transfer problem. Section 10.6 supplies a transported averaging foliation and its moving-wall crossings. Neither connection supplies a new field state, formation mechanism or origin of the regional vacuum values.

10.6. The shared parent evolves into a calculable effective source

Derivation and calculation: normal-clock derivation and compact evolving-parent calculation.

Choose freely falling observers initially normal to the original shared slice. Their proper time defines one simultaneous leaf through both walls and both bridge minima. Its outer boundary follows the original $R=4\ell$ normal ray. This is a declared averaging clock on the same quiet spacetime, with unchanged masses, regional curvatures and membrane law. It provides the previously missing physical history for the composition and source-balance equations in Section 17.12.

In a smooth regional chart write $ds^2=-dt^2+A^2dx^2+R^2d\Omega^2$, with x the initial proper-distance label, $V=R_t$, $B=A_t$ and signed $E=R_x/A$. The complete bulk transport is

$$R_t = V$$
$$V_t = -\frac{M}{R^2} + H_i^2 R$$
$$A_t = B$$
$$B_t = \left(\frac{2M}{R^3} + H_i^2\right)A.$$

Initially $A=1$, $B=V_R$. The signed Killing energy E is conserved on each smooth geodesic segment and satisfies $V^2+f=E^2$. Differentiating the radial equation proves $R_x=EA$, including the regular $E=0$ bridge limit. The clock therefore crosses horizons without dividing by f or treating a spatial minimum as a focal point. Intrinsic curvature, expansion and shear follow independently:

$$^{(3)}R = -\frac{4E_x}{AR} + \frac{2(1-E^2)}{R^2}, \quad \theta = \frac{B}{A} + \frac{2V}{R}, \quad \sigma^2 = \frac{1}{3}\left(\frac{B}{A} - \frac{V}{R}\right)^2.$$

At a material wall put $U=\gamma N+b\ s$ and $b=\gamma\ v$. The signed Israel law implies the outward spatial jumps $[V]=kRb$ and $[E]=-kR\gamma$. A normal geodesic crosses in the direction opposite b, so its future crossing data obey

$$V_{\text{after}} = V_{\text{before}} - kR|b|$$
$$A_{\text{after}} = A_{\text{before}}$$
$$B_{\text{after}} = B_{\text{before}} + \frac{kA}{|b|}.$$

The last impulse follows from the radial Jacobi equation: $R_{NsNs}^{\ sing}=-k\delta(\eta)$ and $N(\eta)=-b$. Omitting it would lose the physical membrane curvature. The inner crossed normals come from the cap; the outer crossed normals come from the exterior. Both enter the host. Their wall clocks satisfy $dt/d\tau=\gamma$; they are not reset independently.

The new strip from crossings at time s has proper-volume element

$$dV(t;s) = 4\pi R(t;s)^2 A(t;s)\frac{|v(s)|}{A(s;s)}\,ds.$$

It is added once, together with each wall's curvature, energy and spatial stress atoms: $C_W=2\kappa\ E_W$, $E_W=4\pi\ R^2\sigma\gamma$, and $S_W=4\pi\ R^2\sigma(\gamma-3/\gamma)$. The analytic normal map has $A>0$ because $B(0)\geq 0$, the smooth Jacobi coefficient is nonnegative, and each impulse increases B. For the smallest host throat, a first-exit estimate gives $R\geq r_b-2Mt^2/r_b^2\geq .0150120120\ell>r_b/2$ through $0\leq t/\ell\leq .01$. The cap centre is exact and regular. The wall rapidity bounds preserve both incoming branches on this interval. The derivation records the piecewise charts at the transported initial wall labels; no additional physical shell is introduced there.

Let C_D denote the proper-volume average of intrinsic three-curvature. Using $L_{ref}=2/k=\ell$, the same scalar effective source gives:

Normal time t/ℓ	$H_D\ell$	$\Lambda_{g,tr}\ell^2$	$(3Q+C_D)\ell^2$
0	.092609049	-.0153506330754	2.11796240047
.005	.094009378	-.0143471868939	2.13982145265
.01	.095427142	-.0134681818918	2.16143093935

Here $\Lambda_{g,tr}=(Q-C_D)/4$ is the geometric trace after retaining the explicit regional and wall sources. Including those sources gives a different total trace, $.455231112462/\ell^2$ at the endpoint. Neither scalar alone specifies a vacuum stress tensor. At the midpoint the actual moment histories give

$$\Lambda_{g,\,\mathrm{tr},\,\mathrm{t}}\ell^3 \simeq .18865034$$
$$I\ell^3 \simeq .0437522915.$$

The independently differentiated history satisfies $Q_t+C_{Dt}+2H_DA_D=2\kappa\ J$, with $A_D=3Q+C_D$. Halving the time-difference step reduces its residual from 7.22×10^{-9} to $1.80\times10^{-9}/\ell^3$. The largest 32/64 change in the trace is $1.28\times10^{-10}/\ell^2$. These are numerical indicators, not observational uncertainties or a certified error enclosure.

The same inner-wall events supply a received signal. For an inward test flash with calibrated wall proper frequency,

$$t_{\mathrm{rec}} = t_e - H^{-1}\log(1 - HR_W(t_e))$$
$$1 + z = \frac{\beta_- + \dot{R}_W}{1 - H^2 R_W^2}.$$

At $t_e/\ell=0$ and $.01$, the central reception times are 1.60943791243410ℓ and 1.64616002849800ℓ, and the redshift factors are 3.74639337610612 and 3.89500911178850. Differentiating reception time with wall proper time independently recovers the frequency ratio. Reception uses the established quiet cap continuation, not an extension of the whole Gaussian foliation to that later time. Thus one history produces both the evolving effective-source table and a clock observable. The redshift alone does not measure the whole average.

There is also a finite-time large-domain conclusion. On the original outer time-symmetric Schwarzschild end, a geodesic starting at rest at areal radius r has

$$R = r - \frac{Mt^2}{2r^2} - \frac{M^2t^4}{12r^5} + O(r^{-8}), \quad A = 1 + \frac{Mt^2}{r^3} + \frac{5M^2t^4}{12r^6} + O(r^{-9}).$$

At fixed bounded regular time, θ, σ^2 and intrinsic three-curvature are $O(r^{-6})$. Their proper integrals converge while volume grows as the cube of the outer cut. All explicit sources remain confined to the interior and its walls. Hence both geometric and total effective trace averages tend to zero as the parent domain grows. The local outer Schwarzschild vacuum contribution also remains zero.

This closes the shared-history-to-effective-source calculation on its finite domain. It does not generate cosmological Lambda: the source fails vacuum form, evolves, and dilutes in this isolated AE parent. The outward connection now requires an actual cosmological domain with a physically determined nonzero regional volume fraction and compatible connector/history, followed by its observer light cone. The cap curvature has not been derived by recovering its supplied value.

10.7. A compact parent and the limits of its global average

Derivation and calculation: compact-parent derivation and complete-domain calculation.

The isolated parent's dilution can be removed by constructing a complete compact domain. Retain its nested centre-to-outer-throat body and attach a second sourced body on the other sheet of the same Schwarzschild bridge. The regular completion uses the original one-level cap and membrane on the right. The nested completion uses another copy of the nested body. Both retain connector mass $.4\ell$, common tension $k=2/\ell$, regional $H\ell=1$ or 0, and $\varphi=0$. The two outer walls begin at 1.5ℓ; each nested interior retains its inner wall at $.8\ell$.

These are different global states in the same piecewise theory. The interiors differ by their structure, while the connected vacuum has one mass invariant. There is no ADM mass at infinity in the compact geometry. No regional vacuum value or measured volume fraction is fitted.

Reflection reverses the right body's radial label and preserves its future normal. Consequently (E,V,A,B) maps to $(-E,V,A,B)$. Reversing the side order at its outer wall also gives

$$\beta^{g}_{-} = -\beta^{loc}_{+}$$
$$\beta^{g}_{+} = -\beta^{loc}_{-}$$
$$b_g = -b_{loc}$$
$$\gamma_g = \gamma_{loc}.$$

The global jumps remain *[E]=-kRγ* and *[V]=kRb*; the angular and temporal Israel jumps remain *-k*. At the regular right wall the global spatial slopes are *2.027786089806* and *-1*, while the future wall areal derivative stays *1.650310950872*. The right body has not been time-reversed.

Both tapers reach the same static vacuum collar at $R=\ell$. In the bridge coordinate $R=2M+y^2$, the initial geometry is even in y, E is odd, and $V=B=0$. The halves agree on open extensions through the throat, adding no seam source. The radial interval's two endpoint spheres collapse regularly to centres, so the underlying manifold is S^3, with three or four material walls and no asymptotic end or arbitrary outer cutoff.

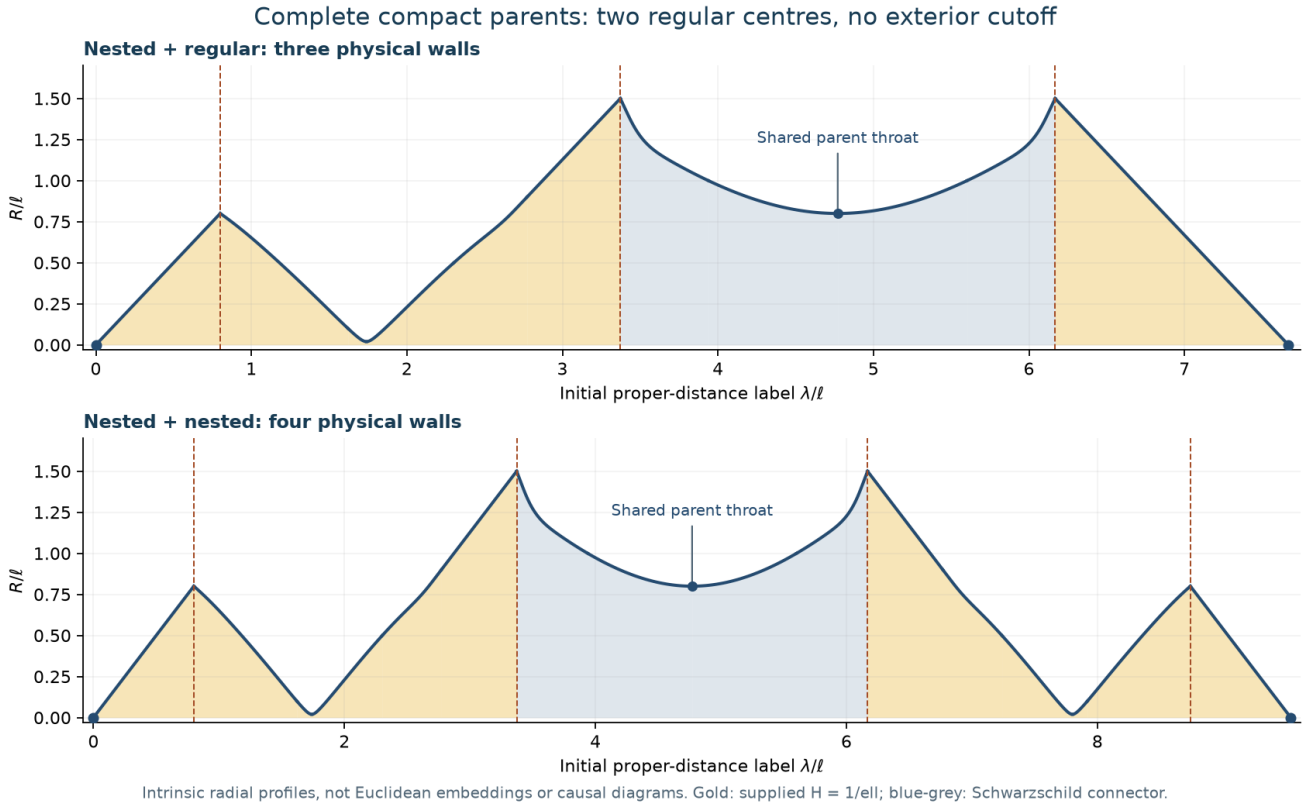


Figure 6. The two complete compact initial profiles. Both use the same connector and local membrane law; the lower state contains an additional right-side descendant. Dashed lines are physical membranes. The radius is plotted against proper distance, so these are intrinsic profiles rather than Euclidean embeddings or causal diagrams.

The normal evolution of Section 10.6 commutes with reflection. Its positive Jacobi scale and radius bounds apply to each half through $0 \le t/\ell \le .01$. Each half contributes its seven moments and its physical wall atoms once. The common throat has no surface source. For the symmetric double, extensive moments double and all means stay unchanged. Different halves require the between-region expansion variance from Section 17.12.

Thus the proper-volume fraction f_H of regions with $H\ell=1$ follows from the geometry and its evolution. At $t/\ell=.01$, $L_{ref}=2/k=\ell$ gives:

Right interior	f_H	$\Lambda_{g,tr}\ell^2$	$\Lambda_{total,tr}\ell^2$
Regular	.48620333	-1.037400004	3.171376758
Nested	.52228729	-1.219330524	3.212316830

The total volumes are $71.1461599485\ell^3$ and $76.5242717224\ell^3$. The total trace has a definite source decomposition,

$$\Lambda_{\text{total, tr}} = \Lambda_{g,\text{tr}} + 3H^2 f_H + \frac{\kappa}{4V_D}\sum_W (E_W - S_W).$$

For the regular completion its terms in ℓ^{-2} units are -1.037400003975, 1.458609994750 and 2.750166766967. The value near three therefore includes the supplied cap vacuum and material-wall stress. The geometric vacuum defects are $6.6134766493/\ell^2$ and $7.3326745903/\ell^2$; both effective geometric sources fail vacuum form. Their midpoint trace derivatives are approximately $1.0245354/\ell^3$ and $1.1843835/\ell^3$. The actual moment histories satisfy the same source-conservation identity, with approximately fourfold residual reduction on halving the time-difference step. The largest 32/64 trace change is below $1.2\times10^{-9}/\ell^2$, a numerical indicator rather than a certified or observational uncertainty.

The comparison also proves a local-observability restriction. Adding the remote nest changes the global total trace by $.040940072190/\ell^2$ at the endpoint. The entire retained left geometry and its inner-cap flash remain unchanged: $1+z=3.895009111788$ at $t_{rec}/\ell=1.646160028498$.

This follows from the actual connector clocks. Set $T=0$ on the original static bridge and use its standard Kruskal coordinates U,W, with future $dU,dW\geq0$ and

$$-UW = (R/(2M) - 1)e^{R/(2M)}.$$

The left outer wall has $U>0,W<0$, and the right one has $U<0,W>0$. Neither can signal the other through the constructed connector. Their initial inward null fronts would require $UW\geq15.6040388088$ to meet in the regular completion (15.6031988754 in the nested one), whereas the physical connector has $UW<1$. This concerns those emission events; it is not a claim that arbitrary future light cones never overlap.

The compact states therefore give different global averages but the same local cap curvature and clock observation. Compactness removes AE dilution; an additional causal/state mechanism is needed before a global average can be identified with a locally measured cosmological Lambda. The integrated constraint is satisfied by the local identities and supplies no extra condition selecting the cap H or absolute scale.

There is a separate continuation question. The small host throat starts with $E=R_t=0$, and its exact quiet inward geodesic obeys $R_t^2=2M_h/R+H^2R^2-1$. With $R=r_b\sin^2\chi$ and $a=H^2r_b^2$, its proper lifetime to the classical endpoint is

$$\tau_b = 2r_b\int_0^{\pi/2} \frac{\sin^2\chi\, d\chi}{\sqrt{1 - a(1 + \sin^2\chi + \sin^4\chi)}}.$$

The result is $.03144345977 96\ell$, with analytic bounds $.031434800637\ell\leq\tau_b\leq.031447397238\ell$. The curvature contains $48M_h^2/R^6$, so this exact quiet geometry does not smoothly continue the whole normal foliation through that event. No physical topology change or claim of a literal infinity in nature follows.

The spherical Einstein equations make the replacement problem precise. For geometric mass $m=R(1-|\nabla R|^2)/2$, normal energy density ρ and radial pressure p_r,

$$\nabla_N\nabla_N R = -\frac{m}{R^2} - \frac{\kappa R}{2} p_r$$
$$\nabla_N\nabla_N R + \nabla_s\nabla_s R = -\frac{\kappa R}{2}(\rho + p_r).$$

At a temporal turn that is also a strict spatial minimum, $E=R_t=0$ and $m=R/2$. A regular bounce requires $\kappa\, p_r<-1/R^2$ and $\rho+p_r<0$. At the current r_b, the pressure threshold is $\kappa\, p_r \ell^2<-2497.9987983973$, compared with the supplied vacuum value -3. This is a necessary source condition at that radius, with compatible density and conservation still required. It does not construct a bounce or prove that one occurs.

The compact atlas connection and its global-versus-local comparison are complete on the stated finite domain. A source-derived regular continuation, cosmological observer history and the origin of the regional vacuum values remain the next physical questions.

10.8. The nested body in an expanding parent

Derivation and calculation: cosmological attachment and observer derivation, compact calculation and independent checks.

An atlas needs both its local rooms and the world they inhabit. The isolated and compact constructions above supply two completions of the same nested body. A third now attaches its right Schwarzschild exterior to an expanding dust universe. The new join has no material membrane. Its geometry is borrowed from Einstein and Straus; the matching formulas are presented in Mars, Mena and Vera, Section 2. The connection here is to the already sourced NSC interior and its observer map.

Keep $M=.4\ell$, the original $L_{ref}=\ell=2/k$, and the increasing-radius exterior beyond the bridge. Choose the initial cosmological join at $R_b(0)=4\ell$. With $G=c=1$, the two metrics are

$$ds_F^2 = -d\tau^2 + a(\tau)^2(d\chi^2 + \chi^2 d\Omega^2)$$

$$ds_S^2 = -f dT^2 + f^{-1}dR^2 + R^2 d\Omega^2$$
$$f = 1 - \frac{2M}{R}.$$

On the comoving boundary $R_b=a\chi_b$, equality of the angular extrinsic curvature requires

$$\sqrt{\dot{R}_b^2 + f_b} = 1$$
$$\dot{R}_b^2 = \frac{2M}{R_b}$$
$$\dot{T}_b = \frac{1}{f_b}.$$

Both induced metrics equal $-d\tau^2+R_b{}^2 d\Omega^2$. The boundary is geodesic, with radial acceleration $-M/R_b{}^2$, so the temporal extrinsic curvature is also zero on both sides. The full Darmois conditions hold. The flat dust Friedmann equation gives

$$M = \frac{4\pi}{3}\rho_d R_b^3$$
$$H_0^2 = \frac{2M}{R_b(0)^3}$$
$$a(\tau) = \left(1 + \frac{3}{2}H_0\tau\right)^{2/3}.$$

This identity supplies the mass accounting. The exterior mass replaces the dust removed to accommodate the attached interior. Its internal proper energy is not another exterior mass to add to that equation. The parent has $H_0\ell = .111803398875$ and $\rho_d(0)\ell^2 = .00149207759149$. The join radius and the dust assumption are supplied; these values are conditional outputs.

The initial surface also needs a connection. Preserve the old static collar through $R=\ell$, then set $dT/dR = w(R)\sqrt{2M/R}/f$. A smoothstep takes w from zero to one by $R=3.5\ell$. The induced radial metric is $(1-w^2\ 2M/R)/f > 0$, becomes one near the join, and satisfies the vacuum constraints. It supplies a spacelike surface connecting the retained interior to the cosmic clock. Keeping the old zero-velocity normal at the join instead would select a different, closed-dust turnaround. The expanding construction changes the outer slicing explicitly; it does not reinterpret the previous normal averages or evolve the compact S^3 into a different topology.

Now follow a light ray. A static emitter at $R_e = 2\ell$ emits at the Killing time of the boundary's $\tau = 0$ event. A comoving observer on the join receives it. With $v_b = \sqrt{2M/R_b}$ and $r_* = R + 2M\log(R/(2M) - 1)$,

$$\int_0^{\tau_r} \frac{d\tau}{1 + v_b(\tau)} = r_*(R_b(0)) - r_*(R_e)$$
$$1 + z = \frac{1 + v_b(\tau_r)}{\sqrt{f_e}}.$$

The calculated reception is $\tau_r/\ell = 3.921887982097$, $R_b(\tau_r)/\ell = 5.602745333858$, and $1+z = 1.778824923298$. An independent null ODE changes the arrival by $1.3\times10^{-12}\ell$; a proper-clock derivative checks the redshift. The experiment lies entirely on the parent side of the bridge. It transmits no signal from the causally separated cap. Its longer exterior history does not assert a regular continuation of the inner throat. Seconds require an independent physical value of ℓ/c.

The parent decelerates with $q = 1/2$ and generates no cosmological vacuum term. Its initial Ricci scalar divided by four is nevertheless $.009375/\ell^2$, supplied by dust. The connected geometry therefore distinguishes expansion, curvature trace and vacuum pressure.

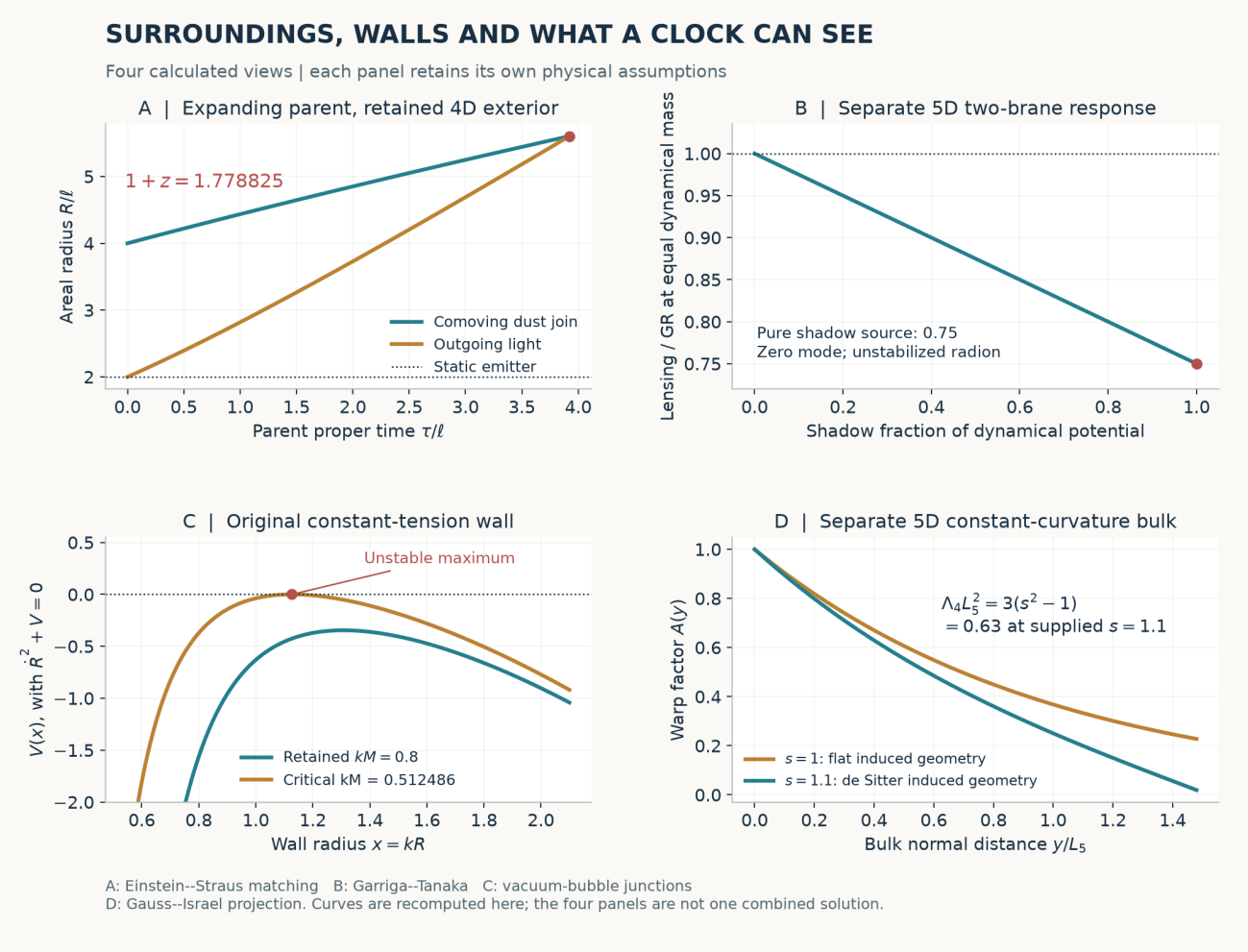


Figure 7. A: the cosmological join and its received light ray. B: the attributed two-brane lensing discriminator. C: the existing constant-tension wall and its unstable critical comparison. D: a separate bulk geometry inducing positive local curvature. The curves are regenerated from the stated equations. Panels A and C belong to the current four-dimensional family; B and D have different five-dimensional backgrounds and have not been combined into one model.

10.9. Two explicit ways surrounding geometry enters local gravity

Derivation and calculation: source conventions and geometric checks, finite examples, Shiromizu--Maeda--Sasaki, Maartens--Koyama and Garriga--Tanaka.

The surrounding region can have a different mathematical role from an ordinary exterior. In a brane construction, an observer occupies a four-dimensional hypersurface of a five-dimensional bulk. The surface curvature then depends on how it lies in that bulk. This is a distinct candidate from the two-sphere membrane of the preceding sections.

For a Z_2-symmetric brane with tension λ, confined matter $T_{\mu\nu}$ and a bulk cosmological term, the Gauss and Israel equations yield

$$G_{\mu\nu} = -\Lambda_4 q_{\mu\nu} + \kappa_4^2 T_{\mu\nu} + \kappa_5^4 \pi_{\mu\nu} - E_{\mu\nu}$$

$$\kappa_4^2 = \frac{\kappa_5^4 \lambda}{6}$$

$$\Lambda_4 = \frac{\Lambda_5}{2} + \frac{\kappa_5^4 \lambda^2}{12}.$$

The full quadratic tensor $\pi_{\mu\nu}$ and the conservation equation are given in the linked derivation. $E_{\mu\nu}$ is the traceless projection of bulk Weyl curvature. Direct contraction of the Gauss equation agrees with this decomposition for the checked tensors, including momentum flux and anisotropic pressure. That algebra does not supply a bulk solution for arbitrary matter data.

A vacuum example does supply the whole surrounding metric. Set $\Lambda_5=-6/L_5^2$ and

$$ds_5^2 = dy^2 + A(y)^2 ds^2_{\mathrm{dS}_4}$$
$$A(y) = \cosh(y/L_5) - s\sinh(|y|/L_5)$$

$$s = \frac{\kappa_5^2 \lambda L_5}{6} > 1$$
$$0 \le |y| < L_5 \operatorname{atanh}(1/s).$$

Both independent bulk sectional curvatures equal $-1/L_5^2$. The junction fixes the first derivative of A at the brane, giving

$$\Lambda_4 L_5^2 = 3(s^2 - 1).$$

For the declared demonstration $s=1.1$, this is $.63$, with $H_4L_5=.458257569496$. It is a positive, locally measurable curvature induced by surrounding geometry and tension. The two inputs s,L_5 have not been selected by inheritance, and L_5 is not identified with the membrane ruler ℓ.

The source channels can be distinguished. In a homogeneous isotropic brane, a separately conserved Weyl contribution behaves as radiation, $p_E=\rho_E/3$ and its density decreases as a^{-4}. For confined dust the quadratic source obeys $\rho_{quad}=p_{quad}=\rho^2/(2\lambda)$. Neither is the constant vacuum term in this explicit example.

Garriga and Tanaka give a second, observationally useful construction: matter on another flat brane affects the observer's metric through a bulk gravitational response. With two branes and an unstabilized massless radion, the static zero-mode source tensor is $T_{\mu\nu}$-$q_{\mu\nu}T/3$. Its nonrelativistic time and spatial components have ratio $2:1$. Use the metric convention

$$ds^2 = -(1+2\Phi)dt^2 + (1-2\Psi)(dx^2+dy^2+dz^2).$$

A pure shadow source then has $\Psi/\Phi=1/2$.

Slow orbits determine the dynamical potential, while light deflection depends on both potentials. The straight-ray integral gives

$$\alpha = \frac{2(1+\Psi/\Phi)G_4 M_{\mathrm{dyn}}}{bc^2}$$
$$\frac{\alpha_{\mathrm{shadow}}}{\alpha_{\mathrm{GR}}} = \frac{3}{4}.$$

This 25% reduction at the same dynamical mass is their published discriminator, recovered here from the tensor and ray integral. It shows what a quantitative outside-matter claim can look like. For positive-brane observers, mixed sources with common profiles and shadow potential fraction f give

$$\gamma_V = \frac{3 - e^{-2d/L_5}}{3 + e^{-2d/L_5}}$$
$$\frac{\Psi}{\Phi} = (1 - f)\gamma_V + \frac{f}{2}.$$

At $d/L_5=5$, the lensing ratios for $f=0,.5,1$ are $.99998487$, $.87499243$ and $.75$. This f is not a cosmological density fraction. A stabilized radion or a different background requires a new response calculation. In particular, these flat-brane perturbations have not been grafted onto the detuned de Sitter example.

10.10. Can the enclosing wall select the cap curvature?

Derivation and calculation: signed wall proof and inheritance map, original wall implementation and symbolic and geometric controls. The compact evidence records the retained parameters and the critical comparison.

The natural equilibrium question is whether the wall itself determines H, making $\Lambda=3H^2$ an output. The relevant building block is the published vacuum-bubble dynamics, credited through Ansoldi, Guendelman and Shilon. First consider the original massless cap-to-Schwarzschild member. For its constant positive tension, $p_s=-\sigma$, $k=4\pi G\sigma$ in $c=1$ units, the signed junction gives the existing radial equation

$$\dot{R}^2 + V(R) = 0$$

with potential

$$V(R) = 1 - \frac{M^2}{k^2R^4} - \frac{M(1 - H^2/k^2)}{R} - \frac{(k^2 + H^2)^2R^2}{4k^2}.$$

Write this as $1-A/R^4-B/R-CR^2$, with $A,C>0$. At a stationary point, $B/R^3=2C-4A/R^6$. Substitution gives

$$V''|_{V'=0} = -\frac{12A}{R^6} - 6C < 0.$$

Every static wall in this family is radially unstable. This exact identity decides the suggested stable-equilibrium closure without a scan. The full static family, with $u=H^2/k^2$, $x=kR$ and $\mu=kM$, can be parameterized by

$$q_* = \frac{u - 1 + \sqrt{9u^2 + 14u + 9}}{8}$$
$$x^2 = \frac{1}{3[(1 + u)^2/4 - q_*^2]}$$
$$\mu_{\text{crit}} = q_* x^3.$$

Its signed factors obey $\beta_->0>\beta_+$ and the temporal Israel condition. Thus the exterior remains on the wormhole branch. The critical mass decreases with $u\geq0$ from $\mu_{crit}(0)=16/27$. Our retained $\mu=.8$ exceeds that ceiling: at fixed M,k it admits no static state at any nonnegative H^2. At the illustrative critical state with $u=.25$, $\mu=.512485700729$, $x=1.126996043952$ and $V_{xx}=-3.881941479272$.

The actual nested outer wall has inner Kottler mass $m=.01\ell$. Its potential replaces A by $(M-m)^2/k^2$ and B by $M+m-(M-m)u$. The same negative-curvature identity applies. More strongly, its dimensionless potential is the massless expression at $k(M-m)=.78$, minus $.04/x$. Since $.78>16/27$, it is everywhere

negative and also has no static root. The massless critical curve is a comparison bound for this nested wall.

The scale-compatible inheritance map

$$(R, M, k, H) \longmapsto (aR, aM, k/a, H/a)$$

preserves kM and H/k for every $a>0$. It transports a dimensionless model without selecting those invariants. The full inherited-operator programme may supply additional conditions; this wall calculation does not yet identify them. The cap radius H^{-1} and its material wall radius R are different quantities. The compact average counterexample does not make them equal.

Gravastars provide a useful next building block. Mazur and Mottola place a material transition layer between a de Sitter core and a Schwarzschild exterior. Visser and Wiltshire make stability depend explicitly on the thin shell's equation of state. For a no-flux shell,

$$\frac{d\sigma}{dR} = -\frac{2(\sigma + p_s)}{R}.$$

The current $p_s=-\sigma$ is one special case. An inherited material or state response could supply another and change stability. That is the connection to derive before using a stable gravastar to select the cap scale. These published results do not select the new material law for us.

The classic Morris--Thorne construction already relates traversable throats to negative radial null stress. Section 10.7 applies a related Einstein-equation condition at the specified dynamical turn; it makes no priority claim for that requirement. Easson and Brandenberger likewise belongs in the black-hole-interior cosmology family. Each construction contributes its established mechanism with its premises.

10.11. The source law behind a growing mass

Derivation and calculation: energy accounting and finite counterexample, calculated balances and conservation checks.

The proposal that black holes supply dark energy makes the source question especially sharp. For fixed comoving number, a mass growing as a^n gives a density growing as a^{n-3}. The balance law is

$$\dot{\rho}_B + 3H(\rho_B + p_B) = Q.$$

A separately conserved effective component has $Q=0$ and $w_B=-n/3$. Pressureless mass receiving energy instead has $p_B=0$ and $Q=nH\rho_B$. The mass law alone does not decide which description follows from the geometry.

For example, take $\rho_B=.1$ and a pressureless reservoir $\rho_X=a^{-3}-.1$, in units of the initial total density. Over $1\le a\le 2$, both are nonnegative and the transfers cancel. The total density is a^{-3}, total pressure zero and $q=1/2$, even though the black-hole component has constant density. This is a homogeneous conservation counterexample, not a microscopic accretion model. It identifies the equation an inherited coupling must supply.

Farrah and collaborators propose a cosmological mass-growth mechanism; Ghodla and collaborators derive binary-merger consequences of such growth. The atlas includes both the proposed connection and an independent way of testing it. Neither supplies an observational validation of the present NSC geometry.

The next selection problem is now explicit: derive an inherited stress/state law or gravitational reduction, obtain its dimensionless parameters, and propagate the same solution to expansion, clustering, lensing and thermal history. The present chapter adds a cosmological connection, an observable clock relation,

useful external mechanisms and a decided equilibrium question. It does not yet predict the cosmological density fractions or matter--antimatter asymmetry.

10.12. Sourced rooms from an occupied wall spectrum

Derivation and calculation: state, stress and signed geometry, finite calculation and independent controls.

The equilibrium question in Section 10.10 asks for a material law. A finite occupied field supplies one: its energy changes when the surface changes, and that derivative determines pressure. This connects the geometric spectrum to a gravitational source before a wall potential is chosen.

The construction introduces a new surface theory. Its Einstein bulk has a de Sitter core, a Schwarzschild collar and a zero classical scalar. The wall carries a positive bare tension and two parity-paired massless Dirac species:

$$S_\Sigma = -\sigma_0 \int_\Sigma \sqrt{-\gamma}\, d^3y + S_D$$

$$S_D = \sum_{j=1}^{2} \int_\Sigma \sqrt{-\gamma}\, \overline{\chi}_j\, i\hbar D_\Sigma \chi_j\, d^3y.$$

Use the conformal reference vacuum and its normal-ordered finite excitation stress, with specified renormalized tension and no additional surface-curvature terms. This is a declared effective material candidate. The original transparent bulk probe has no trapping interaction and is not reinterpreted as this surface field.

Macpherson and Campbell provide the localized-fermion wall precedent. Ogure, Yoshida and Arafune calculate its finite spherical occupation and discuss fragmentation. A finite-thickness realization of the present gravitating surface theory remains a separate problem.

For one two-component spinor, the positive sphere eigenvalues are n/R, with degeneracy $2n$. Abrikosov's spherical Dirac analysis gives the spectrum and its rotational representations, with attribution to earlier Newman--Penrose and related work. Fill all angular states through level q in each of $g=2$ species. The state is isotropic and has the exact finite sums

$$N_q = gq(q+1)$$
$$\mathcal{A}_q = \frac{gq(q+1)(2q+1)}{3}$$
$$E_q = \frac{\hbar c\, \mathcal{A}_q}{R}.$$

The negative-energy reference branch is not counted again as occupied positive particles. Varying the energy with area at fixed occupation gives

$$\sigma = \sigma_0 + \frac{\hbar c\, \mathcal{A}_q}{4\pi R^3}$$
$$p = -\sigma_0 + \frac{\hbar c\, \mathcal{A}_q}{8\pi R^3}.$$

The gas pressure is half its surface energy density. The total obeys $d(\sigma R^2)/dR+2pR=0$. Its occupied part has zero trace, yet changes the full stress tensor and the resulting geometry.

This state is transported exactly within the surface theory. On $-d\tau^2+R(\tau)^2d\Omega^2$, use $d\eta=d\tau/R$ and rescale the spinor by R^{-1}. The resulting Hamiltonian is the same unit-sphere Dirac operator. Its occupied projectors commute with the unitary evolution. Number is conserved and energy scales as $1/R$, without assuming adiabatic motion. This inherits a specified occupation along a wall; it does not select its initial value.

In $c=1$ units put $k_0=4\pi G\sigma_0$ and $d=\ell_P^2 A_q$, with $\ell_P^2=G\hbar/c^3$ when units are restored. The angular junction is

$$\sqrt{\dot{R}^2+f_-}-\varepsilon\sqrt{\dot{R}^2+f_+}=s(R)$$
$$s(R)=k_0R+\frac{d}{R^2}.$$

Here $f_-=1-H^2R^2$ and $f_+=1-2M/R$. The ordinary orientation is $\varepsilon=+1$; the previous wormhole-side orientation is $\varepsilon=-1$. For the latter, at a static point with $a=\sqrt{f_-}$ and $b=\sqrt{f_+}$, implicit differentiation gives

$$V''=\frac{-H^2/a^3-2M/(bR^3)-M^2/(b^3R^4)-6d/R^4}{1/(2a)+1/(2b)}<0.$$

Thus positive occupation does not stabilize the former wormhole branch. The ordinary orientation has a different sign in this test. One exact reference, specified before comparison with observations, is

$$a=\frac{24}{25}$$
$$b=\frac{4}{5}$$
$$H^2R_0^2=\frac{49}{625}$$
$$\frac{M_0}{R_0}=\frac{9}{50}.$$

Both Israel conditions then fix positive components,

$$k_0R_0=\frac{1}{225}$$
$$\frac{d}{R_0^2}=\frac{7}{45}$$
$$4\pi GR_0\sigma=\frac{4}{25}$$
$$4\pi GR_0p=\frac{11}{150}.$$

The centre is regular, the wall lies inside the core's static horizon, and the exterior starts outside $2M$. The resulting room is horizonless. At fixed mass, curvature, bare tension and occupation, its radial potential has

$$V=V'=0$$
$$R_0^2V''=\frac{35159}{7200}>0$$
$$\frac{\omega_{\text{radial}}R_0}{c}=\frac{\sqrt{35159}}{120}=1.562561109916.$$

This establishes linear radial stability in the mean-stress surface model. Finite disturbances carry energy and change the exterior mass at second order. Nonspherical and fragmentation stability are not established; the localized-fermion literature makes that distinction essential.

Fixed couplings, unequal occupied states.

Choose the finite reference $q_0=1000$, with $N_0=2{,}002{,}000$. The spectral sum and the declared gas compactness give

$$\frac{R_0}{\ell_P}=\sqrt{\frac{45\mathcal{A}_{1000}}{7}}=92651.4435937.$$

This is a physical ruler for the conditional example, not a predicted cosmological length. The reference shape and state determine its action parameters once. Thereafter keep $G,\hbar,\sigma_0,H$ fixed and change only the occupation.

For geometric shell mass $m_s=k_0R^2+d/R$, the signed static junction gives

$$M(R,q)=\frac{H^2R^3}{2}+m_s\sqrt{1-H^2R^2}-\frac{m_s^2}{2R}.$$

Equilibrium requires $M_R=0$. The positive reference Hessian $R_0M_{RR}=35159/86400$ gives a locally unique family. Its exact response to spectral energy is

$$\frac{dM_{\mathrm{eq}}}{d\mathcal{A}}=\frac{\ell_P^2\beta_+}{R}.$$

The extra local energy contributes to the mass measured outside with its gravitational redshift. This accounts for the interior through the same mass used in the parent matching.

q	Radius / R0	Mass / R0	Mode / level spacing
999	.9993406734	.1796270290	1.563237585
1000	1.0000000000	.1800000000	1.562561110
1001	1.0006588232	.1803733226	1.561883618

The last column also compares the radial mode to the adjacent surface level spacing c/R. A detector interaction and selection rules are needed before identifying that spacing with an observed spectral line.

10.13. Two physical interiors, one parent, and a central beacon

Derivation and calculation: common parent and null transport, state and clock results and piecewise null-ODE controls.

The same dust matching can now hold two unequal, completely specified interiors. Set the common parent $H_p(0)R_0/c=.075$ and $a_p(0)=1$. The exterior mass of each room fixes its comoving matching sphere:

$$R_{b,q}(0)=\left(\frac{2M_q}{H_p(0)^2}\right)^{1/3}.$$

Place the $q=999$ and $q=1001$ centres twelve R_0 apart in parent comoving coordinates. Their matching radii are $3.997235341585R_0$ and $4.002763442849R_0$, leaving positive clearance. The two excised spheres stay disjoint as the parent expands. Each local Darmois match has the same dust density on its

outside. Together they therefore define one shared piecewise Einstein solution. The wall law and couplings are common; the states, radii and masses differ.

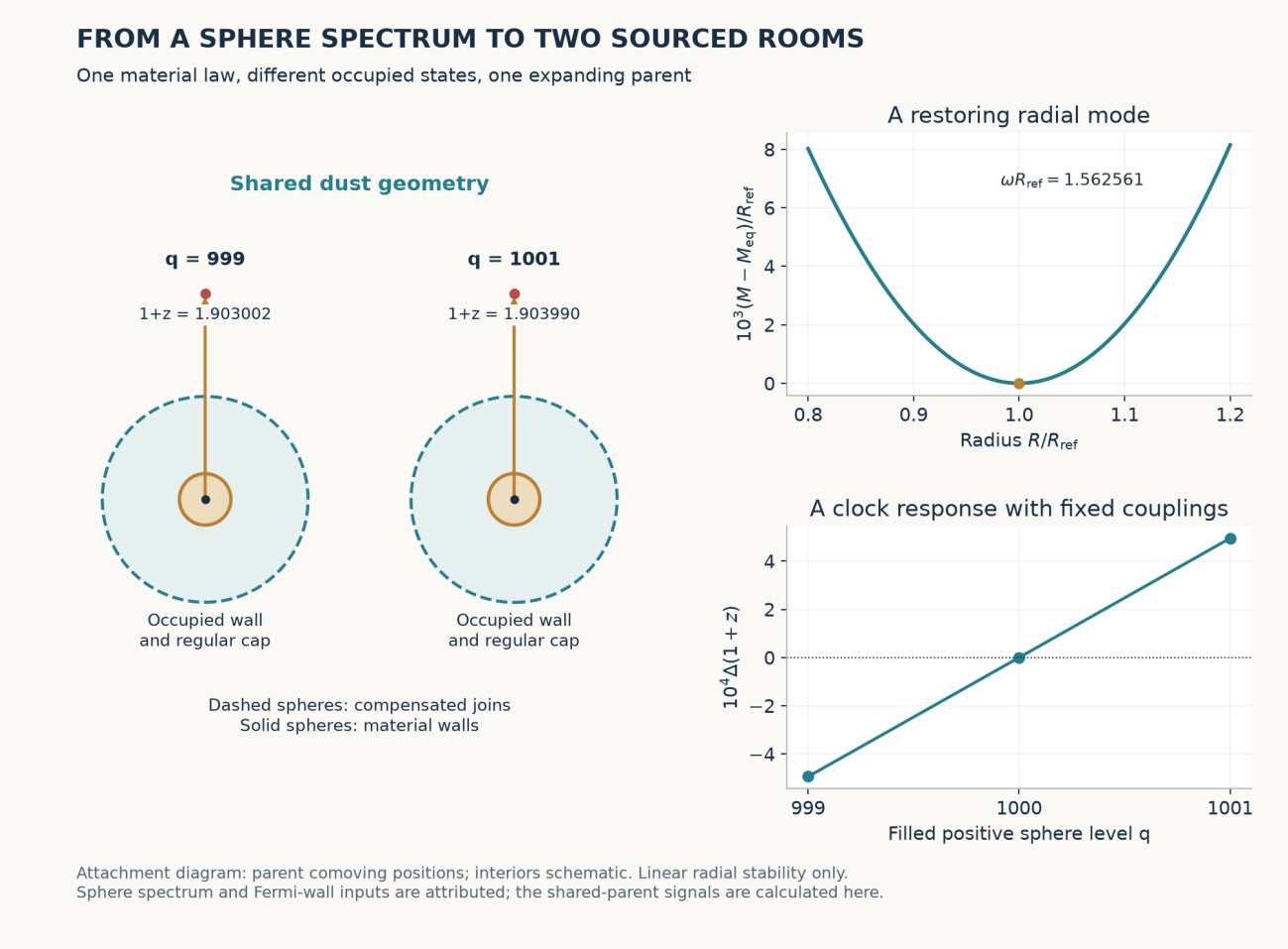


Figure 8. Two disjoint compensated regions in the same expanding dust parent. Parent positions are comoving; the interiors are schematic. The right panels show the reference radial mass minimum and the clock response to three finite states with unchanged couplings. The ordinary horizonless attachment and linear radial result are distinct from black-hole offspring formation and general stability.

A beacon at each regular centre now crosses the entire connection. At the static material wall, proper-time matching gives $dT=(a/b)dt_c$, where t_c is central proper time and T is exterior Killing time. An emission at the Killing time of the join's initial cosmic event spends

$$\Delta T_{\text{core}} = \frac{a}{bH}\operatorname{atanh}(HR)$$

inside the core. With $v_b=\sqrt{2M/R_b}$, reception at the moving dust join satisfies

$$\int_0^{\tau_b} \frac{d\tau}{1+v_b(\tau)} = \Delta T_{\text{core}} + r_*(R_b(0)) - r_*(R).$$

Continue through the dust to an observer eight R_0 away from each centre in the same comoving direction. Those two rays avoid the other matching sphere. The explicit dust conformal time determines the remaining arrival, and the total frequency ratio is

$$1+z=\frac{a}{b}(1+v_b(\tau_b))\frac{a_p(\tau_r)}{a_p(\tau_b)}.$$

q	Reception time / (R0/c)	1+z
999	12.61376979	1.903001657
1000	12.61774703	1.903495521
1001	12.62172978	1.903990397

The parent density and observer comoving offsets are the same in all three calculations. A separate null ODE crosses the core, collar and dust regions; differentiating reception proper time with respect to emission proper time independently recovers the redshift. The saved root comparison is at $3\times10^{-15}R_0$ and the clock-derivative difference at 1.2×10^{-11}. These are numerical indicators, not a microscopic or observational error budget.

This connection reaches from a specified finite state to a conserved surface pressure, a compatible geometry, a common parent and an observer signal. Its inheritance law transports the occupation but does not select it. At fixed fundamental constants, copying that occupation to a scaled radius changes both k_0R and ℓ_P^2A/R^2. A normalized operator relation alone does not preserve gravitational compactness.

The supplied cap curvature is still $\Lambda_{cap}R_0^2=147/625$; the common parent still has zero cosmological Lambda. Formation, occupation transfer, general stability and a collective source law remain physical questions. No cosmic density fraction or matter--antimatter asymmetry is inferred from this finite state-to-observer proof.

11. Complete one reference before assigning its causal arrows

Derivation and calculation: isolated causal completion, observer access, and protected parent interval.

The wall's isolated spherical completion is the flat-exterior limit of solution 13 in Aguirre and Johnson, gr-qc/0508093v2, Figures 7 and 10. Writing $x=R/\ell$, its expanding branch has

$$\dot{R}^2=\frac{25x^2}{16}-1+\frac{3}{10x}+\frac{1}{25x^4}>0.$$

It starts at a finite proper-past $R=0$ boundary in the Schwarzschild white-hole sector, crosses $R=2M$ into the left exterior, and crosses the de Sitter horizon into an expanding interior chart. It reaches arbitrarily large radius at infinite wall proper time. Its Schwarzschild curvature $48M^2/R^6$ diverges at the past boundary. The regular cap on the chosen slice therefore does not imply a regular past formation. The retained reference also contains the usual future Schwarzschild black-hole singular boundary.

Choose three clocks. C is the cap center, with de Sitter flat proper time t_C. W carries wall proper time τ_W, set to zero at $R=1.5\ell$. P is static in the right exterior at $R_P=1.6\ell$, with proper time $\tau_P=\sqrt{1-2M/R_P}\,t_{Schw}$. Their origins coincide with a selected slice convention; this is not an equality of universe ages.

For the flat de Sitter chart,

$$\eta=-\ell e^{-t_C/\ell}$$
$$R=ar$$
$$a=e^{t_C/\ell}.$$

Radial null lines have constant $\eta-r$ or $\eta+r$. At W_0, $\eta=-\ell$ and $r=1.5\ell$. Its inward ray would reach $\eta=0.5\ell$, outside the chart's future $\eta<0$; hence it never reaches C. A central signal emitted at $t_C=0$ likewise cannot catch the future wall.

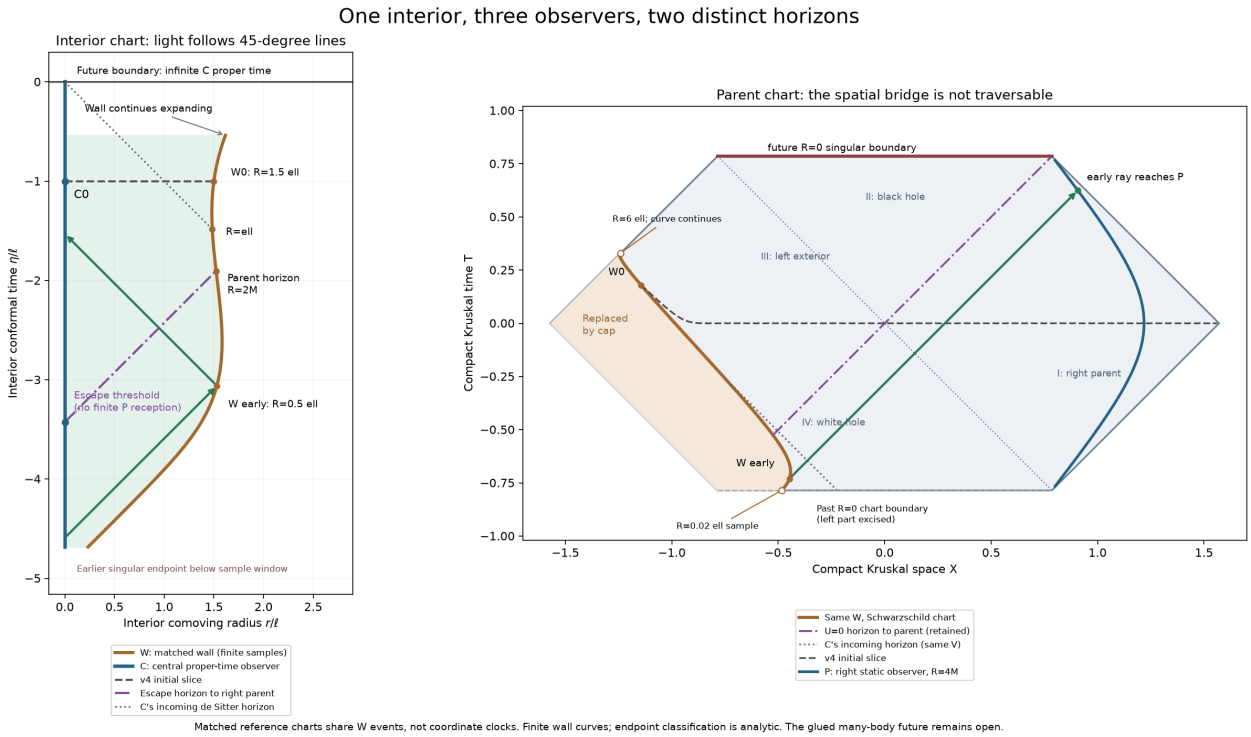


Figure 9. The isolated reference in joined causal charts. Horizons are clipped to retained domains; the central cosmological horizon continues into the Schwarzschild chart along its appropriate constant null coordinate. The faint past R=0 line also shows uncut chart context, part of which is excised. Open wall endpoints at R/ℓ=0.02 and 6 mark finite plot samples, not physical beginnings or endings.

Emission and destination	Result in this reference	Required distinction
New $C\to W$ or $W\to C$ emission at or after the anchor slice	No future reception	Earlier signals may arrive later
Right parent $P\to W$ or $P\to C$	No future causal route	Left and right exterior regions are different
Early $W\to C$, while $R<\ell$	Allowed	Reception time depends on the emission event
Early $C\to W\to P$, crossing while $R<2M$	Allowed	The limiting horizon ray has no finite parent reception
These same arrows in a glued multi-body future	Unresolved	Initial-data gluing does not supply that global future

For a wall event with interior time t_W, the central emission reaching it and the reception of its inward ray satisfy

$$t_e = t_W - \ell \ln(1 + R/\ell)$$
$$t_r = t_W - \ell \ln(1 - R/\ell).$$

The second relation requires $R<\ell$. At the early event $R=0.5\ell$, the recorded central emission is -1.5242168241ℓ and inward reception is -0.4256045355ℓ. An outward early route through that event reaches P at its own proper time $+2.5311656510\ell$. Thus exclusion of **new emissions** after a slice must never be rewritten as exclusion of all later receptions. Emissions approaching the central horizon from below can arrive arbitrarily late.

The critical central emission for escape to the right parent is $t_C=-1.2312944525\ell$. Equality follows the parent horizon and has no finite parent reception. That horizon extends into the cap along η-$r=\eta_W(2M)$-$r_W(2M)$; it is not the surface $R=2M$ everywhere. The central observer's de Sitter horizon

$\eta+r=0$ is a different boundary. Detailed clock domains and finite checks are retained in the observer note.

12. Different exteriors can be locally indistinguishable

There is also a result that does not require the glued parent's full future. Retain the exact right Schwarzschild $K=0$ slice for $2M<R<R_{cut}$, with $R_{cut}=10M$. For a static observer at $R_{obs}=4M$, the exterior cutoff cannot influence it within

$$0 < t_{\mathrm{Schw}} < r_*(R_{\mathrm{cut}}) - r_*(R_{\mathrm{obs}})$$
$$r_*(R) = R + 2M\ln(R/(2M) - 1).$$

The observer's proper duration is the right side multiplied by $\sqrt{1-2M/R_{obs}}$. For $M=0.4\ell$, the bound is $t_{Schw}<3.5090354889\ell$, or $\tau_P<2.4812627896\ell$. The inner optical boundary is at negative infinity, so it introduces no finite inner cutoff.

Shared smooth vacuum data give the same vacuum development in this domain. The owning argument uses standard Cauchy-development uniqueness, with Sbierski (1309.7591v2) as a source. The bound ends a guaranteed interval; it does not predict that a signal actually arrives then. Matching additional fields and their states requires additional data.

This supplies one precise atlas of indistinguishable exteriors. Distinct larger assemblies can share a protected local geometry. The next useful question is which later measurement or extra state datum separates them. Failure to distinguish exteriors can be a result, rather than a reason to add unconstrained parameters until a preferred story appears.

III. One field law from parent vacuum to evolving matter

This is the central source history of the atlas. A charged membrane changes four-form flux; a coupled scalar carries the resulting vacuum floor and excitation. The calculation follows that action through regular preparation, real parent sources, a three-region constrained slice and a protected evolving cosmology. Density growth and the remaining state freedom emerge from the same equations. The conditional initial data are stated once in each construction and carried through its consequences.

13. The shared source and its cosmological domain

The following sections give the worked constructions and their attributed mathematical inputs.

13.1. A field law that changes the inherited vacuum

Derivation and calculation: action, conservation and composition, symbolic and numerical record and scientific checks.

The preceding constructions supply their cap vacuum independently. Brown--Teitelboim's charged-membrane mechanism supplies an alternative: one field law determines the vacuum change at a boundary. We use the fixed-flux action in Bousso--Polchinski, Section 2.1, and the full membrane conditions in Hirano, Section 2.3. This is a new declared matter sector, not an extra term calculated from the original conformal-field Schur response.

With $c=\hbar=1$, $\kappa=8\pi G$, $F=dA$ a four-form, shared membrane tension σ and charge e, the action contains

$$S_{\mathrm{bulk}} = \int \sqrt{-g}\left[\frac{\mathcal{R}-2\Lambda_b}{2\kappa} - \frac{F^2}{2\,4!}\right] d^4x$$
$$S_\Sigma = -\sigma\int_\Sigma \sqrt{-\gamma}\, d^3y + e\int_\Sigma A.$$

Gravitational and fixed-flux boundary terms complete the action. Varying before substituting the solution gives

$$F = f\varepsilon$$
$$T^F_{\mu\nu} = -\frac{f^2}{2} g_{\mu\nu}$$
$$\Lambda(f) = \Lambda_b + \frac{\kappa f^2}{2}.$$

The oriented membrane gives $f_{in}=f_{out}$-e. Repeating that same law gives the composable sequence

$$f_n = f_0 - ne$$
$$\Lambda_n = \Lambda_b + \frac{\kappa}{2}(f_0 - ne)^2$$

$$\Lambda_{n+1} - \Lambda_n = -\kappa e(f_n - e/2)$$
$$\Lambda_{n+2} - 2\Lambda_{n+1} + \Lambda_n = \kappa e^2.$$

The second difference removes the bare vacuum and parent flux. It does not quantize the absolute flux, select which transitions occur, or determine the observed cosmological constant. The action is common; the state changes. Its relation to the earlier inherited operator is open.

Each vacuum region has $F_n(r)=1-2M_n/r-\Lambda_n r^2/3$. On the ordinary outward orientation, $\beta_-\beta_+=kR$, with $\beta_\pm=\sqrt{|}dot\ R^2+F_\pm$ and $k=\kappa\sigma/2$. Both the temporal junction and membrane force must also hold:

$$[K^a{}_b] = -k\delta^a_b$$
$$\sigma\overline{K} = e\,\frac{f_{\text{out}} + f_{\text{in}}}{2} = \frac{\Lambda_{\text{out}} - \Lambda_{\text{in}}}{\kappa}.$$

Our $K_{ab}=+h_a{}^c\nabla_c\, n_b$ is opposite to Hirano's convention. The last equation is also the distributional normal conservation law. Pure vacuum has no tangential flux, and the constant tension obeys intrinsic wall conservation.

Let $A=(M_{out}-M_{in})/k$ and $B=(\Lambda_{out}-\Lambda_{in})/(6k)+k/2$. Then

$$\dot{R}^2 + V(R) = 0$$
$$V = 1 - \frac{A^2}{R^4} - \frac{2(M_{\text{in}} + AB)}{R} - \left(\frac{\Lambda_{\text{in}}}{3} + B^2\right)R^2$$

$$\beta_- = \frac{A}{R^2} + BR$$
$$\beta_+ = \frac{A}{R^2} + (B - k)R.$$

The signs are retained after using this potential. Given the inner mass and initial radius/speed, the unsquared equation fixes the outer mass. That same mass becomes the next wall's inner mass.

13.2. Three vacuum regions in one evolving cosmology

Derivation and calculation: finite continuation and cosmological observer, endpoint values and independent quadratures.

Declare $\Lambda_b L_{ref}{}^2=0$, $\sqrt{\kappa} f_0 L_{ref}=3/2$, $\sqrt{\kappa} e L_{ref}=1/2$ and $kL_{ref}=1/10$. Choose initially resting membranes at $R/L_{ref}=.8,.4$ and a regular massless center. These are benchmark inputs, not measurements.

Region	ΛL_{ref}^2	M/L_{ref}
Parent vacuum collar	1.125	.01523145571612779
Shared child region	.5	.01162657748024113
Regular grandchild core	.125	0

The two vacuum changes are -.625 and -.375 in L_{ref}^{-2}. Both walls satisfy the full junction and charged force law, on one middle-region clock $0 \le T/L_{ref} \le .25$. Timelike coordinate speed bounds keep the inner radius above $.15L_{ref}$ and the outer radius below $1.05L_{ref}$. Positive collars and bounded ODE coefficients establish finite continuation. At the endpoint,

$$R_{\text{outer}}/L_{\text{ref}} = .82656089965304$$
$$R_{\text{inner}}/L_{\text{ref}} = .31785194531276.$$

Join the outer vacuum to flat dust at $R_b(0)=1.2L_{ref}$, with the same four-form flux and no added shell. Both Darmois forms give

$$M_0 = \frac{\kappa \rho_m R_b^3}{6}$$
$$H_p^2 = \frac{\kappa \rho_m + \Lambda_0}{3}$$
$$\rho_m \propto a_p^{-3}.$$

The exact expanding solution, with $h=\sqrt{\Lambda_0/3}$, is

$$a_p(\tau) = \left[\cosh(3h\tau/2) + \frac{H_p(0)}{h}\sinh(3h\tau/2)\right]^{2/3}.$$

Clock matching puts the displayed endpoint at cosmic time $.10872542374877L_{ref}$, with $a_p=1.07034528578242$. This parent accelerates from its declared four-form vacuum. Inner transitions do not change the outside flux or select its value.

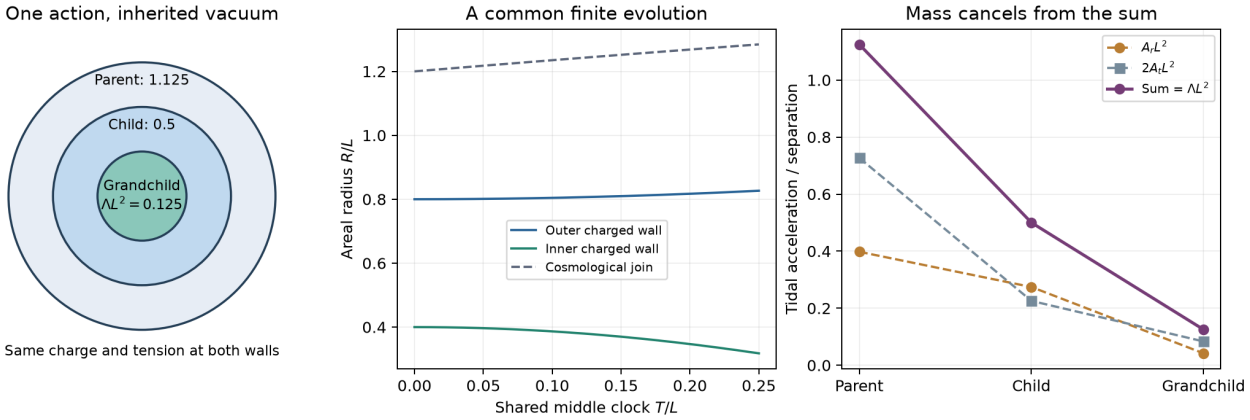


Figure 10. One shared charged-membrane law gives three vacuum values. The middle panel uses one common clock and its synchronized cosmological join. The right panel separates mass and vacuum in free-fall tidal measurements. The initial-region drawing is not a nucleation history.

For locally released freely falling probes in each vacuum collar,

$$\mathcal{A}_r = \frac{2M}{r^3} + \frac{\Lambda}{3}$$
$$\mathcal{A}_t = -\frac{M}{r^3} + \frac{\Lambda}{3}$$
$$\mathcal{A}_r + 2\mathcal{A}_t = \Lambda.$$

The trace removes the Weyl mass. In SI its measured relative acceleration per separation is multiplied by c^2. A second relation reconstructs the same vacuum from the flat dust parent's expansion:

$$\Lambda_0 = H_p^2 + 2\ddot{a}_p/a_p.$$

These conditional relations are registered retrospectively as NSC-P002. Finite instruments and cross-region communication require their own measurement model; these calculated values are not sky measurements.

Six symbolic identities vanish exactly. Independent 40-digit separated quadratures reproduce endpoint radii and clocks within $1.0\times10^{-15}L_{ref}$; two ODE tolerances differ by less than 3.0×10^{-12} in the normalized state. These are numerical diagnostics, not model or observational error bars.

This closes the declared finite source-to-geometry-to-observer connection. Parent flux, charge, bare vacuum, tension and a physical ruler remain inputs. The supplied two-wall Cauchy data have not been derived from nucleation. Selecting a formation history and flux sector, with reheating and observable consequences, is next; refining these trajectories would not select the cosmological scale.

13.3. Preparing the scalar, membrane and geometry together

Derivation and calculation: coupled preparation and scope, regular geometry and checks and scientific controls.

A constant four-form alone supplies vacuum stress. To investigate an evolving component, explicitly add a canonical scalar with

$$U(\phi, q) = \rho_b + \frac{m^2\phi^2}{2} + \frac{(q+\mu\phi)^2}{2}$$
$$\omega^2 = m^2 + \mu^2$$
$$\varepsilon = \frac{m^2}{\omega^2}.$$

The mixing and fixed-flux elimination come from Kaloper--Sorbo, equations (1)--(3). The positive mass term is an additional declared choice; their axion symmetry is not assumed for it. The minimum and residual density are

$$\phi_*(q) = -\frac{\mu q}{\omega^2}$$
$$\rho_{\mathrm{vac}}(q) = \rho_b + \frac{\varepsilon q^2}{2}.$$

A conditional rapid-transfer calculation gave residual density $.125$, excitation $.09375$ and local energy release $.0625$ for $\varepsilon=1/4$, $\omega=20$, $q_p=1.5$, $e=.5$ and $\rho_b=0$. Its conserving homogeneous history did not establish the moving boundary that prepares those data.

There is an exact matching obstruction. At a timelike interface, continuity between a homogeneous child $\varphi_c(t)$ and a constant parent requires $(d\varphi_c/dt)(dt_W/d\tau_W)=0$. Since $dt_W/d\tau_W \geq 1$, the child cannot roll there.

Its constant value also fails the child field equation when $U_{\varphi,c}$-$U_{\varphi,p}$=-μ $e\neq0$. A spatial scalar layer is necessary.

We retain those couplings and constant tension σ=.2, then use regular gravitational preparation in the Coleman--De Luccia construction, with the continuation described by Dong--Harlow, Sections 2.1--2.2. Combined scalar, membrane and gravity preparations also appear in Bandos, Blanco-Pillado, Sousa and Urkiola, Section 5. Their field-dependent membrane coupling gives a scalar derivative jump; our constant-tension example retains a continuous normal derivative. Each cap has a pole-to-wall coordinate ξ_j,

$$ds_E^2 = d\xi_j^2 + a_j^2 d\Omega_3^2$$
$$\chi_j = \phi_j - \phi_*(q_j)$$

$$\chi_j'' + 3\frac{a_j'}{a_j}\chi_j' = \omega^2\chi_j$$
$$a_j'' = -\frac{\kappa a_j}{3}\left(\chi_j'^2 + \rho_j + \frac{\omega^2\chi_j^2}{2}\right).$$

Regularity imposes $a_j(0)=0$, $a_j'(0)=1$ and $\chi_j'(0)=0$. The Hamilton constraint propagates from these poles. The two lengths and scalar pole values are outputs. At the wall,

$$a_i = a_o = R_*$$
$$\chi_i - \chi_o = -\Delta$$
$$\chi_i' + \chi_o' = 0$$

$$a_i' + a_o' = \frac{\kappa\sigma R_*}{2}$$
$$\Delta = \frac{\mu e}{\omega^2}.$$

The parent pole-to-wall derivative points opposite to the common outward normal. Both caps therefore satisfy the same charged force,

$$\sigma\overline{K} = e\left(q_p + \mu\phi_W - \frac{e}{2}\right)$$
$$[K^a{}_b] = -\frac{\kappa\sigma}{2}\delta^a_b.$$

All field gradients and gravitational effects are retained in the solve. The converged ordinary-cap branch gives

Quantity	Normalized value
Wall radius R_*	2.70634531321651
Child pole-to-wall length	2.86731015671418
Parent pole-to-wall length	7.07049444470919
Wall field φ_W	-.0543355679824020
Common normal field derivative	-.215595888249873
Wall normal pressure	.154440177981835

The two collocation tolerances change the radius by 3.81×10^{-12}. The independent Hamilton constraint is below 6.9×10^{-13} and the sampled scalar equation below 2.2×10^{-9}. These are numerical checks of a nonlinear construction, not a validated interval proof of existence or uniqueness.

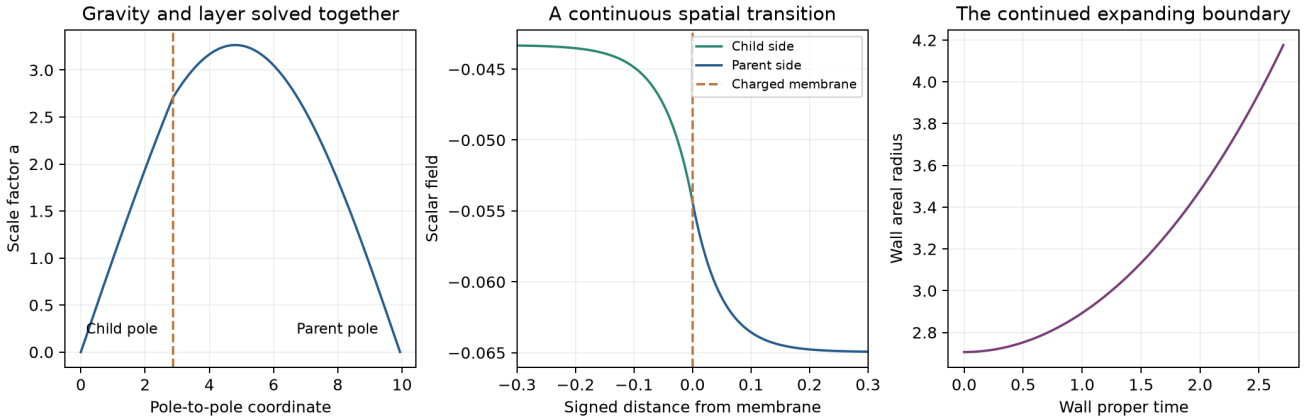


Figure 11. The scalar transition layer and gravitational caps are solved together with the membrane. Both field value and common-normal derivative are continuous. The right panel shows the continued timelike boundary. Lengths and field values use the declared normalized variables.

13.4. Physical boundary reduction and the prepared child

Derivation and calculation: boundary action and continuation, independent action and pole transfer.

At this solved geometry, the regular scalar maps $s_j=\chi_j'(L_j)/\chi_j(L_j)$ are actual Dirichlet-to-Neumann coefficients. Continuity and normal matching determine

$$\chi_i = -\frac{\Delta s_o}{s_i + s_o}$$
$$\chi_o = \frac{\Delta s_i}{s_i + s_o}.$$

Integrating $(a^3\chi\chi')'=a^3(\chi'^2+\omega^2\chi^2)$ gives the scalar excitation action at fixed geometry:

$$S_\chi = \pi^2 R_*^3 \left(s_i\chi_i^2 + s_o\chi_o^2\right) = \pi^2 R_*^3 \Delta^2 \frac{s_i s_o}{s_i + s_o}.$$

The coefficients are $s_i=19.53870489005$ and $s_o=20.30793503714$, with Schur coefficient 9.95794752938. Direct bulk integration and the boundary expression give $S_\chi=.913188612282$, differing by less than 2.8×10^{-13}. This is a physical elliptic preparation reduction. It is not yet a retarded memory kernel or an added bulk Lambda.

Continuing one sphere angle gives

$$ds_L^2 = d\xi_j^2 + a_j^2\left(-d\eta^2 + \cosh^2\eta\, d\Omega_2^2\right).$$

The same scalar and junction equations hold. Wall proper time is $\tau_W=R_*\eta$, and $R_W=R_*\cosh(\tau_W/R_*)$. At $\eta=1$ the areal radius is 4.17610904394739. Scalar energy flux vanishes in the symmetric wall's rest frame; the spatial layer and its stresses are part of the solution.

Near the child pole, $\xi=i\tau$ produces an open cosmology:

$$\ddot{\chi} + 3\frac{\dot{b}}{b}\dot{\chi} + \omega^2\chi = 0$$
$$\dot{b}^2 = 1 + \frac{\kappa b^2}{3}(\rho_i + E)$$
$$E = \tfrac{1}{2}(\dot{\chi}^2 + \omega^2\chi^2).$$

Its regular data are $b(0)=0$, $(db/d\tau)(0)=1$, $(d\chi/d\tau)(0)=0$ and the calculated pole value $\chi(0)=\chi_i(0)$. The coordinate origin has finite curvature. Since $dE/d\tau=-3H(d\chi/d\tau)^2\le 0$ and $\rho_i>0$, comparison bounds b between open de Sitter solutions with densities ρ_i and ρ_i+E_0. The homogeneous future chart therefore continues through every finite proper time.

An independent logarithmic-mode calculation determines the small pole state without subtracting nearly equal field minima:

$$\chi_i(0) = -6.9370429853 \times 10^{-25}$$
$$E_0 = 9.6245130760 \times 10^{-47}.$$

This preparation does not realize the earlier assumed excitation $.09375$. The spatial scalar layer carries energy per membrane area $.00233741846805$, distinct from the bare tension $.2$. The homogeneous child remains vacuum dominated, with $E/\rho_i \le 7.70\times10^{-46}$. The calculation identifies where the excitation resides and prevents importing the rapid-transfer amplitude into a different prepared state.

Regular Euclidean preparation is a declared state choice. No negative-mode count, determinant, preferred contour or tunnelling probability has been derived. A physically specified excited-parent or nonequilibrium preparation is needed if substantial child matter is claimed. Absolute vacuum scale, black-hole collapse formation and infinite physical nesting remain separate questions.

13.5. Real parent sources and received energy

Derivation and calculation: causal and energy derivation, compact calculation and independent checks.

The small regular pole amplitude is not the transmission coefficient for energy already present in the parent. Before choosing an excited parent field, determine which sources can reach the child. Use the same solved scalar/gravity profiles and membrane. An inward radial test body starts at parent distance $.3L_{ref}$ from the wall on $\eta=0$, with supplied local Lorentz factor γ_s. There is no direct material interaction with the membrane in this test-body calculation.

The regular child pole fixes x by $dx=d\xi/a$, with e^x/ξ tending to one. The conserved radial charge is $K=a_s\gamma_s$. An inward trajectory requires K greater than the largest a along its path. Its delay relative to a null ray fixes the label transmitted across the child cone:

$$D(K) = \int \frac{|d\xi|}{a}\left[\frac{1}{\sqrt{1-a^2/K^2}} - 1\right]$$
$$L = \exp(\eta_s + x_s + D).$$

The integral includes the parent cap, membrane and child cap. The metric and geodesic tangent are continuous at the wall. In the open child, $b^2|d\chi/ds|=K$. For the vacuum comparison metric $b=\sinh(h\tau)/h$, $h^2=\kappa\rho_f/3$, the regularized radial integral has the closed form

$$e^{I(\tau)} = \frac{2/h}{\coth(h\tau) + \sqrt{\coth^2(h\tau) + (hK)^{-2} - 1}}$$
$$\chi = \ln L - I(\tau).$$

Hence a finite central arrival exists exactly when

$$L < \frac{2K}{1+hK}.$$

Failing this condition does not prevent entry into the child: a body can remain at positive comoving distance and never reach its center. For light the bound is $N=exp(\eta_s+x_s)<2/h$. On the initial slice, only parent distances below $3.700717257L_{ref}$ can signal the center, out of a parent cap length $7.070494445L_{ref}$. This is a specified observer's causal region, not a universal boundary between different kinds of matter.

At source depth $.3L_{ref}$ the massive threshold is $\gamma_s=1.290485734$. The primary choice $\gamma_s=2$ yields

$$\frac{c\tau_r}{L_{\mathrm{ref}}} = 4.285837228$$
$$\gamma_r = \sqrt{1 + \frac{K^2}{b_r^2}} = 1.546392954$$
$$\frac{E_r}{E_s} = .773196477.$$

The null control arrives at $3.420193817L_{ref}/c$ with $1+z=1.294206955$. The massive received speed is $.762773146c$; arrival is not the production of cold matter. Its energy ratio refers to two local frames, not a conserved global energy.

For a geometric comparison, replace the parent and membrane with a single de Sitter space at the same child h. Fix the source's initial areal radius, inward $\gamma_s=2$ and time-symmetric slice, using the hemisphere nearest the receiver. The embedding control gives reception at $4.014164210L_{ref}/c$ and $E_r/E_s=.811509668$. The nested geometry therefore delays reception by $.271673018L_{ref}/c$ and reduces received energy by 4.721224 percent relative to that control. This distinguishes these two supplied source/geometry pairs; unknown source data or other geometries can retain degeneracies.

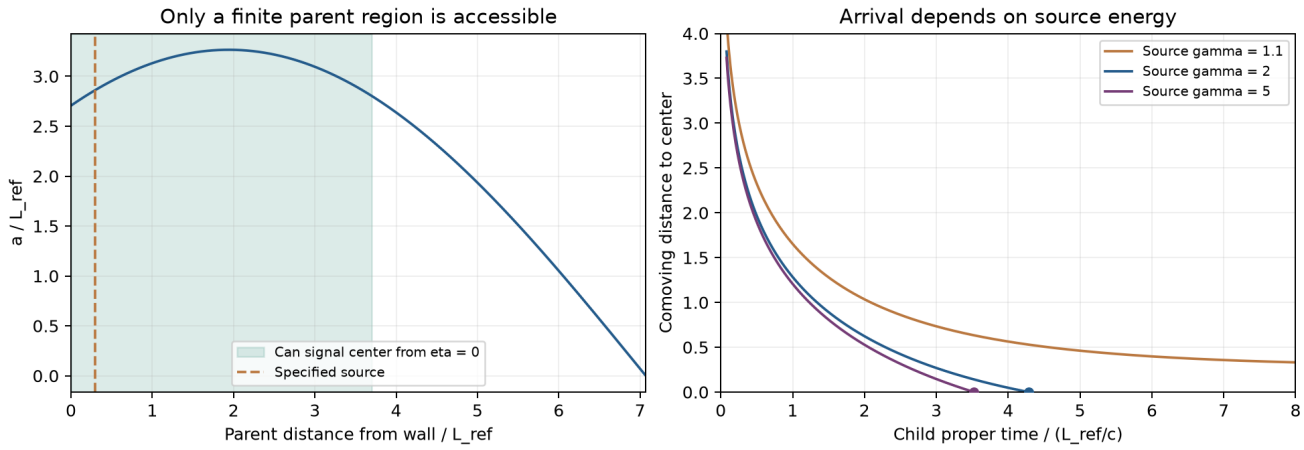


Figure 12. The left panel identifies the parent region that can signal the central child from the selected initial slice. The right panel shows three transmitted massive trajectories. The lowest-energy example enters the child but misses the center; the other two reach it at finite times.

An independent de Sitter embedding recovers the zero-wall limit: for source angle $\theta=h\xi_s$, central access requires $\gamma_s>1/\cos\theta$ and gives $\gamma_r=\gamma_s\cos\theta$. Direct integration of the child trajectory agrees with the closed inverse within $7\times10^{-14}L_{ref}/c$. Two background tolerances change the primary arrival by 1.06×10^{-11} and its energy ratio by 8.74×10^{-13}. These are numerical checks, not observational errors. Section 13.4's two de Sitter comparison metrics bracket the residual homogeneous-background correction; they do not bound backreaction from a new material population.

A simple collision alternative also has an exact diagnostic. The free-passage construction of Giblin, Hui, Lim and Yang motivates adding two scalar profiles. On one prescribed metric, with additive swept flux and continuing prescribed membranes, the present linear scalar equation gives

$$\phi_*(q_p-e_L)+\phi_*(q_p-e_R)-\phi_*(q_p)=\phi_*(q_p-e_L-e_R).$$

Two equal charges $.5$ from $q_p=1.5$ therefore land on the $q=.5$ minimum, with vacuum density $.03125$ and no homogeneous displacement energy there. This control does not solve the interacting gravitational collision or exclude radiation from deflected walls. It prevents assigning the instantaneous quench energy to free passage.

The next source must be a finite, constraint-compatible parent field preparation in the accessible region. Its scalar, metric and membrane perturbations must be matched together before calculating transmitted stress, isotropy or clustering. The smooth-wall bulk framework of Garriga, Montes, Sasaki and Tanaka is useful input; an independently charged membrane requires its own interface equations. Population normalization and the physical ruler remain inputs to be supplied. The present result makes the geometric part of that future source-to-observer calculation explicit.

13.6. A finite parent source with its gravitational response

Derivation and calculation: source, constraint and matching derivation, finite preparations and independent checks.

The next preparation includes the source's gravity. Retain the action, charges and tension of Section 13.3, and prescribe a finite compact scalar displacement in the parent. On the initial slice, the spatial extrinsic curvature and scalar momentum vanish. This is a time-symmetric packet, not a purely incoming wave. Its prior formation history is supplied neither by the regular saddle nor by this Cauchy construction.

Let h_0 and φ_0 denote the regular reference data. Solve $h=\Psi^4h_0$ with $\varphi=\varphi_0+f+g$, where f is the specified parent packet and g is a small wall-layer correction. The scalar Hamiltonian constraint and conformal curvature identity, as in Choquet-Bruhat, Isenberg and Pollack, Section 2, give

$$\Delta_0\Psi=\frac{\kappa}{8}\left[(\phi_0')^2-(\phi')^2\right]\Psi+\frac{\kappa U_0}{4}\Psi-\frac{\kappa U}{4}\Psi^5.$$

Primes use seed proper radial distance. The momentum constraint is zero. Regularity gives $\Psi'=0$ at both poles; the same material tension gives, in the common child-to-parent normal convention,

$$[\Psi]=0$$
$$[\Psi']=\frac{k}{2}(\Psi-\Psi^3)$$
$$k=\frac{\kappa\sigma}{2}.$$

For a second-order smooth initial corner there is an additional scalar requirement. If the old field value and first seed-normal derivative s are retained at the wall, the physical normal derivative becomes s/Ψ_W^2. The scalar wave equation and the full trace jump $[K]=-3k$ require

$$[\phi_{;nn}] - 3k\phi_n - \mu e = 0.$$

Freezing the old layer after changing the metric violates this condition unless $\Psi_W=1$ for the stated couplings. A parent-side correction with $g(0)=g'(0)=0$ and $g''(0)=\lambda$ repairs it precisely when

$$\lambda = \mu e(\Psi_W^4 - 1) + 2ks(\Psi_W^2 - 1).$$

Thus the scalar correction is solved with gravity. It is not an extra surface impulse or an adjustable energy injection. The finite example uses $f=A(1-z^2)^6$ for $|z|<1$, with $z=(d-.3)/.12$, and zero outside. Here d is seed distance into the parent from the wall. The correction is $g=\lambda\, d^2(1-d/.08)^6/2$ for $0\le d<.08$, and zero outside. These profile choices are explicit state data.

The complete instantaneous Israel conditions determine the two normal wall accelerations. With $B_j=\beta_j/R$, $\beta_j=\partial_n R$ and $P=e(q_p+\mu\varphi_W-e/2)$, they are

$$\overline{A} = \frac{P}{\sigma} - (B_i + B_o)$$
$$A_i = \overline{A} + \frac{k}{2}$$
$$A_o = \overline{A} - \frac{k}{2}.$$

The radial Einstein equation then gives the initial areal acceleration:

$$\frac{d^2R}{d\tau_W^2} = -\frac{m_j}{R^2} - \frac{\kappa R p_{r,j}}{2} + A_j\beta_j$$
$$m_j = \frac{R}{2}(1 - \beta_j^2).$$

The two sides agree algebraically using the force and Israel conditions. The scalar acceleration computed from each side agrees only after the second-normal matching condition is included.

Amplitude	Proper scalar excess	Wall radius	Initial areal acceleration
0	0	2.706345313	.369501998
.01	.270536650	2.721705563	.374020762
.02	1.112671383	2.764905208	.386385323

The table uses the declared normalized units. Proper scalar excess includes the quadratic excitation and its interference with the original field, both evaluated on the same sourced metric. It is not a global conserved energy or an ADM mass. For $A=.02$, $\Psi_W=1.010761098$ and $\lambda=.3779028621$. The radius increases by 2.163800 percent and its initial areal acceleration by 4.569211 percent.

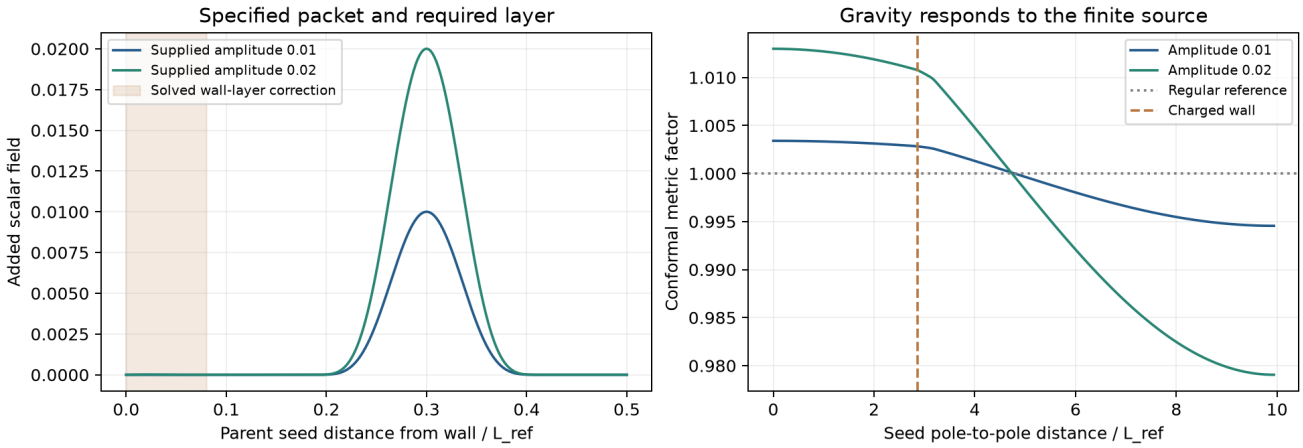


Figure 13. The specified scalar packet and required narrow wall correction are accompanied by a jointly solved spatial geometry. The conformal factor changes on both caps. These are different constrained initial data, not an instantaneous signal travelling from the packet to the child.

For this example the shared initial scalar acceleration at the wall is *.180243894532*. With outward flux *F=-T(n,u)*, its initial value vanishes and its derivative is *.038036802293*. Positive denotes child-to-parent exchange. The initial direction therefore does not establish inward matter deposition; later arrival and transmission of the parent packet must be evolved.

The deliberately frozen-layer control passes the gravitational constraint but leaves scalar compatibility residual *-.362064088*. The solved correction reduces the reported two-sided matching gaps below 3×10^{-15}. Independent mass integration from the two poles gives *.44414721424740* and *.96826983393285*, matching the geometric masses within 7×10^{-13}. Direct Hamiltonian residuals remain below 3×10^{-8}, and two tolerances change the wall radius by 4.36×10^{-10}. Five symbolic identities and the zero-source limit check the construction. These numerical comparisons establish neither interval-certified existence nor uniqueness.

The result supplies finite source gravity, instantaneous wall matching and scalar compatibility through second order. It is not a finite-time evolution or an all-order corner-compatibility theorem. The physical ruler, preparation history and subsequent transmitted stress remain open. The bulk vacuum values are unchanged. The next connected problem is the evolution of these data and the energy delivered through their moving boundary, followed by the stress required for clustering.

13.7. A conditional inward pulse from the excited parent

Derivation and calculation: weak-corner preparation and transfer derivation, exact seed and independent massive-field control.

Smooth second-order compatibility at the initial interface is a choice of regularity. Its failure does not by itself exclude a finite-energy weak transient. A different preparation with the same action makes this distinction useful: put the field uniformly at the child minimum on the entire time-symmetric slice. The child has no initial excitation; the parent has excitation density *3/32* above its own minimum.

The initial densities are $\rho_i=1/8$ and $\rho_p=3/8$. Round Cauchy caps satisfy the gravitational constraints and instantaneous wall laws exactly with

$$B_i = \frac{7}{15}$$
$$B_o = \frac{11}{30}$$
$$R_*^2 = \frac{1800}{467}$$
$$R_* = 1.96325997301.$$

Both poles are regular, and the parent excitation has finite proper integral *38.30744838* in normalized units. The initial wall pressure is P_0*=1/4* and its areal acceleration is *1/*R_*. This is a turning instant, not static equilibrium. The parent is an excited scalar region; its future is not a de Sitter vacuum continuation.

The scalar's initial corner is not twice differentiable in the required piecewise spacetime sense. Suppose a causal local weak development exists with continuous scalar value and normal derivative and bounded one-sided geometric coefficients. Existence of that full coupled development remains a separate question. With wall proper time τ and normal coordinate x, child $x<0$, the tangent equation is

$$(\partial_\tau^2 - \partial_x^2)\chi = -SH(x)$$
$$S = \mu e = 5\sqrt{3}.$$

The initial field displacement and its time derivative vanish. Retarded integration in the child portion of the initial cone gives

$$\chi_i = -\tfrac{S}{4}(\tau + x)^2 \quad (-\tau < x < 0).$$

The corresponding parent expression matches its value and first derivatives at the wall. Spherical terms, finite wall acceleration and the massive-field term enter at higher order under the stated bounded geometry assumptions. Thus the local coupled-development limit is

$$F_{\text{in}} = 18.75\tau^2 + O(\tau^3)$$
$$\tfrac{E_{\text{child}}}{\mathcal{A}_0} = 6.25\tau^3 + O(\tau^4)$$
$$\mathcal{A}_0 = 4\pi R_*^2.$$

Positive flux here means parent-to-child. It is generated by the different potential slopes, with the energy initially stored in the parent. The leading parent-layer energy loss equals the child gain; there is no separately prescribed incoming waveform. The same field controls the membrane pressure and gives a geometric relation:

$$P(\tau) = \tfrac{1}{4} - 18.75\tau^2 + O(\tau^3)$$
$$\sigma\overline{K} = P(\tau)$$
$$\lim_{\tau\to 0^+} \tfrac{P_0 - \sigma\overline{K}}{F_{\text{in}}} = 1.$$

This links transmitted scalar energy and wall curvature from one source and one set of couplings. It is a conditional short-time prediction, not a calculated finite-time cosmological abundance. No remainder constant for the curved coupled evolution has been established.

As a separate check, the exact planar massive retarded solution gives inward flux *.001856325865* and child energy per area *.000006212608434* at *t=.01*. Direct integration of the field energy agrees with the integrated boundary flux and the parent energy loss. Those finite-time numbers belong to that planar control only. The sharp membrane model also leaves finite-thickness corrections outside this limit.

In the initially quiet child, the leading edge cannot reach the center before proper time *2.07981585635*L_{ref}/c, obtained from its de Sitter domain of dependence. This is a causal front time, not a detector threshold or received waveform. The next step is a coupled causal development that tests the

nonzero transfer and pressure relation at finite times. The smooth source of Section 13.6 remains a distinct preparation control, not a compulsory gate on this weak-corner route.

13.8. A one-sided geometric observer and the full mass balance

Derivation and calculation: observer and energy derivation, coupled reference and independent checks.

An observer relation can be derived without first knowing the incident waveform. It combines the charged-wall force with the scalar geometrization approach reviewed by Krongos and Torre. Their full metric conditions include differential integrability; the local tests here do not replace them or prove a bulk spacetime exists.

Use child-side geometry, the normal pointing into the parent, and wall proper time. Write the mixed extrinsic eigenvalues as A,B,B, where $A=n\cdot a$ is normal proper acceleration and $B=n(R)/R$. These are not generally the areal acceleration or the radius's fractional rate of increase. Israel and the material force give

$$P=\sigma\left(A+2B-\frac{3k}{2}\right)$$
$$\chi_W=\frac{P-P_c}{S}$$
$$\Pi=\frac{\dot{P}}{S}$$

where $k=\kappa\sigma/2$, $S=\mu\, e$ and $P_c=e(\varepsilon\, q_c+e/2)$. Thus the field and its wall-time derivative are determined by one-sided geometry once the action parameters are fixed. No parent-side metric observation is needed for this step.

Gauss and Codazzi supply the normal pressure and inward flux:

$$\kappa p_n=-\frac{2\ddot{R}}{R}-\frac{1+\dot{R}^2}{R^2}+2AB+B^2$$

$$\kappa F_{\mathrm{in}}=2\left[\frac{\dot{R}}{R}(A-B)-\dot{B}\right].$$

For the canonical scalar, $Q=2(p_n+U)-\Pi^2$ must be nonnegative and $F_{in}^{\,2}=\Pi^2 Q$. When $\Pi\neq 0$, its signed normal derivative is F_{in}/Π. At $\Pi=0$ only its magnitude is identified by these observables; the inversion is sensitive to noise near that limit.

An independent bulk tidal contraction must then obey

$$Y\equiv R_{uu}-\frac{\kappa}{S^2}\dot{P}^2=-\Lambda_c-C(P-P_c)^2$$
$$C=\frac{\kappa\omega^2}{2S^2}=\frac{\kappa}{2(1-\varepsilon)e^2}.$$

For three distinct pressures on the same wall, the second divided difference of Y is exactly $-C=-8/3$ in the reference model. This relation holds on regular finite intervals, beyond the short-time expansion. Source history has disappeared; the field action has not.

With S and C fixed, two calibration points determine P_c and Λ_c phenomenologically. They must also satisfy the action's relation between flux, mixing and vacuum; the bare term cannot be independently retuned. A withheld third value tests the resulting curve. The synthetic example in the record performs that recovery and detects a deliberate $.001$ discrepancy. It is not an observational comparison.

The actual regular continuation provides a separate physical model check. From its wall geometry the inverse recovers φ_W=*-.0543355679824* and $|\Phi|$=*.215595888250*. An independent metric calculation gives R_{uu}=*-.149351145558*. The potential excitation accounts for *.024351145558*, leaving the declared vacuum *.125*. Interpreting the whole tidal magnitude as vacuum would therefore overestimate it by about *19.48* percent in this example. The continuation has constant P; multiple times repeat one point of the tidal curve rather than supplying several distinct calibration points.

Energy accounting supplies another exact check. For the geometric Misner--Sharp mass,

$$m = \frac{R}{2}\left[1 + \dot{R}^2 - (BR)^2\right]$$
$$\dot{m} = \frac{\kappa R^2}{2}\left[BRF_{\text{in}} - \dot{R}p_n\right].$$

This is the signed wall-frame form of Hayward's energy-supply and work law. Subtracting $\Lambda_c R^3/6$ leaves an excitation-pressure work term. On the existing evolved reference the scalar influx is exactly zero, yet the mass above vacuum grows from *.003668251239* to *.013477983984* over $0 \le \tau \le 2.706345313217$. Independent pressure integration accounts for the increase *.00980973274508*, agreeing within 1.8×10^{-15}. Enclosed mass growth alone is therefore not an accretion measurement, even in this explicit coupled nested geometry.

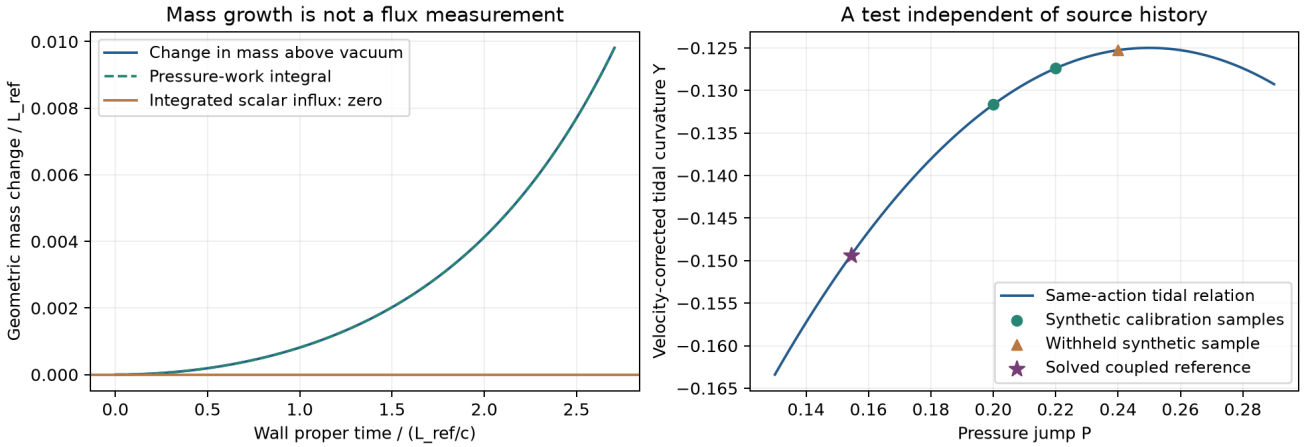


Figure 14. Left: the solved continuation gains mass with zero scalar influx; pressure work explains the change. Right: the star is its one pressure/curvature point. The other markers are explicitly synthetic calibration and withholding checks, not observed universes or new solutions.

The observable test requires normal metric information and independent tidal curvature. Radius or cosmological expansion alone is insufficient. Its identities are necessary conditions for this action, not a complete Rainich theorem, a globally reconstructed parent or an empirically selected cosmology. They supply finite-time diagnostics for the coupled evolution that remains to be constructed.

13.9. What an additional inherited region can identify

Derivation and calculation: coupling equivalence and inverse, exact identity and numerical controls.

There is a precise reason to extend the atlas beyond one interface. Two fixed parameter choices in the same scalar/four-form family can produce identical local classical dynamics while predicting different consequences for the next charge sector. Keep ω=*20*, κ=*1*, σ=*.2* and ρ_b=*0* in both candidates, and choose

$$(\varepsilon, e, q_c)_A = (1/4, 1/2, 1)$$
$$(\varepsilon, e, q_c)_B = (25/28, \sqrt{7}/2, -\sqrt{7}/5).$$

In each case $q_p=q_c+e$. Let $\delta=3\sqrt{3/100}$. Direct expansion gives the exact potential identity

$$U_B\big(\phi+\delta,\sqrt{7}\,(q-6/5)\big)-U_A(\phi,q)=3(q-1)(q-3/2).$$

The difference vanishes for every field value in the original parent and child sectors. A constant field translation preserves the kinetic term, and the two models share both vacuum densities, $S=\mu\ e$, tension and wall-force law. Their reduced classical scalar/gravity equations and matching are identical at that interface, including for mapped time-dependent states. More precision in those observations alone cannot choose between the two parameter sets.

This equivalence excludes independent charge probes, extra matter couplings that fix an absolute field origin and quantum preparation probabilities. It is a statement about the declared classical sector.

One further step of each model's unchanged oriented charge gives

$$(\Lambda_p,\Lambda_c,\Lambda_g)_A=(9/32,1/8,1/32)$$
$$(\Lambda_p,\Lambda_c,\Lambda_g)_B=(9/32,1/8,49/32).$$

These are relaxed potential values, not a constructed grandchild spacetime. Candidate B's next step is upward in ground energy, so the ordinary descending-cap preparation cannot simply be repeated there. The distinct prediction nonetheless identifies information that the first interface cannot supply.

For three comparable ground-vacuum sectors with a common action and equal oriented charge steps,

$$D=\Delta^2\Lambda=\kappa\varepsilon e^2.$$

Combining this with the local tidal coefficient C resolves the mixing and charge magnitude:

$$e^2=\frac{D}{\kappa}+\frac{\kappa}{2C}$$
$$\varepsilon=\frac{D/\kappa}{D/\kappa+\kappa/(2C)}.$$

Both candidates have $C=8/3$, but $D=1/16$ and $D=25/16$ respectively. The inverse recovers both parameter choices. This is a test of an inherited-law hypothesis using additional regional information. It does not prove that constants are inherited, that the third region exists, or that these regions can all be observed in a common normalization. The following construction supplies the third initial region and a protected evolving parent domain.

13.10. Three sourced regions on one compact initial slice

The previous ambiguity can now be tested on a physical third initial region. Keep the scalar/four-form action of Section 13.3, with $\kappa=1$, $\varepsilon=1/4$, $\omega=20$, $e=1/2$, $\sigma=1/5$ and zero bare density. Inside outward, use fluxes $(1/2,1,3/2)$ and one uniform initial field $\varphi_0=-\sqrt{3/80}$, at the grandchild minimum. Its momentum and spatial gradient vanish. The initial and ground densities differ:

$$(U_g,U_m,U_p)=(1,7,21)/32$$
$$(\rho_{v,g},\rho_{v,m},\rho_{v,p})=(1/32,1/8,9/32).$$

The supplied radii are 1 and 1.9, and the middle mass constant is .12; both pole-region constants vanish. These are declared initial data. The Kottler functions with $\Lambda_j=\kappa\ U_j$ represent the instantaneous

constraints. They cannot represent the excited scalar's subsequent vacuum evolution. Reusing the signed Israel equations,

$$\beta_i = \frac{M_o - M_i}{kR^2} + \frac{\kappa(U_o - U_i) + 3k^2}{6k}R$$
$$\beta_o = \beta_i - kR, \quad v^2 = \beta_i^2 - f_i$$

gives two expanding timelike walls. The inner pair of signed beta values is (1.5625,1.4625), with areal speed 1.204916145077. The outer pair is (1.148006694367,.958006694367), with speed .841108986073. An areal speed above one is compatible with timelike normalization. The same middle mass is used on both boundaries.

The caps have slice extrinsic curvature L^i_j=-$H\delta^i_j$, where $H=v/R$ on the corresponding wall. The grandchild sectional curvature is *-369/256*; the parent's is .022776502368. Between them choose a quintic interpolation of H whose first two derivatives vanish at both ends, and put

$$v_s = RH(R), \quad R_\ell = \sqrt{f_m + v_s^2}, \quad L_r = -v_s', \quad L_\theta = -v_s/R.$$

Direct substitution proves both smooth Einstein constraints. The angular jump gives the distributional Hamiltonian source $-4[\beta]/R=2\kappa\sigma$, and the continuous L_θ gives zero surface momentum. Both caps close at regular poles, producing a compact S3 slice of proper length 20.393422999807. This uses the standard Einstein-scalar constraints of Choquet-Bruhat, Isenberg and Pollack (Section 2), together with the material Israel conditions; their smooth theorem is not a shell-evolution theorem.

The temporal jump is also retained. The pressure relation fixes

$$\overline{A} = P/\sigma - (B_i + B_o)$$
$$[A] = -k.$$

A positive lapse collar realizes both one-sided derivatives while taking N=1 on the walls and near the poles. Its global bounds are .95649 and 1.04549. Independent acceleration checks use the full moving mass $m=R(1+v^2-\beta^2)/2$. The detailed construction and exact constraint proof are in the shared-slice owner.

The one-sided observer formula at each wall recovers the common field. After subtracting the excitation, the three ground values are those of candidate A. More strongly, candidate B from Section 13.9 obeys, for every integer sector and every real canonical scalar state,

$$q_n = e_B(3/5 - n)$$
$$\rho_B \geq U_{\min,B}(n) = \tfrac{25}{32}(3/5 - n)^2 \geq \tfrac{1}{8}.$$

The new grandchild geometry instead requires density 1/32 from its Hamiltonian constraint. The unavoidable gap is at least 3/32. Thus changing B's preparation, including its gradients or momenta, cannot reproduce this geometry under its unchanged law. This is a stronger inheritance distinction than the previous potential-only comparison. It is a mathematical rejection on the constructed geometry, not an astronomical measurement or exclusion of all alternative actions.

The scalar corners remain weak: conditional short-time inward flux coefficients are 18.75 and 56.25 at the two separate wall proper clocks. They do not establish a finite interacting evolution. Initial

composition is now explicit; a full weak future remains open. The constraint and sector record contains the independent checks and exact parameters.

13.11. Cosmology in the protected parent domain

The initial parent cap is homogeneous closed-FLRW scalar data with a_0=6.626077076100, H_0=.442688940038, ground density 9/32 and excitation 3/8. Its angular extent from the parent pole is θ_w=2.850764359770. The smooth homogeneous solution determines the domain of dependence of these actual cap data. Any causal completion of the whole atlas must agree there; no completed wall future is required to calculate this domain.

Writing chi for displacement from the parent minimum gives

$$\dot{\chi} = \Pi, \quad \dot{\Pi} = -3H\Pi - \omega^2\chi, \quad \dot{a} = aH, \quad \dot{H} = a^{-2} - \kappa\Pi^2/2$$

$$E = (\Pi^2 + \omega^2\chi^2)/2, \quad 3(H^2 + a^{-2}) = \kappa(\rho_V + E), \quad \dot{E} = -3H\Pi^2.$$

Put $h_{\backslash min}{}^2 = \kappa\rho_V/3$. Because $a_0 h_{\backslash min} > 1$, H cannot vanish on the expanding branch; it stays above .266408516441. The energy decreases, bounding the scalar and H and allowing smooth continuation at every finite future time. Total future conformal distance has the explicit bound

$$\int_0^\infty \frac{dt}{a(t)} \le \arcsin \frac{1}{a_0 h_{\min}} = .515418396612.$$

Hence all comoving points with θ<2.335345963158 remain within the parent domain of dependence for all future proper time. They occupy at least 90.703557% of the initial parent volume. This is a sufficient causal bound, not the exact horizon or a construction of the remaining wall spacetime. The initial lapse collar merely changes coordinates on the smooth development.

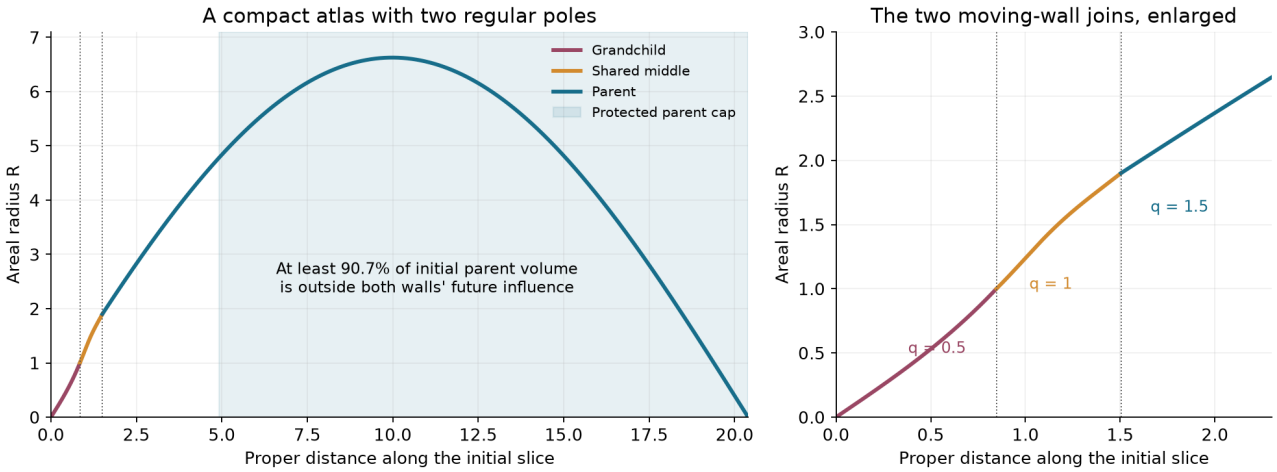


Figure 15. The compact initial geometry, with the two joins enlarged at right. Shading marks the sufficient protected parent cap. The parent contains an evolved scalar source under the same action; the unshaded interacting wall future remains a separate problem.

Taking the initial parent curvature radius as the independently specified reference length gives the conditional inherited value

$$\Lambda_{\text{parent}} L_{\text{ref}}^2 = \kappa\rho_V a_0^2 = 12.348252398928.$$

This depends on the declared couplings and initial data. It does not select the measured cosmic Lambda or provide an SI ruler. Integrating in proper time and independently in expansion e-folds produces the same one-e-fold endpoint:

Quantity	Same-source parent value
Proper time	2.888845513139
Scale factor	18.011544909930
H	.311446897816
Scalar excitation	.018994903671
Integrated scalar expansion work	.356005096325
Comoving scalar energy divided by its initial value	1.017394237405

The excitation is within 1.74% of pressureless dilution across this interval, which motivates a perturbation calculation but does not prove clustering. Maximum Friedmann and work residuals are below 3.3×10^{-11}; independent endpoint differences are below 1.5×10^{-11}. The declared numerical comparison tolerance is 10^{-9} in normalized variables, not an interval enclosure or observational uncertainty.

The compact evidence binds these numbers to the same action, constraints and implementation. The following calculation tests scalar and metric perturbation growth on this evolving domain. Its observer transfer and physical scale remain further connections, while the full wall problem stays explicit.

13.12. A constrained parent perturbation that grows

Near-pressureless homogeneous dilution does not determine whether a field develops gravitational structure. We now perturb the same protected parent, keeping its scalar action and curvature fixed. The linear equations reuse the general curved-background formalism of Hwang and Noh. In Newtonian gauge the canonical scalar has no linear anisotropic stress, so both metric potentials equal Phi. Let X be its field perturbation and Y its proper-time derivative. For S3 scalar harmonics with $\lambda=n(n+2)$ and $n\geq2$,

$$\dot{X} = Y$$
$$\dot{Y} = -3HY - (\omega^2 + \lambda/a^2)X + 4\Pi\dot{\Phi} - 2\omega^2\chi\Phi$$

$$\dot{\Phi} = -H\Phi + \tfrac{\kappa}{2}\Pi X.$$

Here chi and Pi are the existing homogeneous parent field and velocity. Its energy constraint is

$$C = \left(\frac{\lambda-3}{a^2} - \frac{\kappa\Pi^2}{2}\right)\Phi + \tfrac{\kappa}{2}\left[\Pi Y + (\omega^2\chi + 3H\Pi)X\right] = 0.$$

Direct differentiation proves

$$\dot{C} = -3HC.$$

The spatial Einstein equation and perturbed energy conservation are independently verified. No step divides by the oscillating scalar velocity; the numerical evolution passes eighteen ordinary velocity sign changes. The gauge-invariant comoving density contrast is

$$\Delta = \frac{\delta\rho + 3H\Pi X}{E} = -\frac{2(\lambda - 3)\Phi}{\kappa a^2 E}.$$

Density and metric transfer are therefore connected. The local lensing-potential combination is $\Phi+\Psi=2\Phi$; an actual light-cone lensing prediction still needs its observer and source map.

A compact perturbation makes this calculation belong to the actual atlas. Choose an initial radial potential supported at parent-pole angle $\theta<1.4$, with its n=0 and n=1 projections removed by two weighted moments. The explicit C7 profile and normalization are in the calculation owner. The lowest modes are excluded by preparation, not deemed unphysical; their special roles are discussed by Noh, Hwang and Barrow. Since the background starts at zero scalar momentum, choose

$$X_0 = \frac{2(\nabla_\gamma^2 + 3)\Phi_0}{\kappa a_0^2 \omega^2 \chi_0}$$
$$Y_0 = 0$$
$$\dot{\Phi}_0 = -H_0\Phi_0.$$

These data satisfy the linear Hamiltonian and momentum constraints. They vanish on an open neighborhood of both membranes, preserving their initial junction and scalar data. The parent causal bound gives

$$\theta_w - 1.4 - 2\eta_{\text{bound}} = .419927566546 > 0.$$

Thus the entire packet's causal cone fits inside the protected parent. S3 harmonics serve as a computational extension, with no imposed reflecting material boundary. Finite spectral tails are checked explicitly; the exact initial profile has compact support.

Normalize the initial central contrast to one as a linear response coefficient. At one e-fold, proper time 2.88884551314, it becomes

$$\Delta_{\text{center}} = 1.15188159122.$$

This is a 15.188159% increase. Multiplication by an example initial amplitude 10^{-5} gives the corresponding linear forecast. It does not certify the nonlinear remainder at that amplitude. The contrast oscillates during the interval; endpoint growth is not monotonic growth. The same field perturbation on a deliberately fixed metric instead gives .983934130329. That control removes the gravitational response and is not another coupled Einstein solution.

The packet's final sampled density changes by 2.94×10^{-9} between cutoffs 128 and 256. Independent proper-time and expansion-clock formulations differ by 8.07×10^{-11} on the retained packet. The declared numerical comparison tolerance is 10^{-7}, distinct from a uniform error bound or physical state uncertainty. The compact evidence retains the constraints, state, comparisons and producer hashes.

13.13. Inheritance must also supply the structure scale

One density history does not fix the initial scalar velocity. For each harmonic, retain two independent linear columns: a unit initial density contrast with Y0=0, and a unit initial scalar phase with $Y_0=\omega\chi_0$ but zero initial density, field displacement and metric potential. Their one-e-fold density responses are:

Harmonic n	Density column	Phase column
2	1.15091875	.00702776
8	1.15226198	.10801274
32	.92380409	1.36724676
128	-.98302417	-.98772566

The two columns give a transfer map for a specified preparation. They do not invent the covariance or formation mechanism that selects it. Shorter modes can be suppressed or reverse sign in oscillatory evolution, rather than sharing the long-mode growth.

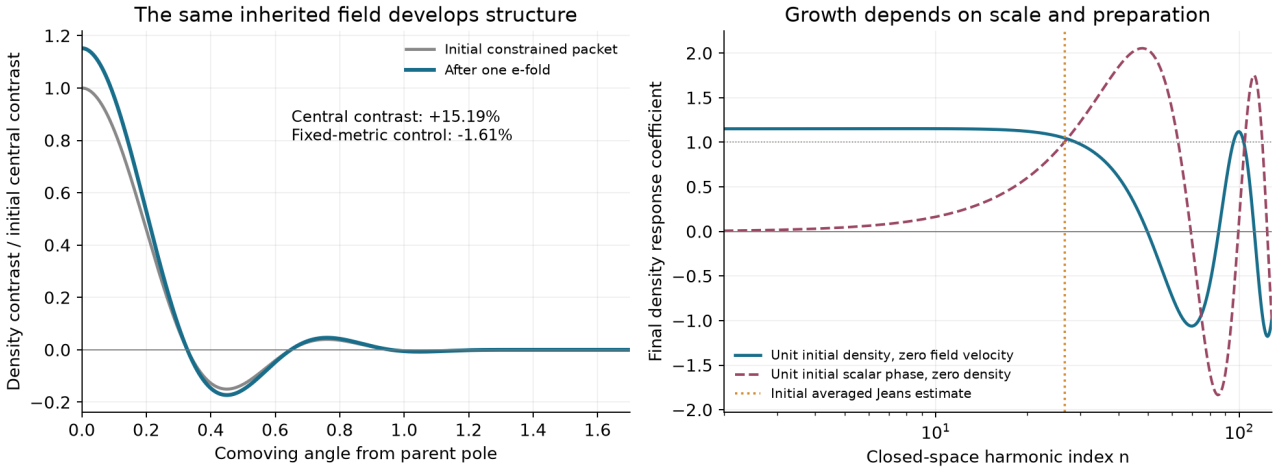


Figure 16. Left: a compact constrained packet develops a larger central density contrast under the same inherited source. Right: the two independent initial-state columns have different scale responses. The dotted Jeans estimate is a separate averaged comparison; the evolution itself retains curvature and oscillations.

Hwang and Noh's massive-scalar calculation provides the leading averaged, short-curvature-scale estimate

$$k^4_{J,\,\mathrm{physical}} = 2\kappa E\omega^2$$
$$\ell_\Lambda = \sqrt{3/\Lambda}.$$

The actual parent preparation has E_0/ρ_v=4/3, giving an initial Jeans wavelength of $.462258806\ell_\Lambda$. Its H_0/ω=.022134447 and curvature-to-Jeans squared-wavenumber ratio .001315002 characterize this comparison's scale separation; they are not a proved approximation error. The unaveraged transfer above does not depend on replacing the field by dust.

For this inherited excitation ratio the estimate implies

$$\frac{\omega}{\sqrt{\Lambda}} = \frac{\pi^2\sqrt{2/3}}{(\lambda_J/\ell_\Lambda)^2}.$$

A smaller pressure cutoff therefore requires a larger scalar frequency relative to the vacuum scale, with a compatible history. The present benchmark's cutoff is a substantial fraction of its vacuum radius; it cannot simply be called cold on every smaller scale. The inherited law has not yet selected the required ratio.

These calculated responses connect geometry, inheritance, expansion and linear structure. An unnested model with identical local field and geometric initial data has the same protected-domain response, so this growth alone does not uniquely demonstrate nesting. The next physical connection is a prepared perturbation state and frequency/scale, followed by the linked density and metric response seen by observers.

13.14. What the inherited law leaves to the state

The two independent perturbation columns cannot be replaced by a unique initial phase without additional physics. After inserting the metric momentum equation, the linear flow on *(X,Y,Φ)* has matrix trace -4H. Its fundamental matrix therefore obeys

$$\det U(t, t_0) = \exp[-4N(t)] > 0.$$

Together with the exact constraint propagation, this makes finite evolution a bijection between the initial and final two-dimensional constraint spaces. Distinct admissible states cannot become the same complete state at finite time. Their covariances transform by U times the initial covariance times U-transpose, rather than being selected by the deterministic equations. Asymptotic attraction, open-system elimination, a boundary condition or an actual formation law remains a separately specified possibility.

The useful next connection can nevertheless proceed: retain the existing compact spatial profile and its two density/phase amplitudes, use two operational observations to infer them, and test a third. This does not select an arbitrary primordial spectrum. It creates a falsifiable observer relation within a declared preparation family. The state-flow identities are retained with the observer calculation.

IV. Concentration, conservation and the surrounding reservoir

A dark-matter account needs a physical source that forms structure and accounts for its surroundings. We first calculate admissible scalar concentrations and their conserved supply, then their persistence and spatial compensation, and finally formation from compact disturbances. The route preserves the inherited parent instead of adding an independent halo mass. Different frequencies and preparations are explicit members of the family; they are not exchanged between examples by a change of units.

14. The energy supply and formation of structure

The following sections give the worked constructions and their attributed mathematical inputs.

14.1. A profile selected by scalar self-gravity

The operational clock family in Section 15.1 uses a specified profile. Here a physical selection mechanism is calculated: a real massive scalar can concentrate through its own gravity and shed radiation, leaving a coherent core. Gravitational cooling is established prior work of Seidel and Suen. The stationary equations are also those studied by Moroz, Penrose and Tod, with time-dependent examples in Harrison, Moroz and Tod.

The weak-field real-scalar reduction with positive vacuum curvature is supplied by Fodor, Forgacs and Mezei. Here its complex envelope represents one classical real field. It introduces neither a new charged species nor a quantum state-reduction law. We independently solve the profile and its finite local evolution, then construct a sourced atlas initial slice.

First distinguish a coupling from a unit convention. At fixed mixing, fluxes and tension, the common field at the grandchild minimum gives

$$\phi_0 = -\frac{\sqrt{1-\varepsilon}\, q_g}{\omega}$$
$$U(\phi_0, q) = \frac{\varepsilon q^2}{2} + \frac{(1-\varepsilon)(q-q_g)^2}{2}.$$

Thus the initial stress, wall jets and shared geometry do not select omega. The later dynamics change with it. For weak amplitude zeta, the local core approximation requires small ζ^2 and

$$h = \frac{H_\Lambda}{\omega\zeta^2}$$
$$H_\Lambda = \sqrt{\kappa\rho_v/3}.$$

With zeta=.1 and the same parent vacuum 9/32, omega 20 gives h=1.53093, outside the small-h approximation. The explicitly different common-frequency member omega 2000 gives h=.0153093. It is chosen for this hierarchy, not fitted to data. P003 retains omega 20; the comparison does not rule out relativistic structures there. The wall radii remain 1 and 1.9, with common middle mass .12. All numbers use those same atlas coordinate units, with no selected SI ruler.

In local coordinates $\rho=\omega\zeta\, r$ and $\tau=\omega\zeta^2 t$,

$$i\partial_\tau\psi = -\tfrac{1}{2}\Delta\psi + (V - h^2\rho^2/2)\psi$$
$$\Delta V = \tfrac{1}{2}|\psi|^2.$$

The h=0 nodeless core, with eigenvalue -1/2, satisfies

$$S'' + 2S'/\rho + sS = 0$$
$$s'' + 2s'/\rho + S^2 = 0$$
$$s(\infty) = -1.$$

Regularity and decay give $S(0)=1.02149303621$ and radial mass norm $I=3.50532970334$. The 99%-mass radius is 5.67468871483 in rho. Its virial identity is checked independently. Scaling the shape fixes an isolated mass-radius relation,

$$\mathfrak{M} R_f \omega^2 = \frac{I\rho_f}{2}$$
$$\mathfrak{M} = GM.$$

The core mass and frequency remain inputs; its dimensionless shape is calculated. In the isolated problem, symmetric decreasing rearrangement lowers the energy at fixed norm; this is the setting of Lieb's minimizing-solution theorem. Norm-preserving dilation gives energy b^2T+bW with a finite minimum at $b=-W/(2T)$. This explains the reference's spherical form and size without establishing nonspherical stability in the full atlas. Gaussian and super-Gaussian clouds with equal initial norm evolve to tau 600. Their late density profiles, averaged over $500\le\tau\le600$ and normalized by remaining mass, differ from the independent coherent shape by 3.77% and 4.11%, compared with initial discrepancies near 80%. Residual oscillations remain.

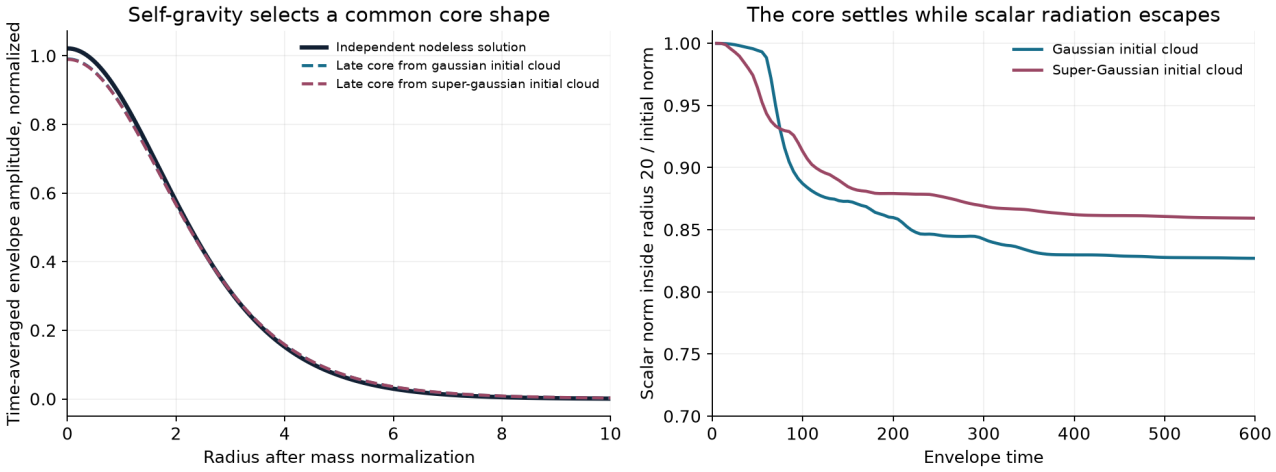


Figure 17. Left: two distinct initial cloud shapes approach the independently calculated coherent profile after mass normalization. Right: scalar norm leaves the central region during relaxation. This is a finite local envelope calculation, not the full evolving parent and membrane history.

The six independent runs use six worker processes. Outgoing current at radius 20 checks loss from the core independently of the outer absorber. Fine-run energy-accounting errors stay below 3.4×10^{-6}; moving the outer boundary changes the late inner mass by about .165%. A free-field control recovers analytic dispersion. The compact record keeps numerical discrepancies distinct from the weak-field, ambient and relativistic approximations. The original cloud and its energy are supplied, so this does not generate a primordial seed from nothing.

14.2. The coherent core and its inherited environment

The Gaussian late inner mass selects a coherent representative with $\zeta_{core}=.0826621713254$. It is the nodeless representative of that mass, not the full radiating snapshot. Add its displacement f to the inherited parent field, and taper its very small tail smoothly to zero before the material wall. Retain zero scalar momentum and homogeneous parent slice expansion v=HR.

The gravitational constraint must include the ambient scalar. Writing $m=M(R)+\kappa U_0R^3/6$, with $U_0=21/32$, gives

$$\beta^2 = 1 + H^2R^2 - \frac{\kappa U_0 R^2}{3} - \frac{2M(R)}{R}$$

$$M' = \frac{\kappa R^2}{4}\beta^2 f'^2 + \frac{\kappa R^2}{2}U'_0 f + \frac{\kappa R^2}{4}\omega^2 f^2.$$

The first and last terms are the gradient and quadratic core contributions. The middle term is coherent interference with the same inherited field. It cannot be discarded as if the core were an independently added fluid.

M(0)=0 gives a regular pole. Outside the core M becomes constant; the unsquared Israel condition then fixes the outer-wall speed and H. Solving that condition with the mass equation rebuilds the large parent cap and common middle. Both wall conditions and the middle seams agree. The positive initial lapse realizes the temporal jets. Removing the core recovers the same-frequency reference and extrinsic curvature.

Coherent turning-point phase	Geometric mass offset	Outer-wall areal speed change
Positive relative displacement	.0000957140113	.000361840526
Opposite relative displacement	.0000494933577	.000187115382

Both have the same isolated leading mass .0000724395411. The exact potential difference explains the main change:

$$U(\phi_0 + f) - U(\phi_0 - f) = 2U'_0 f.$$

The cross contribution has magnitude .0000231104644. The gradient contribution also changes slightly through the metric. The positive-phase proper slice energy excess is .00241200566080; it is distinct from the geometric mass converted to energy units. Canonical absolute energy remains nonnegative even where an excess relative to the background is negative.

Independent field quadrature and proper-distance integration check the mass. Sampled Hamilton residuals stay below 2×10^{-8}, with wall and middle residuals near floating precision. The source derivation and computed admission specify the complete initial construction.

The changed wall speed compares jointly prepared initial slices; it is not a wall response propagated by the local cooling run. The homogeneous reference's future shielding theorem is not asserted for this new core geometry. A single formation history must also contain the outgoing radiation, ambient-field interaction and moving walls. This calculation supplies a profile-selection candidate and its sourced initial admission. It leaves cloud origin, coupling and relative-phase selection, nonlinear global evolution and observed abundance as further physical connections.

14.3. Supplying a local core from the inherited parent

The core-only admission adds geometric mass. A local redistribution instead needs a reservoir and the outgoing scalar field in the same geometry. Keep the previous coherent core, the original parent expansion and the same frequency-2000 action. Surround the core with a depleted scalar annulus and a compact outward-current pulse. Determine the depletion by requiring zero enclosing mass offset. The pulse and reservoir profiles are specified initial data; no formation history is silently imported from the isolated cooling run.

With future normal n, outward unit spatial vector s, define $v=n(R)$, $\beta=s(R)$, $\Pi=n(\varphi)$ and $F=d\varphi/dR$. The exact spherical constraints in the convention $K=-L_n\ h/2$ reduce to

$$K_t = -\frac{v}{R}$$
$$K_r = -v_R - \frac{\kappa R}{2}\Pi F$$
$$M_R = \frac{\kappa R^2}{2}(\rho - U_0 - v\Pi F)$$

where $\rho=(\Pi^2+\beta^2F^2)/2+U$, $U_0=21/32$ and

$$m = \frac{R}{2}(1 + v^2 - \beta^2) = M + \frac{\kappa U_0 R^3}{6}.$$

Here m is geometric mass, equal to G times physical mass for $c=1$. Fix $v=H_0R$. The pulse requires the displayed radial extrinsic-curvature correction; retaining $K_r=-H_0$ there would violate the momentum constraint.

The core ends at $R_c=.120974320414$. A C4 bump depletes the inherited displacement $\chi_0=.000433012702$ on $[R_c,2R_c]$. A sinusoidal pulse with peak envelope $.02\chi_0$ occupies $[3R_c,4R_c]$ and has $\Pi=-\beta\ F$. Its outward current is $j=\beta^2F^2$. The potential is retained, so this instantaneous characteristic choice is not a claim of purely outward massive propagation at all later times.

Solving $M(4R_c)=0$ fixes the maximum displacement reduction to 18.5913687%. The mass contributions are

Region	Geometric excess, atlas units
Coherent core	.0000957140112793
Ambient reservoir	-.0000964548506037
Outward-current pulse	.000000740839324389
Enclosing offset	zero within 2×10^{-11}

The canonical absolute energy obeys $\rho\geq\rho_v=9/32$ throughout. Depletion concerns excitation above that floor. Both original walls retain their initial speed, acceleration and full junction data. Local geometry does change: the maximum compactness excess $2M/R$ is about .00687441, and the parent's proper length increases by .000244008403. Equal enclosing mass does not imply identical internal geometry or proper energy integrals.

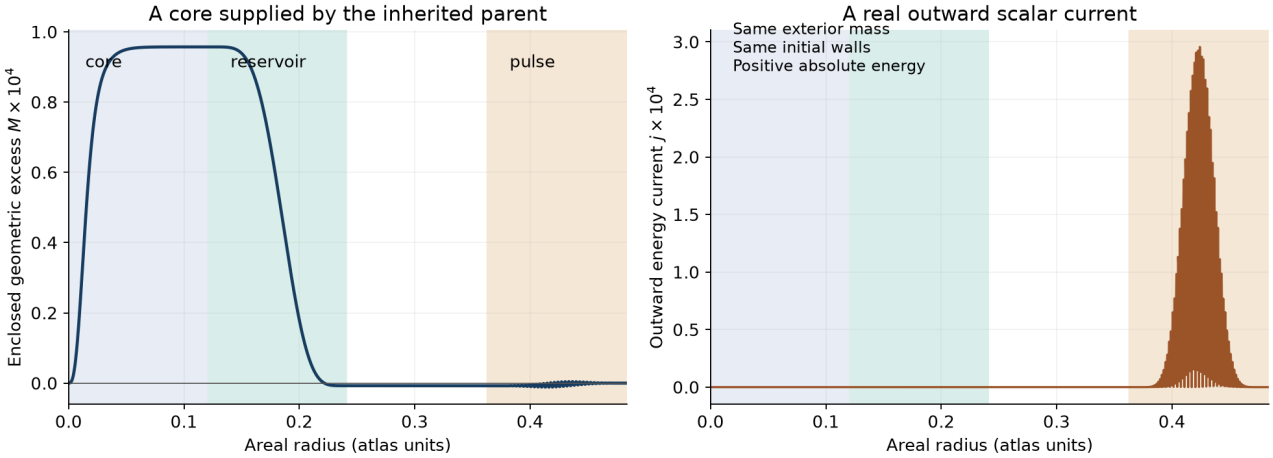


Figure 18. Left: the core's positive geometric mass is supplied by ambient depletion, with the outward-current pulse included in the balance. Right: the pulse carries nonzero outward scalar energy while the total initial exterior mass offset vanishes. These are constrained initial data, not a claim that the previous vacuum-limit cooling run formed this configuration.

Independent proper-distance integration, exact symbolic constraints and zero-source/no-pulse controls check the construction. Tightening integration changes the depletion fraction by less than 10^{-7}. The source and compensation derivation and finite record give the full preparation and numerical comparison. The solved root fixes one compatibility datum; it does not select a primordial seed or abundance.

14.4. A perturbed centre and the same evolving cosmology

At $R_b=4R_c$, the perturbation and its matching derivatives vanish, $\Pi=0$ and $M=0$. Outside this sphere the spatial metric, extrinsic curvature and scalar data equal the original parent exactly. This makes the causal connection stronger than the earlier core-only admission.

Use the homogeneous reference angle θ, with $a_0=6.62607707610$, $\theta_b=\arcsin(R_b/a_0)=.0730942965661$ and $\theta_w=2.85076435977$. Its future conformal-distance bound remains $\eta_*=.515418396612$. Local Einstein--scalar uniqueness gives the reference evolution on $\theta_b+\eta(t)<\theta<\theta_w-\eta(t)$. The fixed annulus

$$.588512693178<\theta<2.335345963158$$

therefore stays in that domain for all future reference proper time. It occupies at least 86.6472% of the **actual initial parent volume**. The bound concerns a smooth exterior domain; it does not assume the same light cones inside the perturbed core or establish a complete moving-wall continuation. Any compatible causal development agrees on this domain.

For $L_{ref}=a_0$ fixed by the retained geometry, its inherited vacuum component remains

$$\Lambda_{\mathrm{vac}} L_{\mathrm{ref}}^2 = \kappa \rho_v a_0^2 = 12.3482523989285.$$

The excited scalar still contributes to the total stress. Rearranging it into local structure does not change the inherited vacuum floor. This establishes coexistence of a nonlinear local source and a connected, evolving surrounding cosmology. It does not determine the observed Lambda or an observed dark-matter abundance.

The dynamical connection must keep flux and pressure work together. Hayward's spherical mass law gives

$$n(m) = \frac{\kappa R^2}{2}(\Pi \beta^2 F - v p_r)$$
$$p_r = \frac{\Pi^2 + \beta^2 F^2}{2} - U.$$

For the supplied outward characteristic, the fixed-areal-radius derivative $D=n-(v/\beta)s$ satisfies

$$D(m) = -\frac{\kappa R^2}{2}\beta F^2(\beta + v)^2 \le 0.$$

Where the fixed-radius trajectory is timelike, normalize D to its proper time. At the sampled peak current its mass rate is approximately -3.81559×10^{-5}. An expanding normal sphere has a different rate because pressure does work. The connected derivation and checks retain both contributions. The next task is to evolve this compensated preparation and calculate a local signal, preserving the source, outgoing field and same-action geometry throughout.

14.5. Evolving the inherited field and its geometry together

The compensated slice supplies a full finite spherical evolution. The scalar, its ambient reservoir and radiation source the local metric throughout. The earlier envelope profile is an initial control; it does not replace the Einstein--Klein--Gordon evolution equation. The action, frequency 2000, regional flux and inherited vacuum remain fixed.

Use areal radius and the parent expansion H(t) in the metric

$$ds^2 = -\alpha^2 dt^2 + \frac{(dR - \alpha H R\, dt)^2}{B} + R^2 d\Omega^2$$
$$B = 1 + H^2 R^2 - \frac{2m}{R}.$$

This fixes angular extrinsic curvature $K_t=-H$, while the radial curvature, mass and lapse respond to the source. With $F=\varphi_R$ and $\Pi=n(\varphi)$, the momentum and Hamilton constraints give

$$K_r = -H - \frac{\kappa R}{2}\Pi F$$
$$m_R = \frac{\kappa R^2}{2}(\rho - HR\Pi F).$$

The angular Einstein equation preserves that slicing by determining

$$B\alpha_R = \dot{H}R + \alpha\left(H^2R + \frac{m}{R^2} + \frac{\kappa R}{2}p_r\right).$$

The lapse is normalized in the untouched parent. Central proper time is calculated from it. The scalar evolves by its unaveraged wave equation on this metric, and the same scalar drives the reference H(t). The equations and implementation retain the shift, lapse gradient, radial momentum and regular-centre limit. The finite record binds the model inputs and numerical controls.

Independent of the radial constraint, the geometric mass obeys

$$m_t = \frac{\alpha\kappa R^2}{2}\left[(B + H^2R^2)\Pi F - HR(\Pi^2 + BF^2)\right].$$

Its time integral is compared with reconstructed mass at six fixed radii. Both expressions subtract the evolving homogeneous density. No absorber deletes radiation, and expansion pressure work is retained.

A smooth moving radial map keeps the core at fixed physical resolution and carries the outer grid with the parent. The boundary's initial angle .648103368884 lies in the permanently protected reference annulus. It can therefore receive the exact homogeneous data without introducing an artificial reflecting wall. This connects the local evolution to an actual parent domain while leaving the complete material-wall future open.

The calculation reuses the spherical oscillaton literature, regular-centre methods and Lawson's integrating-factor approach. The latter integrates the fast scalar rotation exactly within the time step, so numerical damping of that oscillator is not interpreted as cooling. The new connection is the sourced inherited preparation and its observer, rather than a new generic scalar-gravity equation.

14.6. A diffuse cloud develops a compact, breathing concentration

Replace the supplied coherent core by a compactly tapered Gaussian of width .03. Solve its amplitude, .00186251874314, to preserve the initial geometric mass inside the core boundary. This fixes the same total gravitational budget, including the ambient cross term. It does not equate isolated scalar norm between different profiles.

The resulting data are evolved to parent time5. Their final shape is not imposed. The diffuse cloud initially expands, subsequently contracts and then breathes. The same scalar preparation on the homogeneous metric provides a diagnostic of evolution without local gravitational response; that control is not claimed to satisfy the sourced Einstein equations.

Over parent times 4.5--5, the half excess-mass radius inside R=.12 averages .01849207, compared with .03930864 initially: a ratio .470433. The late central excitation averages 148.3729 against 10.53893 initially, a factor 14.0786. With the same field on the homogeneous metric it instead averages .0477285. Thus the concentration depends on the computed local gravitational response. The medium/fine spreads are 1.19e-7 in mean radius and .00232 in mean central excitation.

Resolution multiplier	Maximum integrated mass-flux gap, diffuse run
1	2.55e-7
2	1.27e-8
4	8.42e-10

The finest run keeps the local lapse above .98145. Its final exterior mass offset is 2.04e-9, against the initial core excess 9.57e-5. These finite residuals are numerical diagnostics, not exact-zero assertions. The formation is followed by substantial breathing; no stationary endpoint is imposed or inferred.

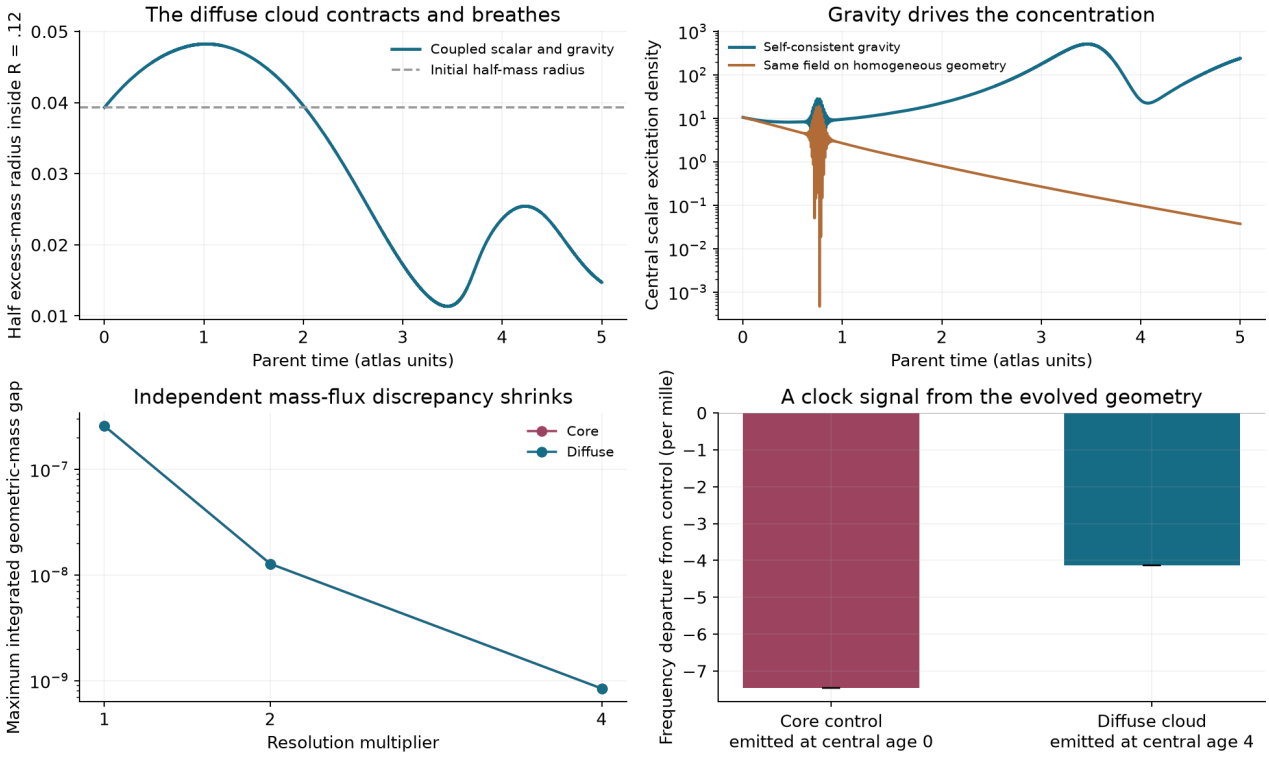


Figure 19. The diffuse preparation develops a compact concentration under its computed gravity. The fixed-geometry field control separates ordinary scalar propagation from that response. Refinement checks the independent mass-flux law. Clock departures use the stated central proper emission ages and a receiver held at the same areal radius in each corresponding homogeneous comparison.

These are finite nonlinear concentration and breathing results. They do not establish an asymptotic stationary remnant, universal formation from arbitrary preparations or nonspherical stability. The formation and refinement record and source owner retain those limits.

14.7. Conservation selects a second local scale

The image-delay relation in Section 15.6 raises a physical question: can the inherited field support a structure much smaller than its cosmological parent? Changing a unit rescales both. Changing the scalar frequency changes the physical model. The frequency family already constructed above holds $\kappa=1$, mixing $\varepsilon=1/4$, fluxes $(.5,1,1.5)$, charge $.5$ and tension $.2$ fixed. Its common initial scalar is at the grandchild minimum. The parent vacuum, excited density and available excitation are then

$$\Lambda_v = \kappa U_v = \frac{9}{32}$$
$$U_0 = \frac{21}{32}$$
$$E_b = U_0 - U_v = \frac{3}{8}.$$

The original initial wall data and parent reference radius are independent of frequency. The later scalar history depends on it. The following preparations use the existing coherent core shape, a depleted ambient annulus and zero scalar normal momentum. They contain no added outgoing pulse and are distinct from the previously evolved diffuse lens.

Let $x=\omega\zeta R$, let $A(x)$ be the retained core shape with its cutoff at $x=20$, and write

$$f = \sqrt{2/\kappa}\,\zeta^2 A(x)$$
$$\mathfrak{M} = \frac{\zeta}{\omega} N(x).$$

The ambient scalar displacement is d/ω, with $d=\sqrt{3}/2$. If C is the original parent spatial curvature, its exact initial mass constraint becomes

$$N' = -\zeta^4 x (A')^2 N + \frac{x^2}{2}\left[A^2 + \zeta^2 (A')^2\right] + \frac{x^2\sqrt{\kappa/2}\,dA}{\omega\zeta^2} - \frac{C x^4 (A')^2}{2\omega^2}.$$

The coefficient multiplying N does not depend on frequency. With a regular centre this proves, for the supplied shape,

$$\mathfrak{M}_c = \frac{\zeta N_0(20)}{\omega} + \frac{\zeta N_1(20)}{\omega^2} + \frac{\zeta N_2(20)}{\omega^3}.$$

This exact decomposition is not a fitted scaling curve. Its coefficients retain self-gravity, interference with the inherited ambient field, and parent curvature. Geometric excess mass is measured above the excited homogeneous parent; in physical units it is GM_c/c^2.

A reservoir cannot supply arbitrary positive core mass. For zero initial scalar momentum, the positive radial chart gives

$$\mathfrak{M}_R = \frac{\kappa R^2}{2}\left(\frac{BF^2}{2} + U - U_0\right) \geq -\frac{\kappa E_b R^2}{2}.$$

Reducing the core mass to zero at the reservoir's outer radius therefore requires

$$R_d^3 - R_c^3 \geq \frac{6\mathfrak{M}_c}{\kappa E_b}$$
$$R_c = \frac{20}{\omega\zeta}.$$

Since $\kappa\, E_b = 4\Lambda_v/3$ for this inherited state, every smooth finite-energy match in the stated domain satisfies

$$\frac{\Lambda_v (R_d^3 - R_c^3)}{\mathfrak{M}_c} > \frac{9}{2}.$$

Equality would require the scalar to remain at its parent minimum almost everywhere through the annulus, contradicting continuous matching to the excited ambient field at its ends. This lower bound does not depend on a chosen depletion profile. Its coefficient belongs to the declared inherited excitation/vacuum ratio, rather than to every possible scalar state.

The core mass falls as ω^{-1}, whereas a reservoir with a fixed ratio R_d/R_c has available excitation falling as ω^{-3}. For $R_d = 2R_c$ and the retained $\zeta = .0826621713$, the resulting necessary frequency ceiling is 6221.94. Above it, even full removal of the excitation from that annulus cannot supply the core. Gradient energy strengthens the restriction. The full derivation also extends the inequality to arbitrary scalar momentum where $B \geq v^2$.

Allowing the reservoir radius to respond resolves the obstruction. Choose

$$\chi(R) = \frac{d}{\omega}[1 - ab(x)]$$
$$x = \frac{R - R_c}{R_d - R_c}$$
$$b(x) = 1024x^5(1 - x)^5$$

with supplied depletion $0<a\leq 1$. The scalar returns smoothly to its original parent value at both joins. Solve the exact constraint for R_d so that the exterior geometric mass offset vanishes. Outside this radius the original initial parent geometry and material-wall matching data are recovered exactly. The independent proper-distance Hamilton equation checks the matching.

For fixed core amplitude and depletion, put $R_d=z\omega^{-1/3}$ and use the positive leading mass coefficient $C_m=\zeta\, N_0(20)$. The scaled endpoint constraint tends to

$$0 = 1 - \frac{\kappa d^2 z^3}{4C_m} J(a)$$
$$J(a) = \int_0^1 x^2[2ab(x) - a^2 b(x)^2]dx > 0.$$

Its positive root has nonzero derivative. The implicit-function theorem continues it to sufficiently large finite frequency. Provided the limiting core retains a positive radial-chart margin, the reservoir also remains regular and lies inside the parent. The finite examples check these conditions. This yields the hierarchy

$$R_c \sim \omega^{-1}$$
$$R_d \sim \omega^{-1/3}$$
$$R_d/R_c \sim \omega^{2/3}$$
$$R_d \ll L_{\mathrm{ref}}.$$

The parent vacuum and original initial wall geometry remain fixed. Each member retains a causally protected parent annulus under the previously derived energy-floor bound. These are initial-constraint and causal-domain results; formation, persistence and the complete coupled wall future are separate calculations.

14.8. A mass--reservoir--vacuum relation and a dimensional example

The same construction produces a dimensionless correlation:

$$\frac{\Lambda_v R_d^3}{\mathfrak{M}_c} \longrightarrow \frac{4\Lambda_v}{\kappa d^2 J(a)}.$$

For the declared half-depletion profile,

$$J(1/2) = \frac{5476736}{66927861}$$
$$\frac{\Lambda_v R_d^3}{\mathfrak{M}_c} \longrightarrow \frac{200783583}{10953472} = 18.3305880546.$$

The finite construction reaches 18.33027699 at $\omega=10^{10}$. Its reservoir/core cutoff ratio is then 40578.7, while $\Lambda_v\ L_{ref}^2=12.34825239893$ is unchanged. The coefficient is specific to the supplied bump and

depletion. The additional quarter- and three-quarter-depletion examples preserve that physical freedom explicitly; this is not a universal constant selected by nesting.

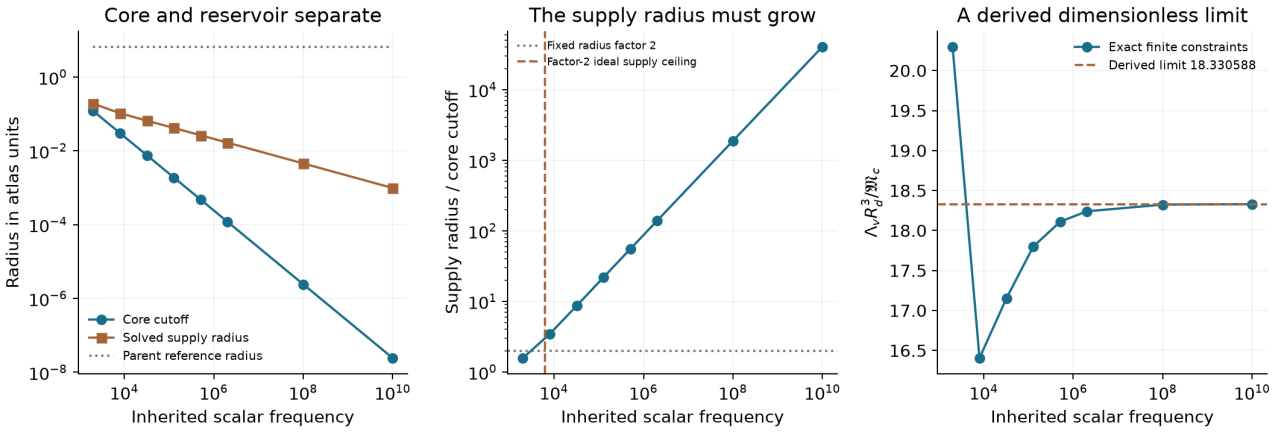


Figure 20. The core shrinks faster than its energy-supply region while the parent reference radius remains fixed. A reservoir constrained to twice the core cutoff becomes insufficient above its ideal supply ceiling. Solving for the reservoir radius restores the initial constraints and approaches the independently derived vacuum-radius-mass relation.

An illustrative member takes $\omega=10^{9}$, $\zeta=.001$ and half-depletion. If the dimensional reference $\Lambda_{v}=10^{-52}$ m^{-2} is supplied, the consistent conversion gives:

Quantity	Conditional value
Core geometric excess mass	6.32×10^{10} nominal solar masses
Half-core-mass radius	3.86 kpc
Compact core cutoff	34.4 kpc
Compensating reservoir outer radius	828 kpc

The unit conversion uses the IAU nominal solar mass parameter. Neither that reference vacuum nor the chosen field/preparation parameters were inferred here. No galaxy observation was fitted. The example proves that compatible initial data can admit these separated scales; it does not establish a stable halo, an observed rotation curve or a cosmic abundance. The exterior total mass offset is zero: the core is supplied by redistribution of the inherited excitation, not by creation of additional total mass.

The compact evidence contains thirteen examples and controls. The exact scaled core identity has zero symbolic residual. Its mass agrees with the original constraint integrator within 2.7×10^{-16} atlas units. The proper-distance Hamilton check leaves less than 4.2×10^{-13} of core mass at the outer join; tighter integration changes reported masses and radii by less than 1.1×10^{-11} relatively. Those are finite integration checks for the supplied profile, not a bound on its uncomputed evolution.

The following section evolves this two-scale preparation in a leading classical envelope with its reservoir retained. The image predictions of Section 15.5 belong to the earlier evolved preparation; they cannot be assigned to this new family by changing units.

14.9. Evolving the core together with its reservoir

Initial compatibility does not establish persistence. The high-frequency member of Section 14.8 has many fast carrier oscillations per binding time, so a direct relativistic grid is costly. We instead derive its leading classical envelope, retain the solved reservoir and excited parent, and test the reduction against the unchanged full Einstein--Klein--Gordon solver on a short, tractable family. The long result belongs to the reduced equations and their stated domain.

For the real displacement $\chi=\varphi-\varphi_{\backslash min}$, define

$$\chi = \sqrt{2/\kappa}\,\zeta^2 A^{-3/2}\mathrm{Re}[\psi(x,\tau)e^{-i\omega t}]$$
$$x = \frac{\omega\zeta R}{A}$$
$$\tau = \omega\zeta^2 t.$$

The normalized parent scale factor starts at $A=1$. The complex amplitude ψ describes the same real field; it is a total amplitude, including its homogeneous value

$$b = \frac{\sqrt{\kappa/2}\,d}{\omega\zeta^2} = .000612372435696$$
$$d = \frac{\sqrt{3}}{2}.$$

For this member $\omega=10^9$ and $\zeta=.001$. The initial field is the previously supplied tapered core plus its depleted ambient annulus. The real scalar value matches the exact constrained initial slice. The initial normal momentum matches only to leading order, since the slow prefactor contributes an $O(H/\omega)$ carrier-quadrature correction. The small initial gradient-mass difference is recorded rather than removed by renormalizing the field.

Averaging the homogeneous carrier retains the parent's excitation, vacuum and positive spatial curvature:

$$H^2 = \frac{\Lambda_v}{3} + \frac{\kappa E_b}{3A^3} - \frac{C}{A^2}$$
$$\dot{H} = \frac{C}{A^2} - \frac{\kappa E_b}{2A^3}.$$

Here $\Lambda_v=9/32$, $E_b=3/8$ and $C=.0227765023676$. Excitation is initially $4/7$ of the parent density; a pure de Sitter replacement would lose a leading contribution. On the local, nearly flat spatial chart, the classical envelope obeys

$$i\psi_\tau = -\frac{\Delta_x\psi}{2A^2} + V\psi$$
$$\Delta_x V = \frac{|\psi|^2 - b^2}{2A}.$$

The complete density contrast is retained, including its coherent cross term. The homogeneous background is already inside ψ, so it is not added a second time. The physical Newtonian potential is $\zeta^2 V$. This uses the established real-scalar envelope framework in the explicit source normalization of this atlas; it introduces no additional field or measurement-probability law.

One exact check clarifies why the homogeneous envelope can be constant. On the averaged parent, a field proportional to $A^{-3/2}\cos(\omega t)$ satisfies

$$(\partial_t^2 + 3H\partial_t + \omega^2)\chi_b = \left(-\frac{3}{2}\dot{H} - \frac{9}{4}H^2\right)\chi_b = \frac{3}{4}\left(\frac{C}{A^2} - \Lambda_v\right)\chi_b.$$

The dust terms cancel. The residual is small relative to the carrier term, about 2×10^{-19} for this member. This is an exact identity on the averaged background, not proof that it equals the full oscillating Einstein-scalar background at every instant.

14.10. Conserved transport, finite persistence and a relativistic check

The envelope's geometric excess mass is

$$\mathfrak{M}_{\mathrm{env}}(R)=\frac{\zeta}{2\omega}\int_0^{\omega\zeta R/A}x^2(|\psi|^2-b^2)dx.$$

The factor of one-half follows from the Einstein mass normalization. The initial envelope core mass differs from its exact GR value by -3.31923×10^{-7} relatively. That difference is retained. Continuity fixes the current; at a fixed physical radius the moving comoving cutoff $x_*=X/A$ adds a necessary term:

$$\frac{dN_*}{d\tau}=-\frac{x_*^2}{A^2}\mathrm{Im}(\psi^*\psi_X)-\frac{A'}{A}x_*^3(|\psi|^2-b^2).$$

Independently integrating this current tests the changing enclosed mass. The energy functional has kinetic and gravitational parts

$$T=\frac{1}{2A^2}\int x^2|\psi_X|^2dx$$
$$W=\frac{1}{2}\int x^2V(|\psi|^2-b^2)dx.$$

With the symmetric Newtonian kernel and a time-independent outer field value $\psi=b$, the reduced equations imply

$$\frac{d(T+W)}{d\tau}=-\frac{A'}{A}(2T+W).$$

Both terms contribute to expansion work. No absorber discards outgoing field. The outer mass current is measured, and a larger domain checks its influence on the core. These finite envelope-domain checks are not a new relativistic causal-shielding theorem.

Eleven numerical cases evolve the retained high-frequency preparation to slow time *600*, with three resolutions and separate time, outer-boundary and no-gravity controls. The parent scale factor reaches *1.27443638723*. Over the late interval $480\le\tau\le600$, mass inside the original fixed physical core cutoff has ratio

$$\frac{\langle\mathfrak{M}_{\mathrm{core}}\rangle}{\mathfrak{M}_{\mathrm{core}}(0)}=1.00374088\pm.00001314.$$

This is a small inward transfer, checked by the mass current. With self-gravity removed, the same late ratio is *.000190975*. The core's late mean physical half-mass radius is *2.23255821* in the scaled length $\omega\zeta R$, with numerical comparison range *.00319767*. Its radius breathes: the peak-to-peak range grows from *.0674* in the first *120* slow-time units to *.1399* in the last *120*. Finite localization therefore does not establish asymptotic stability.

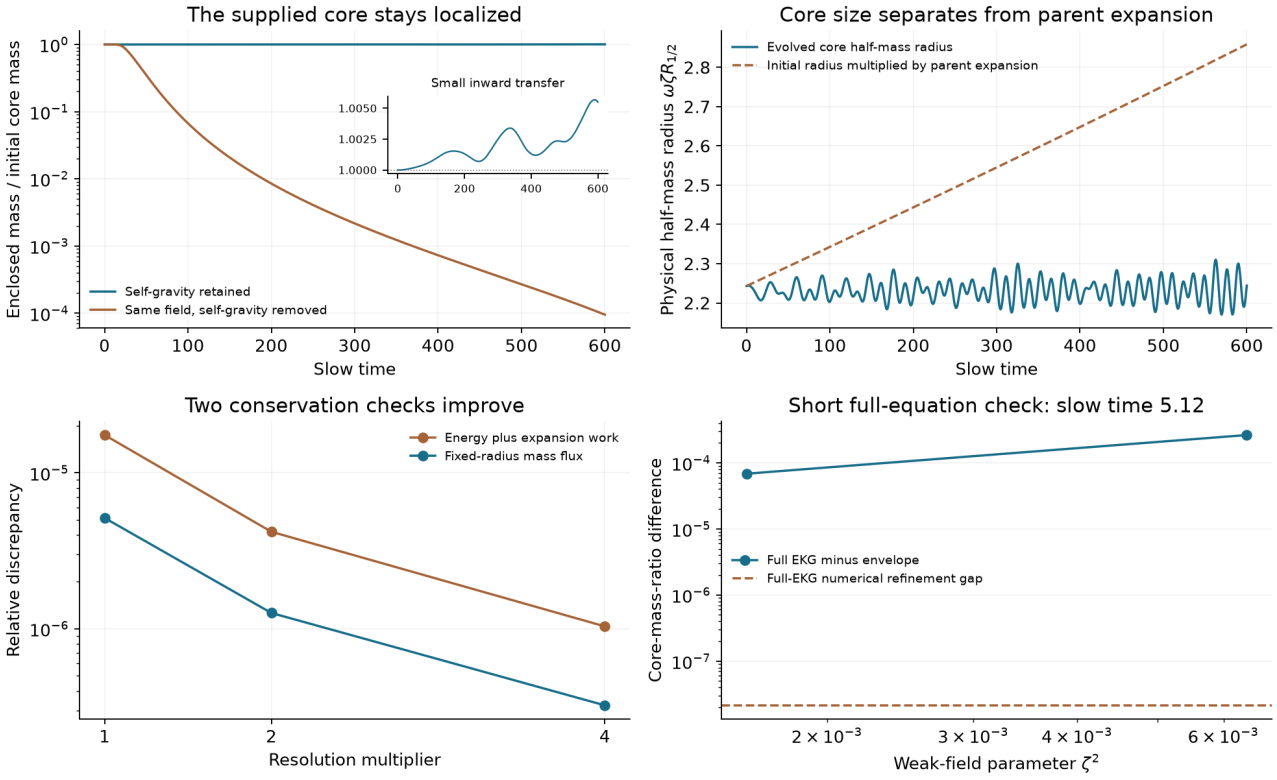


Figure 21. The same supplied field disperses without self-gravity and retains a localized core when self-gravity is included. Its physical size breathes while the parent expands; the dashed size reference is simply initial radius multiplied by the parent scale factor. Mass-current and energy-work checks improve under refinement. The last panel compares the reduced equations with full EKG over a separate short interval.

The mass-current discrepancy falls from 5.13×10^{-6} through 1.26×10^{-6} to 3.24×10^{-7} of initial core mass. The energy-work discrepancy falls from 1.75×10^{-5} to 1.04×10^{-6} of initial energy. The compact calculation records source hashes, finite comparisons and the small net outer transport. The derivation and domain keep numerical error distinct from approximation error.

For the independent short comparison, hold $\omega\zeta^2=12.8$ and evolve to atlas time $.4$, or slow time 5.12. The original full EKG functions are unchanged:

Core amplitude	Frequency	Full-EKG mass ratio	Envelope mass ratio	Difference
.08	2000	1.007402025	1.007139302	.000262723
.04	8000	1.007190375	1.007121981	.0000683932

Halving the amplitude reduces the discrepancy by about 3.84, consistent with weak-field ordering. Refining the smaller full-EKG case changes its ratio by 2.15×10^{-8}, far below that difference. Analytic free Gaussian propagation, an independent Gaussian gravitational potential and moving-cut continuity provide additional checks.

Using the earlier supplied reference $\Lambda_v=10^{-52}\ m^{-2}$, the long run spans 3.36 billion years. Its fixed mass cutoff is 34.4 kpc and its late mean half-mass radius is $3.837\pm.006$ kpc. Those physical values inherit the assumed ruler; their quoted ranges are numerical ranges within the envelope model. No galaxy observations were fitted.

The short full-equation comparison and small expansion, curvature and weak-field parameters do not supply a rigorous full-GR error bound at slow time 600. The supplied initial cloud, growing radial breathing and nonspherical persistence remain physical questions. The completed connection is from

the conserved initial supply construction to finite radial localization in its expanding parent, with a stated approximation and independent transport checks.

14.11. Testing the breathing core beyond spherical symmetry

A changing radius does not by itself diagnose an instability. We now linearize the actual evolving envelope of Section 14.10 and calculate its radial and quadrupole response. The spherical field, reservoir and parent expansion are replayed unchanged. A stationary eigenproblem would answer a different question because this background is time dependent.

Perturbations of Schrodinger--Newton stationary states have prior analyses by Harrison, Moroz and Tod. Zagorac et al. investigate breathing, quadrupolar deformation and time-dependent soliton perturbations. Their stability or profile conclusions are not substituted for this trajectory.

Use real angular functions with angular mean square one. For an axisymmetric quadrupole, $Y_{20}=\sqrt{5}P_2$. With real perturbation parameter ε, write

$$\psi = \psi_0 + \varepsilon \eta_l Y_{lm} + O(\varepsilon^2)$$
$$V = V_0 + \varepsilon v_l Y_{lm} + O(\varepsilon^2).$$

The real-linear equations are

$$i\partial_\tau \eta_l = \left[-\frac{D_l}{2A^2} + V_0\right]\eta_l + v_l \psi_0$$
$$D_l v_l = \frac{\mathrm{Re}(\psi_0^* \eta_l)}{A}$$

$$D_l = \partial_x^2 + \frac{2}{x}\partial_x - \frac{l(l+1)}{x^2}.$$

The potential source is real, so complex phase directions are included explicitly as independent real initial directions. With $d_l=2\mathrm{Re}(\psi_0^* \eta_l)$, regularity and the decaying exterior potential give

$$v_l(x) = -\frac{1}{2A(2l+1)}\left[\frac{1}{x^{l+1}}\int_0^x s^{l+2} d_l(s)ds + x^l \int_x^L s^{1-l} d_l(s)ds\right].$$

Both integrals are retained. The finite source is zero outside L; the potential has its free decaying tail rather than a zero-potential wall. The $l=2$ density has zero angular mean, so its radial coefficient integral is not a monopole mass change.

During the joint potential substep the base field rotates but its density and the density variation remain fixed. The exact tangent flow is therefore

$$\eta_l^{\mathrm{new}} = e^{-ihV_0}(\eta_l - ihv_l\psi_0).$$

This is also the derivative of the nonlinear potential step. In $q=x\eta$ variables, the kinetic operator is $-\partial_x^2/(2A^2)+l(l+1)/(2A^2x^2)$. The radial channel uses the same spectral kinetic step as the base. The quadrupole uses a unitary centrifugal step, checked against an analytic solid-harmonic Gaussian.

The transported bilinear form

$$S_{jk} = \mathrm{Im} \int q_j^* q_k \, dx$$

is conserved. The kinetic and background-potential terms cancel by self-adjointness; the remaining source terms cancel through symmetry of the multipole Green kernel. This symplectic identity remains valid for a time-dependent background. It checks the transport but does not imply norm conservation or stability. A separate passive quadrupole, propagated without its own potential response, supplies the unitary control.

14.12. A finite response theorem for every orientation in a declared span

Choose two compact radial profiles of widths *2* and *4* in the physical scaled radius *Ax*, together with their phase partners. They taper to zero between radii *15* and *20*. Project the radial seeds to preserve initial total mass to first order; quadrupoles have zero angular mean automatically. Real Gram orthonormalization gives four initial directions.

Let N_c be the initial core-field norm. For any real coefficient vector *c* of unit Euclidean length, a fractional field perturbation is

$$\delta q(\tau) = \varepsilon N_c \sum_{j=1}^{4} c_j q_j(\tau).$$

Define $G_{jk}=Re\int q_j^* q_k dx$. The Rayleigh quotient then gives the exact finite-span bound

$$\frac{\|\delta q(\tau)\|_2}{\varepsilon N_c} \le \sqrt{\lambda_{\max}(G(\tau))}\,.$$

A second real Gram matrix uses the density variations inside the fixed physical core cutoff. Normalizing by the current base excess-density norm gives the fractional density response per fractional initial field perturbation. This is not a density-to-density growth factor: some initial phase directions have zero density variation. The density response of the direction with greatest field-norm growth is recorded separately.

Spherical symmetry gives the same radial propagator for all five real quadrupole components. Angular orthogonality gives the direct-sum Gram matrix $I_5 \otimes G$, with the same largest eigenvalue. The four-column calculation therefore covers every coefficient combination in the corresponding twenty-dimensional real quadrupole span. A new quadrupole family is also seeded on the evolved state at slow time *480*.

The normalized core second-moment quadrupole is defined explicitly by

$$\frac{\int R^2 P_2 \, \delta\rho \, d^3R}{\int R^2 \rho_{\mathrm{excess}} \, d^3R} = \frac{\varepsilon N_c}{\sqrt{5}} \frac{\sum_j c_j \int x^4 d_j dx}{\int x^4 d_0 dx}.$$

The factor follows from the angular average of $P_2 Y_{20}$, which is $1/\sqrt{5}$. Direct angular integration independently verifies this moment definition. It is distinct from the coefficient multiplying P_2 and from a Cartesian trace-free tensor component. A core-truncated moment is not inserted into an exterior potential formula; that calculation uses the full multipole source.

These are finite response statements on a specified radial span. Numerical maxima are reported on saved unit-slow-time events, not as rigorous bounds between every pair of samples. The result does not

cover additional radial profiles, reservoir-supported seeds, higher angular multipoles, nonlinear three-dimensional evolution or full-GR stability.

The response calculation reproduces every retained spherical field hash. Its sampled extrema are:

Perturbation family	Field-norm amplification	Density response per initial field perturbation
Radial, seeded at 0	*6.756±.162*	*3.592±.033*
Quadrupole, seeded at 0	*1.08113±.00025*	*1.15675±.00007*
Quadrupole, seeded at 480	*1.0820±.0011*	*1.1720±.0049*

For example, a *1%* fractional field-norm perturbation in the quadrupole span seeded at time zero gives a calculated first-order density response of at most about *1.157%* at the saved events. The quoted ranges compare spatial/time refinement and the outer domain. They are not rigorous interval enclosures, bounds between samples, or nonlinear guarantees at finite amplitude. Different observables can have different maximizing initial coefficient vectors.

Radial sensitivity is larger and more dependent on refinement. At the final event, the direction with greatest field amplification has *99.4%* of its squared norm inside the core; *99.5%* of that core part lies in the local phase direction relative to the base field. This does not make every radial response a phase change. Spatial phase gradients can affect later flow, and the independent density response is nonzero.

The symplectic discrepancy stays below 1.15×10^{-11} in the radial channel and 9.76×10^{-14} in the quadrupole. The passive quadrupole norm remains one within 5×10^{-14}. Independently, centred differences of the nonlinear radial flow converge toward the tangent prediction: halving the initial amplitude from 10^{-4} to 5×10^{-5} reduces the density-derivative error from 2.18×10^{-6} to 5.46×10^{-7}.

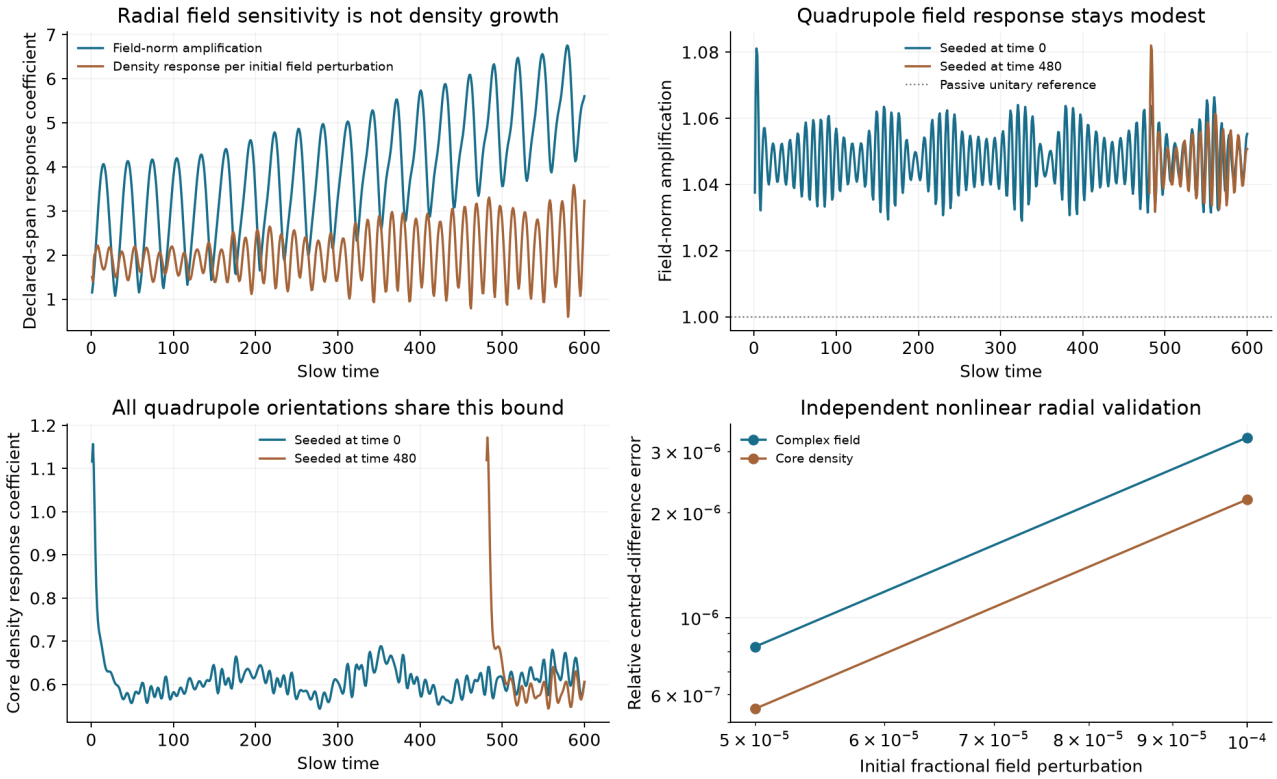


Figure 22. Field sensitivity and density deformation are distinct responses. The quadrupole field response remains modest for seeds at the initial and late events; the same radial matrices cover all five angular components in the declared span. Independent nonlinear radial differences converge to the tangent prediction. These finite sampled results do not establish universal or asymptotic stability.

14.13. The surrounding field must carry the tidal phase

A quadrupole has zero angular mean, so its first-order monopole mass change vanishes. It can nevertheless produce a nonzero external tidal potential. The full multipole kernel was therefore evaluated at fixed physical radius $2R_d$, outside the initial reservoir. Cauchy--Schwarz bounds each response coefficient. The squared angular basis functions sum to five at every direction, so the pointwise all-orientation potential bound is $\sqrt{5}$ times its angular-RMS bound.

The initial external amplitude failed its numerical comparison. At fixed spacing, enlarging the domain changed the coefficient from 3.33×10^{-14} to 9.28×10^{-15}. This is a diagnostic value, not an accepted nonzero tidal prediction. The failure identifies an outer condition that matters for the observer calculation.

In an ideal homogeneous ambient region, with no incoming density disturbance, let the quadrupolar potential be $v_2=C(\tau)/x^3$. It is harmonic: $D_2v_2=0$. The linearized field equation has the particular solution

$$\eta_2(x,\tau) = -ibx^{-3}\int C(s)ds$$
$$2\mathrm{Re}(b\eta_2) = 0.$$

It carries a phase tail without a first-order density change. Therefore

$$q = x\eta_2 \propto x^{-2}$$
$$q_x = -\frac{2q}{x}.$$

A zero-field condition at a finite outer radius suppresses this forced phase and can generate spurious density there. A separate calculation allows the leading harmonic tail through a self-adjoint boundary correction. Its Cayley step remains unitary and agrees with an independent dense-matrix calculation. General outgoing radiative or Jeans responses are not replaced by this particular harmonic condition.

The tail-matching comparisons reduce the outer-domain change from 2.40×10^{-14} to 7.26×10^{-17}, an improvement of about 331. The final core-density response changes by less than 1.8×10^{-8}, supporting the core result against this boundary change.

The external amplitude still fails the refinement requirement: its coarse-to-medium change is 8.80×10^{-15}, so twice that change exceeds the medium coefficient 9.34×10^{-15}. The public record explicitly leaves this outward prediction unresolved. The useful results are the finite core-response bound and the derived tidal-phase matching requirement. An external clock or lens prediction needs further controlled work; it is not inferred from the unresolved coefficient.

The perturbation owner records the definitions, interpretation, independent checks and compute limits. The original response campaign used about 4073 CPU seconds within its 4500-second budget. After the diagnosed boundary failure, the targeted tail calculation used about 615 within its separate 800-second budget. Both retain their completed evidence without enlarging their claims to all-profile, nonlinear or full-relativistic stability.

14.14. A local concentration carries the signature of its surroundings

Derivation and calculation: spatial compensation and retained evolution and derivation and source dictionary.

The dense core and its depleted surroundings are one initial solution. Their relation should therefore survive into a spatial statistic. Let $M(R)$ be excess geometric mass, M_c its value at the core cutoff, and R_d the solved compensation radius. On the retained zero-momentum initial slice,

$$\frac{dM}{dR} = \frac{\kappa R^2}{2}\left(U - U_0 + \frac{BF^2}{2}\right)$$
$$M(0) = M(R_d) = 0.$$

This is the areal mass measure from the Einstein constraint. A proper-volume energy integral is a different object. With $s=R/R_d$ and $K=kR_d$, define

$$W(K) = \frac{1}{M_c}\int j_0(Ks)\, dM$$
$$j_0(x) = \frac{\sin x}{x}.$$

The transform is an exact descriptor of the specified mass profile. Its flat-space density interpretation is local, on scales small compared with the parent's curvature radius. It does not substitute continuous Fourier modes for the compact parent's global spectrum. Compensation gives

$$W(K) = a_2 K^2 + \mathcal{R}_4(K)$$
$$a_2 = -\frac{1}{6M_c}\int s^2\, dM$$

$$|\mathcal{R}_4(K)| \le \frac{K^4}{120M_c}\int s^4 |dM|.$$

To prove this bound, write $j_0(x)$ as the integral of $\cos(xt)$ from zero to one and integrate the fourth-order cosine remainder. The zeroth moment cancels, and spherical symmetry eliminates the dipole. The retained profile gives $a_2=.0608058925116$ and remainder coefficient $.00129586307998$. Integrating the solved mass primitive against $Kj_1(Ks)$ independently reproduces the transform without differentiating stored data.

The simpler initial scalar envelope omits positive gradient mass. Using the same normalization in both descriptions yields an all-wavelength bound:

$$dM - dM_{\mathrm{env}} = \frac{\kappa R^2 BF^2}{4} dR \ge 0$$
$$|W - W_{\mathrm{env}}| \le g = \frac{1}{M_c}\int \frac{\kappa R^2 BF^2}{4} dR.$$

For this source $g=3.31911631\times10^{-7}$. The initial envelope has $W_{env}(0)=-g$; its small residual is retained. Near $K=.00234$ it competes with the quadratic term. Thus the absolute source bound cannot be converted into arbitrary relative accuracy near zero. Nor is it a long-time error bound for the reduced dynamics or an assertion that the instantaneous scalar pressure is dustlike.

Let W_c denote the core-only transform, with the same normalization. The resolved half-power crossing is defined by

$$\left[\frac{W(K_{1/2})}{W_c(K_{1/2})}\right]^2 = \frac{1}{2}.$$

For the unchanged $\omega=10^9$, $\zeta=.001$, half-depletion preparation,

$$K_{1/2} = 4.0522389138$$
$$\frac{\Lambda_v}{M_c k_{1/2}^3} = .2681493031.$$

The second number combines the already solved mass--reservoir--vacuum relation with the new spatial window. Three quadrature orders and a tighter constraint integration give comparison ranges below 9×10^{-12} and 1.8×10^{-12}, respectively. Those ranges concern integration of the fixed numerical core shape, not total model uncertainty.

At fixed weak amplitude and depletion, the established large-frequency hierarchy makes the core pointlike relative to its reservoir. Its window tends to one minus the transform of the normalized positive weight $s^2[2ab(s)-a^2b(s)^2]/J(a)$. Half depletion gives limiting crossing 4.1423205290 and limiting vacuum--mass--turnover ratio $.2578964084$. Finite quarter- and three-quarter-depletion preparations instead give $.47250817$ and $.20537800$. Inheritance has not selected one depletion.

With the previously supplied $\Lambda_v=10^{-52}\ m^{-2}$ ruler, the initial example has $R_d=.827653$ Mpc, $k_{1/2}=4.89606$ Mpc^{-1} and wavelength $2\pi/k_{1/2}=1.28331$ Mpc. These are conditional spatial numbers, not an observed halo population or a fitted cosmological spectrum.

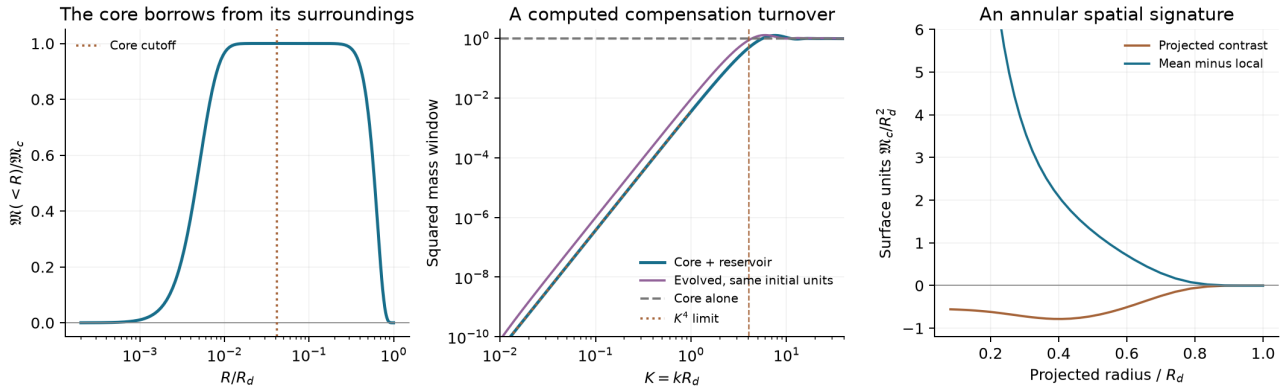


Figure 23. One supplied core and its inherited reservoir, viewed through enclosed mass, spatial frequency and projected contrast. Including the reservoir suppresses long-wavelength response. The purple curve reuses the evolved field in the original mass/radius units. The projected deficit is a spatial statistic; a relativistic lens observation needs its own averaging and light-cone dictionary.

14.15. The age and population are part of the correlation

The inherited vacuum remains fixed while the surrounding field evolves. Postprocessing the five authenticated spherical evolution checkpoints, with physical radius $R=Ax/(\omega\zeta)$, gives at slow time 600:

Spatial quantity	Fine result and comparison range
$k_{1/2}R_{d,0}$	$3.165445\pm.000031$
$\Lambda_v/(M_{c,current}k_{1/2}^3)$	$.559473\pm.000024$
Quadratic coefficient in original mass/radius units	$.0998276\pm.0000072$

The ranges include grid, time and outer-domain comparisons from the saved source; no evolution was rerun. This changes the initial coefficient by much more than numerical error. An age-independent interpretation of $.2681493$ is excluded for this history. Long-time full-relativistic accuracy remains outside the established envelope calculation.

The finite-domain total excess agrees with initial excess minus the outer mass flux to better than 3.5×10^{-9} of the initial core mass. Its small nonzero monopole is not resolved against outer-domain variation. Core accretion is not total mass creation. The turnover is nevertheless resolved, and neither

exact compensation at the original R_d nor a strict evolved zero-wave-number power law has been imposed.

For a spatial density descriptor $h=M_R/(4\pi R^2)$, projection along a line of sight gives

$$\Sigma(b) = 2\int_b^{R_d} \frac{h(R)R\,dR}{\sqrt{R^2-b^2}}.$$

Its planar Fourier transform equals the three-dimensional transform at zero line-of-sight wave number. The same compensation scale therefore appears in projection. Outside the core the initial example has negative surface contrast but positive mean-minus-local contrast, both vanishing outside R_d. A weak-field lens interpretation would connect those two quantities to convergence and tangential shear. It requires the averaged source and null-propagation approximation; the exact instantaneous scalar optics cannot be obtained by renaming this projection a lens map.

The correlation framework is established in Schmidt, 1511.02231v2 and Chen--Afshordi, 1912.04872v5, Section III. The contribution here is a calculated compensated kernel from the inherited source, its mass--vacuum scale relation and its measured age dependence. For identical local profiles with number-density fluctuation Δn,

$$\delta_{\rm red}(k) = \frac{M_c}{\rho_g} W(kR_d)\Delta n(k)$$
$$P_{\rm red}(k) = \left(\frac{M_c}{\rho_g}\right)^2 |W(kR_d)|^2 P_n(k).$$

Here ρ_g is mean geometric-mass density, and P_n is the spectrum of absolute number-density fluctuations. Bounded P_n gives an initial $O(k^4)$ redistribution spectrum; independent centres have $P_n=n_{mean}$. The total matter spectrum also contains the large-scale field and its cross correlations. Suppressing that field would discard the very signal one wants to explain. No abundance, formation law or matter fraction has been calculated by this single-profile construction.

The next connection is consequently formation and age selection in the inherited parent, followed by the observation map for that population. The atlas supplies a joint spatial relation to test when those inputs exist; it does not fix them by matching a desired sky spectrum.

14.16. Does the inherited parent leave enough time to form a core?

Derivation and calculation: formation budget and compact seed and all-wavelength comparison proof.

Persistence of an installed core does not explain its formation. The earlier diffuse-cloud example also began with a nonlinear overdensity. Before another collapse grid, ask how much a small ambient disturbance can grow in the same inherited parent. Keep $\omega=10^9$, $\Lambda=9/32$, $D=\kappa\, E_b=3/8$ and the retained curvature C:

$$H^2 = \frac{\Lambda}{3} + \frac{D}{3A^3} - \frac{C}{A^2}.$$

Linearizing the existing classical envelope gives density contrast δ and phase θ. For local comoving wave number k,

$$\dot\delta = \frac{k^2\theta}{\omega A^2}$$
$$\dot\theta = -\frac{k^2\delta}{4\omega A^2} + \frac{\omega D\delta}{2Ak^2}$$

$$\ddot{\delta} + 2H\dot{\delta} + \left(\frac{k^4}{4\omega^2 A^4} - \frac{D}{2A^3}\right)\delta = 0.$$

The related averaged massive-scalar equation is given by Hwang--Noh, 0902.4738, equations (19)--(28). Here it follows from the already declared local envelope with the actual inherited expansion. This is not a global flat-parent approximation or a new quantum postulate. An initially real envelope has zero current; a growing-mode velocity would be additional preparation data.

There is no need to guess a sufficiently late stopping time. Set $u=1/A$, $\nu=k^2/(2\omega)$ and introduce the rescaled density velocity:

$$p = A^2\dot{\delta}$$
$$\delta_u = -\frac{up}{H(u)}$$
$$p_u = \frac{\nu^2 u - D/2}{H(u)}\delta.$$

The entire expanding future ends at the regular point $u=0$, because H stays positive and tends to the vacuum expansion rate. The transfer matrix in (δ,p) has determinant one, providing an independent check. Regularity of this reduced endpoint does not establish infinite-time agreement with full GR or complete the unresolved wall future.

To bound every wavelength, use kinetic time $d\eta=dt/A^2$. The equation becomes a forced oscillator with nonnegative feedback:

$$\delta_{\eta\eta} + \nu^2\delta = \frac{DA}{2}\delta.$$

The free Green kernel obeys $|\sin[\nu(\eta-s)]/\nu| \le \eta-s$. Iterating the positive Volterra comparison therefore bounds the absolute solution by the zero-frequency density and velocity responses. This proves the bound even when a finite-wave-number solution oscillates or changes sign; simply deleting its pressure term would not prove that statement.

For zero initial current, an elementary Gronwall estimate gives

$$|\delta_k| \le \exp\left(\frac{Du_i^3}{12H_{\min}^2}\right)|\delta_i|.$$

At $A_i=1$, this factor is 1.39616912, below 1.397. Solving the positive comparison gives the sharper evaluated relation

$$|\delta_k(t)| \le 1.26934144123|\delta_i| + 0.68728524662|v_i|$$
$$v_i = \dot{\delta}_i/H_i.$$

The exact comparison solutions define the coefficients; the printed decimals have numerical comparison ranges below 3.4×10^{-12}, not certified interval endpoints. The independent pressureless basis $H(A)$ and its reduction-of-order partner reproduces the result within 8×10^{-15}. The standard basis is attributed to Heath (1977); its governing identity is verified directly here.

Thus a small zero-current disturbance introduced at this late anchor cannot reach nonlinear contrast through the stated linear mechanism, even if evolved forever. This conclusion is specific to its epoch,

initial current and equations. Earlier seeds, forcing or already nonlinear data are different possibilities. The bound also applies to spatial L2 norm; a pointwise statement needs an appropriate Fourier norm.

14.17. A compact earlier seed identifies the nonlinear continuation

Construct a local compensated disturbance without the stationary core:

$$g(r) = (1 - r^2/L^2)_+^4$$
$$\delta_i(r) = \varepsilon \frac{-\Delta(g * g)(r)}{\int |\nabla g|^2 d^3x}.$$

It has support $2L$, zero total density contrast, central value ε and nonnegative Fourier transform proportional to $k^2|G(k)|^2$, where G is the Fourier transform of g. Consequently $|\delta_i(r)| \le \varepsilon$, and the mode comparison gives a pointwise future bound for this entire seed. Its linear Poisson potential is compact as well. These are smooth local-envelope data, not yet a new exact nonlinear shared-wall preparation.

Take $L=.000481562379847$, the previous supply radius interpreted as a comoving length, a supplied peak contrast of 5%, and zero initial phase. At $A_i=1$ the central future gain is 1.26488829; the global pointwise bound keeps the contrast below about 6.35%.

Now place that same dimensionless seed at $A_i=.01$ on the earlier averaged-parent continuation. Its central linear gain is

$$35.609032 \pm .000004.$$

The stated range combines an analytic spectral-tail bound with a numerical integration comparison. No truncated spectrum is renormalized. For $q=kL$, the normalized bump transform is $10395 j_5(q)/q^5$; Parseval fixes the density-weight integral, while the explicit sine/cosine polynomial for j_5 bounds its omitted tail. At $q=60$ the omitted pointwise gain is at most 3.07×10^{-6} for all future times. Quadrature and ODE comparisons add about 1.02×10^{-8}.

The supplied 5% seed reaches linear contrast $.1$ near $A=.0957400$. Its formal linear extrapolation reaches contrast one near $A=1.0190483$. That second event is outside linear validity: it identifies the nonlinear continuation to calculate, not a completed collapse or a formed core. The late-parent bound and the earlier amplification together show why the preparation epoch is a physical part of the proposed formation route.

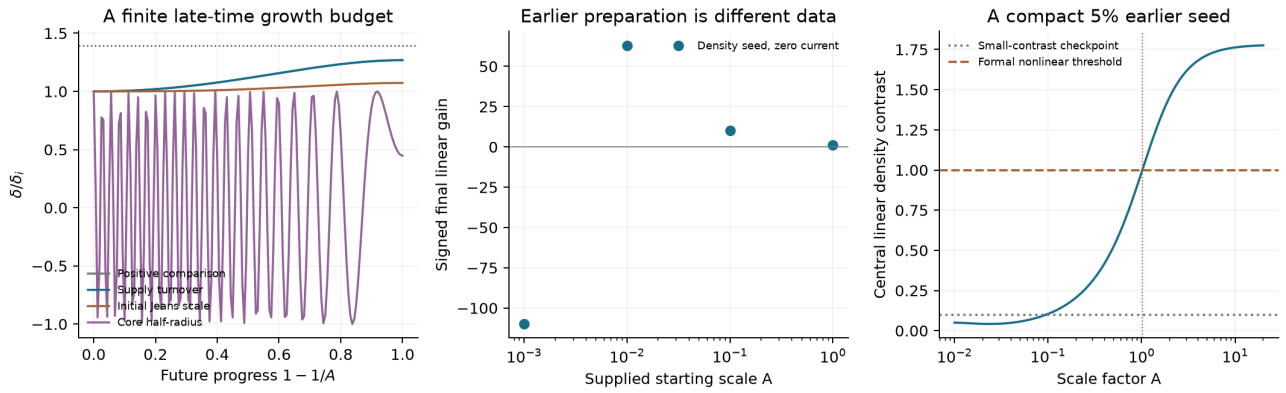


Figure 24. Left: the late parent permits only modest ambient amplification; the small core-scale mode oscillates. Middle: changing the supplied epoch changes both growth and wave phase. Right: a compact 5% seed on the earlier continuation approaches a nonlinear regime. The curve beyond small contrast is a linear extrapolation, not a simulated core.

The earlier benchmark has $H_i/\omega=3.53\times10^{-7}$ and maximum retained $k/(\omega\, A_i)=.01246$. These indicate a useful reduction hierarchy, not a total relativistic error estimate. Its homogeneous past and sufficient local parent domain are explicit assumptions. The existing material walls, radiation history and inherited seed statistics have not been supplied at that earlier epoch. Exactly homogeneous data remain homogeneous; numerical noise cannot establish a physical seed mechanism.

The next calculation should solve the compatible initial source dictionary for this small compact disturbance and evolve it nonlinearly in the same declared earlier parent. Retained mass, concentration, reservoir and the spatial turnover would then be outputs, tested together. No new core profile or desired observed spectrum needs to be inserted to start that experiment.

14.18. Nonlinear formation without installing the final core

Derivation and calculation: nonlinear compact-seed comparison and initial constraints, boundary clock and balances.

The compact disturbance has a finite-amplitude continuation. Keep the same action, inherited vacuum and excitation, comoving length and earlier epoch. The *5%* seed is evolved to $A=16$. A separately declared *10%* preparation is tested through its first compression and rebound at $A=1.2$. This amplitude comparison is not a fit to an observed object or an inferred universal formation threshold. No stationary core is placed in either seed.

The autocorrelation source from Section 14.17 has an exact factored real-space polynomial. Its radial mass primitive is positive inside the support and vanishes at its edge. Normalize the total envelope by its homogeneous value and set

$$f_i(r)=\sqrt{1+\varepsilon S(r)-\lambda W(r)}$$
$$r=\frac{R}{AL}.$$

The square root preserves the intended finite density contrast. The fixed smooth bump W occupies the outer part of the original support. Its coefficient is solved from the full local initial gravitational constraint, not chosen to force a later collapse. Let $M_*=DL^3/2$, $n=M/M_*$, $q=1/(\omega^2\, A_i^2L^2)$, $c=CL^2$ and $\gamma=DL^2/A_i$. Then

$$n_r=r^2\left[\varepsilon S-\lambda W+qB(f_i')^2\right]$$
$$B=1-cr^2-\frac{\gamma n}{r}$$
$$n(0)=n(2)=0.$$

Without the correction, positive scalar gradient energy leaves a small exterior mass offset. The derivative of the zero-gradient endpoint with respect to λ is negative, so a nearby compensating branch exists. The solved values are 9.75930×10^{-12} and 3.85046×10^{-11} for the two amplitudes. An independent proper-distance Hamilton solve checks the endpoint, and the initial radial metric remains positive. This constructs local turning-point Einstein-scalar data in the declared earlier parent patch. It does not construct the earlier material walls.

The envelope omits that gradient contribution, so its small initial mass residual is retained in the subsequent balance. It also retains the stated carrier-quadrature difference from exact zero scalar momentum. These are identified approximation differences, not silently removed inputs.

The nonlinear evolution uses logarithmic scale time $N=\ln A$:

$$if_N = -\frac{a}{2}\Delta f + bVf$$
$$\Delta V = \frac{|f|^2 - 1}{2}$$

$$a = \frac{1}{\omega L^2 A^2 H}$$
$$b = \frac{\omega D L^2}{AH}.$$

The finite boundary field and its gravitational clock must agree. A fixed boundary phase uses $V(L_{box})=0$, implemented by subtracting the free-space potential's boundary value. This leaves bulk spatial forces unchanged and preserves a symmetric Green kernel. An initially inconsistent boundary choice failed conservation before nonlinear growth and is rejected. The corrected kernel passes an independent radial eigenfunction test.

The existing sine kinetic map and full-field potential kicks are retained. Integrated time coefficients follow the early expansion, and a potential phase limit resolves the first rapid compression. With the physical-time Hamiltonian split into kinetic T and self-gravitational W, the exact expansion identity is

$$\frac{d(T+W)}{d\ln A} = -(2T + W).$$

Mass transport is checked at fixed comoving cuts and the outer boundary. The work integral is evaluated every step. Three spatial resolutions, an independent time control and a larger domain test the *10%* history. The *5%* history has its own three resolutions, with no-gravity and small-amplitude controls completing eleven runs.

14.19. Physical contraction separates formation from density contrast

Track a shell carrying a fixed amount of the oscillatory scalar source:

$$M_{\text{shell}} = \frac{DL^3}{2}\int_0^{r_m(N)} r^2 |f|^2 dr$$
$$R_m(N) = AL r_m(N).$$

This is an exact mass label in the envelope continuity equation. It excludes vacuum volume mass and higher-order GR gradient corrections. A decreasing comoving radius alone is insufficient: the physical radius must turn around and contract.

For the *10%* preparation, the shell initially at $r=.25$ turns around near $A=.5852$. At $A=1.2$ it gives

Quantity	Fine value and numerical comparison range
Central density contrast	*59971±1514*
Physical density half-height radius	$(4.22172 \pm .02216)\times10^{-6}$
Selected shell physical radius	$(4.38056 \pm .03788)\times10^{-6}$
Selected shell radius / largest sampled radius	*.109898±.000950*

The selected mass contracts by about *89%* from its largest sampled physical radius. The density profile also contracts and undergoes a first rebound. Ranges use twice the largest fine/medium, time-control or outer-domain difference. They describe numerical uncertainty within the envelope, not a full relativistic or observational error budget.

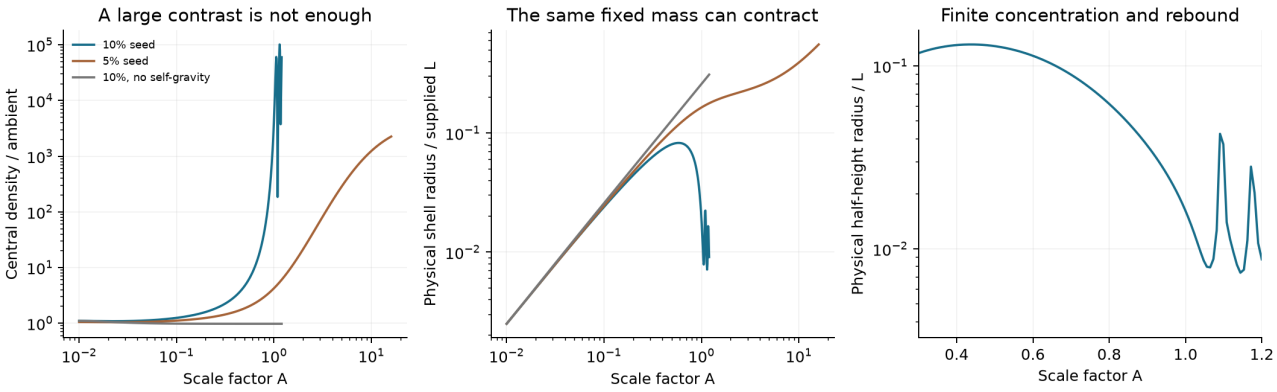


Figure 25. Both supplied seeds develop large contrast, but the sampled fixed-mass shells of the 5% case continue expanding through its interval. The 10% case turns around and contracts. Removing self-gravity removes that contraction. The right panel resolves the forming concentration's first compression and rebound; it does not establish asymptotic stability.

The fixed-cut mass-flux discrepancy decreases from 8.88×10^{-5} through 2.11×10^{-5} to 5.48×10^{-6}, relative to the initial positive excess. Expansion-work discrepancies decrease from *.007108* through *.001689* to *.0004185*. The fine global mass/current discrepancy is 6.69×10^{-10}. The larger domain barely changes the central result. Halving a small seed amplitude also halves its departure from the independently calculated linear response.

The no-gravity control keeps the sampled shells expanding. The *5%* case likewise shows no sampled shell turnaround through *A=16*, despite a final central contrast near *2257*. Contrast can grow while the background thins and the physical region expands. That history is not promoted to a formed bound object. Conversely, finite contraction and rebound of the *10%* case do not prove an asymptotic soliton, nonspherical stability or a population of astronomical objects.

Using the previously supplied $\Lambda=10^{-52}\ m^{-2}$ ruler, the selected shell contains 4.251×10^{9} nominal solar masses and contracts from a largest sampled radius of *68.51* kpc to *7.529±.066* kpc. The final density half-height radius is *7.256±.039* kpc, enclosing a calculated scalar-source mass of $(3.919\pm.069)\times10^{9}$ nominal solar masses. This aperture mass is an output of the forming field. The ruler, seed amplitude, length and epoch remain inputs; no galaxy measurement was fitted.

The general scalar concentration mechanism is established in work including Seidel--Suen, gr-qc/9309015, with expanding-envelope context in Banik et al.,1504.05968. Here the new connection is a specified weak compact seed, compatible local initial gravity and a conserved, converged finite formation history in the inherited parent. Its actual field can now supply an observer calculation. The old installed-core and different-frequency lens histories cannot be substituted for that next signal.

V. The interior observer: motion, light and expansion

An interior observer tests a geometry through clocks, trajectories and light. This part uses the sources already constructed to derive frequency relations, focusing, multiple images and motion-calibrated optical predictions. Endpoint identities connect measurements that belong to one geometry. The final cosmological comparison confronts a quiet limiting state with actual supernova data, identifying the expansion history that the full account must improve.

15. One geometry, several measured consequences

The following sections give the worked constructions and their attributed mathematical inputs.

15.1. Three freely falling clocks and a withheld prediction

Let $L_{ref}=a_0$ and initialize three ideal test emitters at proper separations *(.04,.08,.12)*L_{ref} from the parent pole. They initially rest relative to the slice normal, emit continuously at known rest frequencies, and freely fall without backreaction. The central free-fall receiver observes at fixed proper age 2.251557477076, corresponding to background N=.8. All rays remain inside the protected parent domain.

Observer terms are essential in relativistic measured quantities; see Scaccabarozzi, Yoo and Biern. Here the actual closed-parent metric supplies the calculation. Holding initial proper distance and receiver age fixed gives

$$\delta\theta_s(0) = \int_0^s \Phi(0,\theta)d\theta$$
$$\delta t_o = -\int_0^{t_o} \Phi(t,0)dt.$$

The source's outward peculiar velocity obeys

$$\dot{v}_s + Hv_s = -\frac{\partial_\theta \Phi(t,s)}{a}$$
$$\delta\dot{\theta}_s = v_s/a.$$

Along the unperturbed incoming ray, $\theta=\eta_o-\eta(t)$. The perturbed null equation sets the emission-time correction,

$$\delta t_e = a_e\left[\frac{\delta t_o}{a_o} + 2\int_{t_e}^{t_o}\frac{\Phi}{a}dt - \delta\theta_s(t_e)\right].$$

The measured log-frequency departure from the background is

$$y = \log\frac{(\nu_o/\nu_e)}{a_e/a_o} = H_e\delta t_e - H_o\delta t_o + \Phi_e - \Phi_o - v_e + 2\int_{t_e}^{t_o}\partial_t\Phi\, dt.$$

The last derivative is partial at fixed position. Source motion, potential, propagation and clock contributions all enter; no instantaneous potential difference alone determines the measurement.

Writing $y_i=R_{i1}\alpha+R_{i2}\beta$ for the two retained amplitudes, the computed response is:

Initial distance / L_{ref}	Background frequency ratio	Density column	Phase column
.04	.840672581087	.017861103048	-.000731094396
.08	.721722111018	.030644152339	-.001671366286
.12	.628648035135	.038548233209	-.002804913120

The first two rows have nonzero determinant and condition number 169.341 in this amplitude normalization. Eliminating both amplitudes gives the conditional prediction

$$y_3 = -2.88990465\, y_1 + 2.94232704\, y_2.$$

This is NSC-P003, registered retrospectively without astronomical data. In the synthetic check, amplitudes $(10^{-5},-4\times10^{-6})$ give departures $(1.81535408,3.13126989,3.96701985)\times10^{-7}$. Calibrating only the first two recovers the amplitudes and predicts the withheld third.

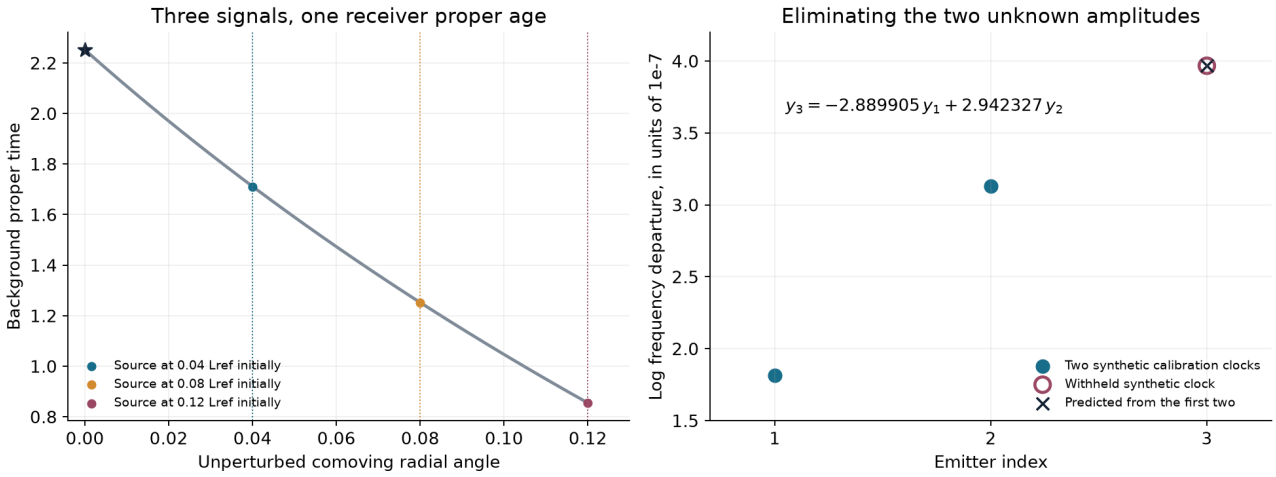


Figure 26. Left: three emitters on the central past radial light cone; their initial proper distances and the receiver's proper age are fixed. Right: a withheld synthetic frequency response agrees with the prediction from two state calibrations. No observational confirmation is inferred.

An independent check integrates timelike sources and radial photons in finite metric deformations, fixing the physical distances and proper age by direct integration. The null Hamilton equation is

$$\frac{d\log(-p_t)}{dt} = \frac{\partial_t A}{2A} - \frac{\partial_t B}{2B}$$
$$A = 1 + 2\Phi, \quad B = a^2(1 - 2\Phi).$$

Symmetric metric variations verify the linear clock response; the mixed-direction derivative differs by 5.23×10^{-11} at variation step .0005. The finite metrics validate the first variation, not a nonlinear Einstein-scalar solution. Refinement changes the null weights by 1.35×10^{-9}; the retained coefficient comparison tolerance is 10^{-7}.

For independent equal frequency-log errors σ_y, the null residual has uncertainty $4.243682044\sigma_y$. A full covariance must replace that example for correlated measurements. Unknown source velocities, an arbitrary spatial profile or extra fields would add parameters and need not obey the three-clock relation. The derivation and compact evidence keep those restrictions explicit. The advance is an operational test surviving the two unknown state amplitudes, while physical formation and an unrestricted primordial spectrum remain open.

15.2. A signal from the concentration's evolving geometry

The central emitter is geodesic. The receiver is actively held at areal radius .6 and starts its proper clock on the initial slice. In the diffuse case the emission occurs at central proper age 4, after the contraction has developed. Reception time is determined by propagation, not chosen as an equal-coordinate-time comparison.

An outgoing photon with frequency E measured by the slice normal obeys

$$\frac{dR_\gamma}{dt} = \alpha(HR_\gamma + \sqrt{B})$$
$$\frac{d\log E}{dt} = \alpha K_r - \sqrt{B}\,\alpha_R.$$

At reception, the held-radius observer's velocity relative to that normal adds a necessary Doppler factor:

$$\frac{\nu_o}{\nu_e} = \frac{E_o}{E_e}\sqrt{\frac{\sqrt{B_o}+H_oR_o}{\sqrt{B_o}-H_oR_o}}$$
$$\frac{d\tau_o}{dt} = \alpha_o\sqrt{1-\frac{H_o^2R_o^2}{B_o}}.$$

An independent canonical null-geodesic calculation transports p_R with Hamiltonian $\alpha(HR+\sqrt{B})p_R$. It is checked against this normal-frame formula. A separate unaveraged homogeneous integration gives the corresponding control signal analytically along its FLRW light cone.

For the diffuse run emitted at central proper age 4, the fractional frequency departure from the homogeneous comparison is

$$\frac{(\nu_o/\nu_e)}{(\nu_o/\nu_e)_{\mathrm{hom}}} - 1 = -0.0041375586 \pm 1.1\times10^{-8}.$$

This is approximately -0.413756 percent, with a finite numerical error estimate of .0000011 percentage points. The receiver's additional proper arrival time is .01703546, with numerical comparison error 2.2e-7 in atlas time units. Both the emitter's accumulated proper clock and propagation contribute to this delay. The independent canonical frequency differs by less than 4.7e-11 on the finest run.

The coherent control, emitted at central age 0, has frequency departure -.007456287 with numerical comparison error 2.1e-7 and proper arrival delay .000476466. These use different declared emission ages; they are not a same-epoch comparison of the two states. All errors here concern numerical resolution and event calculation, not physical preparation uncertainty.

The frequency departures are dimensionless. Delays are in atlas time units; an independent physical length would convert them to seconds. These conditional observables do not import P003's earlier linear coefficients or select a primordial population. The complete protocol and numerical comparisons give the emission events, receiver motion, error estimates and model scope.

The connection now runs from a declared inherited scalar preparation, through self-consistent local evolution, to a received signal. The parent vacuum remains $\Lambda_{vac}a_0^{\,2}=12.3482523989285$. Selecting its observed scale, explaining the population of structures and testing their nonradial persistence remain further physical questions.

15.3. The same concentration bends and focuses light

The radial clock is followed by a nonradial optical test on the same evolved scalar geometry. The field stepper and its physical inputs remain unchanged; final-state hashes must reproduce the preceding source calculation at every retained resolution. The lens is not replaced by a fitted density profile or a static metric.

Scalar lenses have published precedents in Dabrowski and Schunck, and Aimar et al. study time-dependent real-scalar optical signatures. The contribution here is the connection to the computed inherited parent and its formation history. The exact lens-map approach follows Frittelli and Newman, with the observer-screen formulation made explicit below.

The canonical bulk equation gives an exact null contraction:

$$R_{ab} = \kappa(\nabla_a\phi\nabla_b\phi + Ug_{ab})$$
$$R_{ab}k^ak^b = \kappa(k\phi)^2.$$

The potential, including the inherited vacuum, drops out of this direct focusing term. It still affects the field and spacetime through which the ray travels. Its absence from this contraction is not invariance of the whole optical experiment under a change of Lambda.

For normal energy E, radial momentum $q=\sqrt{B}p_R$ and conserved angular momentum L, the two principal tidal eigenvalues are

$$T_\perp = \tfrac{\kappa}{2}(E\Pi + q\sqrt{B}F)^2 + \tfrac{3WL^2}{R^2}$$
$$T_\parallel = \tfrac{\kappa}{2}(E\Pi + q\sqrt{B}F)^2 - \tfrac{3WL^2}{R^2}$$

where

$$W = \tfrac{m}{R^3} - \tfrac{\kappa}{6}\left(U + \tfrac{\Pi^2 - BF^2}{2}\right).$$

Ricci curvature focuses both directions. Weyl curvature changes them with opposite signs, producing optical shear. The spherical Einstein Hessians and warped-product Riemann tensor independently give these factors and signs. In the homogeneous parent W vanishes. At a pure vacuum extremum both tidal eigenvalues vanish, although expansion and distances still depend on the vacuum scale.

The principal Jacobi scales obey

$$\tfrac{d^2D_i}{d\lambda^2} = -T_iD_i$$
$$D_i(0) = 0$$
$$D_i'(0) = 1.$$

This is the attributed optical transport framework of Seitz, Schneider and Ehlers, applied to the current source. The curvature derivation and compact calculation separate the exact identities from finite numerical lensing evidence.

15.4. A proper-clock lens map during contraction and breathing

Hold a source and a spherical family of receivers at areal radius .6. Their clocks start on the initial slice. The source emits transparent test light at proper ages 2.8 and 3.4, with inward rest-frame angle ψ. The primary declared angles are .02, .04 and .08 radians. Unit emitted energy fixes affine normalization; beam stress is neglected as in the previous clock calculation.

The null Hamiltonian is

$$\mathcal{H} = -p_t = \alpha(E + HRp_R)$$
$$E = \sqrt{Bp_R^2 + L^2/R^2}.$$

Each ray is followed to its first outward recrossing of the receiving sphere. Proper emission time, the changing metric and the receiver's velocity relative to the slice normal are retained. A separate normal-frame frequency equation checks canonical photon transport.

The full landing map gives an independent computation of both beam scales:

$$D_{\perp} = \frac{R_s \sin\varphi_o}{\sin\psi}$$
$$D_{\parallel} = -R_s \mu_o \frac{d\varphi_o}{d\psi}.$$

Here μ_o is the measured outward radial cosine. The derivative includes the angle-dependent arrival event. These screen identities and the curvature equations calculate the same beam by different routes. Flat space gives both scales equal to the chord length; a separate closed-FLRW great-circle solution checks the evolving homogeneous control.

For inward launch angle .02 rad, the same source gives the following finite-screen results. Errors are numerical comparison envelopes, not uncertainties in the physical preparation.

Source proper age	Landing-angle departure	Beam flux / homogeneous	Principal-axis ratio
2.8	0.01928843 rad	1.648532 +/- 0.000021	2.19831
3.4	0.00666950 rad	1.381442 +/- 0.000018	1.03582

The early landing-angle departure is about 1.105 degrees. The beam is focused and distorted by different amounts at the two epochs, with no change to the scalar action or initial preparation. The six source replays match the prior field-state hashes exactly.

The independent rotation-Jacobi discrepancy at the primary probes falls from 1.35e-5 to 8.80e-7 to 5.16e-8 under refinement. The in-plane angular-map discrepancy falls to 1.82e-6. The independent homogeneous area distance agrees within 1.42e-8; its frequency check agrees within 2.05e-8.

The early fan brackets a tangential caustic between source angles .006 and .008 rad at all three resolutions. On the finest run the signed perpendicular scales at those endpoints are -.00118667 and .0802128; their independent rotation checks differ by less than 1.7e-6. No tangential sign change is found in the sampled late fan. This is finite numerical evidence for the optical caustic, not a certified interval root or a prediction of infinite physical brightness. The calculated metric remains regular; a light-bundle caustic is not a spacetime singularity.

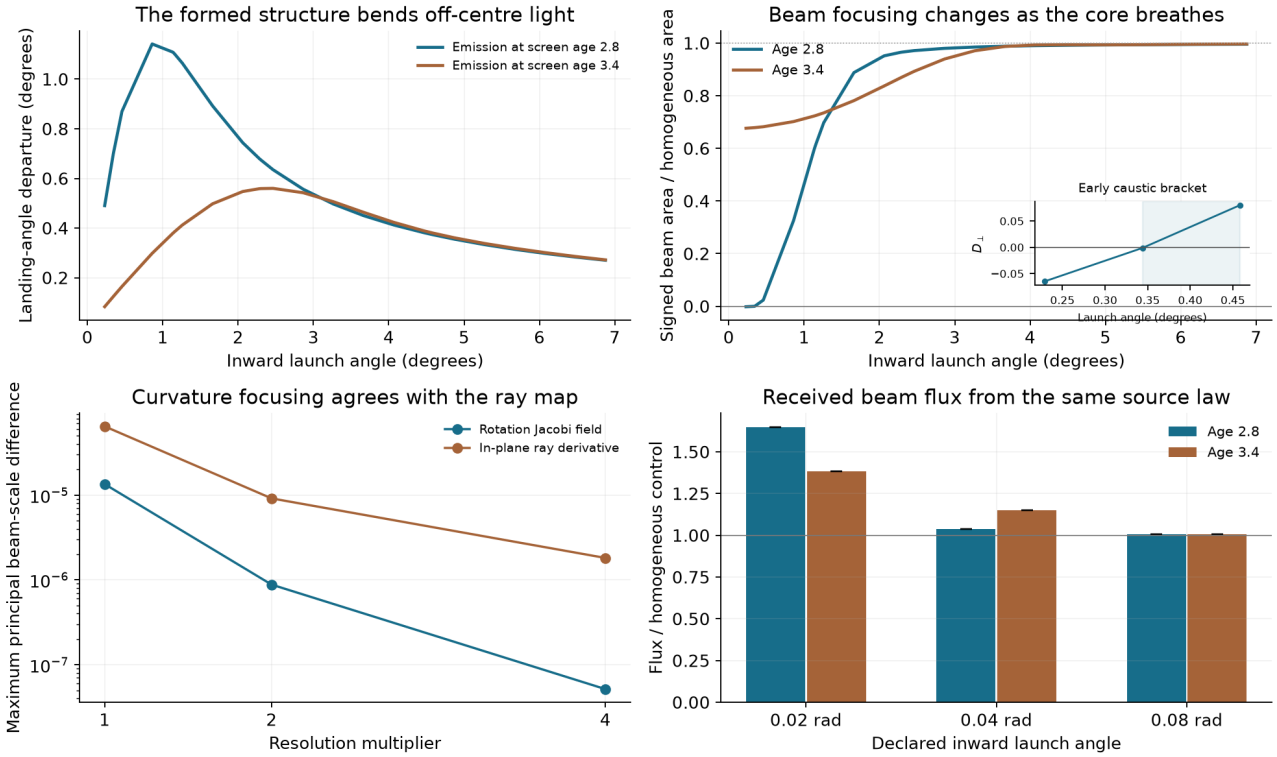


Figure 27. Two proper emission ages probe the same evolving concentration. The landing-angle map and signed beam area change as the core breathes. The inset shows the sampled transverse-scale sign change near the early caustic. Curvature-based Jacobi scales converge toward their independent ray-map values. Relative received beam flux uses the fixed source luminosity, measured frequency and projected beam area.

For fixed isotropic emitted luminosity, the received bolometric beam flux is proportional to

$$\frac{\nu_o^2}{|D_\perp D_\parallel|}.$$

The absolute area enters flux; its sign retains parity relative to the homogeneous beam. Comparisons identify the same source proper age and launch angle on the two geometries, so receiving worldlines can differ. This finite-screen experiment is not yet a fixed-source/fixed-observer astronomical image fit. Optical caustics, when bracketed by the fan, do not establish infinite physical brightness; finite-source and wavelength effects would require a further calculation.

The observer protocol and limits and numerical comparisons bind these optical numbers to the same source that produced concentration and the radial clock signal. The inherited law has not thereby selected a cloud population, a physical SI scale or an observed dark-matter fraction.

15.5. Two paths, one receiving clock

A lens map becomes an image prediction when the emitting and receiving worldlines are fixed. Keep the scalar-parent solution of the preceding sections. Hold both worldlines at areal radius $R_s=.6$, put the emitter at azimuth zero and the receiver at π-b, with $b=.0002$. A pulse leaves at emitter proper age 2.8. If $\Phi(\psi,\tau_e)$ is the positive-orientation landing map, the two selected branches satisfy

$$\Phi(\psi_+,\tau_e)=\pi-b$$
$$\Phi(\psi_-,\tau_e)=\pi+b.$$

Reflection sends the second ray to $-(\pi+b)$, which is the same receiving worldline modulo 2π. The images arrive at different proper times on that one observer's clock. This improves on a fixed-launch-angle comparison, where the landing observer itself can change. The two selected angular patches do not constitute a census of all possible images.

A null geodesic's endpoint variation relates these image measurements. Normalize its emitted energy to one. The null action gives $p_a \delta x_o^a - p_a \delta x_e^a = 0$. The emitter is spatially fixed; the held receiving sphere contributes the local energy and conserved angular momentum L. Hence

$$d\tau_o = \frac{d\tau_e}{\nu_o} + \frac{L}{\nu_o} d\varphi_o.$$

This equation does not assume a static lens or a conserved photon energy. It uses the actual evolving metric and endpoint observer tetrads. At a fixed receiving worldline, each image obeys $d\tau_i / d\tau_e = 1/\nu_i$. For the relative delay $\Delta = \tau_- - \tau_+$,

$$\frac{d\Delta}{d\tau_e} = \frac{1}{\nu_-} - \frac{1}{\nu_+}$$
$$\frac{d\Delta}{d\tau_+} = \frac{\nu_+}{\nu_-} - 1.$$

The second expression requires only the receiver's clock and the ratio of frequencies of matched source features. Image delays and frequency shifts are linked consequences of one geometry. The general endpoint and lens-map ideas are established mathematics; their evaluated connection to the forming source in this atlas is the present result.

Conserved angular momentum also fixes the received image directions: $\sin|\xi_i| = \sin\psi_i / \nu_i$. Choose positive sky direction opposite increasing azimuth, so that $\xi_+ > 0$ and $\xi_- < 0$. For a common, infinitesimal isotropic pulse, its observed bolometric fluence is

$$\mathcal{F}_i = \frac{E_e}{4\pi} \frac{\nu_i}{|D_{\perp,i} D_{\parallel,i}|}.$$

A continuous source instead gives a flux proportional to $\nu_i^2 / |D_{\perp,i} D_{\parallel,i}|$, because reception rate adds a second frequency factor. The signed determinant retains image parity. These relations assume transparent geometric optics and a detector perpendicular to the arriving ray. Finite source size, duration and wavelength require separate modeling.

The fixed-worldline derivation gives its endpoint proof, angular inverse and independent checks. Source ages $2.8 \pm .0002$ and $2.8 \pm .0001$ test the delay derivative; curvature and angular resampling test the two image inverses. All source evolution files remain unchanged.

The completed pair is resolved at three grid spacings, with an additional optical-time comparison. Its conditional observables are:

Observable	Calculated value	Numerical comparison range
Image separation	0.682670 degrees	0.000493 degrees
Relative proper arrival delay	1.44125317e-6 atlas units	9.54e-10 atlas units
Negative/positive frequency	1.00000922695	8.52e-9
Negative/positive pulse fluence	1.401958	0.006483

The negative-parity image is brighter and arrives later. The rotation/Jacobi error decreases from 5.65×10^{-4} to 3.35×10^{-5} to 1.92×10^{-6}. The angular inverse and its curvature-based check agree to better than 1.31×10^{-10} radians on the fine patches. All four field replays match the previously retained

solutions exactly. The compact evidence records the fixed inputs, individual images and numerical comparisons.

The directly differentiated delay agrees with the inverse-frequency identity to 1.55×10^{-8} per source proper time, within its 1.67×10^{-7} numerical comparison envelope. That small derivative residual is not monotone between the middle and fine grids. The result is a finite checked prediction with numerical ranges, not a rigorous interval enclosure or an asymptotic stability statement.

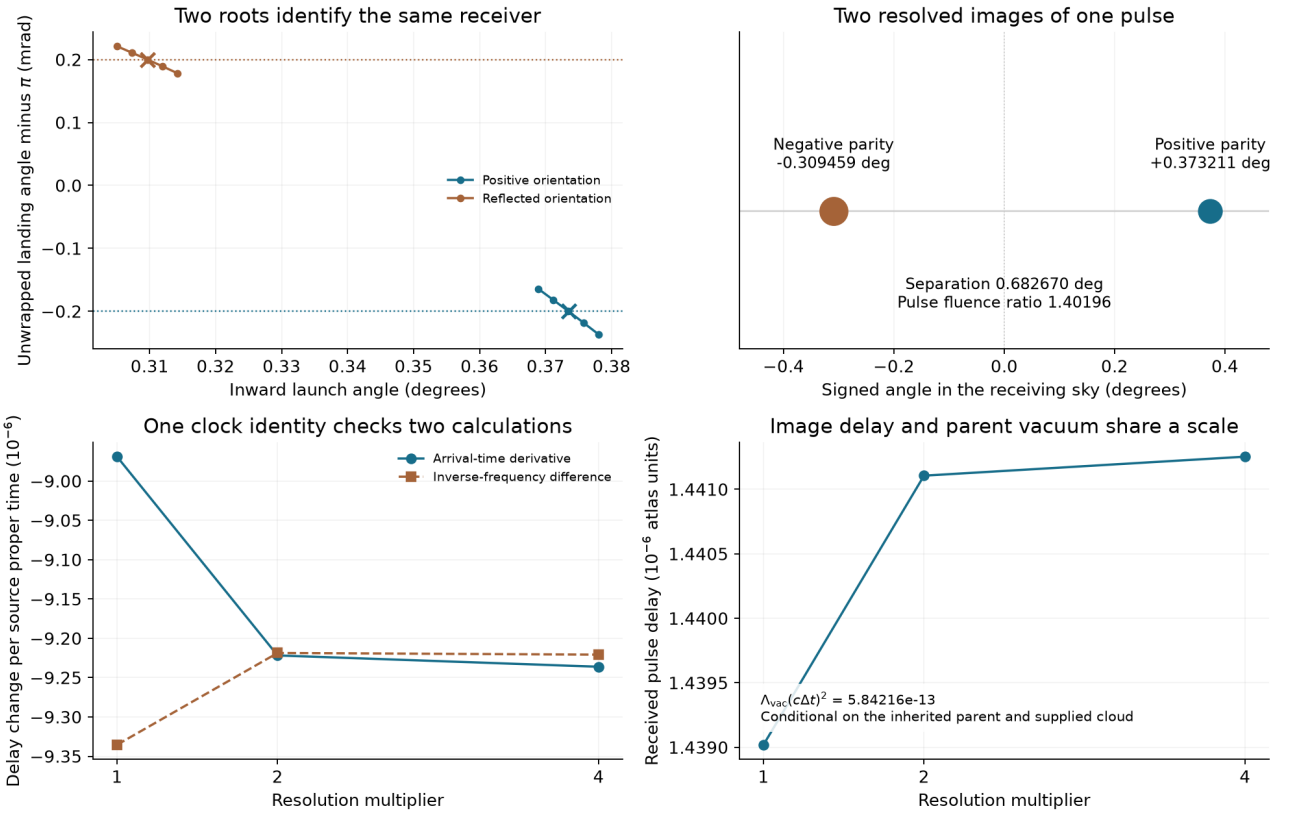


Figure 28. The two angular roots reach one receiver after reflection of the negative branch. Their sky positions, pulse fluences and arrival times are computed from the unchanged scalar-parent evolution. Two independent calculations check the delay-frequency identity. The final panel links the resolved delay to the inherited parent vacuum through a scale-free product; it does not identify this benchmark with an observed lens.

15.6. A local image delay and the parent's vacuum must share one scale

The local concentration belongs to the same sourced parent whose protected annulus retains the inherited vacuum. For this declared action and flux, the vacuum and reference radius in atlas units are

$$\widehat{\Lambda} = \kappa U_{\mathrm{vac}} = 9/32$$
$$\hat{L}_{\mathrm{ref}} = 6.62607707610.$$

Thus the earlier conditional result $\Lambda\, L_{ref}^{\,2}=12.34825239893$ still applies.

Let L_{unit} be the physical length assigned to one atlas unit and let δ be the computed dimensionless image delay. Any consistent dimensional embedding must obey

$$\Lambda_{\mathrm{phys}} = \frac{\widehat{\Lambda}}{L_{\mathrm{unit}}^2}$$
$$\Delta t = \frac{L_{\mathrm{unit}}}{c}\,\delta$$

and therefore

$$\Lambda_{\mathrm{phys}}(c\Delta t)^2 = \widehat{\Lambda}\,\delta^2.$$

For the computed pair,

$$\Lambda_{\mathrm{phys}}(c\Delta t)^2 = (5.8422 \pm .0078) \times 10^{-13}.$$

The quoted range propagates the numerical delay envelope while holding the declared parent vacuum fixed. It does not include uncertainty in a physical interpretation or an unspecified preparation law.

The free unit cancels. Calibrating a physical scale with an image delay leaves the vacuum value, angular image separation and frequency/fluence ratios as linked tests. Matching each by an independent scale choice would break this connection. This does not select the physical scale or prove that the inherited vacuum equals the value inferred from cosmology. It creates a precise joint requirement on that proposed identification.

The cloud, emitting age and receiver position remain specified inputs. Population selection and nonspherical persistence remain necessary for an astronomical application. The atlas provides the mathematical route from inherited vacuum, through formed structure, to a fixed observer's measurements without an independently fitted lens mass.

15.7. The forming field supplies one connected observer map

Derivation and calculation: fixed-observer response and metric, mass-current and endpoint derivation.

The finite concentration can now be used as a source for light. Start from the authenticated $A=1.2$ fields of the preceding formation calculation. Short backward and forward steps reconstruct a local interval around that event using the same field operators; no formation grid or fitted halo profile is substituted. The source's time dependence remains in the ray.

The declared averaged weak metric is

$$ds^2 = -(1+2\Phi)dt^2 + A^2L^2(1-2\Phi)\left(\frac{dr^2}{B_0} + r^2 d\Omega^2\right)$$
$$B_0 = 1 - CL^2r^2$$
$$\Phi = \frac{DL^2}{A}V.$$

It retains the parent spatial curvature and equal leading scalar potentials. Null propagation is solved in this metric. Omitted higher-order Einstein-scalar, carrier and stress terms remain approximation errors; the numerical ranges below are not bounds on the full relativistic model.

There is a direct connection between the field current and the potential's time derivative. Let $N=\ln A$, $j=\mathrm{Im}(f^{*}f_{,r})$ and $m(r)=\int_0^r s^2(|f|^2-1)ds$. The existing envelope continuity equation gives $m_{,N}=-ar^2j$, with $a=1/(\omega L^2A^2H)$. Since the boundary-clock potential is an integral of m/r^2,

$$\Phi_r = \frac{DL^2}{2A}\frac{m(r)}{r^2}$$

$$\Phi_N = -\Phi + \frac{DL^2 a}{2A}\int_r^{L_{\text{box}}} j(s)ds.$$

The potential, its spatial force and its time variation therefore come from the same formed source. Spectral field derivatives and exact radial polynomial integrals evaluate them; no derivative of a stored plot is used. A Hermite interpolation in time preserves a consistent metric and its derivatives. Pure-phase data give an independent analytic check of the current term.

Place test clocks at comoving radius r_s=*.25*, with the emitter at azimuth zero and receiver at *Θ=π-.08*. They are held at these coordinates. They do not follow the scalar flow or the contracting mass shell that began at *r=.25* in the earlier preparation. Their proper-clock rate is

$$\frac{d\tau_s}{dN} = \frac{\sqrt{1+2\Phi(r_s,N)}}{H(N)}.$$

Set both clocks to zero on the declared initial slab slice. The emission proper age equals the homogeneous reference clock's age at *A=1.2*. Solve its actual emission event in the perturbed metric, then shoot the inward angle to the fixed receiving worldline. This matches proper age and receiver position, rather than comparing different arrival locations. It resolves one weak-image branch in the declared angular bracket.

With *dq=dt/(AL)*, the null Hamiltonian is

$$\mathcal{H}_q = F\sqrt{B_0 p^2 + J^2/r^2}$$

$$F = \sqrt{\frac{1+2\Phi}{1-2\Phi}}.$$

The emitted frequency is normalized to one. Its received value obeys

$$\log\frac{\nu_o}{\nu_e} = N_e - N_o + \frac{1}{2}\log\frac{1+2\Phi_e}{1+2\Phi_o} + \int\frac{2\Phi_N N_q}{1-4\Phi^2}dq.$$

The small time-dependent term is integrated directly, avoiding cancellation between nearly equal photon energies. An endpoint energy identity checks it independently. The receiver's proper flight time is also integrated; it differs from a coordinate Shapiro delay.

15.8. Direction, delay and frequency are derivatives of one arrival map

Let $T(\tau_e,\Theta)=\tau_o$ be the arrival map. Null endpoint variation gives, for the specified worldlines,

$$\nu_o\, d\tau_o = \nu_e\, d\tau_e + J_{\text{physical}}\, d\Theta.$$

For unit emitted frequency this implies

$$\frac{\partial T}{\partial \tau_e} = \frac{1}{\nu_o}$$
$$\frac{\partial T}{\partial \Theta} = R_o \sin \beta_o$$
$$R_o = A_o L r_s \sqrt{1 - 2\Phi_o}\,.$$

Here β_o is the observed ray angle relative to the radial axis. Subtracting the homogeneous reference gives two independently testable relations for the excess delay. They belong to the general spacetime Fermat/endpoint framework described by Perlick and Fleury--Larena--Uzan. The contribution here is their evaluated application to the actual forming source. The identities check one connected prediction; they are not three independent confirmations of nesting.

For the primary fixed observer, the result is:

Readout relative to the matched homogeneous parent	Fine result and numerical comparison range
Image-direction change	*21.20±.20* milliarcseconds
Proper flight-time excess, with the supplied ruler	*27.5906±.0120* days
Log received-frequency change	*(-1.44088±.01399)×10^{-11}*

The delay conversion uses the previously supplied *Λ=10^{-52} m^{-2}* ruler; the angular conversion does not need an SI length. Source parameters, clock protocol and geometry are specified, and no astronomical data were fitted. The same calculation also fixes a scale-independent relation:

$$\Lambda_{\text{physical}} (c\,\Delta\tau)^2 = (5.10730 \pm .00441) \times 10^{-23}.$$

This connects the inherited vacuum to a local flight-time excess within the declared model. It does not determine the observed absolute vacuum.

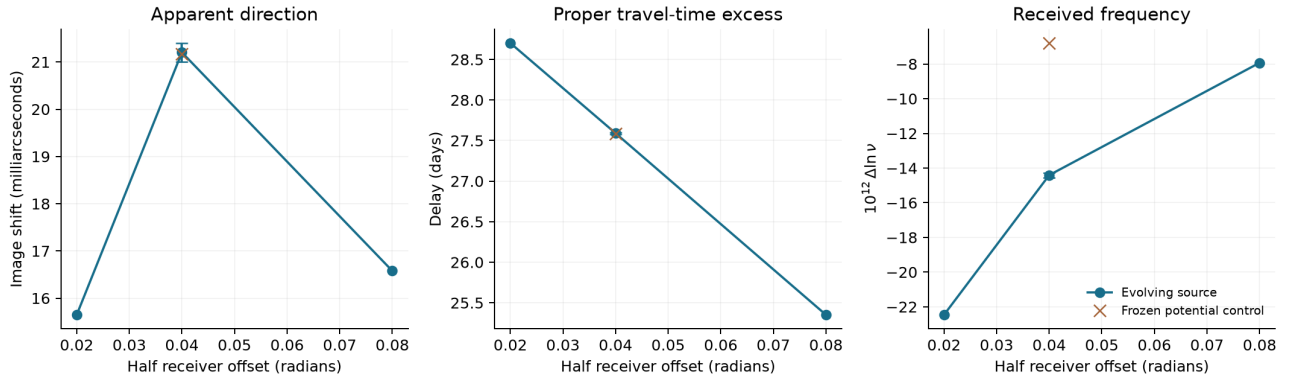


Figure 29. The actual forming source supplies direction, proper delay and frequency for specified receiver offsets. The middle offset has the full quoted source and integration controls; the other two illustrate the map. The cross freezes the comoving potential under the same clock protocol. Its frequency discrepancy exposes the source's time dependence.

The time derivative of excess delay is *1.44090×10^{-11}*, agreeing with the independently calculated frequency relation within *1.36×10^{-15}*. The smaller-step angle derivative agrees with the direction relation within *6.59×10^{-16}*. Source-grid, source-time, outer-domain, short-slab and ray-tolerance controls determine the reported numerical ranges. A curved homogeneous ray and an independent Plummer Born calculation check the geometry and optical normalization.

Freezing the comoving potential removes its expansion-corrected frequency term. The total log-frequency contrast becomes *-6.79268×10^{-12}*, while the direction and delay change much less. Thus

an almost unchanged image can accompany a distinguishable dynamical frequency contribution. The frozen potential is an optical approximation control, not a second self-consistent formation history.

This is a finite forward prediction for held test clocks in the stated weak metric. Freely falling clocks or emitters moving with the scalar flow need their own trajectories, Doppler effects and aberration. Those effects can exceed the small gravitational frequency contrast. The next link to observations is therefore physically prepared or measured kinematics and the assessment of omitted metric terms, using this same formed source.

15.9. Motion calibrates a prediction before the new ray is evaluated

Derivation and calculation: frozen NSC-P004 forecast, separate validation and derivation and preparation.

The preceding result supplied light propagation from a known forming source. The next test reverses part of that inference: calibrate its force from finite timelike motion and predict a new optical response before computing the corresponding null ray. The source, weak metric and action remain unchanged.

Preparation first checks whether an equilibrium rotation curve is justified. For the source acceleration g at fixed physical radius, compare its variation rate with $\Omega^2=g/R$. The instantaneous ratios $|\partial_t \log g|/\Omega$ are about $.611$ near $r=.0073$ and $.457$ near $r=.015$. They do not support slow variation over an orbital period. This is a timescale diagnostic, not a theorem excluding every orbit. The selected method therefore uses short freely falling motions.

For positive lapse $\alpha^2=1+2\Phi$, spatial scale $s^2=A^2L^2(1-2\Phi)$ and $B=1-CL^2r^2$, a radial unit-mass test body has

$$\mathcal{H}_t = \alpha\sqrt{1+BP^2/s^2}\,.$$

Its trajectory and proper clock follow the timelike Hamilton equations. At release with zero peculiar velocity, the local radial velocity relative to the stated cosmological frame obeys

$$-\frac{dv}{d\tau}\Big|_0 = \frac{\sqrt{B}\,\Phi_r}{AL\sqrt{1-2\Phi}\,(1+2\Phi)}.$$

The test body's proper four-acceleration is zero. This is a relative velocity change, measured in the specified reference frames. Centered before/after velocities reconstruct the leading force:

$$\Phi_r^{\mathrm{rec}} = -\frac{AL}{\sqrt{B}}\frac{v_+ - v_-}{\tau_+ - \tau_-}.$$

The refined calibration uses 96 probes on the incoming and outgoing halves of the retarded background chord. Their central release events, proper-time intervals and initial velocities are specified. The initial source and receiver themselves are another pair of radial geodesics, released at the slab start. They have the same trajectory at a common time, but different radii at emission and reception.

The new receiver is at $\Theta=\pi-.12$, absent from the earlier optical fan. Let its reference chord have impact parameter b, length d and coordinate endpoints z_e, z_o. A fixed-endpoint Born calculation gives

$$\Delta\beta_{\mathrm{lens}} = \frac{2}{d}\int_{z_e}^{z_o}(z-z_e)\Phi_r^{\mathrm{rec}}(N(z),r(z))\frac{b}{r(z)}dz.$$

The endpoint weight follows by integrating the transverse displacement equation with both positions fixed. It differs from the total deflection of a freely specified incoming ray. The retarded times come from the background chord, not from the withheld optical path.

Calibrated endpoint positions fix the geometric direction. The measured receiver velocity supplies the exact radial aberration:

$$\cos\beta_{\mathrm{obs}} = \frac{\cos\beta_{\mathrm{normal}} - v_o}{1 - v_o\cos\beta_{\mathrm{normal}}}.$$

The prediction function consumes these motion measurements and geometric inputs only. It does not read the source potential or interpolate the old optical fan, and it does not fit an independent lens mass. Equal leading metric potentials remain a premise of this model. Combining dynamics and lensing is established practice, illustrated in a different statistical setting by Reyes et al.,1003.2185.

15.10. NSC-P004 passes its frozen synthetic cross-observable check

The forecast and comparison tolerance were committed and pushed in e626f2f before evaluating the new null boundary problem. The forecast is creation-only and remains byte-identical after validation.

Stage	Image shift, milliarcseconds
Motion-calibrated forecast	19.43948287
Frozen comparison tolerance	.18327870
Withheld moving-endpoint null calculation	19.43946116
Null result minus forecast	-.00002171
Null-grid comparison range	.08049

The check passes without changing its prediction, tolerance, source or lens normalization. Adding the null-grid comparison range to the residual still leaves it inside the frozen window. The small closure residual is not a new absolute accuracy claim: both channels share the prescribed metric and underlying source. This is prospective synthetic validation, not astronomical confirmation or a unique signature of nesting.

The frozen window contains a numerical comparison envelope, a probe-release sensitivity estimate and a predeclared Born allowance of *0.5%* of the lens contribution. It is a test tolerance, not a certified error bound on the complete relativistic theory. Numerical refinement of the impulse and quadrature is performed before the optical target is evaluated.

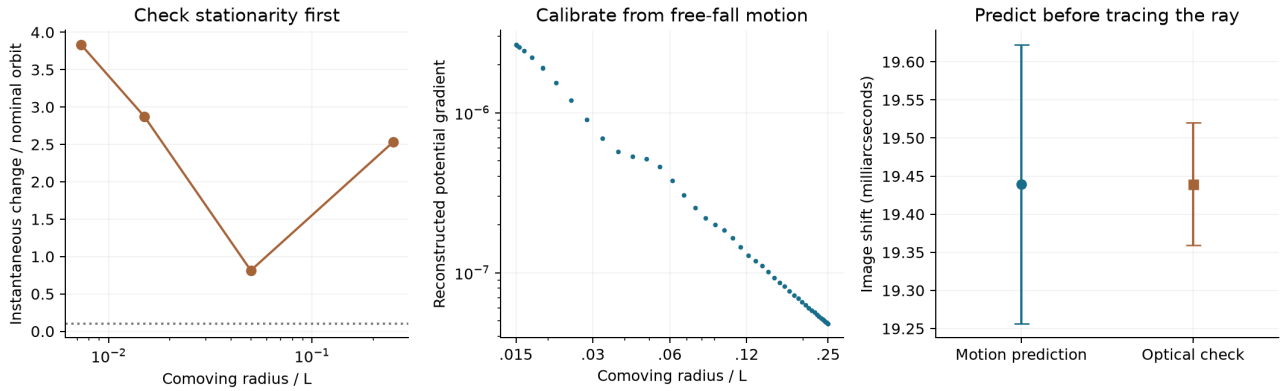


Figure 30. The force-timescale precheck selects finite free-fall calibration. The middle panel reconstructs the force from motion along the retarded chord. The right panel compares the committed forecast with the later null calculation; its bars represent the stated comparison ranges, not empirical confidence intervals.

Observer motion is not a negligible relabeling. At the emission and reception events, radial velocities are approximately -4.27921×10^{-8} and -6.67110×10^{-8} in units of c. Their Doppler contribution to log frequency is 1.09306×10^{-7}, while the expansion-removed normal-frame gravitational contribution is -2.93250×10^{-12}. The exact endpoint boost is

$$\log\frac{\nu_o}{\nu_e}=\log\frac{\nu_{o,\,\mathrm{normal}}}{\nu_{e,\,\mathrm{normal}}}+\log\frac{\gamma_o(1-v_o\mu_o)}{\gamma_e(1-v_e\mu_e)}.$$

Here μ is the local photon direction cosine. The moving-clock arrival identity remains valid and is checked by emission-age differences. Related frequency and viewing-direction effects are treated by Korzynski--Kulejewski,2601.13912v2; the radial boosts and worldlines here are evaluated for this source.

The calibration assumes known endpoint kinematics. Near the measured angle, velocity error changes the direction by approximately $\sin\beta\ \delta v$. Keeping that error alone below the frozen tolerance requires roughly 4.4 m/s receiver-velocity precision with other quantities fixed. Position, distance and release-state errors also need propagation. No such astronomical calibration is claimed here.

The new connection is a motion profile predicting a withheld light measurement, with its preparation and possible failure specified first. The calibration probes the inner chord. An alternative geometry agreeing there would pass the same test. The next outward comparison should ask whether the surrounding compensation and inherited scale relation break a specified same-core degeneracy.

15.11. Two observer distances, on two metrics

Scalar volume averages do not by themselves give an observer a distance. On the exact zero-source quiet cap the luminosity shape is

$$\frac{D_L}{c/H}=z(1+z).$$

The comparison retains 596 released Pantheon+ light curves of 545 distinct supernovae, excluding calibrators and retaining correlated repeat light curves. The low band is $0.025\le z_{HD}<0.075$ (335 curves); the high band is $0.15\le z_{HD}\le0.25$ (261 curves). Both lie inside the declared quiet-cap causal domain. For each curve define

$$y_i=m_{b,\,i}^{\mathrm{corr}}-5\log_{10}\left[z_{\mathrm{HD},\,i}(1+z_{\mathrm{HEL},\,i})\right].$$

Generalized least squares fits the two band intercepts jointly using the selected full statistical and systematic covariance. Their high-minus-low contrast cancels a common brightness and Hubble scale. The quiet prediction is zero; the measured contrast is

$$\Delta=-0.1046180474751224\pm0.014373986469318333\ \mathrm{mag}.$$

The estimator was frozen before this evaluation. The catalogue itself was already public, so the test is a retrospective comparison of one quiet state, not a blind test and not a falsification of a completed Nested-Space cosmology. No ΛCDM parameter was fitted, and no cosmological constant was supplied to improve the residual. The uncertainty is conditional on the released covariance and standardization, including its peculiar-velocity treatment. The distance-contrast record stores the estimator specification, selected-row identity, zero prediction and this residual.

The separate evolving metric also has an explicit null cone. Receiver, emitters and ideal reflectors sit on fixed-coordinate normal worldlines. Those worldlines may be accelerated. They are not introduced as cosmological comoving matter, and a slope of redshift against lookback is not a measurement of H_0. For

transparent, photon-conserving geometric optics, with $N=rQ$, reception $(T_o,x_o)=(0.3,1)$ and past rays $T_e=T_o-\chi$, $x_e=x_o+\eta\chi$,

$$1+z=\frac{N_o}{N_e}$$

$$D_A=\frac{r_o r_e}{N_o}\int_0^{\chi} Q(T_o-s,x_o+\eta s)^2\,ds$$

$$D_L=(1+z)^2 D_A$$

$$D_{\text{radar}}=\frac{1}{2}\int_{T_o-2\chi}^{T_o} N(T,x_o)\,dT.$$

The lapse in these formulae is the full product rQ. The three predeclared lookbacks are $\chi=0.025$, 0.075 and 0.125, in both directions. The longest radar trip begins at $T=0.05$, inside the saved time interval. At $\chi=0.125$ both directions have the same radar distance $D_{radar}/L_{ref}=0.031382863175331$.

Side at $\chi=0.125$	Redshift	D_L/D_{radar}
Smaller x	0.0016677447278268254	1.0000739824887968
Larger x	-0.007306104080127329	0.991108506097534

The ratios carry no unit. The largest 32/64-point difference is 3.34×10^{-16}. Across the four stored cases the largest deviations are 2.94×10^{-10} in redshift and 4.84×10^{-10} in the ratio. Those figures do not bound interpolation or the total model uncertainty. The method was frozen at `f0f639d` before the rays were evaluated, with no new gravitational or field solve. The light-cone record contains all six rays. Figure 31 shows the three declared lengths; the lines only connect those samples. This is a finite observer map on one saved metric, not a full-sky prediction and not a match to the Pantheon+ residual.

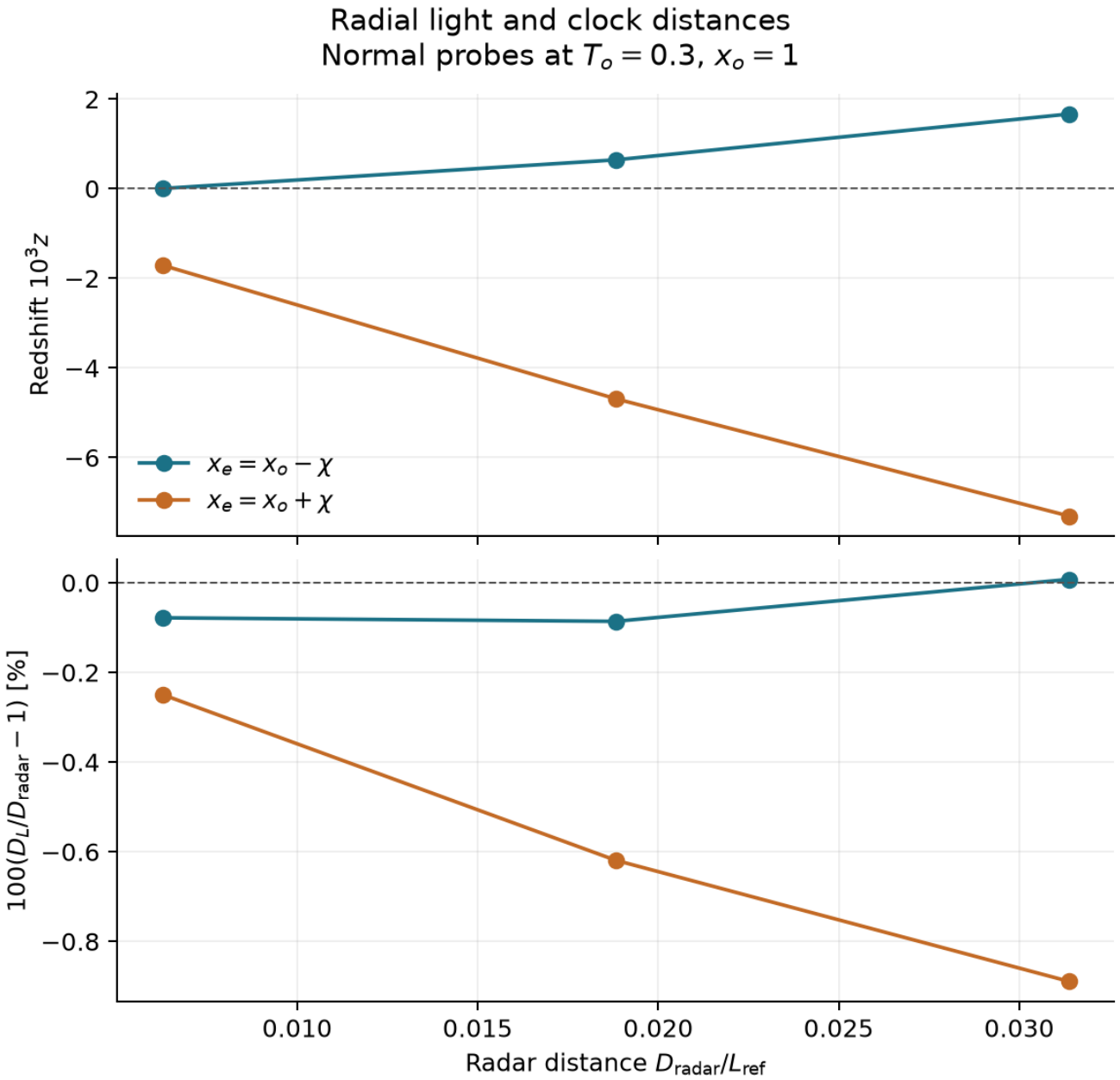


Figure 31. Three predeclared lookbacks in each radial direction, for the fixed-coordinate normal probes of the saved conformal metric. Markers are the declared rays. Lines connect samples and do not fill intermediate evolution.

VI. Gravitational exchange, fluid motion and quantum state

The same geometric question extends to gravitational waves, unresolved motion and quantum fields: which boundary and state determine the measured response? A separately specified stable material geometry gives an actual gravitational Schur action and return contrast. Fluid correlations provide a conserved stress mechanism in their own domain. Physical Dirac propagation and the geometry-derived cosmohedron exchange then connect explicit quantum states to sourced geometry. Each mathematical bridge retains its field content and approximation.

16. Responses across matter, gravity and quantum descriptions

The following sections give the worked constructions and their attributed mathematical inputs.

16.1. Flow geometry and a common boundary stress

The flow dictionary connects several published constructions through explicit operations. PG coordinates expose radial null directions as $dr/dt=v\pm1$. Linear potential disturbances in a prepared barotropic inviscid medium instead define an acoustic metric. Specified near-horizon gravitational perturbations have a fluid limit. Their source, state and domain determine which parts can be joined. Hamilton--Lisle supplies the river construction, Visser the acoustic dictionary, and Bredberg et al. the Einstein/Navier--Stokes limit. The dictionary checks the latter's shear coefficient directly from its perturbed Rindler metric, including the proper-clock conversion.

One connector is the Brown--York boundary stress. With a common spacelike normal from minus to plus,

$$t^{\mathrm{BY}}_{ab} = \frac{Kh_{ab} - K_{ab}}{8\pi G}$$
$$[t^{\mathrm{BY}}_{ab}] = S_{ab}$$
$$D_a(t^{\mathrm{BY}})^a{}_b = -T_{\mu\nu}n^\mu h^\nu{}_b.$$

The jump is precisely Israel's material-interface law. Different material actions can therefore occupy this mathematical interface. A horizon fluid uses an additional limit and constitutive response on an appropriate surface; the current pure-tension wall is one candidate in this larger map.

Douglas's circulating-scale hypothesis asks whether nested vortical regions and transfer across scales can supply a physical continuation. Published holographic turbulence provides a gravity/cascade block in its own asymptotically AdS setting. The spherical river alone has no ordinary circulation. A rotating cosmological state, its energy-transfer budget and its light-cone observations remain calculations to construct. The next example supplies a finite gravitational response with an explicit material action, rather than assigning an arbitrary reflector to a horizon.

16.2. Two stable material interfaces in one static geometry

Derivation and calculation: background, force and response record. The derivation distinguishes this candidate from the earlier formed scalar history. It reuses the Visser--Wiltshire shell dynamics and Pani et al. gravitational junctions. Jampolski--Rezzolla supplies the explicit nested-gravastar antecedent; neither nesting nor gravastar perturbations are claimed as discoveries here.

Use $G=c=M=1$ and one time normalized at infinity. The three regions have

$$ds_i^2 = -A_i^2 f_i dt^2 + f_i^{-1}dr^2 + r^2 d\Omega^2$$
$$f_i = 1 - \frac{2m_i}{r} - H_i^2 r^2.$$

The exterior has $m=1,H=0,A=1$, the parent annulus has $m=.1$, and the central region has $m=0$. The radii are $R_o=2.5$ and $R_i=.25$. On each surface a canonical massless scalar has homogeneous stress $p=\sigma=s^2/2$, where s is its proper-time derivative. The bulk four-form action, with its fixed-flux boundary term, supplies the regional vacuum stress $\Lambda_i=4\pi\ G\ q_i^2=3H_i^2$; surface charges are the oriented flux jumps. The material law and geometric ratios are shared declared inputs.

For ordinary increasing-radius normals, the two static Israel conditions are

$$\sigma = \frac{\sqrt{f_-} - \sqrt{f_+}}{4\pi R}$$
$$p = \frac{1}{8\pi}\left[\frac{f'}{2\sqrt{f}} + \frac{\sqrt{f}}{R}\right].$$

Putting $p=\sigma$ solves the interior curvature at each interface and its matched clock. The full algebra is given in the owner; in this member,

$$A_-^2 = A_+^2 \frac{f_+}{f_-}$$
$$H^2_{\text{parent}} = \frac{3(29-\sqrt{265})}{1000}.$$

Quantity in exterior M units	Parent / outer wall	Center / inner wall
H squared	.0381635382117	4.41709564871
A	.541737767916	.283041476224
sigma equals p	.0120417630513	.129330203316
Radial potential second derivative	.442766777226	51.2125679208

All retained lapse factors are positive. The central bulk is nonsingular; the two interfaces carry the specified thin material sources. Surface conservation gives $\sigma R^4=constant$. The signed wall potentials obey $V=V'=0, V''>0$ with this law, and the normal force balance agrees independently. These are radial linear stability results. Polar and nonlinear stability are additional questions.

The same construction returns the conditional curvature products $\Lambda_{parent}M^2=.114490614635$ and $\Lambda_{child}(.1M)^2=.132512869461$. The supplied radius/mass ratios and four-form sectors have not been selected by astronomical data or by an absolute vacuum-selection mechanism.

16.3. The gravitational action now composes through the interfaces

Derivation and calculation: common-clock wave and junction controls. For linear axial modes $\ell\geq 2$, use the Fourier convention below and the common-time Regge--Wheeler variable

$$\psi(t,r) = e^{-i\omega t}\psi(r)$$
$$\psi(r) = \frac{Afh_1}{r}$$
$$\frac{dx}{dr} = \frac{1}{Af}$$
$$\psi_{xx} + [\omega^2 - U(r)]\psi = 0$$

$$U = A^2 f\left(\frac{\ell(\ell+1)}{r^2} - \frac{6m}{r^3}\right)$$
$$[\psi] = 0, \quad [\psi_x] = q_s\psi, \quad q_s = 4\pi\sigma\sqrt{F_s} > 0.$$

Here $F_s=A^2 f$ is the common wall lapse squared. The shell response is derived from Pani et al.'s general conditions $[h_0]=0$ and $[\sqrt{f}\, h_1]=0$. Applying the zero-density simplification from their special gravastar directly would give the wrong matching here. The remaining temporal Israel condition is also checked against the canonical surface-scalar stress. This is a coupled gravitational perturbation, not a test field assigned a spin-two potential. The standard Kottler bulk potential is also given by Cardoso--Lemos, equation 11. Their subsequent near-extremal approximation is not used here.

Let T propagate (ψ,ψ_x) through a region. Its determinant is 1, and a shell adds $q_s\psi$ to the second component. Both preserve the Wronskian flux. If $Z=\psi_x/\psi$, the regional response composes as

$$Z_{\text{out}} = \frac{T_{21}+T_{22}Z_{\text{in}}}{T_{11}+T_{12}Z_{\text{in}}}.$$

Away from a Dirichlet pole the boundary action has $K_{11}=T_{11}/T_{12}$, $K_{12}=K_{21}=-1/T_{12}$ and $K_{22}=T_{22}/T_{12}$. Adding adjacent actions and eliminating their shared boundary by a Schur complement gives exactly the propagated result. This supplies a physical gravitational realization of the atlas's composition method in this sector.

The bulk potentials and shell strengths are nonnegative, so

$$\int \left(|\psi_x|^2 + U|\psi|^2\right)dx + \sum_s q_s|\psi(x_s)|^2 \geq 0.$$

With a regular center and radiative exterior, integration by parts excludes finite-energy exponentially growing axial modes. This covers a propagating gravitational sector across both material levels. It does not establish polar stability or the nonlinear formation of a rotating hierarchy.

16.4. The same exterior mass permits different gravitational returns

Derivation and calculation: two-frequency contrast and numerical controls. Keep the outer wall, parent annulus and exterior fixed. Replace only the regular child and its inner interface by the same Kottler region extended to its horizon, with an ingoing state there. Drive the finite worldtube $r_o=10M$ in the $\ell=2$ channel. Define

$$a_{\text{in}} = \tfrac{1}{2}\left(\psi - \tfrac{\psi_x}{i\omega}\right)$$
$$a_{\text{out}} = \tfrac{1}{2}\left(\psi + \tfrac{\psi_x}{i\omega}\right)$$
$$\mathcal{R} = \tfrac{a_{\text{out}}}{a_{\text{in}}}.$$

These are the stated finite-boundary canonical energy channels. They have not been extrapolated into an astronomical waveform at infinity.

M omega	Absolute difference of complex returns	Horizon absorbed fraction
.4	.00000160391406	.000000288534441
2	.258116878344	.442777762923

The regular child returns all incident power in this lossless sector. At the higher frequency the horizon control absorbs about 44.2778%. Its numerical indicator in absorbed fraction is 3.69×10^{-9}. Lower-frequency waves are strongly screened from the changed interior. Three numerical levels, an independent hypergeometric cap, another integrator and flux controls resolve both contrasts. The indicators are not physical confidence intervals or rigorous enclosures.

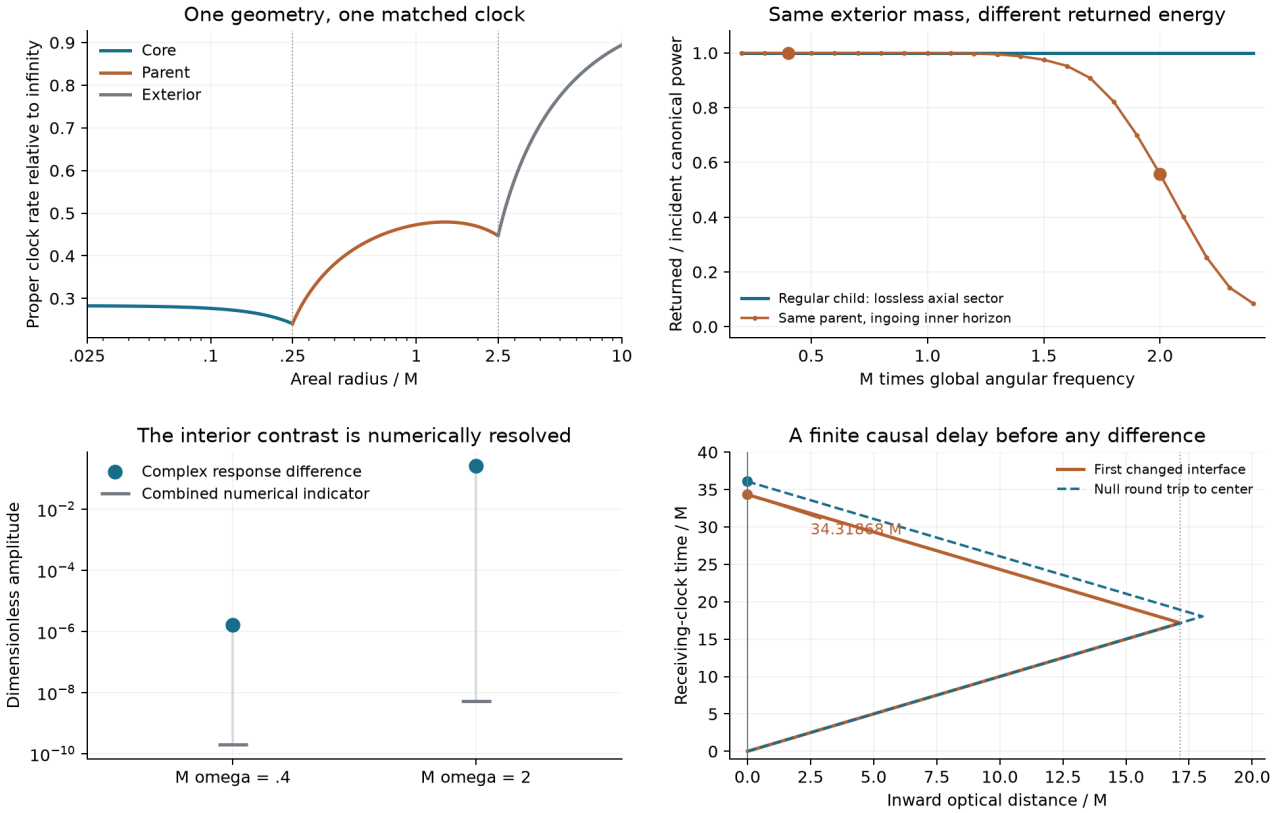


Figure 32. The matched clocks are continuous across both sourced interfaces. The same parent/exterior yields a different return when its inner termination changes. The lower panels compare the contrast to numerical indicators and show the principal-characteristic return times. The curves use 23 fixed plotting frequencies, not a fit or a search over object parameters.

For initially quiet perturbations, the first possible difference follows a round trip from the worldtube to the inner interface. On the receiving clock,

$$\Delta\tau_{\mathrm{first}}/M = 34.3186793341.$$

A null round trip to the center is *36.0742605915M*. Neither is a fitted echo peak. With the independently supplied example mass of 10^6 nominal solar masses, the first bound becomes 169.036344 seconds and the higher proper drive frequency .0722529542 Hz. The IAU nominal mass parameter sets the unit conversion.

The calculator and derivation retain the exact action, finite domain and thirteen new controls. The calculation uses 145106 RHS calls and about 1.6 solver CPU seconds in four workers, with no wave history stored. The result connects a nested source, a common gravitational evolution equation, Schur composition and a receiving clock. A direct retarded pulse, polar response and a specified rotating or dissipative state are the next physical connections.

16.5. A published nonlinear vortex mechanism enters the atlas

Reference and reading boundary: source identity and finite calculation. OpenAI's Finite time blowup for Navier--Stokes is linked from its 8 September 2026 announcement and accompanied by a Lean formalization. Theorem 1.1 constructs a three-dimensional incompressible flow for every positive viscosity, starting from rest under a smooth compactly supported force. Its velocity becomes unbounded in finite time while kinetic energy stays bounded. OpenAI's theorem is incorporated as an attributed mathematical input; the independent atlas calculation follows in Sections 16.6--16.7.

The relevant mechanism appears in Sections 2.2 and 3.2--3.3: zero-mean oscillatory velocities have nonzero quadratic momentum fluxes. The source uses those stresses to close a concentrating vortex's momentum balance. Its full profile construction and singularity argument have not been independently reproduced here. The formalization is version-pinned; it and its dependencies were not rebuilt in this task.

For time remaining tau and *0<h<.01*, the source's core scales are

$$\ell_r \sim \tau^{1/2}, \quad \ell_z \sim \tau^{1/2-h}, \quad U_{\text{core}} \sim \tau^{-1/2-h}.$$

Direct power counting gives

$$E_{\text{core}} \sim \tau^{1/2-3h}$$
$$D_{\text{core}} \sim \tau^{-1/2-3h}$$
$$\|u\|_{L^3(\text{core})} \sim \tau^{-4h/3}.$$

The shrinking volume permits small core energy and time-integrable loss alongside large pointwise velocity. These scaling checks illuminate the result; they do not replace its global proof or the force-regularity estimates.

The new finite atlas entry extracts the momentum-transfer mechanism for an independent calculation. Its prepared initial velocity is nonzero, its initial mean is zero, and the force is zero after release. Those assumptions differ from Theorem 1.1.

16.6. Fine motion produces an exact resolved circulation

Derivation and calculation: initial rate, covariance and conservation checks. In cylindrical coordinates choose the smooth bump

$$g(s) = \exp\left(1 - \frac{1}{1-s^2}\right) \quad (|s| < 1)$$
$$g(s) = 0 \quad (|s| \geq 1).$$

Set *a(r)=g(2(r-1))*, *b=-a/2* and *χ(z)=g(z)*. For an integer angular mode m and kappa=-2, the vector potential and initial velocity are

$$\phi = m\theta + \kappa m \log r$$
$$\mathbf{A} = \frac{\chi(z)\cos\phi}{m}(0, b, a)$$
$$u_0 = \nabla \times \mathbf{A}.$$

The field is smooth and compact, with support away from the axis, and its divergence is zero exactly. Let U be its rotation-group mean. At the initial instant U=0, but the covariance is

$$R_{ij} = \langle u_{0i} u_{0j} \rangle = \tfrac{1}{2}\mathrm{Re}(W_i \overline{W_j}) = R_{ij}^{(0)} + \frac{R_{ij}^{(2)}}{m^2}.$$

Here u_0=*Re(W* $e^{i\varphi}$*)*; the owner gives each component. Both covariance terms are positive semidefinite. Their nonzero radial--tangential correlation transfers angular momentum.

For the local classical unforced Navier--Stokes solution at any constant positive viscosity, the initial mean viscous term vanishes because the mean velocity is zero. Mean pressure has no tangential gradient. The actual initial equation is therefore

$$\partial_t U_\theta|_0 = -\left(\partial_r R_{r\theta} + \frac{2R_{r\theta}}{r} + \partial_z R_{z\theta}\right).$$

The cylindrical factor two is necessary. At the symmetry plane z=0, the envelope derivative vanishes, and substitution proves

$$\partial_t U_\theta(r, 0, t)|_{t=0} = -\frac{2aa'}{r^2}.$$

For every prepared m and positive viscosity, the rates at r=.75 and 1.25 are

$$-\frac{1024}{81}e^{-2/3} = -6.49060654184413$$
$$\frac{1024}{225}e^{-2/3} = 2.33661835506389.$$

These are rates in reference units U squared over L. Opposite resolved rotating bands begin to develop even though the initial mean velocity vanishes. Direct angular averaging of the original convection term, including its basis contribution, reproduces the stress result. This is an initial derivative of the unforced equation, not a chosen residual force.

There is also a finite composition law. Around any admissible axisymmetric parent B, add distinct angular modes with amplitudes ε_j. Cross covariances average to zero, as do the linear cross terms with B. Relative to the solution prepared from B alone,

$$\delta(\partial_t U_\theta)_{t=0,z=0} = \left(\sum_j \varepsilon_j^2\right)\left(-\frac{2aa'}{r^2}\right).$$

This composes the initial effects of prepared resolutions while retaining their covariance state. An infinite sequence additionally needs regularity and dissipation summability; it is not supplied by this finite identity.

16.7. The correlations retain their energy and momentum budget

Derivation and calculation: quadrature controls and physical unit example. With $I=\int \chi^2 dz=.983380812913$, the outward angular-momentum and axial-momentum fluxes through a cylinder are

$$\mathcal{J}_L = 2\pi r^2 \int R_{r\theta} dz = 2\pi a^2 I$$
$$\mathcal{J}_z = 2\pi r \int R_{rz} dz = \frac{\pi a^2 I}{r}.$$

They vanish at the radial support edges, so the global initial momentum budgets cancel. The circulation comes from internal redistribution. The energy and initial dissipation have the exact form

$$E_m = E_0 + \frac{E_2}{m^2}$$
$$D_m = \nu\left(D_2 m^2 + D_0 + \frac{D_{-2}}{m^2}\right)$$
$$E'(0) = -D_m.$$

Angular mode	Initial kinetic energy	Initial loss, viscosity .001
4	5.476541766802	1.722601258802
8	4.892129540839	4.237090234552
16	4.746026484348	14.757095523724

Energy approaches a positive 4.697325465517 while the viscous cost grows quadratically in mode number. For a finite sum the added energies and losses are their amplitude-squared weighted sums. A fixed energy budget therefore bounds the mean-force increment for this fixed envelope. Keeping the mean force at the initial instant does not establish a sustained cascade or uniform lifespan.

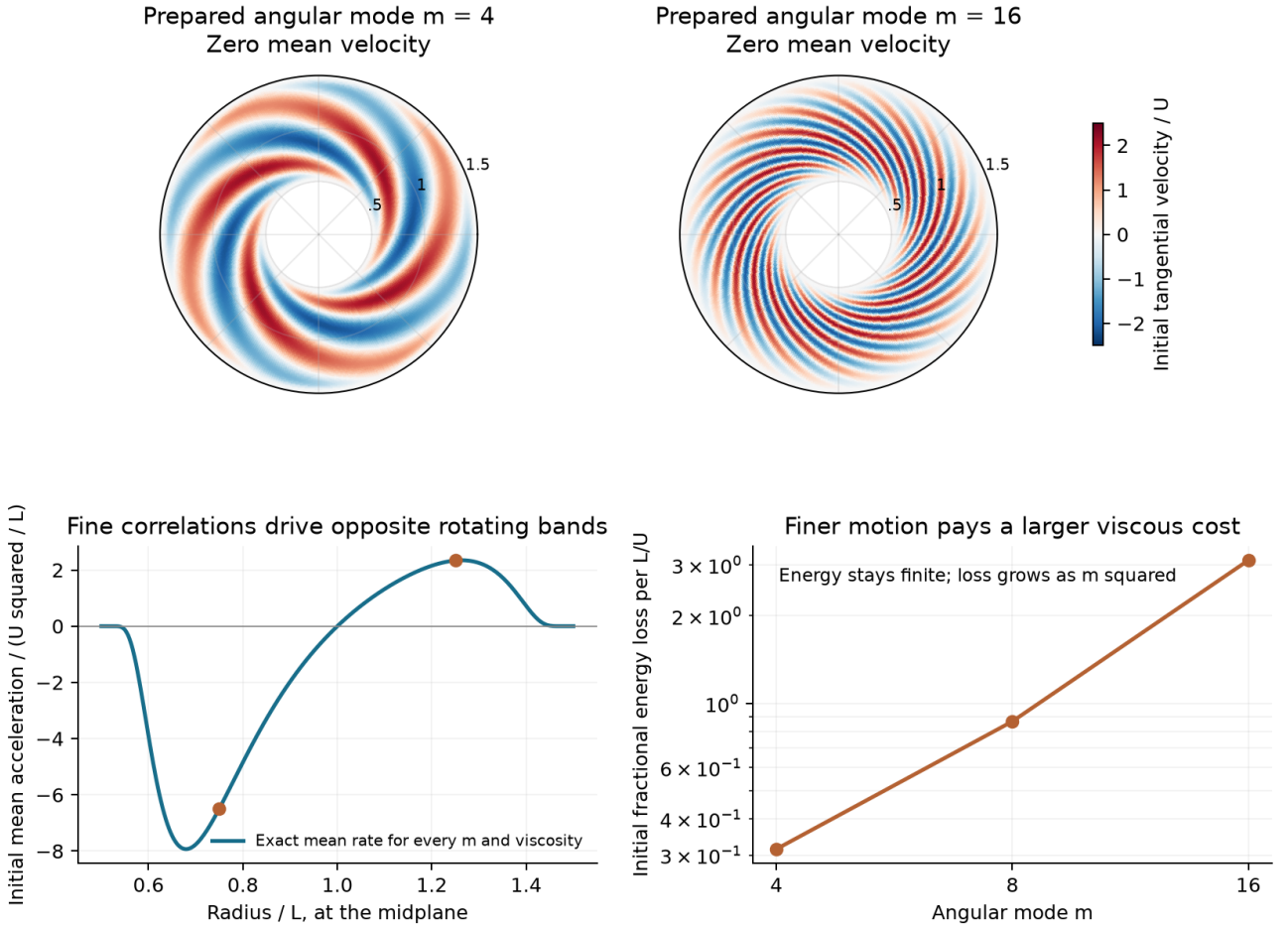


Figure 33. The upper panels show actual tangential-velocity patterns of the analytic initial data, not a simulated singularity or an image of the universe. Both patterns average to zero velocity. The lower left gives the exact common initial mean acceleration; the lower right shows the increasing fractional viscous loss. The preparation, transfer and budget share one velocity field.

In the supplied example L=1 m, U=.1 m/s, density 1000 kg/m³ and kinematic viscosity .0001 m²/s, the two initial accelerations are -.0649060654 and +.0233661836 m/s². These are conditional measurable rates for the stated preparation, not experimental observations. Independent 64/128/256 quadratures,

curl/gradient energy identities and direct convection checks accompany the exact formulas. Fourteen field/budget controls and an independent parent/composition check pass.

This adds a nonlinear source to the atlas's boundary dictionary. A coarse description that retains only U loses a real momentum flux; retaining R keeps its force and energy cost. Coupling that tensor to gravity requires the selected relativistic stress and geometry. In Bredberg et al.'s map, a three-spatial-dimensional fluid has a five-dimensional gravitational lift; the earlier four-dimensional Rindler example used a two-dimensional fluid. The singular endpoint also leaves a fixed hydrodynamic expansion regime. Thus the source paper contributes a defined mathematical block, while the gravitational attachment and a sustained rotating hierarchy remain further connections to calculate.

16.8. A material action now supplies the interaction vertex

Derivation and calculation: actual-mode normalization, vertices and controls. The interaction owner turns the cosmohedra comparison into a finite calculation on the sourced body. Arkani-Hamed, Figueiredo and Vazao organize a cubic scalar wavefunction through nested energy factors; Ardila-Mantilla and collaborators prove the associated Matryoshka face combinatorics. Here the low-point kernel is tested with a vertex calculated from a selected material/gravity action. No cosmohedron in physical spacetime is assumed.

The supporting wall scalar has a nonzero background and cannot be treated as a free probe. Explicitly add a second canonical component chi_s with zero background on each wall,

$$S_\chi = -\tfrac{1}{2}\sum_s \int_{\Sigma_s} \sqrt{-\gamma}\,\gamma^{ab}\partial_a\chi_s\partial_b\chi_s\,d^3y.$$

Its background and linear stress vanish. The classical geometry, original wall sources and axial operator are preserved. This adds stated probe matter; quantum vacuum loops and their backreaction are not calculated. For wall lapse $N_s=A\sqrt{f}$ and real unit spherical harmonics, the canonical modes use the shared global time:

$$\chi_s = \frac{\sqrt{N_s}}{R_s}\sum_i Q_{si}Y_i$$

$$L_{2s} = \tfrac{1}{2}\sum_i (\dot{Q}_{si}^2 - \omega_{si}^2 Q_{si}^2)$$

$$\omega_{s\ell} = \frac{N_s}{R_s}\sqrt{\ell(\ell+1)}\,.$$

Select the ell=1,m=1 cosine and ell=2,m=1 sine harmonics on each wall. For the axial metric harmonic $X^A=\varepsilon^{AB}D_BY_{20}$ and $\delta\gamma_{tA}=h_0X_A$, direct variation yields

$$L_{3s} = -\frac{h_0}{R_s^2}\sum_{ij} C_{ij}\dot{Q}_{si}Q_{sj}$$

$$C_{ij} = \int Y_i X^A D_A Y_j\,d\Omega = -C_{ji}.$$

The selected real harmonics give $C_{12}=3/(2\sqrt{\pi})$. The sign convention is fixed in the owner. Its opposite reverses both vertices, leaving their exchange product unchanged. This angular integral and the proper-clock normalization determine the vertex without a cubic fit.

16.9. An actual gravitational mode carries the cosmohedron seed

Derivation and calculation: mode, selected exchange and numerical controls. Retain the first regular axial ell=2 mode, with Dirichlet master at r=10M. This is an explicitly conditional quantum domain. It differs from the earlier open characteristic experiment at that radius and adds no physical spacetime wall. The retained scalar harmonics have positive frequencies; the scalar shift zero mode is excluded. Quantum theory and the positive Gaussian reference state are imported explicitly.

The existing spatial operator and real material jumps determine the mode. A trial Rayleigh quotient brackets its first eigenvalue. A phase and eigenvalue-sensitivity calculation gives the norm through a Wronskian identity; direct spatial quadrature and ordinary Regge--Wheeler propagation check it independently. The result is

$$M\omega_g = .392804111132$$

with no interior node. Martel--Poisson, the master relation and radiation-flux normalization, fix the canonical Cunningham--Price--Moncrief field:

$$\alpha_\ell = \frac{(\ell-1)\ell(\ell+1)(\ell+2)}{64\pi G}$$
$$\Psi_C = \frac{e(r)Q_g}{\sqrt{\alpha_\ell}}$$
$$h_0 = \frac{Af\Psi_C + r\Psi_{C,x}}{2}.$$

Here the spatial mode has unit square integral in x. Legendre transforming the scalar action therefore supplies the selected Hamiltonian vertex

$$H_3 = Q_g \sum_s c_s (P_{s1}Q_{s2} - P_{s2}Q_{s1})$$
$$c_s = \frac{C_{12}}{2R_s^2\sqrt{\alpha_2}}(Afe + R_s e_x)_{\Sigma_s}.$$

Both sides of the actual material junction give the same h0 and coupling. The vacuum pair numerator is $n_s=c_s(\omega_{s1}-\omega_{s2})$. Equal-frequency pairs have zero numerator. In exterior M units and with G factored out, the calculated inputs are

Quantity	Inner wall	Outer wall
First scalar frequency	1.36230331533	.252982212813
Second scalar frequency	2.35957855748	.438178046004
Current coupling divided by sqrt(G)	.001441720226	.028346822842
Pair numerator divided by sqrt(G)	-.001437791887	-.005249713475

For the two distinct wall pairs, a direct perturbative Schrodinger calculation gives the mixed connected term in log(Psi):

$$\Gamma^{(g_1)}_{1234} = -n_{\text{in}}n_{\text{out}}K_4$$
$$K_4 = \frac{1}{E_t E_L E_R}$$

$$E_t = \omega_1 + \omega_2 + \omega_3 + \omega_4, \quad E_L = \omega_1 + \omega_2 + \omega_g, \quad E_R = \omega_3 + \omega_4 + \omega_g.$$

The energy denominator is the seed of Cosmohedra, equation 49. The imaginary first-order phase and derivative numerator follow from this current interaction. An independent number-state perturbation calculation recovers the same kernel and selected normalized connected moment.

This is one propagating-mode tree contribution in the specified basis. Other radial modes, even parity, nonpropagating constraint/contact terms and higher orders are excluded. It is not the complete Einstein four-point function. Mode frequencies replace plane-wave momentum magnitudes; the full momentum-polytope realization has not been identified with this geometry. The result is a calculated graph-kernel connection.

16.10. Interior geometry changes the interaction and fixes its scale law

Derivation and calculation: two preparations and fixed-G scaling. Move the inner wall from .25M to .30M and re-solve its stiff junction. Its radial potential curvature stays positive. The parent parameters, outer wall and exterior are unchanged. These are separately prepared static states, not an instantaneous change or a new signal claim.

Quantity	Child radius .25M	Child radius .30M
M times gravitational frequency	.392804111132	.392804157226
Selected coefficient times M^4/G	-3.83479234e-7	-6.32140173e-7
Selected normalized moment times M^2/G	-3.21208795e-7	-4.91767335e-7

The selected coefficient changes 64.84%, while this one gravitational frequency changes about 1.17e-7 relatively. Inner scalar frequencies also change. This does not assert degeneracy of every linear observation. The numerical contrast is resolved by three solver tolerances and independent mode/action checks; it is not an astronomical measurement.

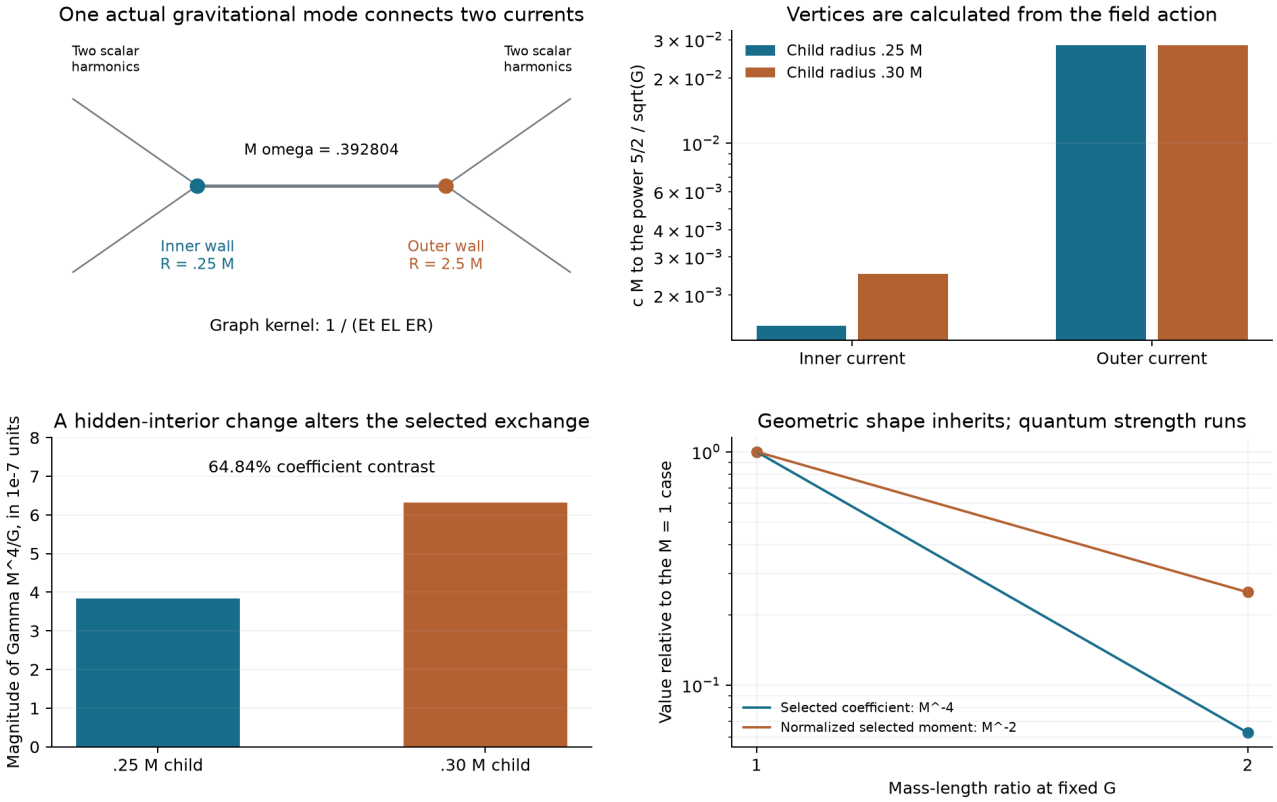


Figure 34. The upper-left panel is the selected interaction graph, not a spacetime diagram. Its vertices are calculated from the same material action. The lower-left panel compares the two inner geometries. The scale control at fixed G preserves geometric shape while changing quantum strength. The panels concern the declared one-mode exchange contribution.

At fixed dimensionless geometry, unit-normalized mode shape scales as M to the power minus one-half. The action consequently gives

$$c_s \sim \sqrt{G}M^{-5/2}, \quad n_s \sim \sqrt{G}M^{-7/2}, \quad \Gamma_{\text{selected}} \sim GM^{-4}, \quad C_{\text{selected}} \sim G/M^2.$$

Thus the M=2 calculation preserves the geometric quantities Gamma M^4/G and C M^2/G. The dimensionless quantum strength is G/M^2 in hbar=c=1 units; restoring units gives squared Planck length over squared geometric mass length. A physical inherited-law map must carry that parameter and the state. A fixed quantum coupling does not follow from classical geometric covariance. Shapes tabulated per G are not assertions of perturbative control at a Planck-sized mass.

The completed finite link is: material action, common clocks, actual modes, derived vertices, selected quantum exchange and an interior-dependent coefficient. Completing the constrained/pole-summed response and extending the state to a time-dependent parent remain further work. No black/white transition or observed Lambda value has been deduced from this graph kernel.

16.11. The actual cap--wall Dirac operator and the missing nested slip

The cap--wall calculation adds a declared massless, minimally coupled probe with no direct surface spinor interaction to the original classical scalar candidate. It uses the known quiet reference; P001's unconverged endpoint supplies no controlled corrected background. In its static cap, let $A=1-H^2r^2$ and $z=\operatorname{atanh}(Hr)/H$. Rescaling the physical spinor by $\psi=A^{-1/4}r^{-1}u$ gives

$$
\begin{aligned}
\mathrm{H}_k &= -i\sigma_2\partial_z + \sigma_1 W \\
W &= \frac{k\sqrt{A}}{r} \\
V_\pm &= W^2 \pm \partial_z W \\
\partial_z W &= -\frac{k\sqrt{A}}{r^2}.
\end{aligned}
$$

At $r/\ell=.5$, the lowest angular channel has $W\ell=\sqrt{3}$ and squared potentials $V_+\ell^2=-.4641016151$, $V_-\ell^2=6.4641016151$. Flat space at the same radius gives 0 and 8 for those potentials. These are curvature effects. The smooth control continues the same cap and retains its identical bulk operator.

Nesting changes the material domain. For the common cap-to-exterior normal, $[K]=-6/\ell$. Continuity of the spinor and its Dirac image gives $[\partial_n\psi]=-[K]\psi/2=3\psi/\ell$ in a common normal frame. The smooth-cap control has no jump. The reference record and ten independent controls verify these operator inputs. The exterior's usual static coordinate is spacelike at this cut, so it cannot supply a static joined spectral problem.

No nested Ψ/Φ or slip difference has yet been calculated. The loop requires a Dirac two-point state on the joined causal domain, renormalized bulk/interface response, and a conserved metric source with its wall displacement and gauge-invariant observer readout. Those inputs are absent from the classical scalar preparation and the flat profile. For free fermions the retarded bilinear response already depends explicitly on its initial covariance through $-iTr(C[V(t),V(0)])$; geometry alone does not select it. The completed result here is the curved operator/domain connection, not a loop polarization or an NSC slip number.

16.12. Local Dirac trace on the cap and material wall

The free four-component Dirac determinant and general heat-coefficient framework are imported; the spin/statistics contractions and actual curvature are calculated. In the two quiet smooth regions, where $\Box R=0$,

$$
\frac{\langle T^{\mu}{}_{\mu}\rangle}{\hbar c} = \frac{7R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} + 8R_{\alpha\beta}R^{\alpha\beta} - 5R^2}{360(4\pi)^2}.
$$

The one-sided evaluation therefore gives $-11H^4/(240\pi^2)$ in the cap and $7M^2/(120\pi^2r^6)$ outside. A flat reference has zero local curvature trace. These are actual-background terms of the declared quantum probe, distinct from the flat nonlocal polarization in Section 17.11.

At the material wall, integrate a smooth normal curvature jet before taking a thin limit. The declared class smooths the kinked conformal factor while the conformal product curvature remains bounded. With one normal pointing from cap to exterior, put $t=K^{\tau}{}_{\tau}$, $b=K^{\Omega}{}_{\Omega}$, $A=R_{\tau\tau}/R$ and $B=(1+R_{\tau}{}^2)/R^2$. The Chern scalar is $B_3=-8tb^2+8Bt+16Ab$. Direct Lorentz curvature contraction gives the Euler surface coefficient

$$
\mathcal{E}_\Sigma = -[\mathcal{B}_3] = 4k_{\mathrm{wall}}(k_{\mathrm{wall}}^2 + 3H^2) = 56/\ell^3.
$$

The exterior mass and wall radius cancel through the signed junction equations. Pure-trace leading singular terms cancel in both Euler and Weyl curvature squares; no delta square is evaluated. The Euler-wall reference then gives

$$
\frac{\langle T^{\mu}{}_{\mu}\rangle_{\Sigma,E}}{\hbar c} = -\frac{77}{720\pi^2\ell^3}\delta_\Sigma.
$$

A smooth continuation of the cap has no such interface coefficient. The scheme-dependent total derivative, finite interface terms, traceless stress and quantum preparation remain separate inputs. This one Euler term is not the complete wall stress. Trace-only projections do not establish vacuum stress or replace the supplied cap $\Lambda_{0,-}\ell^2=3$.

16.13. An actual wall radar chart and a finite clock prediction

The wall-clock construction uses the same $H\ell=1$, $k_{wall}\ell=2$, $M/\ell=.4$ and $.45\le R_{wall}/\ell\le.55$. In each static region take optical coordinates $u_i=t_i-r_i^*$, $v_i=t_i+r_i^*$. Their wall derivatives are

$$\frac{du_i}{d\tau}=\frac{1}{\beta_i+\dot R}$$
$$\frac{dv_i}{d\tau}=\frac{1}{\beta_i-\dot R}.$$

Inside the exterior horizon the second derivative is negative. Label both null rays by their wall proper times U,V, preserving that orientation. Then

$$T=(U+V)/2,\quad x=(V-U)/2,\quad ds_2^2=C_i(-dT^2+dx^2),\quad C_i=f_i(r)u_i'(U)v_i'(V)>0.$$

The cap occupies $V\le U$ and exterior $V\ge U$. The proper-clock range $0\le U,V\le S$, $S/\ell=.1256873632901$, gives the finite causal diamond in Figure 35. An analytic conservative bound gives $N=\sqrt{C}\ge.3851891$ throughout that domain. The material wall remains $x=0$, $T=\tau$, $C=1$; both Israel jumps are retained. This construction uses no spatial cutoff profile and has no reflecting timelike ends. The earlier wide coordinate chart's fold remains an unsuccessful construction, without changing the underlying quiet geometry.

The actual Dirac half-density has constant characteristic speed:

$$\xi=C^{1/4}r\psi$$
$$H_\kappa=-i\sigma_2\partial_x+\sigma_1W_\kappa$$
$$W_\kappa=\kappa\sqrt{C}/r.$$

Both W_κ and its first derivatives are continuous at the wall. The density spinor and its spatial derivative are continuous in this frame, while the physical parallel-normal jump remains $3k_{wall}\psi/2$.

For a finite clock protocol, emit a radial optical phase at wall $U=\tau(.475)$ and receive at $(U,V)=(\tau(.475),\tau(.525))$ on a fixed wall-radar-position observer. The signed radar position is $x/\ell=.0314673818165$ and $T/\ell=.0584746895099$. The null phase $-U$ and receiver proper clock give $\omega_{rec}/\omega_{wall}=1/N_{rec}$.

Quiet geometry with the same radar protocol	r_{rec}/ℓ	N_{rec}	$\omega_{rec}/\omega_{wall}$	$W_1\ell$
Nested cap--wall--exterior	.5068911	.8668323	1.153626	1.7101
Smooth cap continuation	.5364941	.9253385	1.080686	1.7248

For the independently supplied $\ell=10$ km, this radar range is 314.674 m. The smooth continuation is a geometric control with the same observer worldline and clock labels; it is not another matched membrane solution. These are conditional quiet-background predictions, with no observed frequency used to tune the geometry. They are not a quantum slip or an astronomical identification. The optical-map comparison with independent quadrature differs by 2.11×10^{-14} or less; finite mode propagation and the full quantum error account are not supplied by this geometry calculation.

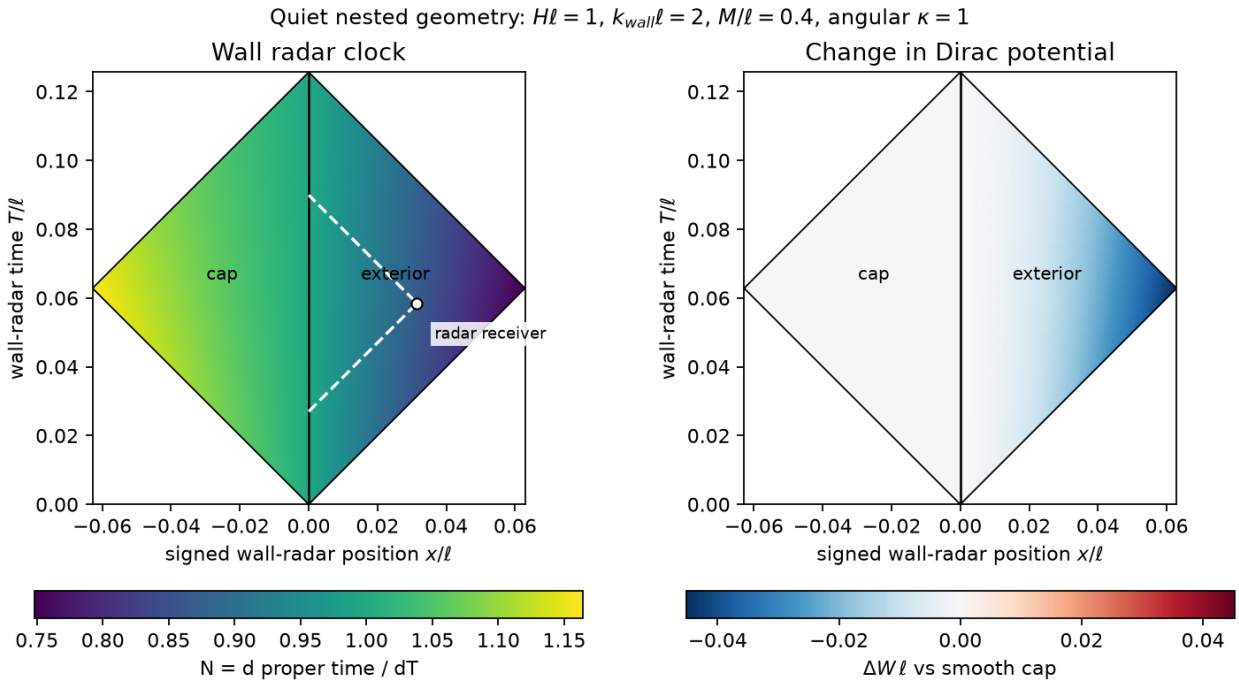


Figure 35. The finite quiet wall-clock diamond. The material wall is the vertical line; null edges are causal limits. Left: proper-clock lapse, with the emitted/reflected radar ray and receiver. Right: the radial Dirac-potential difference from a smooth continuation of the same cap, using the same wall clock and probe angular sector. These colors describe the background geometry, not an evolved quantum state or a metric-loop response.

16.14. Stress vertices and the inherited quantum preparation

The same canonical radial Dirac action supplies its gravitational sources. For a full magnetic multiplet let $d=2|\kappa|$, $f=d/(4\pi)$, $\xi=\sqrt{a}\, r\psi$, $q_A=-i(\xi^\dagger\, A\xi_y-\xi_y^{\,\dagger}\, A\xi)/2$ and $s_1=\xi^\dagger\sigma_1\xi$. Its real Hamiltonian density is $h=d[N(q_\alpha/a+\kappa\, s_1/r)-\beta\, q_l]$. Metric variation at fixed canonical half-density gives

$$\rho = \frac{f}{ar^2}(q_\alpha/a + \kappa s_1/r)$$
$$p_r = \frac{f q_\alpha}{a^2 r^2}$$
$$p_t = \frac{f\kappa s_1}{2ar^3}$$
$$T^{\hat{0}\hat{1}} = \frac{f q_l}{a^2 r^2}.$$

The stress and contact calculation checks the Hilbert energy/flux, complete angular normalization, four-dimensional covariant conservation and the first/second metric variations. Finite-mode stress has zero classical trace and transmits continuously; the anomaly in Section 16.12 is retained separately. Radial ports and first-order time-end variations remain in an integrated response. No scalar source or flat polarization is substituted.

The radar characteristic equations define the transfer of prepared incoming data to the middle Cauchy segment. A parent covariance must include both incoming branches and their cross-correlations. Existing Gaussian boundary elimination can preserve those terms once the actual transfer is supplied. Neither the chart nor the common-law relation chooses a root state. Section 16.17 supplies one additional preparation axiom and establishes its local short-distance class; it is not inferred from a null-line frequency label alone.

The characteristic transfer and a finite prepared-state mean response are evaluated next. A compatible baseline preparation, bulk/interface renormalization conditions and the complete mixed response to a physical source still separate them from a nested Ψ/Φ prediction.

16.15. Prepared Dirac propagation, mean metric and clock

The actual transfer propagates the canonical chiral field on the wall-clock diamond. With $P=\eta_+$ and $M=\eta_-$, the equations are $2P_V=-iW_\kappa M$ and $2M_U=-iW_\kappa P$. The characteristic cell map is a two-component Cayley unitary in positive null-flux weights. Direct metric variation gives the ordered matter response $-i[J_A,J_B]$ for the declared exterior source and later cap insertion. For one specified first-channel occupation difference its magnitude changes by about .3774% relative to the smooth-cap control. This is one matter-response component.

The mean-clock reference then supplies the mean metric needed before a full mixed source/tadpole calculation. The incoming mode has $P(U,0)=0$ and $M(0,V)=A\ b(V)\exp(-12iV/S)$, where b is the stated smooth peak-one bump centered at $.6S$ with half-width $.3S$: put $z=(V-.6S)/(.3S)$ and $b=\exp[1-1/(1-z^2)]$ for $|z|<1$, and zero otherwise. The constant A normalizes its continuum null flux to one. For a compatible supplied parent covariance C_H, define $\Pi=I-|f\rangle\langle f|$, $C_{empty}=\Pi\, C_H\Pi$ and $C_n=C_{empty}+n_{occ}|f\rangle\langle f|$, $0\le n_{occ}\le 1$. This declared preparation changes one mode's occupation and removes its cross terms; it selects no root vacuum. The parent must make the finite-rank modification smooth on this region; its global preparation remains an assumption. Both magnetic states of the $\kappa=1$ multiplet have the same radial data. Other inherited modes cancel between the two states.

Analytically differentiated characteristic equations supply first derivatives of the propagated field. The same stress vertices then determine the nonvacuum Misner-Sharp mass one-form. In units $c=\hbar=1$, with $\kappa_g=8\pi G$ and the cap vacuum mass evaluated at the actual areal radius,

$$m_{\text{total}} = M_i + \Lambda_i r^3/6 + m_i$$
$$(m_U, m_V) = \frac{\kappa_g r^2}{C}(t_{UV}r_U - t_{UU}r_V,\ t_{UV}r_V - t_{VV}r_U).$$

The two wall masses drive the existing signed radius-response equation. In outgoing EF coordinates, the radial lapse follows from $\psi_V=\kappa_g\, r\, t_{VV}/(2r_V)$ and its proper-wall normalization. Analytic sensitivity of the ingoing null ray gives $s=\delta\, r$ and $z=s_V$, hence $q=\delta\ln C=\psi+z/r_V$. The specified radial phase and a receiver at fixed proper-wall radar position give $\delta\ln(\omega_{rec}/\omega_{wall})=-q/2$. No separate bulk metric grid is needed for this first-order supplied source.

The proper-clock labels remain $U=\tau(.475)$ and $V=\tau(.525)$ of the quiet reference. Define

$$\varepsilon_q = \frac{8\pi\hbar G}{c^3\ell^2}$$
$$\mathcal{C}_{\text{clock}} = \frac{\delta\ln(\omega_{\text{rec}}/\omega_{\text{wall}})}{\varepsilon_q n_{\text{occ}}}.$$

Geometry and observer protocol	C_{clock}, 128 cells	Absolute 64/128 spread
Nested cap-wall-exterior	-.41780	1.12e-5
Smooth cap, shared observer motion	-.39929	9.95e-6

The smooth cap has no membrane. Its observer worldtube is explicitly driven along the computed nested wall's areal-radius history. The coefficient difference is $-.01851197$, about 4.63625% in magnitude, with a correlated numerical spread 1.23×10^{-6}. This compares the two geometries at common observer motion; it is not an independently evolving alternative wall model.

The normalized nested mass-path discrepancy decreases from 1.15×10^{-6} to 2.92×10^{-7}; the smooth control also improves. Orders 8/12 of the piecewise quadrature differ by less than 5.34×10^{-15} in the

clock coefficient. Exact charged-wall and nonzero-lapse fluid families independently check the reconstruction. These indicators support the finite calculation; they are not a rigorous continuum remainder or an observational confidence interval.

For the independently supplied $\ell = 10\ km$, the nominal prepared-state fractional frequency correction is

$$\delta \ln(\omega_{\mathrm{rec}}/\omega_{\mathrm{wall}}) \simeq -2.7430 \times 10^{-77} n_{\mathrm{occ}} \left(\frac{10\ \mathrm{km}}{\ell}\right)^2.$$

The nested resolution spread converts to 7.34×10^{-82} at unit occupation and that scale. The CODATA 2022 uncertainty in G is separate. The length and mode preparation are inputs, and no astronomical datum selects them.

This finite state-difference result supplies the mean h_q needed in the later mixed Einstein and source variations. It does not determine the baseline vacuum, full retarded Ψ/Φ or a cosmological Λ. The finite-amplitude scalar P001 remains unconverged; its original solver and grids were not changed by this calculation.

16.16. A conserved compact source and local gravitational tides

A central mass source can be supplied on the same finite annulus, with its body or inner horizon below the causal domain. The cap and exterior vacuum functions and dimensionless source mass are

$$f_- = 1 - 2b/r - H^2 r^2$$
$$f_+ = 1 - 2M_p(b)/r$$
$$\hat{b} = b/\ell.$$

Hold the initial physical wall radius and proper speed fixed, as well as H and tension. Both signed junction constraints then require $dM_p/db|_0 = \beta_{+,0}/\beta_{-,0} = 2755/8587$. Holding the parent mass fixed would change that speed. No regular body interior or formation history is supplied by this annular construction.

The same continuum-normalized Dirac occupation difference is propagated on each member of the mass family. The mean Einstein reconstruction, stress, wall and observer vary together. Its two mass-flux laws and wall displacement have an exact triangular solution: since $d\ln v/d\tau = acc/v$,

$$\frac{\rho(\tau)}{v(\tau)} = \frac{\rho(\tau_0)}{v(\tau_0)} + \int_{\tau_0}^{\tau} \frac{\beta_+(\mu - \nu)/(k_{\mathrm{wall}} R^2) + \mu/R}{v^2}\, d\tau'.$$

Piecewise Gauss integration on the existing field-cell breakpoints evaluates these primitives. The prefactor v retains the acceleration term; no constraint residual is canceled. The independent exact charged and fluid families check this reconstruction.

The compact-source reference uses $b/\ell = 0{,}5\times10^{-5}, 10^{-4}$, the same 64/128 field cells, and joint wall/bulk quadrature orders 8/12. Its receiver is inside the cap at $U=.8S, V=.7S$, at fixed proper-wall radar position. An inward phase -V is emitted at the earlier wall time V. The smooth SdS continuation shares the nested observer's classical and mean areal-radius history, without another material wall.

Define

$$\mathcal{C}_B = \frac{\partial}{\partial \hat{b}} \frac{\delta \ln(\omega_{\mathrm{rec}}/\omega_{\mathrm{wall}})}{\varepsilon_q n_{\mathrm{occ}}} \bigg|_{\hat{b}=0}.$$

This derivative uses a source present before the window, not a retarded mass-switching experiment. It includes the background Einstein variation that a matter-only insertion on fixed geometry would omit.

A second readout uses actual orthonormal curvature. In the specified radar observer's frame put $A=R_{0r0r}-R_{0\theta0\theta}$ and $B=R_{r\theta\, r\theta}-R_{\theta\varphi\theta\varphi}$. Direct Ricci/Weyl decomposition gives

$$w=\frac{b+m_q}{r^3}-\frac{\kappa_g}{6}(\rho_q-p_{rq}+p_{tq})$$
$$A=-3w-\frac{\kappa_g}{2}(p_{rq}-p_{tq})$$
$$B=-3w+\frac{\kappa_g}{2}(p_{rq}-p_{tq}).$$

The fixed vacuum term cancels. At fixed radar labels its first-order Weyl variation also retains $-3b\,\delta\, r/r^4$. For nonzero b, $A_0=B_0=-3b/r^3$. Define the occupied-minus-empty local ratio response by $\delta(B/A)=\varepsilon_q n_{occ}C_{tid}$. The 128-cell results are:

Geometry and observer	Clock derivative C_B	Tidal coefficient at $b/\ell=10^{-4}$
Nested cap-wall-exterior	-.002329216	-540762.871
Smooth SdS cap, shared motion	-.002329519	-540765.571

The clock-derivative difference is $+3.03553\times10^{-7}$, with a correlated grid spread 2.58×10^{-9} and a source-step indicator 5.95×10^{-10}. The tidal-coefficient difference is +2.7001584, about 4.9932 parts per million of the prepared correction, with a correlated grid spread .0011855. Individual tidal-coefficient grid spreads are about 14.59. For the supplied $\ell=10$ km, the occupied-minus-empty changes in B/A are $-3.55031184\times10^{-71}$ and $-3.55032957\times10^{-71}$. Their nominal nesting difference is $+1.77276\times10^{-76}$. The length, source and state are inputs.

All prescribed reference checks pass. The maximum normalized nested mass-path gap falls from 8.08×10^{-7} to 1.93×10^{-8}; joint quadrature changes the clock coefficient by less than 4.45×10^{-15}. These spreads are diagnostics, not rigorous remainder bounds or observational confidence intervals. Earlier failed precision/execution records remain in the scientific owner.

The local ratio is a curvature observable in the stated frame. In a flat-FLRW Newtonian metric its comoving counterpart compares the radial trace-free Hessians of Ψ and Φ. Reconstructing those potentials also requires radial zero-mode and background-split data; below the exterior horizon one cap FLRW pair does not cover the joined spacetime. This finite prepared-state result therefore supplies no absolute empty vacuum ratio, full vacuum Ψ/Φ or cosmological Λ.

16.17. A correlated continuum parent and local vacuum regularity

Specify the parent before evaluating a baseline vacuum. One explicit alternative is the Dirac ground state on a static product of a radial line with a sphere of radius $R_0=.45\ell$. In its chiral canonical frame,

$$\mathrm{H}_\kappa(p)=p\sigma_3+\mu\sigma_1$$
$$\mu=\frac{\kappa}{R_0}$$
$$\omega=\sqrt{p^2+\mu^2}\,.$$

Fill its negative band for every signed angular sector, including each magnetic multiplet once. The Fourier measure is $dp/(2\pi)$. On the two incoming null legs put $a=(\omega-p)/2$ and $b=(\omega+p)/2$. Their plane phases are $\exp(iaU)$ and $\exp(ibV)$. The same normalized spinor supplies both amplitudes, so the state has Hardy diagonal kernels and the cross-leg kernel

$$C_{+-}(U,V) = -\frac{\mu}{2\pi} K_0\left(|\mu|\sqrt{UV}\right)$$
$$U, V > 0.$$

Here K_0 is the modified Bessel function. Its normalization follows independently from the equal-time cylinder Fourier integral. Restricting the global projector to a finite diamond gives a generally mixed CAR covariance. Its exact finite compressions retain this cross term; filling the negative spectrum after truncating an energy matrix selects a different state.

Use an explicit auxiliary preparation, with fixed physical duration S and a smooth switch $\chi(s)$ supported in $(.1S,S)$, equal to one on $[.2S,.9S]$. Define

$$q = \frac{N}{r}$$
$$q_{\text{aux}} = \frac{1}{R_0} + \chi(U)\chi(V)\left(q - \frac{1}{R_0}\right).$$

The auxiliary product metric has $r_{aux}=R_0$ and $N_{aux}=a_{aux}=R_0 q_{aux}$. Blending q directly preserves its continuous first wall jets. Where the switches are one this metric is conformal to the physical metric and the canonical massless spinor is unchanged. This neighborhood includes the cap detector of Section 16.16. The construction assigns a state; it is not a physical formation history or an Einstein source model. Stress and clocks still use the physical metric, including its anomaly.

For every angular sector the auxiliary Hamiltonian differs from the cylinder by a bounded time-dependent potential on a common radial domain. Its convergent interaction series gives unitary evolution. The angular direct sum is unitary as well. Moreover,

$$\{H_\kappa, \sigma_1\} = 2\kappa q_{\text{aux}}$$

gives a gap at least $|\kappa| q_{\min}$ when $q_{aux} \geq q_{\min} > 0$. The prescribed covariance is $C(T)=U(T)C_0U(T)^\dagger$.

Local regularity proposition. On an admitted regular patch, this state is Hadamard on any relatively compact smooth interior neighborhood inside the matched operator region. Both geometries therefore have a smooth physical two-point difference on their common cap neighborhood.

The proof and calculated boundary symbols make the calculus input explicit. Fold the two sides onto a half-line. The transparent condition is the allowed space of a constant Hermitian involution that anticommutes with the normal matrix. It has maximal zero flux. At real spacelike frequency the decaying-root map is invertible, uniformly away from glancing, including both angular polarizations. This verifies the complementing condition for the standard finite-bundle first-order boundary parametrix. One-sided smoothness supplies its jets.

Normal regularity follows by pairing the unitary field with smooth tangential tests and transferring tangential derivatives through its first-order equation. The resulting map from the initial Hilbert space to every one-sided normal Sobolev space gives smooth normal traces. Boundary elliptic regularity confines singularities to the compressed causal cone. Splitting its two orientations then makes the unwanted component satisfy a smooth inhomogeneous boundary problem with absent past data; energy uniqueness makes it smooth. Glancing remains inside the causal cone and does not join its opposite time orientations. The local CAR and the regular physical conformal map finish the argument.

The existing Weyl projector and transport algebra now receives the actual radar metric jets through fourth order. Here local p is conjugate to radar x, with $\omega=\sqrt{p^2+W^2}$; it is not the incoming cylinder momentum label. In the canonical Pauli frame the first subleading term is $c_1=-pW_T\sigma_3/(4\omega^3)$. The first discontinuous wall coefficient enters c_2, through $[W_{xx}]=-6\kappa(M_p-b)/R^4$. This curvature contribution is

retained. At a cap detector the two models share all local subtraction terms. For a single covariance the fourth-order remainder is qualitatively of order minus five; the two-state difference is smooth there.

This establishes existence of the local relative stress for this stated preparation. It does not evaluate it or provide a numerical tail constant. The full vacuum Ψ/Φ, wall renormalization and physical selection of the preparation remain separate work. No astronomical measurement sets this root radius, switch, state or resulting numerical coefficients.

16.18. Inherited preparation and the cap's local late state

Derive the state transport before choosing another parent benchmark. A complete ancestral covariance is inherited as $C_c=K\,C_a K^\dagger$, with $KK^\dagger=I_c$ for current-normalized restriction and evolution. Both incoming null branches, their cross-correlation and $I-C_a$ are transported. A change of null clock carries a square-root Jacobian on each kernel argument. Finite unitary transport does not select C_a.

The actual quiet cap nevertheless selects a local asymptotic state. Keep $M_-=0$, $H\ell=1$, $M_+/\ell=.4$, $k_{wall}\ell=2$, and the transparent massless Dirac probe. In ℓ units its signed wall obeys

$$\beta_- = \frac{3R}{4} + \frac{1}{5R^2}$$
$$v^2 = \dot{R}^2 = \frac{25R^2}{16} - 1 + \frac{3}{10R} + \frac{1}{25R^4}.$$

This positive expanding root reaches $R>1$ in finite wall and central cap time. In the expanding flat cap chart its regular clock is $dt/d\tau=(1+v^2)/(\beta_-+Rv)$; at $R=1$ it equals $761/760$. After that crossing choose a Cauchy ball of physical radius $H^{-1}<R_B<R_w$. Its conformal future contains a neighborhood of the central endpoint at $\eta=0$, with a positive spatial margin. The future timelike wall remains outside that null domain.

Inherited-state proposition. Suppose the inherited field state is Hadamard on a neighborhood of this closed ball. For Gaussian detector correlations take a gauge-invariant quasifree state. In each fixed translated central proper-time window, its full four-component two-point function approaches the conformal de Sitter one. Parent-dependent stress has a decaying amplitude, rather than a new freely chosen root state.

Proof. Set the cut to $t=0$, $a=e^{Ht}$ and $\eta=-H^{-1}e^{-Ht}$. The conformal field $\psi_{flat}=a^{3/2}\psi$ satisfies the flat massless Dirac equation. Subtract the conformal Minkowski two-point function as a proof reference. Its Cauchy-data difference is smooth. Cut off both arguments inside a smaller ball still larger than H^{-1} and evolve with the flat Dirac propagator. Finite propagation and the strict margin extend this smooth difference through a neighborhood of the virtual endpoint. No condition at a physical infinity is added.

The physical two-point difference is $a^{-3/2}a'^{-3/2}F(\eta,\eta')$, with F smooth and bounded there, so it decays as e^{-3HT} under a common time translation T. The same-action tetrad stress difference is

$$\Delta T_{\hat{a}\hat{b}} = e^{-4Ht}Q_{\hat{a}\hat{b}}(-H^{-1}e^{-Ht})$$
$$\left|\Delta T_{\hat{a}\hat{b}} - e^{-4Ht}Q_{\hat{a}\hat{b}}(0)\right| \le \frac{M_1}{H}e^{-5Ht}.$$

Here M_1 bounds the corresponding conformal-time derivative; anomaly and local curvature ambiguities cancel in the state difference. The amplitude $Q(0)$ and its bound are inherited data. This proves a local limit and a finite conditional error, without asserting equal finite-time states or loss of global unitary information. The complete proof and calculator give the wall positivity and domain argument.

On the central geodesic the limiting Dirac kernel's scalar factor is $[H/(2\sinh(H(s-s'-i0)/2))]^3$. Its imaginary-time antiperiod is $2\pi/H$. A weak scalar-density detector coupled to $:\bar{\psi}\psi:$ has a bosonic bilinear kernel with power six, giving, up to one positive detector constant,

$$\mathcal{R}(x) = \frac{x(x^2+1)(x^2+4)}{e^{2\pi x}-1}$$
$$x = \frac{E}{\hbar c H}$$
$$\frac{\mathcal{R}(x)}{\mathcal{R}(-x)} = e^{-2\pi x}.$$

The polynomial follows independently by shifting the Fourier contour to a nonsingular $cosh^{-6}$ integral. Late-state translation precedes the long smooth observation limit; finite switching does not have the exact stationary ratio. Detector statistics and the conformal/no-hair mechanism have prior literature. The project result connects them to this sourced wall's causal future.

Finally the existing cap trace, $A=-11\hbar\, cH^4/(240\pi^2)$, has a full asymptotic stress interpretation. Homogeneity, isotropy and conservation give $-\rho+3p=A$ and $d\rho/dt+4H\rho=-HA$. The radiation constant decays as a^{-4}, leaving, at fixed renormalized classical Λ and G,

$$\langle T_{\mu\nu}\rangle_{\text{late}} = -\frac{11\hbar c H^4}{960\pi^2} g_{\mu\nu}$$
$$\delta\Lambda\ell^2 = \frac{11}{120\pi}\frac{\ell_P^2}{\ell^2}.$$

For the supplied $\ell=10$ km ruler this is $7.62221759052\times10^{-80}$; the temperature is $3.64446440565\times10^{-8}$ K and the stress e-folding interval is 8.33910237995 microseconds. At $x=1$ the detector ratio is 0.00186744273171. These are conditional model scales, with no observational calibration. Seven new independent controls pass, including the separate detector integral and the FLRW Ward identity.

The same smooth cap has the same late state and source: the late nesting-specific vacuum offset is zero at fixed H. This does not generate the supplied classical cap scale or solve quantum wall backreaction. Regularity of a fully transmitted inherited state on the later ball remains a hypothesis. The compact-source $b>0$ cap is not conformally flat; its finite-time state and Ψ/Φ require their own compatible source history and retarded response.

VII. Effective gravity, vacuum form and the physical scale

A nonzero response is not automatically a cosmological constant. This part varies the relevant action or transported geometry to determine energy, pressure, memory and curvature together. The calculations decide which proposed reductions have vacuum form, which behave as other sources, and which require a different state or action. Those decisions delimit the routes by which the larger geometry can contribute to interior expansion.

17. What the eliminated geometry contributes

The following sections give the worked constructions and their attributed mathematical inputs.

17.1. The current scalar's classical vacuum projection

The current cap-wall candidate allows a direct first calculation of this target. Its action fixes a real massless conformal scalar and independent regional vacuum terms. Contracting the complete improved scalar stress gives

$$T^{\phi\mu}{}_{\mu} = \phi(\Box - \mathcal{R}_J/6)\phi = 0$$
$$\mathcal{R}_J = 4\Lambda_0.$$

Consequently its additional smooth classical trace projection is $\Delta\Lambda_{classical}L_{ref}{}^2=0$, including nonzero on-shell pulses. The scalar still carries stress, mass and flux. The cap baseline $\Lambda_{0,}\ell^2=3(H\ell)^2=3$ is a

supplied input, while $\Lambda_{0,+}=0$ outside. The scalar calculation does not derive that baseline. Distributional wall stress and geometric averaging are separate operations.

The exact calculation record binds the symbolic expressions and their producer. The calculation and frame check also recover the same Jordan curvature from the Einstein-frame potential. Independent metric-jet checks evaluate the implemented nonlinear equations at zero and nonzero amplitude. The scientific owner gives the complete derivation.

17.2. A conditional quantum contribution and its physical scale

As an explicit extension, import one quantized real four-dimensional massless conformal scalar with a Hadamard bulk state and locally covariant renormalization. This uses quantum field theory; it does not derive quantum statistics from the atlas. The local trace coefficient follows from the scalar heat-kernel formula, equation (4.28), and conformal anomaly, equations (15)-(16). Substituting $E=-R/6$ cancels its R^2 term. On either smooth quiet region,

$$\langle T^{\mu}{}_{\mu}\rangle = \frac{\hbar}{2880\pi^2}(\mathcal{R}^2_{\mu\nu\rho\sigma} - \mathcal{R}^2_{\mu\nu}).$$

The scheme-dependent $\Box R$ term vanishes because the quiet scalar curvatures are constant. Finite volume and Newton counterterms remain supplied renormalized coefficients. Direct curvature contractions give $(Riemann^2, Ricci^2)=(24H^4, 36H^4)$ in the cap and $(48M^2/R^6, 0)$ outside.

Set $h=H\ell$, $m=M/\ell$, $r=R/\ell$ and $\zeta=\kappa_4\hbar/\ell^2$. With $\Delta\Lambda_{tr}=-\kappa_4\langle T^{\mu}{}_{\mu}\rangle/4$,

$$\Delta\Lambda_{\mathrm{tr},-}\ell^2 = \frac{\zeta h^4}{960\pi^2}$$
$$\Delta\Lambda_{\mathrm{tr},+}\ell^2 = -\frac{\zeta m^2}{240\pi^2 r^6}.$$

At the supplied $h=1$, $m=2/5$, $r=11/20$, the coefficients of ζ are 0.0001055428996274352 and $-0.0024402417804070 79$. For the independently declared P001 fiducial $\ell=10000\ m$, restoring SI units gives $\zeta=8\pi G\hbar/(c^3\ell^2)$ and

Quiet trace projection	$\Delta\Lambda_{tr}\ell^2$	Standard uncertainty from G alone
Cap	6.92929×10^{-81}	1.5573×10^{-85}
Exterior at the reference wall radius	-1.60211×10^{-79}	3.6006×10^{-84}

These use CODATA 2022 constants. The length is a fiducial input; the quoted uncertainty is not a total theoretical or observational error band. The exterior term varies with radius. A constant cap trace alone does not establish a stress proportional to its metric, since the material wall need not preserve state symmetry.

There is a necessary test of whether this term can generate the cap scale without supplying a volume term. Put $X=(H\ell)^2$ and $\lambda_0=\Lambda_0\ell^2$. The semiclassical constant-curvature trace equation requires

$$3X = \lambda_0 + \frac{\zeta X^2}{960\pi^2}.$$

At $\lambda_0=0$ its roots are $X=0$ and $X=2880\pi^2/\zeta$. The nonzero root has $H^2\ell_P{}^2=360\pi$, outside the controlled small-curvature semiclassical regime. Thus this one-scalar contribution does not generate a nonzero small-curvature de Sitter cap by itself. The same exact record retains the coefficients, SI inputs and necessary trace condition. No cosmological observation selected these values.

17.3. The quantum interface and the remaining exterior reduction

The wall contributes its own local operator data. The existing scalar matching $[\varphi]=0$, $[n\varphi]=k\varphi$ identifies the transmittal coefficient $U_\Sigma=k$. In the inward-normal convention of the transmittal heat-kernel theorem, $L^+=-K_+$ and $L^-=K_-$. Their sum is kI, so the second surface coefficient is proportional to $tr(kI)/3-k=0$.

Keeping every curvature term in the fourth coefficient gives the local surface polynomial

$$a_4^\Sigma(1)=\int_\Sigma \sqrt{|\gamma|}\, q_\Sigma\, d^3y$$
$$q_\Sigma \ell^3 = -\frac{b(b^2+3h^2)}{1440\pi^2} = -\frac{7}{720\pi^2}, \quad b=k\ell=2$$

at $h=1$. This evaluates the imported Riemannian polynomial on the quiet wall's Lorentzian contractions. A joint Lorentzian quantum state and its interface stress have not been constructed. The Gauss and Euler-boundary checks agree with the recorded coefficient. It is a surface invariant; obtaining its metric variation requires the full off-shell operators and their renormalized coefficients.

The physical Schur connection must use the exterior conformal wave operator with this transmission law. For boundary field $\varphi=\varphi|_\Sigma$, let $N_\pm$ map it to the corresponding outward normal derivative. The homogeneous matching equation is $(N_-+N_++U_\Sigma)\varphi=0$; incoming initial data add a drive. In real time these are causal history operators. Their retarded response and the joint state's symmetric two-point function are needed for the effective stress. A coefficient multiplying a boundary field quadratic form alone does not supply a metric volume term.

17.4. The exterior scalar response on the quiet wedge

The physical exterior reduction begins with the quiet Schwarzschild wedge and the prepared source already fixed by the characteristic problem. The outward Robin response on the exterior side of the wall is

$$B_+\phi = \mathcal{N}_+\phi + \frac{K_+}{3}\phi.$$

For the original unit-amplitude source, zero bottom data and the prepared wall trace, the boundary pairing is

$$\frac{1}{\epsilon^2}\int \phi\, B_+\phi\, d\Sigma = 0.3703318455512892.$$

The response at the wall is therefore nonzero. A compact Weyl variation supported in the smooth cap, with the interface, exterior geometry and characteristic data held fixed, does not move that boundary integrand. The on-shell improved scalar trace remains zero, so this variation adds

$$\Delta\Lambda_{\mathrm{Schur,\,cl,\,local}} L_{\mathrm{ref}}^2 = 0.$$

The retarded map is causal. It is not in general equal to its adjoint, so the single-history pairing is not by itself a variational action for a metric stress. The exterior response record binds the pairing, the zero bulk variation and the quadrature diagnostics. The Schur-memory owner states the domain: a quiet finite wedge, not a whole-geometry reduction, a quantum state, or a cosmological value. The supplied cap input $\Lambda_0 \ell^2=3$ is unchanged.

17.5. Parent response, slice fit and a separate contracting geometry

Eliminating the whole quiet body, as seen from its right Schwarzschild end, is a different operation from the cap-centre scalar map. On the static tube $R/\ell=4$, with $M/\ell=0.4$, proper boundary time and the outward normal, Minkowski subtraction gives, in geometrized units $G=c=1$,

$$E = R\left(1-\sqrt{1-\tfrac{2M}{R}}\right)$$
$$E/\ell = 0.4222912360003365.$$

The ADM mass is M, not this finite-radius energy. The accompanying surface pressure is a stress on the tube. It is not a bulk pressure and not a four-dimensional vacuum density. For dilute, slow, conserved bodies whose separations are large compared with their tubes, the leading parent action is

$$S_{\mathrm{body}} = -\sum_i M_i \int d\tau_i.$$

Its homogeneous metric variation has zero bulk pressure and density scaling as a^{-3}. A general on-shell energy $E(V)$ at fixed particle number and entropy obeys $\rho=E/V$ and $p=-dE/dV$. Vacuum form would require E proportional to V as a consequence of that action. Prescribing a growing population or a time-dependent mass by hand is exchange through the tube, not that variation. No parent population, source state or controlled interaction correction is supplied. The gravitational reduction record contains the tube value and the dust-versus-extensive identities.

The isolated initial slice from the regular centre through the wall and bridge to $R_{cut}/\ell=4$ can be averaged in the hypersurface-normal clock. Use the sign convention

$$K_{ij} = -\tfrac{1}{2}\mathcal{L}_n h_{ij}$$
$$\theta = -h^{ij}K_{ij}.$$

The scalar fit absorbs the variance Q and the mean three-curvature into

$$\kappa_4\rho_g = -(Q+\langle {}^{(3)}R\rangle)/2$$
$$\kappa_4\rho_g = -Q/2 + \langle {}^{(3)}R\rangle/6.$$

The trace projection and vacuum-form residual are

$$\Lambda_{g,\mathrm{tr}} = \frac{Q-\langle {}^{(3)}R\rangle}{4}$$
$$\Lambda_{g,\mathrm{tr}}\ell^2 = 0.030775890732$$

$$(3Q+\langle {}^{(3)}R\rangle)\ell^2 = 1.80095618027.$$

The fit is therefore not vacuum. It is a scalar reduction of one slice, not the average of a four-dimensional stress tensor, and the domain is not the causal past of the cap centre. A $K=0$ connector added to repeated copies increases the vacuum volume and does not remove the defect inside the retained bodies. The slice-average record keeps the wall distribution, both bridge branches and the normal clock. The shared parent geometry and its observer domain remain absent.

The two-level initial geometry in Section 10.2 makes this test at an actual additional boundary, with the same outer mass and retention sphere. Both membranes obey $k\ell=2$; cap and host have the supplied $H\ell=1$. Including both wall curvature distributions gives

$$\Lambda_{g,\mathrm{tr}}\ell^2 = -0.0153506330754$$
$$(3Q + \langle {}^{(3)}R\rangle)\ell^2 = 2.11796240047.$$

The geometric density and pressure again fail vacuum form. Gauss 64-minus-32 differences in these quantities are -1.28×10^{-10} and -3.91×10^{-10}; the integrated Hamiltonian residual is -3.33×10^{-16}. These are numerical indicators, not total error bounds. Adding the level changes the physical state and its finite projection; this comparison is not solely a change of observational resolution of one fixed state. The two-level record and average derivation close this finite calculation. An evaluated parent and physical observer reduction remain necessary for the local cosmological target.

A second, already evolved geometry supplies a different finite average. Its metric and action are not those of the cap-wall scalar. In the saved conformal gauge

$$ds^2 = r^2\{Q^2[-dT^2 + dx^2] + d\Omega^2\}$$

with $r,Q>0$, zero shift and periodic $S^1\times S^2$ topology. The action contains the Einstein term, a Weyl auxiliary and a Dirac source. The named clock is the normal worldline at $x=1$, $d\tau=rQ\,dT$. The reference length $L_{ref}=4.80028044552352$ is that observer sphere's areal radius at $T=0.05$. Write $\nu=N/N_{observer}$ and use the weighted moments Q_ν,R_ν defined in the observer-clock derivation. The clock term is

$$P_\nu = \langle(\partial_{\tau_o}\nu)\theta\rangle + \langle \nu D^2\nu\rangle.$$

The geometric trace and vacuum defect at $T=0.3$ are

$$\Lambda_{g,\mathrm{tr}}L_{\mathrm{ref}}^2 = -0.5725162917234052$$
$$(3Q_\nu + \mathcal{R}_\nu + 2P_\nu)L_{\mathrm{ref}}^2 = 2.177444644645245.$$

The domain contracts. The largest four-case deviations, 4.65×10^{-9} and 1.92×10^{-8}, compare stored resolutions. They are refinement indicators, not a continuum or total-error bound. The metric identity used for E and S reconstructs an Einstein source from the geometry. It does not independently verify the matter constraint: the primary Hamiltonian defect remains about 9×10^{-6} in its source normalization. No new evolution was run. The negative trace is not a positive cosmological vacuum response, and the periodic shell is not the parent of the quiet cap. The conformal average record binds the clock, the reference length and these two numbers.

17.6. The interface required by a common action

The two actions remain alternatives. Importing the literal quiet tension wall into the Weyl-squared chart is a compatibility question, not a derivation of either model from the other. With the $(-+++)$ convention, spacelike normal and $[K_{ab}]=-kh_{ab}$, every singular coefficient in the linear Weyl tensor cancels. Thus the Weyl-squared term has no squared-delta coefficient from this pure-trace jump at that algebraic level. The regular Weyl tensor still steps from the conformally flat cap, where it vanishes, to the Schwarzschild value. In the orthonormal wall frame, and after a radial boost,

$$[C_{anbn}] = \frac{M}{R_w^3}\mathrm{diag}(-2, -1, -1).$$

At $M/\ell=0.4$ and $R_w/\ell=1.5$,

$$\ell^2[C_{anbn}] = \mathrm{diag}(-32/135, -16/135, -16/135).$$

The components have dimension inverse length squared. In a thin-layer limit that keeps the metric, connection and Weyl curvature bounded and approaching these one-sided values, the leading Bach double-layer strength is the jump $[C_{\mu\,n\nu\,n}]$. Ordinary tension is a delta-order source and does not supply that derivative of a delta. The quiet scalar vanishes. A continuous scalar with bounded one-sided first derivatives does not supply it either. The Euler and Box-R ledger completes divergences on smooth periodic metrics and finite time caps. It is not an adjustable source on this timelike wall.

Lower-order products of discontinuous curvature, and a complete junction, remain unproved. A smooth transition, or an effective interface between resolutions whose action is actually derived, is still required. The check is necessary under the stated thin-layer control. It is not a universal impossibility, and it does not remove the separate P001 Einstein model. Reina, Senovilla and Vera, Sections 3.1--3.2, discuss the hazards of formal quadratic cancellations. Their generic distributional class is not applied here as a theorem. The direct tensor calculation is in the wall-connection note, with the normal-Weyl components exposed by the gravitational reduction and checked in its tests.

17.7. One state at two physical resolutions

The inherited law and a state's resolved structure are different mathematical objects. To calculate a resolution effect, hold the separate conformal candidate's full state fixed and choose a retained space from its physical radial Laplacian. On the reference slice $T=0.05$,

$$\mathcal{L} = -\Delta_h = -\frac{1}{r^3 Q}\partial_x\left(\frac{r}{Q}\partial_x\right)$$
$$\mu L_{\mathrm{ref}}^2 \le 1.$$

The reference ruler remains $L_{ref}=4.80028044552352$. The mass and stiffness integrals use the existing 511 geometry variables and their 2048-point quadrature. Orthonormalizing the selected eigenspace in the canonical coordinate measure gives a fixed symmetric projector P. Apply it to all six log-canonical variables $a=\log r$, $\pi_a=rp_r$, $b=\log Q$, $\pi_b=Qp_Q$, and χ,p_χ. The fermion columns and occupations remain fully retained. This rule keeps one geometric mode. The first omitted eigenvalue is 0.4795862384, above the fixed cutoff 0.04339770651. No observed target selects that cutoff or the ruler.

At $T=0.3$, evaluate the same Hamiltonian and its vector field on the full and filtered states. The difference at fixed canonical environment is

$$H_{\mathrm{corr}}(z_c; z_e) = H(z_c + z_e) - H(z_c)$$
$$\delta F_c = PX_H(z) - PX_H(Pz).$$

Here P acts as the identity on the fermions. The exact change of canonical variables includes the momentum product rule. The four-dimensional vacuum volume supplies an independently derived comparison direction,

$$H_\Lambda = 8\pi A\Lambda \int r^4 L Q\, dx$$
$$A = \frac{1}{16\pi G}.$$

The owned action uses R_E=-R_L and -A R_L-2$A\Lambda$; its existing r^2 potential is sphere curvature. The displayed r^4 weight follows from the physical volume. Varying before choosing L=Q fixes the lapse constraint and both momentum forces. This potential changes no coordinate or fermion rate. Adjoint pullback precedes the nodal canonical change.

The calculation gives H_{full}=-1.363×10^{-11}, H_{coarse}=1.015478298556, and unit-volume generator 231.9284384454. Their energy projection is

$$\Lambda_E L_{\mathrm{ref}}^2 = -0.100890404750047.$$

Its force does not reproduce the correlation. Each row below uses that block's canonical units; the RMS values are not combined across blocks.

Log-momentum rate	Correlation RMS	Vacuum candidate RMS	Residual RMS
π_a	0.21206304	0.50773915	0.29567611
π_b	0.01318283	0.25386957	0.24068675

The residuals are 58.23% and 94.81% of the larger RMS in their respective rows. The retained a_T and χ_T correlations are 0.00098884455 and 0.67276495 RMS per coordinate time, where a vacuum potential gives zero. The unchanged input fermions also acquire a rate difference. The b_T difference, 5.54×10^{-18} RMS, is treated as roundoff.

The original Hamiltonian-constraint maximum is 9.005×10^{-6}; the filtered state's is 25.5817 in that density normalization. No constraint reset is performed, and the filtered metric with the retained fine matter is not a second physical solution. The missing correlation contains constraint, kinetic and matter response beyond a single volume term. Fifteen controls include the analytic spectrum under unit rescaling and independent energy differentiation. The relative eigen residual is 1.65×10^{-15}; replay of the archived prolonged rates differs by at most 8.72×10^{-9} in a momentum component. These controls do not bound the continuum or physical model error. Rates use the same conformal coordinate generator; a matched proper clock, causal elimination, repeated scale composition and a fractal dimension remain open. The fixed-state record binds the original state and frozen method `cd00b35`. Its energy coefficient is a failed vacuum-volume representation of this instantaneous response, not a derived cosmological constant.

17.8. A prepared-state change propagated to a photon measurement

The same conformal episode carries an initial matter variation through the coupled geometry to two received frequencies. Before prolongation, apply $U_\varepsilon(x)$=exp[$i\varepsilon$ $f(x)\sigma_2$] to every original spinor, with $f(x)$=cos(2π x/8) and unchanged occupations. This preserves the initial Gram matrix and paired current cancellation. The radius direction is obtained by differentiating the original projected constraint level. The original full constraint defect, 9.97845×10^{-6}, remains; a zero root is not assumed. Initially at x=1, $d\log N/d\varepsilon$=-0.27203771374589336.

Evolve the complete state and its derivative together, z_T=$X_H(z)$ and u_T=$DX_H(z)u$, at the same four RK4 stages. All six geometry fields and both complex column blocks evolve. The fixed spatial grid is n_f=512,n_g=511,n_q=2048, period 8. The two original time steps 0.001 and 0.0005 take 300 and 600 steps to T=0.3. Both baseline states reproduce their saved histories at T=0.05 and 0.3 within the declared field-scaled 10^{-8} tolerance.

Reception is fixed at $(T_o,x_o)=(0.3,1)$; emission occurs at $(T_e,x_e)=(0.175,0.875)$ or $(0.175,1.125)$. These are exact step and quadrature events. The normal observers measure $1+z=N_o/N_e$, $N=rQ$, so the propagated response is

$$K=\frac{d\log(1+z)}{d\varepsilon}=\frac{u_{r,o}}{r_o}+\frac{u_{Q,o}}{Q_o}-\frac{u_{r,e}}{r_e}-\frac{u_{Q,e}}{Q_e}$$
$$\frac{dz}{d\varepsilon}=(1+z)K.$$

Emitter x_e	Fine K	Fine $dz/d\varepsilon$	Absolute time-step change in $dz/d\varepsilon$
0.875	-0.0009129945410	-0.0009145171829	1.22×10^{-13}
1.125	0.0027692088369	0.0027489767089	1.71×10^{-13}

These differences are empirical temporal indicators. The inherited spatial and constraint errors, physical source mechanism and finite-ε range are not bounded by them. At the fine endpoint, the full Hamiltonian constraint maximum is 8.91×10^{-6} and the weighted constraint variation norm is 1.09×10^{-6}. The Gram and tangent-Gram errors are 1.63×10^{-13} and 1.68×10^{-14}. The sampled Hamiltonian variation drifts by 8.59×10^{-12}, versus 2.76×10^{-10} on the coarse time step; no uniform-in-time estimate is inferred.

The initial response and propagated response bind the original data, frozen method and reproduction command. A stopped central-difference preflight is preserved; the independent initial Fourier force check agrees to 6.35×10^{-9} relative error without changing the force equations. The calculation uses 1736.1875 process-CPU seconds including that stopped attempt. One 477146-byte ignored checkpoint stores the final states and derivatives; the published summary stores its hash. This closes one finite prepared-direction observer calculation. The metric-dependent reduction still needs to retain the corresponding causal response and physical state before identifying an effective vacuum term.

17.9. A constrained gravitational exterior and its validity test

The same local auxiliary action supplies a linear vacuum exterior response. This reference has zero Maxwell flux and Dirac field, physical metric $ds^2=N^2dt^2-a^2dR^2-R^2d\Omega^2$, and flat $N=a=1$, $\chi=0$. It is not an attachment to the charged periodic episode. Write $N=1+\varepsilon\ n$, $a=1+\varepsilon\ b$, $\chi=\varepsilon\ c$, and $k=4\pi\ A/\alpha$, $\alpha=-4\pi\ C_W/3$. Varying before imposing areal gauge retains both constraints. They fix $kRb+c_R+c/R=kM$, with constant mass perturbation M; the full radius equation then vanishes. Linear Schwarzschild has $n=-M/R$, $b=M/R$, $c=-12M/R$.

For zero initial perturbation and $M=0$, set $\psi=c/R$. Independent variation of all five original fields and elimination of the constraints give

$$\psi_{tt}-\psi_{RR}+(6/R^2+m^2)\psi=0$$
$$m^2=\frac{A}{2C_W}$$

$$S_{2,\mathrm{bulk}}=K\int[\psi_t^2-\psi_R^2-(6/R^2+m^2)\psi^2]dt\,dR$$
$$K=-\frac{8\pi C_W^2}{3A}.$$

The action weight K is distinct from the normalization of the wave equation. Radial total derivatives are retained in deriving the face response; no boundary energy normalization is selected.

On $R\geq R_0$, impose a compatible causal curvature history at R_0 and no incoming perturbation at infinity. For a Laplace parameter s with real part greater than $\gamma=\sqrt{max(0,-m^2)}$, the decaying branch $q=\sqrt{s^2+m^2}$, with positive real part, yields

$$\frac{\hat{c}_R(R_0)}{\hat{c}(R_0)} = -\frac{z^3+2z^2+3z+3}{R_0(z^2+3z+3)}$$
$$z = qR_0.$$

The outward normal of this exterior points toward decreasing R. The lapse and radial metric histories obey the same constraints, so this is one compatible response column. It is not a choice of three independent boundary histories or a bulk vacuum density.

The scalar evaluation keeps the saved A,C_W and independent ruler $R_0=L_{ref}=4.80028044552352$. Frozen method `6c68d00` uses one declared Laplace point, no archive, trajectory, root or fit, and 0.015625 process-CPU seconds:

Reference quantity	Value
$m^2L_{ref}^2$	-614.4717961513472
γL_{ref}	24.788541630183637
Growth time divided by L_{ref}/c	0.04034121954082023
K/L_{ref}^2	$-5.754230595887687\times10^{-6}$

For $m^2=-\gamma^2$, a compact sine beyond $\max(R_0,4/\gamma)$ with width $4\pi/\gamma$ has spatial quadratic-form quotient at most $-9\gamma^2/16$. Every finite inner radius therefore leaves growing modes in this infinite flat exterior. The retarded map exists in the stated half-plane, while a stable stationary reduction about this reference is unavailable. Nine controls include the exact radial solution, pre-gauge constraints, Schwarzschild mass mode, reduced action and physical-clock face response.

This tests the local action used nonperturbatively. The spherical action owner retains a conditional curvature expansion and an incomplete full spectral/state contribution. An effective-theory reading needs its cutoff, order reduction or causal completion specified before promoting this additional mode to a physical exterior state. The comparison ruler is not a derived ultraviolet cutoff. The finite periodic computations retain their stated domain; the actual nested exterior geometry and its physical state remain necessary for a cosmological Lambda calculation. No coupling was changed to remove the reference instability.

17.10. The inherited spectral profile and its gravitational response

The local coefficients have an existing spectral owner. For the fixed warped compact interval, its matched profile is $H(u)=h_\Lambda(u)-E_1(u/\mu_B^2)$, where $\Lambda_{UV}=\Omega/r_e$ and $\mu_B=\sqrt{|q|}/r_e$. They are distinct from the observer ruler. The coefficient account gives $Q_1=\int_0^\infty H(u)\,du$, $Q_0=H(0)$, $A=Q_1/[6(4\pi)^2]$ and $C_W=-Q_0/[40(4\pi)^2]$. Compact interval, warp and matching scales are fixed during this external metric variation.

The flat Euclidean transverse-traceless heat response of Alkofer--Saueressig--Zanusso, 1410.7999v2, Section II, supplies the linear spectral-profile dependence. With the existing zero-volume-term normalization, exchanging its integrations gives

$$\mathcal{P}_H(x) = x\left[1-\frac{x}{4Q_1}\int_0^1 (1+2v)\sqrt{1-v}\,H(xv/4)\,dv\right]$$
$$x = p^2 \geq 0.$$

For the owned half-trace action, the quadratic density is $(A/4)h_{TT}P_H(-\partial^2)h_{TT}$. Its first two terms are $x+(2C_W/A)x^2$, exactly matching the local coefficient account. The derivation and controls retain the spectral-calculus assumptions, unit scaling and an independently integrated finite proper-time check. The earlier exponential example is not substituted for this compact profile.

The single evaluation keeps the saved A,C_W, $\Lambda_{UV}=3.973074368754331$, $\mu_B=2$ and $L_{ref}=4.80028044552352$. Frozen method 9b68122 uses $p^2L_{ref}^2=1$:

Finite reference quantity	Value
p/Λ_{UV}	0.05243324000377066
$P_H(p^2)/p^2$	0.9983741797605584
Local polynomial divided by p^2	0.9983725860059594
Full-profile minus local ratio	$1.5937545990718505\times10^{-6}$
Extra local root divided by Λ_{UV}^2	1.6893333026263349

Both fixed Jacobi rules, of orders 16 and 32, round to the same normalized response; their raw correction integrals differ by about 10^{-15}. This is an integration indicator, not a zero total error or a bound on inherited compact spectral errors. Seven controls pass, and the profile's zero limit recovers C_W within 5.42×10^{-19} absolute. The scalar application takes 0.1875 process-CPU seconds and stores 4720 bytes without a raw cache, scale root, trajectory or fit.

The local polynomial's additional root lies above the stored ultraviolet cutoff. It therefore cannot by itself diagnose the full spectral theory's stability. This calculation does not locate the full kernel's other poles or replace the constrained spherical causal map. Curved real-time metric dependence and the exterior's physical state remain necessary before identifying an effective cosmological volume term.

17.11. Flat-reference curvature channels and the relation between motion and lensing

The same fixed-profile functional has a covariant quadratic-curvature connection. Reusing Codello--Zanusso, 1203.2034v2, Section 5, and the owned Dirac Clifford traces gives Weyl and scalar functions $F_C(-\nabla^2)$ and $F_R(-\nabla^2)$ in the action $[CF_C\ C+RF_R\ R]/[2(4\pi)^2]$. The complete derivation works to second order in curvatures with compact weak perturbations of Euclidean R^4, fixed topology and fixed compact data. Euler and boundary terms are removed only within that domain; a material wall or strong-curvature attachment needs its own terms.

This is an imported Dirac fixed-profile calculation about flat space. The numerical slip below is not a consequence of NSC's cap--wall nesting.

The Weyl function recovers Section 17.10 through $a_2(x)=P_H(x)/x=1+6xF_C(x)/Q_1$. The scalar function is

$$\mathcal{F}_R(x)=\frac{1}{72}\int_0^1\left(1-\frac{5v}{2}\right)\sqrt{1-v}\,[H(xv/4)-H(0)]\,dv.$$

The constant subtraction changes nothing because its weight integrates to zero. Thus $F_R(0)=0$, while a nonzero-momentum scalar response remains. Its trace factor is $a_0(x)=1-18xF_R(x)/Q_1$. This correction adds no zero-momentum volume term; the previous vacuum normalization is still a premise.

For a conserved weak static density perturbation $T_{00}=\rho$, $T_{ij}=0$, write the physical metric as $ds^2=(1+2\Phi)dt^2-(1-2\Psi)dx^2$. Direct metric and source variation, with source action $+\int\rho\Phi$, gives the ratios to $\Phi_{GR}=\Psi_{GR}=-\rho/(4Ax)$:

$$G_\Phi=\frac{4}{3a_2}-\frac{1}{3a_0}$$
$$G_\Psi=\frac{2}{3a_2}+\frac{1}{3a_0}.$$

Slow-motion acceleration measures $-\nabla\Phi$; the null optical index is $1-\Phi-\Psi$. Consequently the lensing ratio is $(G_\Phi+G_\Psi)/2=1/a_2$, and gravitational slip is $\Psi/\Phi=G_\Psi/G_\Phi$. The density normalization cancels from these

relations. This is an external static probe, not a supplied astrophysical source or the earlier nested-body preparation.

The one application keeps the old profile, couplings and reference scale, using frozen method 9796add at $p^2L_{ref}^2=1$:

Conditional weak-field quantity	Value
Scalar trace factor a_0	*0.9999996814199501*
Motion-potential ratio G_Φ	*1.0021711842538266*
Spatial-potential ratio G_Ψ	*1.0010857514169889*
Lensing-potential ratio	*1.0016284678354077*
Gravitational slip Ψ/Φ	*0.9989169187321566*

Eight controls pass, including independent curvature contractions, pure gauge, the Einstein density and its flux, heat/profile integration and direct source variation. The fixed 16/32 rules round to the same response floats and recover the old TT factor. Their scalar kernels differ by 3.95×10^{-16}; these are integration indicators, not bounds on the inherited spectrum or physical model. The application takes 0.140625 CPU seconds and retains 5797 bytes without a trajectory or scale fit.

A real-time response requires its state and closed-time-path variation. Putting a retarded kernel in an ordinary quadratic action instead produces its symmetric retarded--advanced average. The static relation therefore does not supply the actual curved causal exterior, physical length calibration or a cosmological Lambda. Those remain the connection needed between the larger geometry and its observer's effective volume response.

17.12. Composition and conservation of a nested geometric response

The next connection runs from a nested geometry to its effective scalar gravity and expansion test. On one common slice/clock, retain each region's volume and integrals of theta, theta squared, shear squared, intrinsic curvature, normal energy and spatial stress trace. Add these seven moments, with every material interface counted once. If f_i is its volume fraction and H_i its mean theta/3, the variance identity gives

$$H_D = \sum_i f_i H_i$$
$$Q_D = \sum_i f_i Q_i + 6\sum_i f_i (H_i - H_D)^2.$$

Moment addition is associative. The same result therefore follows from direct integration or successive child-to-parent composition. With the surface relation C_surface=2*kappa*E_surface retained, summing the local Hamiltonian preserves the averaged scalar constraint at every level. Dropping the inter-region variance would violate it. This applies established multidomain averaging to the actual signed atlas and its material walls; it is not a new spacetime gluing theorem.

For the existing two-level slice, grouping its cap, Kottler host and Schwarzschild exterior gives

$$Q_D\ell^2 = 0.280192255373 + 0.233947711669 = 0.514139967042.$$

The required between-region term is 45.50% of the total for this grouping. Its 32/64 quadrature change is -5.02e-11; this is a numerical indicator, not a certified bound. All wall atoms remain explicit. The initial material-boundary fluxes in ell^2 units are -2.25902368244, 6.03328027292 and -3.77425659048; they cancel globally. Reynolds transport adds these fluxes to the individual volume rates. Their contribution

to the derivative of the H weights is .008489878632/ell^2; local H derivatives remain separate physical inputs. Mean local expansion cannot silently be used as the expansion of a moving material volume.

For a regular history in the same clock, let R be its mean3-curvature, P its complete lapse acceleration, and E,S its correspondingly weighted explicit energy and spatial stress trace. The averaged Hamiltonian and Raychaudhuri equations give

$$3H^2 = \kappa E - (Q+R)/2$$
$$3(H_{,t} + H^2) = -\kappa(E+S)/2 + Q + P.$$

Set A=3Q+R+2P and Lambda_trace=(Q-R+2P)/4. Define the explicit-source balance J=E_dot+3HE+HS. Differentiation now proves

$$Q_{,t} + R_{,t} + 2HA = 2\kappa J$$
$$\Lambda_{\mathrm{tr},t} = A_{,t}/4 + HA - \kappa J.$$

The corresponding source is

$$\kappa\rho_g = \Lambda_{\mathrm{tr}} - A/4$$
$$\kappa p_g = -\Lambda_{\mathrm{tr}} - A/12.$$

Pure constant vacuum requires A=0 and J=0 on an interval. A constant vacuum component plus radiation is also possible: for J=0, the trace is constant exactly when A_dot+4HA=0. A non-vacuum result at one instant does not distinguish those histories. Neither missing derivative is chosen to satisfy this equation.

The same trace and vacuum condition follow independently from expansion:

$$\Lambda_{\mathrm{tr}} = 3H^2 + \tfrac{3}{2}H_{,t} - \kappa(E-S)/4$$
$$A = 6H_{,t} + \kappa(3E+S).$$

On the declared instantaneous normal continuation, H*ell=.092609048739 and H_dot*ell^2=.278302667627 recover the recorded trace -.0153506330754 and vacuum condition 2.11796240047 in ell units. H here is domain volume expansion; its identification with sky-inferred expansion requires the missing light-cone calculation. The full derivation and calculator state the common-clock, source and infinite-resolution conditions.

These proofs establish a composable scalar response and its required time balance. They do not supply a collective parent metric, a closed history or a generated cosmological constant. The existing quiet massless model still has no demonstrated nesting-induced bulk Lambda. The compact proof calculation binds the source hashes and the finite application; full fields need not be stored to reproduce it.

VIII. Physical inheritance and the completed cosmological account

The preceding parts supply a connected set of mathematical constructions. This final part states what they establish collectively, how their physical parameters must be inherited, and the remaining closures needed for one tested cosmological account. Quantum probabilities, matter asymmetry and successive generations have explicit mathematical targets alongside the dark-sector and observer programme.

18. Physical inheritance carries law, scale and state

Physical inheritance transports more than a picture. It specifies the fields, dimensionless couplings, state and clocks that constitute a region. The normalized operator theorem supplies one exact notion of a shared law. The scalar/four-form construction supplies a physical charge-sector map, and its three-region energy floor distinguishes laws that one interface cannot separate. These are complementary steps toward the same account.

A single parameter table must therefore accompany the connected history: fundamental couplings, ancestral flux/state data, object properties, calibration quantities and predicted quantities. Local mass and spin can differ between objects. The fundamental source law is shared across the observations used to test the cosmology.

The quantum interaction gives a concrete additional requirement. At fixed G, geometric rescaling preserves its shape while its normalized strength varies as G/M^2. A physical inheritance map carries that dimensionless strength and the state. Absolute frequency units, occupations and local environment are not determined by relabeling the same operator.

The formation calculations likewise distinguish a frequency coupling from an arbitrary unit. The mass--reservoir relation and a light-travel-time relation can eliminate an overall ruler, but their preparation dependence still has to be constrained. A scale is calibrated once; the remaining measurements become joint predictions of that calibration and common law.

The foundation's scalar-link construction retains the Dirac benchmark E^2=|p|^2+Phi^2 in its invariant sector. Extending this to observed fields requires their representations, interactions and detector observables. The next physical inheritance result should determine descendant data from one stated ancestral preparation and prove that those data admit the next transition. That is the composition needed by the proposed hierarchy.

19. Geometric spectra, quantum observables and causal scales

The atlas uses fields and geometry before assigning particle or wave language. A localized scalar concentration is a dynamical field profile; a transmitted pulse is a propagated field configuration; a detector couples to a stated operator and state. Those definitions specify what the words refer to in each calculation.

The calculated quantum links import quantum theory openly. The Dirac analysis supplies physical propagation, stress vertices and a conditional late-state detector spectrum. The surface-probe analysis supplies positive mode frequencies and an action-derived current interaction whose exchange has the cosmohedron energy kernel. These are constructive connections between geometry and quantum observables.

A derivation of quantum behaviour beneath the imported theory would need amplitudes and measurement probabilities, spin and charge representations, statistics, interference and entangled correlations in one dynamical model. Discrete cavity modes alone address only a spectral part of that problem. The selected four-point contribution also needs the remaining constrained terms and physical state before it becomes a complete observer prediction.

The familiar combinations

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}$$
$$t_P = \frac{\ell_P}{c}$$

set dimensional scales, as tabulated by NIST CODATA. A minimum observable wavelength requires a probe and backreaction model. Hossenfelder's review distinguishes the different possible resolution bounds. A horizon, a cutoff, an initial clock origin and a thermal state perform different operations in the model.

The present atlas keeps the conventional local null cone. Its proper-clock conversions and memory delays are calculated without changing that causal law. A proposed change in regional constants would require operational dimensionless measurements, following the distinction developed by Ellis and Uzan. This gives a precise route from geometric intuition to a quantum or causal claim: define the state and detector, derive their response, and compare the dimensionless result.

20. The connected dark-sector mechanism and its remaining physical inputs

The dark-sector proposal is that interior gravitational observations can encode inherited fields and unresolved surrounding geometry. Its physical content is the effective stress and the measurements generated by that stress. The scalar/four-form branch supplies the clearest worked route: one potential contains a vacuum floor, while its excitation evolves, redistributes energy and forms local concentrations.

For that action the minimum density is rho_v=rho_b+epsilon*q^2/2. The protected parent evolves the excitation under the same potential. The later local calculations retain both the ambient field and the concentration, including their interference, mass current and pressure work. The optical calculations then use that actual evolving source. The local mass--vacuum--turnover relation and vacuum--delay products make expansion and structure share a physical scale.

These connections define a candidate mechanism. Their remaining physical inputs are concrete: ancestral flux and excitation, scalar frequency, seed spectrum and phase, formation epoch and population. The regular Euclidean preparation gives very little homogeneous child excitation; the more strongly excited Cauchy preparations are different states whose physical origin must be supplied. Deterministic transport carries a chosen state and does not select an arbitrary initial covariance.

The source history must next generate both the measured expansion and the observed clustering from one parameter account. That means calculating past-light-cone distances, growth and the projection of the surrounding compensation for the same population. An arbitrary halo-by-halo depletion law would erase the predictive relation that the atlas has constructed.

The quiet-cap supernova contrast already supplies a concrete test: that limiting history disagrees with the selected covariance-weighted distance statistic. The scalar cosmology has to be propagated to the same observable before agreement can be claimed. Early radiation and thermal evolution also have to be included before using a sound horizon inferred from early-universe observations. The density fractions are outputs of that completed history or parameters independently constrained within it; the number "95%" is not imposed as a geometric boundary condition.

21. Charged matter and the asymmetry question

Matter asymmetry is another physical inheritance question. A parent--child transition could prepare correlated charge sectors, but its transformation must act on actual charged fields and conserved currents. The current real scalar has no particle--antiparticle charge degree of freedom.

The atlas's literature library includes Poplawski's torsion-based asymmetry mechanism alongside CPT and black-mirror proposals. Each supplies a particular action or state assumption to examine. A radial orientation reversal alone is not the charge-conjugation operation of those field theories.

The constructive next result would specify a charged matter sector, its parent and child state map, the interactions permitting an asymmetry and the relevant current balance. It would then calculate a matter--antimatter observable and its connection to the same thermal history used for cosmology. A baryon-to-photon ratio requires both the baryonic charge and the photon/entropy history. This is the route by which the matter chapter can join the inherited vacuum and scalar-structure chapters.

22. Formation, continuation and successive generations

The atlas establishes finite compositions and several controlled continuations. The quiet two-wall geometry has a shared history; the scalar/four-form slice has a protected evolving parent; local scalar concentrations form and produce measurable signals. The static material branch separately has radial and axial linear stability. These results specify domains in which the proposed architecture actually works.

A continuing hierarchy also needs the transitions between those domains. A regular spatial centre is not a collapse formation history, and the quiet wormhole-side reference has a singular classical past and a finite throat lifetime. Its replacement requires compatible matter and conservation, using the existing regular-interior literature where its assumptions apply. The stable ordinary-interface branch provides another route; it is not identified with that wormhole-side history.

Proper times belong to named worldlines and matching conventions. The thermodynamic arrow requires the state, entropy and exchange of those regions. A clock rescaling alone selects neither an entropy direction nor a matter asymmetry.

Repeated spherical inheritance requires one transition's output to be admissible data for the next. A physical composition rule must carry energy, charges, state correlations and entropy. The unlimited-depth operator theorem proves its own inherited law; a spacetime realization needs this additional source and evolution statement.

For a proposed offspring population, established branching mathematics can then be used. Altenberg's finite-type treatment shows how a specified offspring law supplies a mean matrix and expected population evolution. The physics to derive is the transition law and its inherited parameters.

Penrose and Meissner's conformal continuation offers a separate geometric comparison. Conformal aeons, black-hole offspring and communicating mouths have different matching assumptions. The atlas can contain all three and connect them where an explicit map and state law warrant the connection.

23. The connected result and the route to a tested cosmology

The book's connected result is a route from specified geometry and inherited state to a calculable interior world. Normalized inheritance, sourced attachments, scalar supply, structure formation and observer relations all have explicit realizations. The physical gravitational boundary action and quantum vertex extend the same method to further channels. Their common feature is that geometry and state determine a response that can be calculated and challenged.

The central remaining work is concentrated in the following closures.

Physical closure	Concrete result needed	Connection it completes
One compatible source history	State the ancestral preparation and common couplings, transport them through the required interacting boundaries, and retain the descendant energy budget	Parent, child and inherited matter belong to one history
Cosmological evolution	Carry that history through the epochs used by the selected observations; calculate distances and growth on the observer's light cone	Vacuum and excitation become expansion and clustering predictions
Structure population	Derive or independently constrain seed statistics, formation ages and abundance; project the resulting compensation and lensing	The single-source relation becomes a local-universe correlation
Compact-object response	Prepare a causal wave/state, include the angular and rotation sectors needed by the observation, and calculate the received signal	The finite return contrast becomes an astrophysical test
Quantum and charged matter	Complete the chosen quantum response and detector statistics; specify the charged-sector and entropy evolution required for asymmetry	The quantum and matter chapters share a physical interpretation with the geometry

Physical closure	Concrete result needed	Connection it completes
Joint observation	Calibrate declared inputs once, freeze the remaining cross-observable predictions, and evaluate independent data with their covariance	One mathematical account faces multiple tests without hidden retuning

The causal domain required by an observation sets the evolution to calculate. A protected parent domain can be used directly where it applies; a full remote future is not required just to label an already determined local measurement. Conversely, a claim about transmitted matter or a new generation requires the actual relevant transition.

The strongest first observational route is the connection already built between the forming source, its surroundings and its interior observer. Motion constrains the force; light tests the metric; the compensation and inherited scale constrain the environment. Compare against a specified alternative with the same calibrated inner observables. The test is whether the outer relation remains distinguishable after source, age, kinematic and model uncertainties are propagated.

The theory is complete at its advertised scope when every claimed connection has compatible geometry, sources, state and evolution; its observable numbers have controlled errors; shared parameters survive independent observations; and its main calculations can be reproduced. The manuscript and matching source/evidence release then present the same scientific version. A later withheld observation outside the declared error range rejects the tested claim even after publication.

Three failure modes deserve direct attention. An incompatible source or boundary prevents the claimed common history. Unrestricted preparation can let different geometries reproduce the same measurements, defeating identification. Uncontrolled approximation error or disagreement between independent datasets can reject the proposed physical interpretation. Each is a mathematical or observational decision to resolve, rather than a reason to add an unrelated calculation.

Spherical inheritance remains the organizing hypothesis of this atlas. The completed constructions show how that hypothesis can be made calculable: define a region, derive its admissible connection, carry its state, account for its energy and ask what the observer measures. The remaining closures carry that same argument to a tested account of our local universe.

Appendix A. Characteristic fields, finite-amplitude response and memory

This appendix retains the detailed field-transfer and nonlinear-development calculations. They provide mathematical tools, boundary diagnostics and finite-domain results for the original conformal-scalar branch. Their source, gauge and error statements remain explicit. The main scalar/four-form history uses its own stated action and preparation; the appendix is not an additional mandatory calculation queue for that history.

A.1. A first field that reaches the interior

Derivation and calculation: field-transfer record and field-transfer owner.

On the fixed de Sitter/Schwarzschild wall, retain $H\ell=1$, $k\ell=2$, $M/\ell=0.4$ and a real classical massless scalar in its spherical mode, with conformal coupling $\xi=1/6$ and no additional scalar wall interaction. For $\chi=R\varphi$, the distributional scalar equation gives

$$
\begin{aligned}
&[\phi] = 0\\
&[n\phi] = 6\xi k\phi\\
&[\chi] = 0\\
&[n\chi] = (6\xi - 1)k\chi = 0.
\end{aligned}
$$

The conformal interface barrier cancels; the smooth exterior potential $f_+ 2M/R^3$ remains. A regular center has $\chi=F(\eta+r)-F(\eta-r)$. For incoming wall characteristic $I=(d/d\tau_W+n)\chi$, its exact conditional response is

$$\phi_C[T(\tau_W)] = \frac{I(\tau_W)}{T'(\tau_W)}$$
$$T = t_W - \ell \log(1 - R/\ell)$$
$$T' = \frac{\gamma(1 + v_{\mathrm{pec}})}{1 - R/\ell} > 0.$$

The prescribed wall-clock calibration has half-width 0.03ℓ, duration stretch 2.63230219 and normalized extrema $+23.84144800$ and -24.46533478. Its central field vanishes after the received support through $t_C=0$, conditional on zero incoming data afterward and the proved no-return interval. These are field amplitudes, not energy fractions. The calibration does not generate its exterior input.

A.2. Where the early signal can originate

Derivation and calculation: preparation record and preparation owner.

Independent data can instead be placed in retained white-hole sector IV. The ordinary right-parent observer remains causally excluded. Let $x=R/\ell$, $F(x)=(1-x/0.8)e^{x/0.8}=UV$, $V_a=V_W(0.45)$, $V_b=V_W(0.55)$ and $U_0=F(0.4)/V_a$. Define D by $V_a\le V\le V_b$ and $U_0\le U\le U_W(V)$; there $0.4\le R/\ell\le 0.55$. Prescribe zero bottom and cap data and $g(V)=\varepsilon\, B[4(V-V_m)/(V_b-V_a)]$ on $U=U_0$, where $B(z)=(1-z^2)^4$ for $|z|<1$ and zero otherwise. This is a compact C^3 input, independent of any desired central waveform.

With $L(u)=V_a$ below U_a and the inverse wall graph above it,

$$\chi = g - \int_{U_0}^{U} du \int_{L(u)}^{V} dv\, K\chi$$
$$K = \frac{16M^4}{R^4} e^{-R/(2M)}$$
$$\chi_U|_W = 0.$$

The integral operator has analytic uniform-norm bound

$$q_* = 16e^{-1/2}[\sqrt{F(0.4)} - \sqrt{F(0.55)}]^2 = 0.13882099234357653 < 1.$$

It therefore determines a unique finite transfer. The central arrival window is $-0.581758951595344\le T/\ell\le -0.250230700560700$; the no-return null-label gap is 2.634591912177392ℓ. Remote physical generation and data beyond that window remain unspecified.

A.3. A prepared pulse leaves a finite scattering tail

Derivation and calculation: prepared-response record and response owner.

For this unchanged input, let A denote the positive integral operator above. After the compact input ends, its Neumann series gives $-Ag\le\chi\le-(1-q_*)Ag<0$. Hence the determined wall derivative

$$J = g'(V) - \int_{U_0}^{U_W(V)} K\chi\, dU > 0$$
$$(1 - q_*)\mathcal{B}(V) \le J(V) \le \mathcal{B}(V), \quad \mathcal{B} = \int KAg\, dU$$

produces a positive central tail on $-0.340772594886934<T/\ell\le -0.250230700560700$. The sampled normalized extrema are 47.29833694 and -45.69595941; the final field is 0.00129049806. At the final

point, $J/\varepsilon=0.0008020390045$ lies in the independently evaluated bracket $[0.0006965878198,0.0008088769160]$. The 257-to-513 spatial and 129-to-257 geometry comparisons change the central field by at most 2.79286258×10^{-7} and 3.40622250×10^{-8}. The calculation uses 5,597,264 kernel values within its ten-million/60-CPU-second limits. These numerical controls support the finite waveform; the sign follows from the analytic argument. No permanent memory, continuum enclosure or later decay law is established.

A.4. What the signal determines when preparation is unknown

Derivation and calculation: inverse record and inverse owner.

In this fixed geometry, put $f=\chi/\varepsilon$, $h=f(U_W(V),V)$, $\theta=T/\ell$ and $Y=\ell\varphi_C/\varepsilon$. The zero cap fixes the integration constant:

$$h(V)=\tfrac{1}{2}\int_{\theta_a}^{\theta(V)}Y(s)ds$$
$$f=(I-\mathrm{B})^{-1}Eh$$
$$g(V)=f(U_0,V)$$

$$(\mathrm{B}f)(U,V)=\int_U^{U_W(V)}dp\int_{L(p)}^{V}dq\,K(p,q)f(p,q).$$

The operator-norm bound $q_*<1$ gives a bijection between initially quiet C^2 source traces and compatible C^1 signed central histories. For nonzero nonnegative compact sources, the post-input restrictions are $h<0$, $Y>0$ and $\int_{\theta_a}^{\theta(V)}Yds=2h<0$. An arbitrary fitted trace need not satisfy them. This analytic source inverse neither identifies geometry nor reconstructs a source numerically from sampled or intensity-only data.

A.5. The transmitted probe and the wall's stress

Derivation and calculation: stress record and wall-stress owner.

Fix canonical kinetic normalization, $\kappa_4=8\pi G$ and $f(\varphi)=1/(2\kappa_4)-\varphi^2/12>0$ in the weak-field interpretation. The full stress includes the nonminimal Einstein term once:

$$T^{\phi}_{\mu\nu}=\nabla_\mu\phi\nabla_\nu\phi-\tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2+\xi[G_{\mu\nu}\phi^2+(g_{\mu\nu}\Box-\nabla_\mu\nabla_\nu)\phi^2].$$

For the original sufficiently regular traces, $d=[n\varphi]=k\varphi$ and $C_{ab}=[K_{ab}]-\gamma_{ab}[K]=2k\gamma_{ab}$ give $S^{\varphi}{}_{ab}=\xi(2\varphi d\gamma_{ab}-\varphi^2C_{ab})=0$ and $[T^{\varphi}{}_{na}]=(1-6\xi)k\varphi D_a\varphi=0$. Thus no additional scalar surface source or tangential flux jump is required at quadratic order. The individual fluxes and smooth bulk gravitational source need not vanish.

A.6. The prepared pulse requires gravitational initial data

Derivation and calculation: initialization record and initial-data owner.

On the source leg, keep the reference lapse, radius 0.4ℓ, both corner expansions and affine normalization. Use the affine background radius $x=R_0(U_0,V)/\ell$, $V(x)=F(x)/U_0$ and $b(x)=B(4[V(x)-V_m]/(V_b-V_a))$; primes here and in section A.7 mean d/dx. With $p=b(x)/x$, $\varphi=(\varepsilon/\ell)p+O(\varepsilon^3)$ and $\alpha=\kappa_4\varepsilon^2/\ell^2$, write $R/\ell=x+\alpha\,h+O(\alpha^2)$. The intrinsic constraint forces

$$h'' = -\tfrac{x}{3}p'^2 + \tfrac{x}{6}pp''$$
$$h(0.4) = h'(0.4) = 0.$$

After the compact trace, $h'<0$ and $h<0$ by the integrated expressions; h' becomes constant. At $x_b=0.4815042082576364$, $h=-6.12283354819419$ and $h'=-162.18055362491776$. Direct integration, integration by parts and the transformed affine Einstein-frame equation agree. These are initial radius coefficients, not a mass shift or an evolved wall; no finite amplitude or perturbative remainder bound is selected.

A.7. An invariant mass response on the same initial surface

Derivation and calculation: source-mass record and mass owner.

With $m=M/\ell=0.4$, the fixed transverse corner datum and scalar equation supply $d=(a/x-1)/4$, $a=0.8$, and $s'=-(s+dp')/x$, $s(0.4)=0$. For the geometric mass length, use

$$\mathfrak{m} = \tfrac{R}{2}[1 - g^{ab}\partial_a R \partial_b R]$$
$$\frac{\mathfrak{m}_J}{\ell} = m + \alpha\mu_J + O(\alpha^2).$$

The borrowed flux and full conformal frame conversion yield

$$\mu_E = -\tfrac{1}{4}\int_{0.4}^{x} t(a-t)p'(t)^2 dt$$
$$\mu_J = \mu_E + \frac{mp^2}{12} + \frac{x^2 p}{3}(s + dp').$$

After the input, the frame masses agree at the constant negative coefficient $\mu_J=\mu_E=-28.88319705406449$. Their largest sampled difference during input is 4.265459834842238. Independent mixed-radius and improved stress laws agree. This is a finite source-leg response relative to the reference mass, not a negative total mass, vacuum future or asymptotic energy.

A.8. The scalar adds no normal-force jump

Derivation and calculation: normal-force record and normal-force owner.

The Gauss jump $[G_{nn}]=-2k\bar{K}$ and complete scalar stress give

$$[T^{\phi}_{nn}] = 2\xi(6\xi - 1)k\bar{K}\phi^2 = 0$$
$$S^{ab}\bar{K}_{ab} = [T_{nn}].$$

The cap contribution $G^{-}_{nn}=-3H^2$ is essential. Fixed vacuum and material tension therefore require $\delta\bar{K}=0$ at order ε^2. This mean-curvature variation includes the bulk metric, induced metric and wall displacement. It does not require zero displacement or an unchanged trajectory.

A.9. Exterior mass closes from the known wall field

Derivation and calculation: analytic closure record and wall-mass owner.

On the smooth exterior wall limit in the same $0.45\le R/\ell\le 0.55$ window, put $s=\tau/\ell$, $x=R/\ell$, $v=dx/ds$, $\beta=\beta_+$, $w=\chi_W/\varepsilon$, $j=dw/ds$ and $p=w/x$. The two known derivatives are $A=(j-vp)/x$ and $B=(j-\beta p)/x$. Both null mass fluxes and the complete conformal endpoint term give

$$L = \frac{x^2}{2}\left[\beta AB - \frac{v}{2}(A^2 + B^2)\right]$$
$$Q = \frac{mp^2}{12} + \frac{x^2 p}{6}(vA - \beta B)$$
$$\mu_J(s) = \int_{s_a}^{s} L(s')ds' + Q(s), \quad Q(s_a) = 0.$$

The original-frame improved stress independently gives $d\mu_J/ds=L+dQ/ds$. Neither expression divides by β. This analytic conditional functional is evaluated with the coupled quadratic geometry in Sections A.11--A.13. The wall tail alone supplies no late mass sign or plateau. Physical preparation selection, the next scalar amplitude, stability and a finite-amplitude validity bound remain open. Detailed scalar-chain derivations belong to their owners; historical supporting calculations are cited at Zenodo.

A.10. Cap mass and a regular wall displacement functional

Derivation and calculation: cap-wall record, model specification snapshot and cap-wall owner. The common model interface separates the fixed action, prescribed preparation and unit scale from derived responses. Its parameters $h=H\ell=1$, $b=k\ell=2$ and $m=M/\ell=2/5$ remain declared inputs. We retain the same canonical real massless scalar, conformal coupling, scalar-independent vacuum terms and constant material tension in the Jordan frame.

Let $r=x+\alpha\rho+O(\alpha^2)$ be the actual wall radius in units of ℓ, with $\alpha=\kappa_4\varepsilon^2/\ell^2$ and proper wall time $s=\tau/\ell$. The two geometric masses have expansions

$$\frac{\mathfrak{m}_+}{\ell} = m + \alpha\mu_+ + O(\alpha^2)$$
$$\frac{\mathfrak{m}_-}{\ell} = \frac{h^2 r^3}{2} + \alpha\nu_- + O(\alpha^2).$$

Here μ_+ is the exterior coefficient μ_J of Section A.9. The cap vacuum is evaluated at the actual radius: its total quadratic mass coefficient on the reference wall is $\nu_-+3h^2x^2\rho/2$. Holding that vacuum mass constant would omit a term required by the angular junction. The shared quiet corner fixes $\mu_+(s_a)=\nu_-(s_a)=\rho(s_a)=0$.

Using the signed normals $\beta_-=\beta_++bx$, define $B_-=(j-\beta_- p)/x$ with the same $p=w/x$, $j=w_s$ and $A=p_s$ as above. The cap's vacuum-subtracted mass closes as

$$L_- = \frac{x^2}{2}\left[\beta_- AB_- - \frac{v}{2}(A^2 + B_-^2)\right] + \frac{h^2x^2vp^2}{2}$$

$$Q_- = \frac{x^2 p}{6}(vA - \beta_- B_-) - \frac{h^2x^3p^2}{12}$$
$$\nu_-(s) = \int_{s_a}^{s} L_-(s')\,ds' + Q_-(s).$$

Direct differentiation reproduces the Jordan-frame improved-stress law $\nu_{-,s}=x^2(\beta_- q_- - vP_-)/2$. Here $q_-=t_{uN,-}$ and $P_-=t_{NN,-}$ are the flux and normal pressure of $t_{ab}=\ell^4 T^{\varphi}_{ab}/\varepsilon^2$. The extra vacuum terms in L_- and Q_- are essential; the exterior's zero-vacuum shortcut cannot be carried into the cap. The evaluation uses first scalar derivatives, while covariant Hessians enter the independent transport check. No derivative of sampled waveform data is used.

For this conformal scalar, $f+3f'^2=1/(2\kappa_4)$ reduces the scalar-tensor pure-tension junction to the same signed angular law. With $D=\mu_+-\nu_-$, its linearization gives

$$\delta\beta_{\pm} = \beta_{\pm,x}\rho + \frac{D}{bx^2}$$
$$\rho_s = v_x\rho + \frac{\mu_+/x + \beta_+ D/(bx^2)}{v}.$$

The cap expression replaces the numerator by $v_-/x+\beta_- D/(bx^2)$ and agrees exactly. Because $v=x_s>0$ on $x\in[0.45,0.55]$, the equation stays regular through $\beta_+=0$. Its homogeneous solution is $C\ v(s)$, a relative proper-time origin shift removed by the shared quiet event.

The existing scalar junction identities give equal one-sided fluxes and pressures, $q_+=q_-=q$ and $P_+=P_-=P$. Subtracting the two mass laws yields $D_s=-bx^3q/2$. Together with the displacement equation, this gives equal corrections to the dimensionless normal accelerations $a_\pm$ and zero change in the dimensionless mean extrinsic-curvature trace:

$$\delta a_+ = \delta a_- = \frac{6m\rho}{bx^4} - \frac{2D}{bx^3}$$
$$\delta\overline{K} = 0.$$

Thus the temporal jump and normal-force condition close with the angular condition and mass transport, without setting $\rho=0$ or adding a freely chosen cap mass or wall force. These are formal quadratic identities, checked by independent algebra, not an evaluated displacement curve or a completed sourced spacetime.

The next connection is the smooth bulk metric on both sides; the exterior characteristic calculation below supplies its first evolved domain. Geometric mass alone does not determine the null lapse, the wall's null-coordinate embedding or the central observer's clock. Their reconstruction must use the shared wall data and characteristic initial data before calculating the corrected arrival map and subsequent scalar response. A nonzero allowed amplitude, a controlled remainder, formation and stability remain separate questions.

A.11. The prepared field drives a smooth exterior metric response

Derivation and calculation: metric refinement, field refinement, derivative-quadrature comparison and bulk-response owner, with separate geometry, source quadrature and time tolerance comparisons.

The gravitational data of Sections A.6--A.7 determine a computed smooth exterior response at quadratic order on $[U_0,U_a]\times[V_a,V_b]$, whose two past null sides are specified. This rectangle lies inside the retained wedge. Its final transverse leg $U=U_a$ supplies data for a later extension toward the moving wall; no exterior metric is continued through the excised cap.

Keep the original null coordinates and fixed Jordan source lapse. With $r_0=R_0/\ell$, write $r_E=r_0+\alpha\ H$ and $log\ C_E=log\ C_0+\alpha\ c$ to the retained order. The leading scalar is $\varphi=(\varepsilon/\ell)p$, with both first derivatives obtained from its characteristic integral equations. The existing smooth exterior Einstein equations simplify under $y=r_0H$, $z=c+H/r_0$ to

$$y_{UV} = \frac{C_0}{4r_0^2}y - \frac{C_0}{4}z$$
$$z_{UV} = -\frac{3C_0}{4r_0^4}y + \frac{C_0}{4r_0^2}z - p_Up_V.$$

Integrating their right sides in U gives a linear evolution in V. The quiet bottom has $y=z=0$. At $U=U_0$, let $x=r_0(U_0,V)$ and let h_{init} denote the original Jordan radius coefficient from Section A.6. The source values in the unchanged coordinate gauge are

$$y_0 = x h_{\mathrm{init}} - \frac{x^2 p^2}{12}$$
$$z_0 = \frac{h_{\mathrm{init}}}{x} - \frac{p^2}{4}.$$

The frame and affine-parameter corrections must both be retained; the Einstein-affine radius coefficient alone is not the source value at fixed original coordinates. Returning to the Jordan metric gives $h_J=H+r_0p^2/12$ and $c_J=c+p^2/6$, where $r_J=r_0+\alpha\ h_J$ and $log\ C_J=log\ C_0+\alpha\ c_J$.

At (U_a,V_b), the 65-node metric calculation with 257 field nodes reports the following dimensionless quadratic coefficients:

Quantity	Recorded coefficient
Radius, h_J	*-5.7131513827*
Logarithmic null lapse, c_J	*-1.5461215575*
Geometric mass, μ_J	*-28.8833375991*

Changing the metric grid from 33 to 65 nodes reduces the largest sampled mass-transport residual from 1.65×10^{-6} to 4.31×10^{-7}. On the common sampled grid, the changes in h_J,c_J,μ_J are respectively at most 1.15×10^{-7}, 3.82×10^{-6} and 1.23×10^{-6}. A separate 129/257 field-grid comparison changes these coefficients by at most 2.14×10^{-8}, 9.55×10^{-8} and 5.78×10^{-7}; the 32/64 derivative-quadrature comparison changes them by less than 7.2×10^{-15}. Both null constraints are checked in integral form, and geometric mass is compared with transport along both null directions. The radial lapse reconstruction checks the same U constraint in different variables.

Changing geometry interpolation and source quadrature changes these three coefficients by less than 7.2×10^{-15} and 1.5×10^{-13} respectively; tightening the time tolerance tenfold changes them by less than 1.21×10^{-11}. Each complete two-solve comparison took between 17.16 and 21.91 process-CPU seconds within its 120-second budget.

These are finite numerical comparisons, not rigorous error enclosures or a selected nonzero physical amplitude. They establish a sourced exterior metric response on the rectangle and provide its transverse radius/lapse data. Sections A.12--A.13 continue to the moving wall and cap using the same source and material junction. The full finite-amplitude sourced configuration, corrected observer signal and finite-amplitude remainder have not yet been established.

A.12. The same field displaces the wall and changes its exterior clock

Derivation and calculation: field refinement, radial quadrature, clock integration and the wall-evolution owner.

The leading field supplies a computed response on the moving boundary, with the action, source and quiet corner of Sections A.6--A.11 unchanged. Extending its first-derivative evaluator to the retained wedge requires the correct lower endpoint: a column with $U>U_a$ begins where it meets the wall. At that event $\chi_U=0$, while $p_U=-p\ r_{0,U}/x$ is generally nonzero. Both derivatives are obtained from the integral equations of the original field solution.

Define the areal-null lapse $Z=C/(2r_U)$. The Einstein-frame mass and logarithmic lapse coefficients follow radial constraints at fixed V:

$$\mu_E = \mu_{E,0} - \int_{U_0}^{U} \frac{r_0^2 r_{0,V}}{C_0} p_U^2\, dq$$
$$\zeta_E = \zeta_{E,0} + \int_{U_0}^{U} \frac{r_0}{2r_{0,U}} p_U^2\, dq.$$

Their source constants are already fixed by the characteristic data. At the wall, frame conversion gives $\mu_+^{radial}=\mu_{E,W}+Q_+$ and $\zeta_{J,W}=\zeta_{E,W}+p^2/4$. The independent timelike mass functionals of Section A.10 determine μ_+ and ν_-, and hence the quiet-anchored displacement ρ. The radial and timelike mass values are compared, rather than equated by the numerical implementation.

The exterior embedding also follows from these data. Write $V_W=V_0(1+\alpha\eta)$, with s the original Jordan wall proper time, $d=v-\beta_+>0$, and $Z_0=-2a/V_0$, $a=2m$. Expanding $V_{W,s}=1/[Z_J(r_{W,s}-\beta_+)]$ gives

$$\eta_s = \frac{1}{2ad}\left[\zeta_{J,W} + \frac{\rho_s - \delta\beta_+}{d}\right]$$
$$\eta(s_a) = 0.$$

The background lapse is evaluated at the moved V_W in this expansion. The equation stays regular through the zero of the signed β_+.

At the final reference radius $x=0.55$, the 257-node field calculation gives:

Dimensionless quadratic coefficient	Recorded value
Exterior mass, μ_+	*-28.8834223256*
Cap excess mass, ν_-	*17.7590784205*
Wall displacement, ρ	*-5.2868390417*
Exterior clock ratio, $\eta=V_1/V_0$	*-0.8178125677*

For example, the radius expansion is $r_W=0.55+\alpha\rho+O(\alpha^2)$; the tabulated number is its coefficient, with no finite α selected. The cap excess subtracts its vacuum mass at the actual wall radius.

Changing the leading field from 129 to 257 nodes reduces the largest sampled mass-path discrepancy from 6.49×10^{-7} to 1.63×10^{-7}. The maximum changes at nine common time cuts in μ_+,ν_-,ρ,η are respectively 3.38×10^{-7}, 2.09×10^{-7}, 3.64×10^{-8} and 3.57×10^{-9}. Doubling the clock panels from 8 to 16 changes η by at most 2.41×10^{-8}. The radial 2/4-point comparison changes it by less than 1.35×10^{-13}. The angular jump, two forms of the displacement equation and clock normalization are checked on the same trajectories.

The 32/64 field-derivative comparison leaves the eight reported comparison quantities identical in this run. Separate geometry, source and ODE tolerance comparisons change them by less than 5.64×10^{-11}, 2.85×10^{-14} and 8.25×10^{-13} respectively. All seven comparisons completed within their individual 120 process-CPU-second budgets. An independent rectangle overlap finds mass and lapse differences of at most 4.08×10^{-7} and 1.30×10^{-6} against the mixed metric solver. Different radial quadratures expose error that a shared-rule constraint residual can cancel; that residual alone is not a total metric error estimate.

These computations connect the sourced exterior and the two-sided mass functionals to an actual wall displacement and exterior null embedding at quadratic order. Their numerical differences are refinement diagnostics. Section A.13 supplies the cap metric and central arrival correction. The next scalar waveform and finite-amplitude remainder remain to be calculated.

A.13. The cap geometry changes the central arrival clock

Derivation and calculation: field refinement, radial integration, further radial refinement, ray integration, further ray refinement, wall-time integration, geometry, derivative quadrature, centre quadrature, centre threshold and the cap derivation.

The same prepared wall trace fixes the leading regular field in the cap. With $A_0=1-r^2$, $r_*=\operatorname{atanh} r$ and $f(q(V))=\chi_W(V)/\varepsilon$, where q is the existing central arrival map,

$$p(T,r)=\frac{f(T+r_*)-f(T-r_*)}{r}$$
$$p(T,0)=2f'(T).$$

The reflected term remains in the interior. Both profile derivatives and stable centre evaluations come from the existing field's continuum integral equations. The fixed cap vacuum transforms to a quadratic potential $H^2\varphi^2$ in the Einstein frame; it is retained in the metric.

For the polar-areal metric $ds_E^2/\ell^2=-e^{2\alpha\delta_E}A\,dT^2+A^{-1}dr^2+r^2d\Omega^2$, write $A=A_0-2\alpha\nu_E/r$. The radial constraints give

$$\nu_{E,r}=\frac{r^2}{4}\left(\frac{p_T^2}{A_0}+A_0p_r^2\right)+\frac{r^2p^2}{2}$$
$$\delta_{E,r}=\frac{r}{2}\left(p_r^2+\frac{p_T^2}{A_0^2}\right).$$

Regularity fixes $\nu_E(T,0)=0$. Taking T to be Jordan central proper time fixes $\delta_E(T,0)=-p(T,0)^2/12$. These constants require no new source history. At the wall the Jordan excess is $\nu_J=\nu_E-r^2A_0pp_r/6-r^3p^2/12$; its endpoint term agrees with Q_-, while the independent radial ν_E agrees with $\int L_-ds$. The scalar equation gives the corresponding mass-conservation identity.

Let the cap wall time be $T_W=T_0+\alpha\,T_1$ and its Einstein radius be $x+\alpha\,e$, where $e=\rho-xp_W^2/12$. The material normalization fixes T_1 from the same mass, lapse and wall displacement. The inward null equation then gives the central arrival coefficient at fixed wall proper time:

$$\Delta_C=T_1+\frac{e}{A_0(x)}+\int_0^x\left[\frac{2\nu_E}{rA_0^2}-\frac{\delta_E}{A_0}\right]_{T=q_0-r_*(r)}dr$$
$$q_0=T_0+r_*(x).$$

The integral is regular at the centre. At a fixed exterior null label, the moved emission event adds the correction

$$\Delta_C|_V=\Delta_C-q_{0,s}\frac{V_1}{V_{0,s}}$$
$$T_C(V)=q_0(V)+\alpha\Delta_C|_V+O(\alpha^2).$$

At the final reference wall event, the primary 257-node calculation with 16-point radial/ray rules and 16 cap clock panels gives:

Dimensionless quadratic coefficient	Recorded value
Cap Einstein excess mass, ν_E	*17.7590647726*
Cap lapse, δ_E	*103.4370285720*
Wall cap-time shift, T_1	*-8.4727761614*
Central arrival at fixed wall proper time, Δ_C	*-128.0670033577*
Central arrival at fixed exterior label	*-130.3664151052*

The maximum radial/timelike cap-mass discrepancy is 2.52×10^{-9} over the recorded cuts and cap clock quadrature nodes. Changing the field from 129 to 257 nodes changes the fixed-label arrival coefficient

by at most 6.42×10^{-7}. The clock convention matters: the final exterior-label shift alone contributes 2.2994117476 to the difference between the two arrival coefficients.

The 8/16 ray refinement changes the arrival coefficient by at most 2.31×10^{-5}; the further 16/32 ray comparison with 129 field nodes changes it by 1.84×10^{-9}. At that fixed field resolution, 129/257 geometry knots change it by 1.06×10^{-8}, and moving the stable-centre threshold from 10^{-4} to 10^{-3} changes it by 9.53×10^{-10}. Doubling the centre quadrature changes it by 7.11×10^{-15}; doubling the differentiated-integral quadrature leaves the saved coefficients equal at floating-point precision. The coarser 8/16 radial comparison changes the arrival coefficient by 6.30×10^{-3}; a separate 16/32 radial comparison at 129 field nodes reduces that sensitivity to 1.97×10^{-9}. The 8/16 cap time-panel comparison changes it by 1.92×10^{-5}, exceeding the field and finer radial/ray sensitivities. Its two time integrations reuse identical reference rays, with fresh and reused work recorded separately. These differences do not form a rigorous total error bound.

The metric ray integrand has both mass and lapse terms. With $D=\delta_E-2\nu_E/(rA_0)$, the radial constraints independently give $D_r=2\nu_E(1-3r^2)/(r^2A_0^2)-rp^2/A_0$ and the integrand is $-D/A_0$. Positive cap mass does not fix its sign. The records preserve signed quadrature contributions and sampled extrema; at the final primary cut the ray integral is -112.0145129711.

The comparisons retain explicit fixed settings. Geometry, cap derivative, centre and further-radial/ray controls use 129 field nodes to fit their complete pairs within the declared budget; the main field comparison uses both 129 and 257. Refinement differences are numerical sensitivity estimates. No finite α is selected, and this metric null-arrival correction does not include the order-$\varepsilon\alpha$ scalar amplitude. Its perturbed wave equation and source continuation are specified in the next section. Propagating that correction and establishing finite-amplitude control remain the next physical connection.

A.14. Geometry supplies an outgoing scalar boundary correction

Derivation and calculation: field refinement, radial integration, ray integration, further radial refinement, further ray refinement and the scalar-metric derivation.

Keep the same prescribed reduced-field input exactly on the original anchored source and write $\chi/\varepsilon=c_0+\alpha\, c_1+O(\alpha^2)$. Conformal covariance preserves χ between the two frames. In cap null coordinates normalized to the central clock, $c_0=f(q_+)-f(q_-)$ and the computed metric gives

$$\partial_-\partial_+\widetilde{c}_1=-K_1c_0$$
$$K_1=\frac{A_0}{4}\left[\frac{2\nu_E}{r^3}-\frac{p^2}{3}-\frac{p_T^2}{6A_0}+\frac{A_0p_r^2}{6}\right].$$

The retained wall interval has $q_{-,W}<q_a$. Its outgoing homogeneous correction is quiet, so the returning derivative is determined before the incoming correction is solved:

$$I_-=-\int_{q_a}^{q_{+,W}}K_1(q_{-,W},q)f(q)\,dq.$$

Combining the reduced-field transmission laws with the moved wall event then fixes the exterior areal derivative,

$$c_{1,r,W}=\frac{2q_{-,0,s}}{d}I_--\left(e-\frac{vV_1}{V_{0,s}}\right)\frac{2mw}{x^3d^2}$$
$$d=v-\beta_+$$
$$e=\rho-\frac{xp_W^2}{12}.$$

Here $w=c_{0,W}$ and $q_{-,0,s}=(\beta_{-}-v)/(1-x^2)$. The two terms are cap backscatter and the displacement pullback. At $x=0.55$, the primary 257-node field calculation gives $J_{-}=-1.8223975002$, a backscatter term -2.9736230746 and a displacement term -0.5062222284, hence $c_{1,r,W}=-3.4798453030$. The outgoing correction is therefore nonzero for this prepared pulse; the leading zero-outgoing boundary condition cannot be reused at this order.

Over nine common wall cuts, changing the field from 129 to 257 nodes changes $c_{1,r,W}$ by at most 1.73×10^{-6}. Radial order 8/16 changes it by 3.99×10^{-7} and ray order 8/16 by 3.26×10^{-3}. Separate 16/32 controls at 129 field nodes reduce these changes below 1.17×10^{-10} and 1.31×10^{-10} respectively. In the primary calculation, independent integrations in incoming label and radius agree in J_{-} within 5.59×10^{-11}. These are numerical sensitivity checks, not a rigorous total error enclosure.

The boundary samples provide a determined input for the exterior forced equation. The following calculation resolves its interpolation and propagates the exterior correction. No new physical preparation mechanism or nonzero finite-amplitude validity range is asserted.

A.15. The exterior scalar correction propagates on the computed geometry

Derivation and calculation: mass-moment comparison, clock-resolved boundary, forcing cache, scalar evolution and the conservative equation.

Pull the exterior areal coefficient back to the reference null coordinates: $y(U,V)=c_1(r_0(U,V),V)$. The same metric and first scalar jets determine the known flux F and density G in $y_{UV}+K_0y=F_U-G$. The original prescribed source has the determined areal-coordinate pullback $g_1=-H_{E,src}c_{0,r}|_{src}$. It is retained after the compact leading input; setting it to zero would change the preparation continuation.

Let $B_U=b_{out}$, where $b_{out}=r_{0,U}c_{1,r}|_W$ is the outgoing boundary derived above, and choose $B=0$ below the initial wall coordinate. Then $z=y-g_1-B$ has zero source and bottom data and $z_U|_W=0$. Its conservative evolution is

$$z_V = F(U,V) - F(U_0,V) - \int_{U_0}^{U} [K_0(z+g_1+B)+G]\,dU.$$

The fixed-U discretization clips inactive nodes to the moving wall. The terminal z state has no extra advection term, while the reconstructed wall derivative includes the lift transport:

$$\frac{dy_W}{dV} = g_1'(V) + z_V|_W + U_W'(V) b_{\mathrm{out}}(U_W(V)).$$

A cumulative pre-arrival cap-mass moment agrees with independent radial integration within 2.50×10^{-9} at the retained samples and makes the boundary evaluation economical. Its local Hermite lift initially exposes a clock interpolation error. Refining the exterior clock from 16 to 32 panels reduces the withheld boundary discrepancy from 1.73×10^{-6} to 5.75×10^{-8}, below the unchanged target. The lift uses 41 training knots and 40 withheld queries. The exterior forcing cache independently compares 17/33 time knots against 32 withheld times, including radial midpoint queries.

Four evolutions then hold the physical source, 129-node leading field and 257-knot reference geometry fixed:

Numerical comparison	Maximum y_W change	Maximum dy_W/dV change
17/33 forcing times, 129 correction nodes	1.755×10^{-3}	1.326×10^{-4}
65/129 correction nodes, 33 forcing times	6.984×10^{-6}	9.317×10^{-5}
Relative tolerance $2\times10^{-6}/2\times10^{-7}$, fine cache/grid	3.728×10^{-9}	3.956×10^{-8}

The finer sampled peak is 0.2679995064; the terminal coefficient is 0.2194607242. The cache-time change is 0.655% of that peak, the grid change 0.00261%, and the ODE change much smaller.

Source-value, source-rate and wall-flux defects vanish. Manufactured moving-wall solutions independently check the integration signs, source subtraction and material derivative. The four evolutions take 5.9375 process-CPU seconds and 950,926 scalar kernels using the saved inputs.

This is a formal exterior cubic scalar response for the declared preparation. Sampled refinement differences are not uniform bounds. The central readout additionally requires reflected cap forcing and the observer-clock pullback; no finite amplitude or remainder is fixed here.

A.16. Reflected cap transport gives the central scalar correction

Derivation and calculation: central derivation, radial comparison and incoming-ray comparison.

Write the dimensionless Jordan central proper time as $T=\tau_C/\ell$ and put $u=T-\operatorname{atanh} r$, $q=T+\operatorname{atanh} r$. The leading reduced field is $c_0=f(q)-f(u)$. Unlike the outgoing wall ray, an incoming ray to the centre can contain both profile terms. Its metric and forcing therefore use the full reflected field.

Write c_N for the central-clock null correction. Let $a(q)$ be the outgoing label of the wall event with incoming label q, and write $J(q)=c_{N,u}|_W$. Its wall value is

$$H(q) = y_W(V(q)) - C(q)f'(q)$$
$$C = \Delta C_{\text{fixed } V}.$$

The forced cap equation is $c_{N,uq}=-K_1c_0$. Differentiating the wall value gives $c_{N,q}|_W=H'-a'J$; integrating inward and imposing regular reflection at $u=q$ then yields

$$p_{1,C}(T) = 2H'(T) - 2a'(T)J(T) - I_C(T)$$
$$I_C(T) = \int_0^{x(T)} W_1(T-\operatorname{atanh} r, r)[f(T) - f(T-2\operatorname{atanh} r)]\,dr$$

$$W_1 = \frac{2\nu_E}{r^3} - \frac{p^2}{3} - \frac{p_T^2}{6(1-r^2)} + \frac{(1-r^2)p_r^2}{6}.$$

Here $x(T)$ is the radius of the wall event whose incoming label is T. The factor two follows from regular reflection. The wall derivative uses $y_{W,s}=V_{0,s}\,dy_W/dV$ and the differentiated optical constraint for C_s. These include the arrival-rate and arrival-shift contributions at fixed central time; adding another shift would count that pullback twice. The cap's cubic frame term cancels its central lapse pullback. This last cancellation is an algebraic check, not a second evolved solution.

The formal observable is

$$\frac{\ell\phi_C(T)}{\varepsilon} = p_0(T) + \alpha p_{1,C}(T) + O(\alpha^2)$$
$$p_0 = 2f'(T).$$

Five cuts lie strictly inside the retained arrival interval. With incoming radial and ray orders both 32, they give:

Central time T	Leading p_0	Cubic coefficient $p_{1,C}$
-0.54363418	0	3.49×10^{-137}
-0.46546305	42.21372894	3395.219
-0.42507962	-0.265835912	-34413.869
-0.38358521	-40.14663419	4369.585

Central time T	Leading p_0	Cubic coefficient $p_{1,C}$
-0.29640937	*0.00127167605*	*1454.355*

The 16/32 radial comparison changes the coefficient by at most 1.098×10^{-5}; the independent 16/32 incoming-ray comparison changes it by 5.285×10^{-5}. Passing the existing exterior cache, grid and ODE comparisons through the same cap algebra changes it by 1.155×10^{-4}, 1.667×10^{-4} and 5.958×10^{-8} respectively. These are separate sampled sensitivities, not a combined uncertainty enclosure. The leading field uses 129 nodes and the reference geometry 257 knots. Compatible saved cap and wall data are reused; no new wall, cap-clock or exterior evolution is performed. The radial pair takes 27.9375 process-CPU seconds and the ray successor 50.046875 seconds, each within its own 120-second and 30-million-kernel budget.

The quiet sample remains below 3.49×10^{-137}. Positive-radius centre samples retain the potential separately from its analytic $K_1=O(r^2)$ limit. The radial-32/ray-16 travel integral reproduces the previous cap calculation within 1.43×10^{-14}; changing to ray order 32 moves it by 5.21×10^{-10}. Manufactured controls check the optical derivative, integration sign, reflection factor and clock conversion independently of this physical run.

The coefficient is large for this preparation, particularly near the leading field's zero crossing. Its size neither selects a finite amplitude nor bounds the omitted term. The exact nonlinear equations now specify the next connection. The same action fixes the canonical Einstein potential and wall tension:

$$\widehat{V}_{\pm} = 3h_{\pm}^{2}\cosh^{4}(\Psi/\sqrt{6})$$
$$\sigma_E = \sigma\cosh^{3}(\Psi/\sqrt{6}).$$

An exact source-leg constraint admits a nonzero small-amplitude interval with positive radius and conformal factor. Extending that control through the bulk, moving interface and regular centre, with a quantitative remainder for this observable, remains open. The following calculation tests the coupled equations on the original window without assuming that estimate.

A.17. A nonlinear coupled window and its unresolved accuracy

Derivation and calculation: six-case evolution, source and runtime manifest, preceding transport comparison and figure record. The finite-amplitude owner specifies the equations, gauge, source and discrete representation.

The original compact preparation now drives a simultaneous cap, moving wall and exterior calculation at $\alpha=0.00012983127142551443$ and at its normalized zero-amplitude limit. The source label spans $0.4\le x\le0.4815042083$, corresponding to reference wall radii $0.45\le R\le0.55$. This window ends before the signal reaches the centre. The action, source, wall tension and reference gauge are held fixed; the amplitude is selected by the source-only bound, not an observed output. Each amplitude uses 16 cells with 256 steps, 32 cells with 256 steps and 32 cells with 512 steps, retaining 129 common history cuts.

The cap carries a reflected characteristic background

$$\chi_{\mathrm{can},0}(T,r) = F_{\mathrm{bg}}(T+\operatorname{atanh} r) - F_{\mathrm{bg}}(T-\operatorname{atanh} r)$$

and evolves corrections to its conservative cell integrals. Here $\chi_{can}=r\Psi/\sqrt{\alpha}$ is the normalized canonical reduced field; the transmitted invariant field c differs from it at finite amplitude. One cap primitive determines the common wall field. The full incoming correction remains evolved, and total fields, including all cross terms, source the cap metric. The C^2 quintic profile uses 65 alternating cuts of the previous zero-case wall history and its first jets. The 64 withheld cuts give maximum value and first-jet discrepancies 1.808×10^{-5} and 0.01439. Interpolation curvature is a computational choice, not an independent physical second derivative.

An independent characteristic pullback from each new zero-case wall trace tests cap propagation. Its 32-cell primitive discrepancy falls from *0.380945* to *0.000890490* with linear interpolation, or from *0.381936* to *0.000836671* with PCHIP. The remaining discrepancy is comparable to the *0.00133* interpolation sensitivity. Zero-case outgoing *p-q* remains exactly zero, and its spurious pre-arrival centre field is below 3.9×10^{-15}. These checks distinguish accurate incoming transport from an identity imposed by the cap primitive.

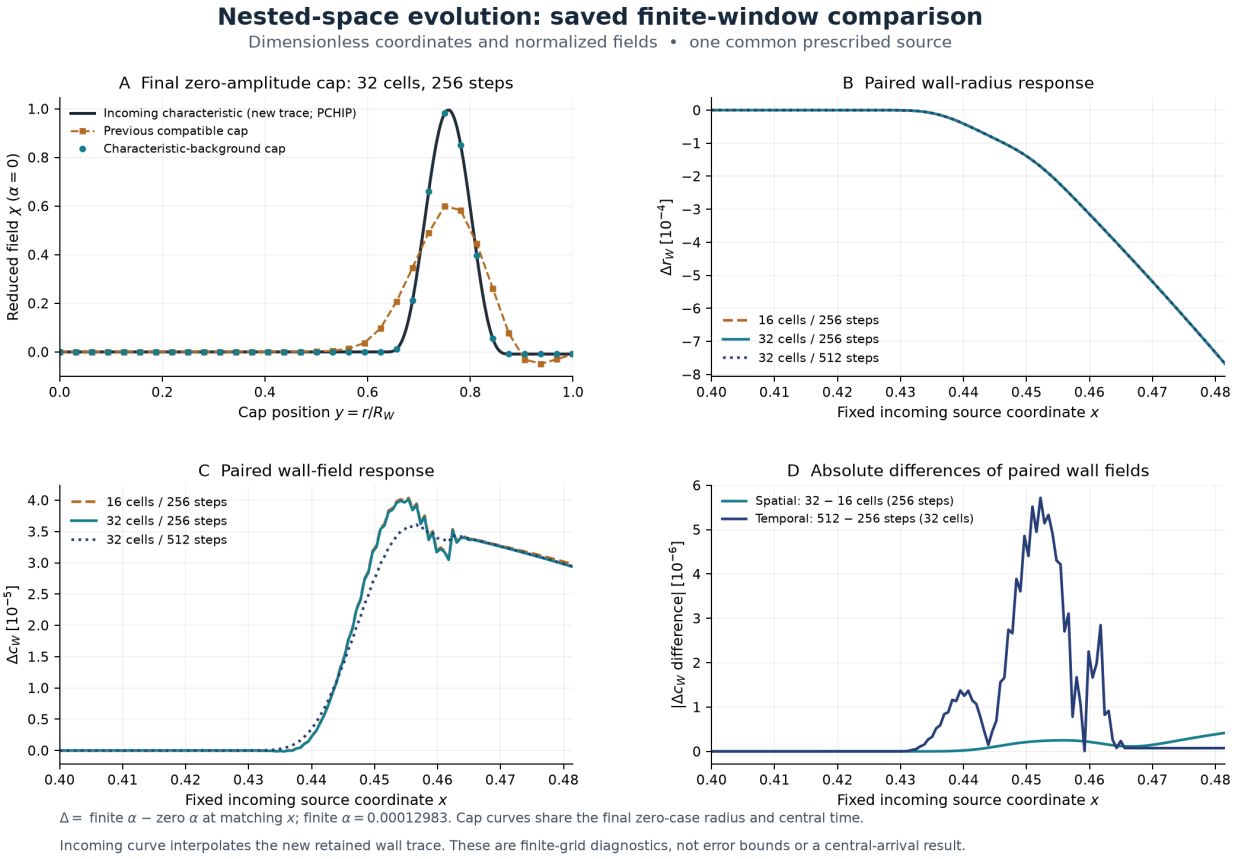


Figure 36. Saved finite-window comparisons. A: the previous and current zero-case cap primitives, with an independent incoming characteristic pullback from the new wall trace. B and C: finite-minus-zero wall radius and scalar histories. D: absolute spatial and temporal changes of the paired scalar response. The comparisons neither bound the continuum error nor extend to central arrival.

Define each response as the finite-minus-zero history at the same source label. In the table, spatial differences compare 16/32 cells at 256 steps; temporal differences compare 256/512 steps at 32 cells. Endpoint values use the 32-cell, 512-step pair. Here *L=log C*, *T* is the central coordinate clock, *s* is Einstein proper wall time and *c* is the normalized invariant wall scalar.

Quantity	Endpoint response	Maximum spatial difference	Maximum temporal difference
Radius *r*	-7.67297×10^{-4}	1.57484×10^{-7}	7.56646×10^{-9}
Log lapse *L*	-6.48536×10^{-3}	1.92145×10^{-6}	5.58556×10^{-8}
Central clock *T*	-1.27928×10^{-3}	8.00326×10^{-7}	1.10009×10^{-8}
Wall clock *s*	-1.19742×10^{-4}	4.54211×10^{-8}	6.59243×10^{-10}
Wall scalar *c*	2.94167×10^{-5}	4.18041×10^{-7}	5.71668×10^{-6}

Geometry and clock spatial differences are below 0.063% of their fine response peaks, and time differences below 0.001%. The scalar time difference is instead 15.83% of its fine peak 3.61049×10^{-5}. Its small endpoint time difference, 7.42×10^{-8}, therefore does not characterize the history. The finite

outgoing characteristic also changes by 12.41% of its fine peak under spatial refinement. Frozen cap radial order-4/8 comparisons change the wall mass coefficient by at most 3.00×10^{-8} and the lapse by 2.29×10^{-11}; this quadrature sensitivity does not explain the scalar time difference.

Independent exterior constraint maxima (E_U,E_x) decrease from $(7.88\times10^{-4},0.01312)$ to $(5.48\times10^{-4},0.008783)$, but remain nonzero. The source-array E_x defect remains 0.001303, and final 32-cell, 256-step wall kinematic defects in (r,L,c) are approximately $(-0.0003125,0.007739,-0.000119)$. Constructed scalar flux agreement near roundoff does not remove these independent derivative defects. Thus the integrated geometric responses are reproducible within the tested comparisons, while scalar time resolution, exterior constraints and wall slopes still require correction.

Creation and authentication use 228.9375 process-CPU seconds and 43,551,872 charged points within a shared 600-second, 100-million-point allowance. The manifest binds 66 inputs and producers. Authentication reconstructs saved diagnostics without source integration or time evolution. No continuum error enclosure, converged nonlinear spacetime, central-arrival waveform, formation mechanism or stability result follows from this window.

A.18. Profile-knot phase and a finer temporal comparison

Derivation and calculation: 128/256-step histories, 512-step comparison, source-bound manifest and figure record. The time-contribution derivation retains the background chain, physical wall flux and frame conversion separately.

The same spatial operators and physical inputs now use the three-stage SSPRK3 method at 128, 256 and 512 steps, always with 32 cells. The last comparison adds only the missing midpoint data: the original 513 rows are preserved exactly, and the same source snapshot supplies 512 new midpoints. Direct reference-gauge quadrature at orders 32 and 64 differs by at most 1.99×10^{-13} across its values and jets. The source geometry still uses its inherited Hermite approximation; its residuals are retained. Neither the physical preparation nor the 65-knot computational cap profile is refitted.

The latter profile is C^2, with possible jumps in its third derivative. An exact local model explains why the position of a knot inside a step matters. For a prescribed path with $B(x)=A(x-x_*)_+^3/6$ and $\theta=(x_*-x_0)/h\in[0,1]$, the SSPRK3-weighted background-chain defect is

$$\mathcal{R}_B = B(x_0+h) - B(x_0) - \tfrac{h}{6}\{B'(x_0) + 4B'(x_0+h/2) + B'(x_0+h)\}.$$

$$\mathcal{R}_B = \tfrac{Ah^3}{12}(1-2\theta)\min(\theta,1-\theta)^2.$$

It vanishes at an endpoint or midpoint and has largest magnitude $|A|h^3/324$ at phases $1/3$ and $2/3$. For a smooth increasing wall label $\zeta(x)$ crossing a profile knot, the leading coefficient is $A=[F_{bg}''']\zeta_x^3$. This local quadrature identity does not prescribe the coupled stage trajectory or identify its entire error. In particular, shifting the crossing phase on refinement can change the sign and size of a difference without a fixed observed convergence order.

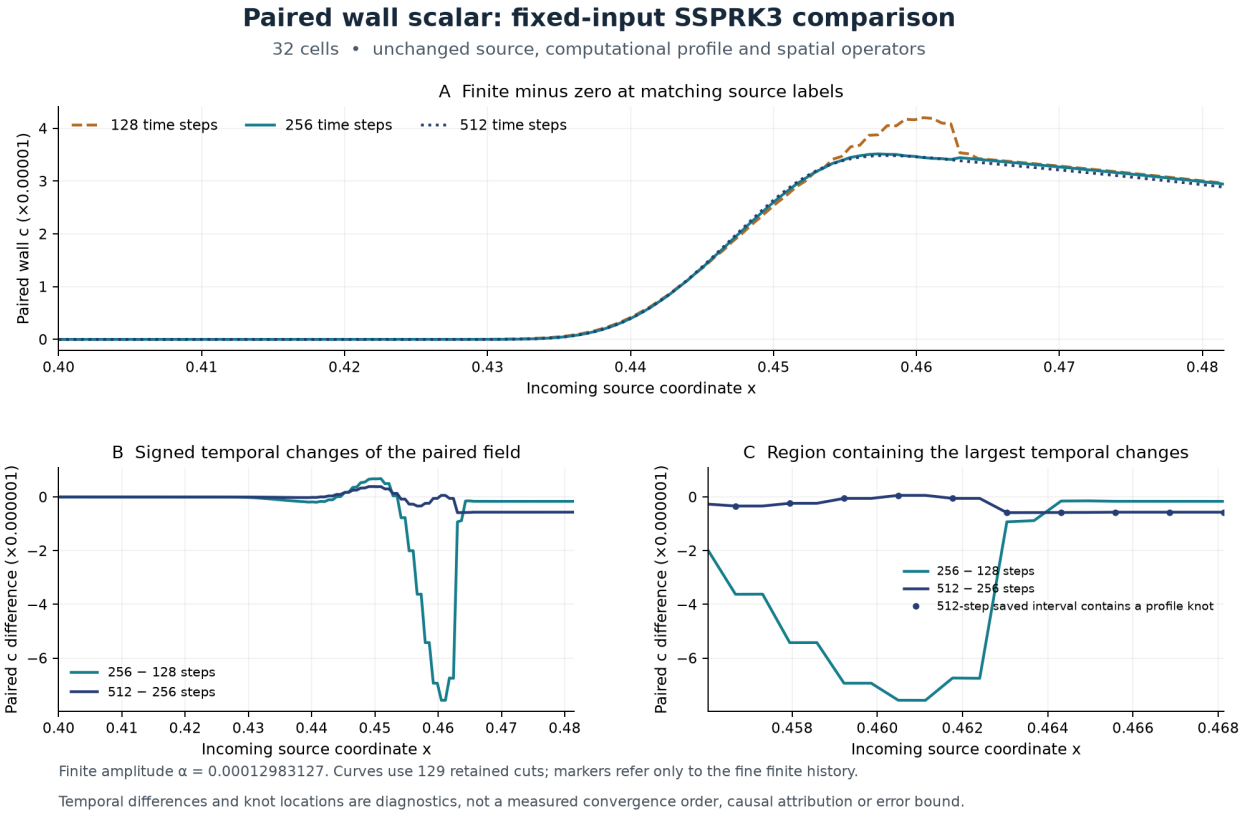


Figure 37. A: finite-minus-zero wall scalar at 128, 256 and 512 time steps. B: signed changes between successive histories. C: the region containing their largest changes; dots mark fine finite-history saved intervals containing a profile knot. These endpoint brackets do not reconstruct internal-stage paths or assign causation.

The paired scalar maximum time difference falls from 7.57345×10^{-6} for 128/256 steps to 5.84100×10^{-7} for 256/512 steps. The latter occurs at saved cut 99 and is 1.67616% of the fine peak 3.48476×10^{-5}. The difference ratio, 12.966, is a measured quotient, not a convergence order. The final scalar change is -5.66081×10^{-7}, so an endpoint alone does not characterize the history. All entries below use finite minus zero at common labels.

Quantity	512-step endpoint response	Maximum 256/512-step difference
Radius r	-7.67296891×10^{-4}	6.62412×10^{-11}
Log lapse L	-6.48536218×10^{-3}	6.54539×10^{-10}
Central coordinate clock T	-1.27927881×10^{-3}	1.38604×10^{-10}
Wall proper clock s	-1.19741847×10^{-4}	1.09942×10^{-11}
Wall scalar c	2.88661471×10^{-5}	5.84100×10^{-7}

The earlier SSPRK2 extrapolant $(4\Delta c_{512}-\Delta c_{256})/3$ agrees with the new 256-step history within 5.60381×10^{-8}, but with the new 512-step history only within 5.92862×10^{-7}. The closer earlier agreement therefore does not certify the true error. Independent finite constraint maxima remain approximately $(5.55\times10^{-4}, 0.008769)$ for (E_U, E_X). The unchanged computational profile's pre-support contribution also remains a separate representation issue: fine cap-integral leakage is about 1.66×10^{-7} even though the wall scalar leakage falls to 1.25×10^{-9}. Spatial defects, profile support, central arrival and a uniform finite-amplitude remainder remain open.

The two new histories and their validation use 174.90625 process-CPU seconds and 33,604,170 points, within 300 seconds and 50 million points. Their manifest binds 86 inputs and producers. A separate

authentication uses 1,418,330 points, reconstructs midpoint data and two final stages, and replays no source integration or trajectory.

A.19. A common exterior derivative and a mixed accuracy result

Derivation and calculation: fixed-input spatial comparison, same-grid operator comparison and source-bound manifest. The common-derivative construction specifies the discrete change and its limited consistency and stability controls.

The preceding 32/64-cell comparison at 512 SSPRK3 steps changes the paired wall scalar by at most 2.69431×10^{-7}, or 0.775% of the fine peak. This does not resolve the whole field: adjacent-summed cap integrals still differ by about 7% of the corresponding old 32-cell paired peak, and their physical radial cells move differently. The successor holds 64 cells, 512 steps, both amplitudes, source data, reference gauge and computational cap profile fixed. It replaces the first-order exterior departure derivative with a quadratic stencil. One derivative supplies both the radial first jets and transport on the moving coordinate grid. No boundary derivative is replaced by a value demanded by the wall law.

The independent constraints use saved field differences and analytic vacuum jets, rather than the evolution stencil. Write $E_a=r_{aa}-L_a\,r_a+r\Psi_a^2/2$ for $a=U,x$. The outgoing diagnostic retains the actual source curvature through $R_x=E_x-r_{src}E_{x,src}/r$. All entries below concern the finite-amplitude history on the same 8,001 interior diagnostic points.

Diagnostic	First-order derivative	Quadratic derivative
Maximum absolute E_U	3.53518×10^{-4}	3.64417×10^{-4}
Root-mean-square E_U	6.20959×10^{-5}	2.24521×10^{-5}
Maximum absolute R_x	4.93833×10^{-3}	3.21118×10^{-3}
Root-mean-square R_x	1.14164×10^{-3}	2.98901×10^{-4}

At the final wall event, the kinematic mismatches in (r,L,c) change from $(-1.54418\times10^{-4},3.75664\times10^{-3},-6.02063\times10^{-5})$ to $(4.70092\times10^{-6},-1.74363\times10^{-4},2.84633\times10^{-6})$. The maximum method change in the paired wall scalar is 1.81989×10^{-7}, or 0.524% of its new peak 3.47257×10^{-5}. Paired geometry and clock changes are each below 0.018% of their respective new peaks. These are sensitivities to the chosen derivative, not bounds on the continuum solution.

The improved average residual and wall slopes do not settle wall curvature. Independent backward-stencil estimates of the wall E_U peak increase from 4.87×10^{-4} and 4.49×10^{-4} to 6.67×10^{-4} and 7.73×10^{-4} at native width. The first estimate uses three nodes for first derivatives and four for second derivatives; the other uses five for both. Their maxima of the native-minus-double-width differences are 3.58×10^{-4} and 6.06×10^{-4}. The source Hermite curvature remains a one-sided trace at saved knots; its nonzero constraint is retained, not replaced by the source ODE target. The cap's characteristic-pullback discrepancy remains comparable to its interpolation sensitivity, and pre-support cap-integral leakage remains about 3.34×10^{-7}. The wall responses are comparatively stable under this method change, while local derivative accuracy remains unresolved. Neither a junction failure nor a converged coupled spacetime follows. Central arrival and a uniform finite-amplitude remainder remain open.

The two new histories and authentication use 256.25 process-CPU seconds and 55,398,966 charged points within 450 seconds and 80 million points. The manifest binds 104 files. Separate authentication uses 4,040,624 points and reconstructs saved fields, diagnostics and two final stages without source integration or trajectory replay.

A.20. A finite-amplitude onset solution

Derivation and calculation: exact reference gauge, quiet initial anchor, actual source bounds, and conditional tube receipt. The complete construction and continuation argument supplies the mathematical connection between those inputs and existence.

Finite-cell result. With the same action, signed material-wall laws, original compact C3 source and exact quiet initial data, the cap, wall and exterior admit a compatible regional solution on

$$x_L = \frac{475452113650795}{1125899906842624}$$
$$x_R = \frac{3809352257986645}{9007199254740992}$$

$$0 \le \alpha \le \frac{4789928473501569}{36893488147419103232} \simeq 1.2983 \times 10^{-4}.$$

Here x is the fixed affine source label. The cell is approximately *[0.4222863,0.4229231]*; it is not the complete earlier wall-radius window. Native fields are C3, native radius is C4, and the polar cap metric is C2. The centre remains a regular quiet de Sitter core. Local uniqueness holds in the stated regular class and fixed gauges.

The construction first integrates the outgoing mixed equations, then solves the existing wall ODE, transports incoming jets from their actual entry, and reconstructs each field from the same frozen forcing. This preserves field primitives and common wall values at every iteration. Exponential weights control the long exterior prefix; a cap strip based on its spacelike initial data has short characteristic lengths. Weighted derivative bounds and moving-entry estimates give a local contraction. The resulting wall laws and exact initial/source data propagate both incoming Einstein constraints.

Two ordered Volterra/transport derivative pairs give C2 first jets from the actual C3 source. Their finite bounds and continuity moduli remain uniform on the declared interval. The physical cap clock is constructed from the normalized radius-gradient flow. Its Jacobian stays nonzero; the transformed lapse is fixed by that flow and the quiet central clock. Restart slices keep their actual curved shape in the original null coordinates. Separate estimates for old radii and radii newly enclosed by the wall keep these slices inside the constructed domains.

On every initial subinterval, the same interval majorants improve all ten assumed departure bounds strictly; their largest ratio is below 0.516. The positive travel gap keeps the core quiet. A first exit through the field bounds, regularity, coordinates or quiet overlap is therefore excluded, and the positive source-label rate carries the solution to x_R.

For the bounds below, $f=\Psi/\sqrt{\alpha}$ denotes the normalized canonical scalar, with its continuous normalized limit at $\alpha=0$. The reduced canonical fields are $X=rf$ outside and $\chi=rf$ in the polar cap; these differ from the invariant reduced field $c=r\sinh(af)/a$ used in the native construction, where a is the nonnegative root of $6a^2=\alpha$. In the cap, $z=a\tanh r$, $\beta=Ae^{\delta}/(1-r^2)$ and $E^{\pm}=\chi_T/\beta\pm\chi_z$. The quantities J_r,J_L,J_X are integrated transverse-jet departures. All departures use the exact vacuum reference and the same fixed exterior wall and source labels. The resulting dimensionless bounds include:

Quantity	Strict upper bound
Integrated radius-jet departure J_r	5.662×10^{-9}
Integrated log-lapse-jet departure J_L	7.183×10^{-8}
Integrated canonical-field-jet departure J_X	1.474×10^{-10}
Absolute wall-to-central-clock departure ΔT	7.061×10^{-9}
Absolute incoming cap field E^+	0.002061
Absolute outgoing cap field E^-	3.639×10^{-19}
Absolute cap reduced field χ	3.129×10^{-6}
Absolute exterior radius-jet departure Δr_U	1.451×10^{-6}
Absolute exterior log-lapse-jet departure ΔL_U	4.224×10^{-5}
Absolute exterior canonical-field jet X_U	2.098×10^{-9}

The exact interval endpoints are retained in the receipt. That immutable record reports a conditional bound, not existence by itself; the separate construction and continuation argument supplies existence. Symbolic controls check the field primitives, wall tangents, differentiated equations and coordinate identities. No new physical trajectory is used in this proof. These bounds do not certify the saved numerical histories, a perturbative remainder, central arrival, formation or observational agreement.

A.21. Amplitude dependence on the onset cell

Derivation and calculation: source and comparison-chart regularity, coupled linear inverse, first quotient with one coordinate derivative, and second quotient and Taylor estimate. The source-bound record and conditional tube record retain the numerical input bounds; the linked derivation supplies the parameter theorem. Neither record evaluates its Taylor constant.

Amplitude result. On exactly the source cell and coupling interval of Section A.20, the normalized solution has a continuous first amplitude derivative into C1 comparison jets and a continuous second amplitude derivative into C0 comparison jets. At the interval endpoints, derivatives are one-sided. The wall radius and anchored central clock are included.

The comparison uses the original source label x. The exterior has the fixed interval $0 \le U \le W(x)$; the complete cap uses $r = r_c + \eta(R(x,\alpha) - r_c)$, $0 \le \eta \le 1$, with a fixed radius r_c in the quiet core. This keeps all radii enclosed by the wall, including those already present at the initial event. Each solution retains its own clock $T(x,\alpha)$ and the actual coordinate Jacobians. In particular, T labels the wall's polar slice by the anchored central proper clock; it is not the arrival time of a later signal at the centre.

Let $U(\alpha)$ collect the normalized comparison state and write $U_1 = \partial_\alpha U$ and $U_2 = \partial_\alpha^2 U$. Exact finite differences first give Lipschitz dependence into C0. Uniform spatial C2 bounds and continuity moduli then give continuity into C2 by second differences. The first parameter quotients converge into C1 in the order outgoing exterior derivative, cap gradients, incoming exterior derivative. The cap's radial constraints express the radial derivatives of its propagation coefficients without extra field-gradient unknowns. Actual inherited initial data and the wall corner identity are retained at each restart.

For a cap transport equation $(\partial_x - \sigma\partial_\eta)w = R$, its second variation has the same coupled linear operator as the first. Put $v_1 = (U_1, 1)$, including variation of the explicit parameter. The known second-order forcing is

$$2D\sigma[v_1]\,\partial_\eta w_1 + D^2\sigma[v_1, v_1]\,\partial_\eta w + D^2\mathcal{R}[v_1, v_1].$$

The factor two comes from varying the transport speed and the field gradient separately. Convergence of the first quotients in C1 controls this forcing in C0. The bounded coupled inverse and a finite sequence of compatible time slabs give convergence of the actual second quotients, not just a formal differentiated equation. The exterior, radial mass, lapse, wall and clock equations participate in the same limit. No fourth source derivative is used, and no C1-valued second derivative of all jets is asserted.

For a specified normalized state component, wall radius or anchored clock O, define the output-specific finite constant

$$M_{2,O} = \sup_{0 \le \alpha \le \alpha_{\max}} \|\partial_\alpha^2 O(\alpha)\|_{C^0}.$$

Twice integrating in amplitude gives

$$\|O(\alpha) - O(0) - \alpha O_1(0)\|_{C^0} \le \tfrac{1}{2} M_{2,O}\,\alpha^2.$$

This is an analytic quadratic remainder on the onset domain. The general state constant is finite; Section A.22 supplies explicit remainder bounds for selected geometric outputs by a positive comparison. It also identifies their first coefficients by matching the source, gauge and quiet initial data. Errors in the older numerical histories require their own bounds. The normalized fields have the stated amplitude derivatives. The physical scalar $\Psi=\sqrt{\alpha}f$ is not asserted to be twice differentiable in α at zero. The quiet core, original source and coupling interval are unchanged. Continuation through central arrival and an error bound for the observer waveform remain separate steps.

A.22. Uniform geometric remainders and signed endpoint response

Derivation and calculation: first geometric coefficients, geometric, canonical scalar and normalized cap first-difference bounds, quadratic source remainders, complete geometric remainder, and Jordan material-radius conversion. The comparison and frame derivation connects these compact certificates.

Keep exactly the source cell, action, quiet data and coupling interval of Section A.20. All quantities here are dimensionless. The outputs are the Einstein areal radius R_E, the material Jordan areal radius R_J, and the wall's centre-anchored polar-slice label T_W. Their first coefficients are the actual derivatives of the constructed solution at $\alpha=0$. They coincide with the continuum source, mass, lapse, wall and clock functionals after the source-label event shift is included. This identification does not assign an error bound to a saved numerical history.

Uniform comparison. For each retained geometric component G, the calculation supplies an explicit constant C_G such that

$$\|G(\alpha)-G(0)-\alpha G_1(0)\|_{C^0}\le\alpha^2 C_G.$$

The radius estimate covers the retained exterior, including the wall; the clock and material-radius estimates cover their wall histories on this cell. The source supplies four quadratic remainders. Uniform first differences of the canonical scalar and normalized cap mass and lapse bound the remaining nonlinear terms. Half second derivatives of the exact bulk and signed wall maps are enclosed over the full certified state box, with the radius-kernel cross term and scalar kinetic product retained.

After division by α^2, the three coupled geometric bounds obey $y\le My+f_2$, with the same positive matrix as the first-difference comparison. Its weighted norm is below 0.146, so $y\le(I-M)^{-1}f_2$. The remaining outputs follow from its dependent rows. This uses one rational three-by-three solve and interval algebra, with no new field evolution or grid scan.

Material frame and event identification. With the normalized canonical field $f=X/R_E$ of Section A.20, the exact conversion is

$$R_J=R_E\cosh\left(\sqrt{\alpha/6}\,f\right)$$
$$\mathcal{F}=\operatorname{sech}^2\left(\sqrt{\alpha/6}\,f\right)>0.$$

The invariant Jordan trace $c=R_E\,f\operatorname{sinhc}(\sqrt{\alpha/6}\,f)$ is a different variable at finite amplitude. For a fixed target α, join $(0,R_0,X_0)$ to $(\alpha,R_\alpha,X_\alpha)$ by an affine segment. Its directions are the actual difference quotients, bounded by the certified first-difference constants. At the expansion base the frame map has state partials one and zero, and explicit alpha derivative $X_0^2/(12R_0)$. Therefore the Einstein radius remainder enters with coefficient one; the frame half-Hessian adds at most 1.604501×10^{-23} to C_{R_E}. The conformal factor has certified lower bound 0.9999999999999999 on the whole comparison.

The first Jordan coefficient at fixed source label is

$$R_{J,1}|_x = \rho + 2a_{\mathrm{ref}} v (v - \beta_+) \eta_{\mathrm{emb}}$$
$$a_{\mathrm{ref}} = 4/5.$$

Here ρ is the Jordan radius coefficient at fixed wall proper time, v is the reference proper-time radial velocity, and η_{emb} is the exterior embedding coefficient defined by $V_W=V_0(1+\alpha\eta_{emb})+O(\alpha^2)$. Its interval includes both signs and is retained in the conversion; η_{emb} is distinct from the cap chart coordinate of Section A.21.

Endpoint result. The following are outward-rounded bounds. The last column gives the shift from the vacuum reference at x_R and $\alpha=\alpha_{\max}$, divided by 10^{-18}.

Output	Upper bound for C_G	Endpoint shift / 10^{-18}
Einstein wall radius R_E	6.067430×10^{-17}	[-2.138954,-1.008091]
Material wall radius R_J	6.067431×10^{-17}	[-1.235768,-0.09856682]
Anchored wall clock T_W	1.145597×10^{-15}	[-2.635641,-1.410894]

For a first-coefficient upper endpoint $U_G<0$, the criterion $\alpha_{\max}C_G<-U_G$ makes the sign strict at every positive admitted amplitude. The exact ratio is below 1.038×10^{-5} for the material radius and 1.369×10^{-5} for the clock. Hence

$$R_J(\alpha, x_R) - R_J(0, x_R) < 0$$
$$T_W(\alpha, x_R) - T_W(0, x_R) < 0 \quad (0 < \alpha \le \alpha_{\max}).$$

The clock retains the same Einstein-polar slicing and equals physical Jordan central proper time on the quiet core. Its value assigned to the wall event is neither wall proper time nor the arrival time of a received central signal. The reference wall is expanding: these inequalities are sourced differences at the same input label, not absolute contraction or an evaporation law. The preparation is still prescribed characteristic data; no physical emitter, SI scale or observational calibration is selected.

The geometric and material-frame certificates retain exact rational endpoints, producer commits and input hashes. Seventeen targeted controls, separate rational audits and independent Grok derivation/code reviews check the identities, finite boxes, matrix closure, frame conversion and strict inequalities. The audits authenticate the saved results without rerunning a field solution. Continuation to central arrival, numerical trajectory error, preparation selection and observational comparison retain their separate requirements.

A.23. The completed full-label numerical trial

All three original fixed-grid runs reach the common incoming-label event $x=0.4815042082576364$, $V=1$, at $\alpha=0.00012983127142551443$. The observable is the Jordan material-wall areal radius, measured by intrinsic area or equatorial proper circumference on that cross-section. The ruler $\ell=10$ km is a declared input; it is not an inferred astrophysical scale. This event differs from the fixed wall-proper-time event used by the historical perturbative coefficient.

Cap/exterior cells	Jordan radius (m)	Actual mass-gap maximum	Wall-flux maximum
2/32	5492.322036563843	0.00232832534	2.24045303
4/64	5492.322058752734	0.00237798075	2.19447899
8/128	5492.322035897825	0.00240631809	2.14212668

The fine value is 5492.322035897825 m. The full grid spread is 22.8549 micrometres; the fine-centred resolution interval is [5492.322013042916,5492.322058752734] m, with the same spread as its

half-width. Common input error and errors beyond this grid family are not included. This is not a confidence interval or a continuum error bound. The successive pair differences, *22.19* and *22.85* micrometres, do not contract. The grids are nonuniform and their time steps change too, so no convergence order is inferred.

The wall-flux maximum decreases 4.39%, while the actual mass maximum grows 3.35%. Each residual uses its recorded normalization. The matched-mass maximum rises slightly then falls on the fine grid. These mixed results keep the finite-amplitude extension of NSC-P001 an **unconverged numerical estimate**. The earlier onset theorem retains its shorter domain; the full endpoint solution is not certified. The coarse evolution completed, but its auxiliary accumulated-transport summary failed and remains unavailable.

The completed comparison and saved-cut decomposition bind the original producer, three inputs and all nine common cuts. The fine cut at $x=0.45094013016102275$ has physical mass residual -0.0007620291372472, split into strong/weak trace contribution -0.0005505985471954 and balance contribution -0.0002114305900518. Their sum agrees to 1.62×10^{-17}. The latter is the explicit background time-projection remainder up to its stored identity defect. At the endpoint the fine flux is -0.5466500011, compared with -0.00965372325 on the coarse grid: a smaller all-stage maximum does not mean every cut improves. The saved cuts cover only 19.3%, 25.1% and 31.7% of the actual-mass envelopes. No maximizing state was saved, so this split does not locate or uniquely explain that unseen peak. The remaining connection is the represented field's strong/weak and time-projection coupling to the wall. The radius, local measurement protocol and qualified range are recorded in NSC-P001.

The active-region diagnosis uses the original onset label q_a and history time $T_h=T+T_{offset}$. Before central arrival the retained field is zero for $z<q_a-T_h$. At every active saved cut that front lies in the final cell on all three levels. The full outer widths are $.80,.73,.66$; the clipped endpoint widths are $.56458,.52678,.48898$. The active layer is never bisected. Added inner cells lie below the retained background's front; this is not a statement about the unarchived evolved departures. The spaces change inner knots and do not form a halved-h cap refinement. Their unresolved wall/time projections need not improve monotonically. The sampled absolute integral of the actual mass residual also grows 5.11%, excluding denser maximum sampling as the sole explanation. Seven front/partition controls pass; the read-only calculation uses .015625 CPU seconds with no field evaluation or evolution. The missing maximum states still prevent unique attribution of those peaks. Check the active-region wall trace and affine transport before another grid.

The subsequent direct PDE check uses a separately archived evolved stage of the same numerical method. The complete weak bulk loads agree with the direct field equation to 2.4×10^{-9}, while its pointwise density-force defect is 5.09915. The boundary-lift decomposition supplies 5.15013 of that defect, with the remaining projection contribution $-.0509851$. At the same frozen geometry the three endpoint evaluation gains grow $72.6963\to79.7052\to87.2537$. Only the coarse evolved load was archived; the other gains are operator controls, not new evolutions or a reconstruction of the missing maxima.

This isolates a finite boundary-projection mechanism at an actual evolved state. The production split coupling remains unresolved. A shared-action manufactured control now recovers a represented scalar and its proper-time velocity over the quiet wall strip, including original GHY and canonical conversion. Its normalized field/velocity errors are 6.71×10^{-12} and 7.11×10^{-11}; energy plus metric/input work closes to 1.75×10^{-15}. Its prescribed quiet geometry and manufactured forcing do not repair or certify the finite-amplitude P001 evolution.

A.24. A separate finite model with geometric feedback

Derivation and calculation: focused companion snapshot, finite episode, local response, and memory controls.

The finite companion supplies another part of the programme: fields transfer between selected regions while the same declared action evolves the geometry. It uses a periodic radial chart, positive chart variables and antiperiodic spinors. In the present signature convention its metric is

$$ds^2 = r^2[-L^2 dt^2 + Q^2(dx + \beta\, dt)^2 + d\Omega^2].$$

The physical lapse is rL and radial scale is rQ. Conformal gauge has $L=Q$, $\beta=0$; the reduced field Hamiltonian is $H=\sigma_2 P+\kappa\ Q\sigma_1$. Here $\kappa=1$ labels the retained angular block and is independent of the wall-tension parameter used in the atlas. Six prepared Gaussian fermion columns determine

$$C = \Phi\, \mathrm{diag}(0.75, 0.75, 0.5, 0.5, 0.25, 0.25)\Phi^\dagger$$
$$\Phi^\dagger \Phi = I_6.$$

This is a finite admissible state with $0 \le C \le I$ and $Tr\ C=3$. Unoccupied retained bands are part of the preparation, not a derived physical vacuum. The action coefficients, magnetic sector, clock and cutoff are recorded inputs.

Four matched discretizations, using 256/512 field modes and time steps 0.001/0.0005, reach coordinate $T=0.3$ without resetting the state. The reported transfer and curvature comparisons have sub-percent refinement indicators in their stated observables. The finite evolution supplies a same-action example, not a continuum error theorem, an eternal engine or the creation of new universes. Collars of this periodic geometry are regions of one evolving model.

Its local reduction keeps the original first two mode columns and eliminates their Hilbert-space complement. That complement includes modes with support inside and outside the spatial observation window. Consequently it is not the parent end of Section 10, and its Hamiltonian cannot be transferred to the wall model without a field-theoretic identification.

A.25. Memory is a consequence of elimination, not an added reservoir

For a time-dependent Hermitian block Hamiltonian with retained block $A(t)$, coupling $B(t)$ and complementary block $D(t)$, let $U_E(t,s)$ propagate D. Eliminating the complementary amplitude gives

$$y(t) = U_E(t, t_*)y_* - i\int_{t_*}^{t} U_E(t, s)B^\dagger(s)x(s)\, ds$$

$$\dot{x}(t) = -iA(t)x(t) - iB(t)U_E(t, t_*)y_* - \int_{t_*}^{t} K(t, s)x(s)\, ds$$

$$K(t, s) = B(t)U_E(t, s)B^\dagger(s).$$

The memory kernel, actual initial complementary drive and initial cross correlations all matter. If R_A,R_E denote the corresponding solution maps, the retained covariance is

$$C_A = R_A C_{AA} R_A^\dagger + R_E C_{EE} R_E^\dagger + R_A C_{AE} R_E^\dagger + R_E C_{EA} R_A^\dagger.$$

The companion uses the actual evolved state at $t_*=0.2$, then compares the reduced and full histories through $t=0.3$. The selected occupations change from approximately $(0.10557448,0.10557464)$ to $(0.03214496,0.03226900)$. The maximum reduced/full occupation discrepancy is 6.35×10^{-6}, compared with a physical-model change of about 0.07343.

Conditional omission control	Maximum occupation effect	Its separate numerical reference discrepancy
Omit the memory term	0.0168185	2.20×10^{-6}
Omit the actual initial complementary drive	0.0485412	1.66×10^{-5}
Omit initial cross correlations	0.0920614	1.23×10^{-5}

These controls hold the generated geometry fixed. They are not separately coupled physical evolutions; in particular, deleting the memory term is a diagnostic alteration, not a new Hermitian theory. They demonstrate why an interior description must retain the appropriate history and state information. The next bridge to the atlas would derive such a regional Hamiltonian and state from a particular matched spacetime and field, rather than labeling these finite modes as universes.

A.26. The retained local incoming gate

Derivation and calculation: majorant v7 record and gate budget v5 record.

The local incoming source-insertion majorant v7 is a separate controlled application. For the fixed switched state law $C_{\Sigma}[g]=U_g C_{up} U_g^{\dagger}$ on $I=S(1)+[.12,.18]$, family $(14,1)$ covers **1168/1168 signed rows**, with original weights counted once and no remaining row keys. Its continuous insertion bounds are

$$\varepsilon_N \le 364667405445736602902445249676942015176418810542611887660 9\,2^{-229}.$$

$$\varepsilon_\beta \le 579334325084668618351501392806714145470733192952896733996 9\,2^{-230}.$$

These are approximately $4.226958072607968\times10^{-12}$ and $3.357610065481873\times10^{-12}$ for this family's source-error insertion. The physical local gate remains **OPEN**: `physical_upstream_budget_component` is null. Source quadrature, numerical field error, value-assembly arithmetic and every other source family are excluded from these bounds. Complete row coverage in this family does not supply those missing errors.

The v5 gate budget names nine distinct components, with total error allocation 2×10^{-11} and residual reserve 10^{-11} against the physical tolerance 3×10^{-11}. Its applicable component bounds remain null; 60 positive and 120 signed families are declared requirements rather than authenticated coverage. The gate is neither an existence nor a non-existence certificate and does not determine Λ_{eff}. Its paused campaign does not invalidate the finite inheritance theorem or the scalar-chain results. Retired supporting records are cited at Zenodo.

Appendix B. Evidence, attribution and reproducible mathematics

This living manuscript draws from source-bound atlas records without changing their historical bytes. Record residuals and refinement comparisons are reported at their declared scope; floating-point quadrature agreement is not a rigorous error bound. The overview producer reads existing records and renders their relationships without launching a new physics calculation.

Result family	Owning derivation and preserved evidence
Finite inheritance and nested reduction	Proof; snapshot
Two-hole parent and radiation attachment screen	Atlas v2; manifest
Signed moving wall and gluing literature	Atlas v3; manifest
Sourced slice and vacuum observer domain	Atlas v4; manifest
Unequal sourced bodies and finite grandchild composition	End assembly; unequal bodies and two-level geometry/observer
Fixed-state physical resolution	Canonical split and vacuum-volume comparison; finite record

Result family	Owning derivation and preserved evidence
Full-label three-grid trial	Observable and status; comparison and saved-cut split
Independent mass/lapse diagnosis on the older saved stage	Full equations and scope; independent saved-stage check, distinct from the unsaved maximum of the later full-label trial
Isolated causal completion and clocks	Atlas v5; manifest
Classical incoming state and scalar transfer	Atlas v6; manifest
Finite exterior preparation and research dependencies	Preparation; programme; v7 manifest
Evaluated exterior preparation and finite tail	Response; v8 manifest
Fixed-model source inverse and preparation exclusions	Proof; record
Conformal probe surface stress and tangential flux	Derivation; record
Characteristic gravitational initialization	Constraint and frame check; record
Geometric mass on the prepared null surface	Mixed geometry and stress checks; record
Conformal scalar normal-force compatibility	Projection and force sign; record
Exterior wall-mass functional	Both fluxes and full frame conversion; record
Cap mass and wall displacement functional	Vacuum subtraction and junction closure; record
Smooth exterior metric response	Characteristic evolution and radial checks; metric, field and derivative comparisons
Moving wall and exterior clock	Mass paths, displacement and clock; field, radial and clock-time comparisons
Smooth cap and central null clock	Regular centre, metric and arrival map; field, radial and further ray comparisons
Exterior cubic scalar correction	Conservative evolution; clock lift, forcing cache and propagated response
Central cubic scalar correction	Reflected transport and clock pullback; radial comparison and incoming-ray comparison
Nonlinear coupled pre-arrival window	Exact model and characteristic background; six-case result and manifest; scalar time sensitivity and pointwise constraints remain unresolved
Fixed-input SSPRK3 temporal refinement	Profile-knot and chain identities; 128/256 histories, 512 comparison and manifest; finite temporal differences, without an error enclosure
Same-grid exterior derivative comparison	Common derivative and independent diagnostics; 64-cell baseline, operator comparison and manifest; smaller average residuals and wall kinematic mismatch, with unresolved constraint maxima and wall curvature
Finite-amplitude onset solution and parameter dependence	Construction and continuation, parameter theorem; original source bound and strict tube
Uniform geometric remainder and material-radius sign	Full comparison, frame and clock interpretation; geometric certificate and material-radius certificate
Finite spherical transfer, feedback and memory	Companion; frozen build/evidence
Foundation boundary, spectral and source constructions	Archived v0.26.0; research inventory

The archived foundation retains, among other strands, its full-spinor boundary maps, scalar-link Dirac benchmark, source and work ledgers, curved spectral constructions and explicitly limited black-universe benchmarks. Their theory, state and domain remain with their owning notes. An RN pressure diagnostic is not promoted to mechanical balance, a reached inner horizon is not promoted to central regularity, and stopped numerical runs without completed summaries supply no new physical conclusion. These distinctions preserve useful work without making historical routes mandatory.

The arXiv library records paper versions, reading depth, reusable equations, assumptions and unresolved connections. It includes the sources for the supplementary multiple-daughter, multi-mouth, CPT, torsion and regular-interior branches. A bounded search cannot certify that no one has already closed an apparent gap. The paper guide locates the unchanged atlas working drafts and the focused companion. Calculations, programming and drafting for these finite records and for this synthesis used OpenAI Codex/ChatGPT and xAI Grok. The author's responsibility includes the interpretations and synthesis here; borrowed results retain their source attribution and are not presented as new laws.

The following preserved dependency diagram records the original finite field-development programme. Its evidence remains useful within that domain; Part VIII states the current physical closures of the full atlas.

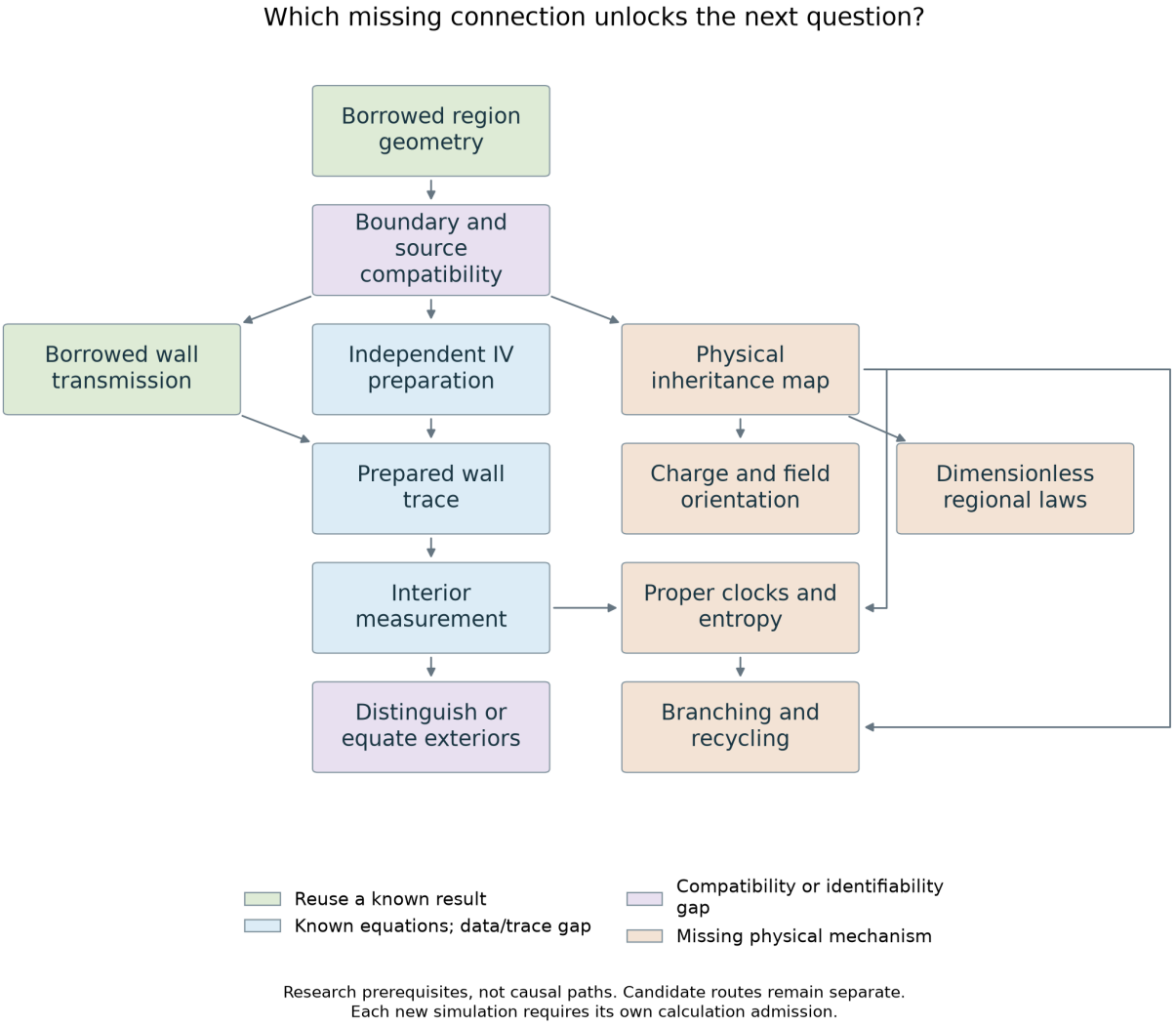


Figure 38. Research dependencies, distinct from spacetime causal arrows. The v8 programme record names prerequisites, decisions, budgets and stop conditions for eight themes. Each calculation retains its own domain; arrows do not declare questions solved.